

Chapter 1. Introduction to Project Management

Learning Objectives

After reading this chapter, you will be able to:

- Articulate the growing need for better project management, especially for information technology (IT) projects
- Explain what a project is, provide examples of IT projects, list various attributes of projects, and describe constraints of project management
- Define project management and discuss key elements of the project management framework, including project stakeholders, the project management knowledge areas, common tools and techniques, and project success
- Discuss the relationship between project, program, and portfolio management and the contributions each makes to enterprise success
- Summarize the role of project managers by describing what they do, what skills they need, the talent triangle, and career opportunities for IT project managers
- Recall key aspects of the project management profession, including important components of its history, the role of professional organizations like the Project Management Institute (PMI), the importance of certification and ethics, and the advancement of project management software

Opening Case

Anne Roberts, the director of the Project Management Office for Information Technology at a large retail chain, stood in front of 500 people in the large corporate auditorium to explain the company's new strategies during a monthly all-hands meeting. She was also streaming live video to thousands of other employees at other locations, suppliers, and stockholders throughout the world. The company had come a long way in implementing new information systems to improve inventory control, sell products online, streamline the sales and distribution processes, and improve customer service. However, a recent security breach had alarmed investors and the stock price plummeted. People were anxious to hear about the company's new strategies.

Anne began to address the audience, "Good morning. As you know, competition is fierce in our industry. We have made a lot of progress the last few years to become a more agile organization, especially in valuing people over process and responding to change over

following a plan. We all have to work together to overcome recent problems.

Our two most important goals include providing the best computer security possible and improving online collaboration tools for our employees, suppliers, and customers. Our challenge is to work even smarter to deliver solutions that provide the most benefit for the company by leveraging the power of information technology. If we succeed, we'll continue to be a world-class corporation."

"And if we fail?" someone asked from the audience.

"Let's just say that failure is not an option," Anne replied.

Introduction

Many people and organizations today have a new—or renewed—interest in project management. Until the 1980s, project management primarily focused on providing schedule and resource data to top management in the military, computer, and construction industries. Today's project management involves much more, and people in every industry and every country manage projects. Project management is a distinct profession with degree programs, certifications, and excellent career opportunities.

New technologies have become a significant factor in many businesses. Computer hardware, software, networks, and the use of interdisciplinary and global work teams have radically changed the work environment. The following statistics demonstrate the significance of project management in today's society, especially for projects involving information technology (IT):

- Worldwide IT spending was \$3.5 trillion in 2017, a 2.4 percent increase from 2016 spending. Communications services accounted for 40 percent of the spending.*
- The Project Management Institute reported that the number of project-related jobs reached almost 66 million in 2017, and demand continues to increase. "By 2027, employers will need 87.7 million individuals working in project management-oriented roles."*
- The unemployment rate for IT professionals is generally half the rate of the overall labor market in the United States. The U.S. Bureau of Labor Statistics estimates the rate to be only 2 percent, and project management is one of the ten hottest tech skills.*
- In 2017, the average annual salary (without bonuses) for someone in the project

management profession was \$112,00 per year in the United States and \$130,866 in Switzerland, the highest-paid country. Salaries of survey respondents across 37 countries were 23 percent higher for those with the Project Management Professional (PMP®) credential than those without it.*

- The top skills employers look for in new college graduates are all related to project management: team work, decision making, problem-solving, and verbal communications.*
- Organizations waste \$97 million for every \$1 billion spent on projects, according to PMI's Pulse of the Profession® report. Excelling at project management definitely affects the bottom line.*

The complexity and importance of IT projects, which involve using hardware, software, and networks to create a product, service, or result, have evolved dramatically. Today's companies, governments, and nonprofit organizations are recognizing that to be successful, they need to use modern project management techniques, especially for IT projects. Individuals are realizing that to remain competitive in the workplace, they must develop skills to become good project team members and project managers. They also realize that many of the concepts of project management will help them in their everyday lives as they work with people and technology on a day-to-day basis.

What Went Wrong?

In 1995, the Standish Group published an often-quoted study titled "The CHAOS Report." This consulting firm surveyed 365 IT executive managers in the United States who managed more than 8,380 IT application projects. As the title of the study suggests, the projects were in a state of chaos. U.S. companies spent more than \$250 billion each year in the early 1990s on approximately 175,000 IT application development projects. Examples of these projects included creating a new database for a state department of motor vehicles, developing a new system for car rental and hotel reservations, and implementing a client-server architecture for the banking industry. The study reported that the overall success rate of IT projects was *only* 16.2 percent. The surveyors defined success as meeting project goals on time and on budget. The study also found that more than 31 percent of IT projects were canceled before completion, costing U.S. companies and government agencies more than \$81 billion. The study authors were adamant about the need for better project management in the IT industry. They explained, "Software development projects are in chaos, and we can no longer imitate the three monkeys—hear no failures, see no failures, speak no failures."* Although this study was done 20 years ago, it was significant in making senior executives pay attention to the importance

of IT project management.

In another large study, PricewaterhouseCoopers surveyed 200 companies from 30 different countries about their project management maturity and found that *over half of all projects fail*. The study also found that only 2.5 percent of corporations consistently meet their targets for scope, time, and cost goals for all types of projects.*

Although several researchers question the methodology of such studies, the results have prompted managers throughout the world to examine ways to improve their practices in managing projects. Many organizations assert that using project management techniques provides advantages, such as the following:

- Better control of financial, physical, and human resources
- Improved customer relations
- Shorter development times
- Lower costs and improved productivity
- Higher quality and increased reliability
- Higher profit margins
- Better internal coordination
- Positive impact on meeting strategic goals
- Higher worker morale

This chapter introduces projects and project management, explains how projects fit into programs and portfolio management, discusses the role of the project manager, and provides important background information on this growing profession. Although project management applies to many different industries and types of projects, this text focuses on applying project management to IT projects.

What Is a Project?

To discuss project management, it is important to understand the concept of a project. A **project** is “a temporary endeavor undertaken to create a unique product, service, or result.”* Operations, on the other hand, is work done in organizations to sustain the business. It

focuses on the ongoing production of goods and services. Projects are different from operations in that they end when their objectives have been reached or the project has been terminated. It is important to note that people focusing on operations and projects must work together for a smooth transition. For example, in software development, [DevOps](#) is a fairly new term used to describe a culture of collaboration between software development and operations teams to build, test, and release reliable software more quickly.

Examples of IT Projects

Projects can be large or small and involve one person or thousands of people. They can be done in one day or take years to complete. As described earlier, IT projects involve using hardware, software, and networks to create a product, service, or result. Examples of IT projects include the following:

- A large network of healthcare providers updates its information systems and procedures to reduce hospital acquired diseases.
- A team of students creates a smartphone application and sells it online.
- A company develops a driverless car.
- A college upgrades its technology infrastructure to provide wireless Internet access across the whole campus as well as online access to all academic and student service information.
- A company implements a new system to increase sales force productivity and customer relationship management that will work on various laptops, smartphones, and tablets.
- A television network implements a system to allow viewers to vote for contestants and provide other feedback on programs via social media sites.
- A government group develops a system to track child immunizations.
- A large group of volunteers from organizations throughout the world develops standards for environmentally friendly or green IT.
- A global bank acquires other financial institutions and needs to consolidate systems and procedures.
- Government regulations require monitoring of pollutants in air and water.
- A multinational firm decides to consolidate its information systems into an integrated

enterprise resource management approach.

Gartner, Inc., a prestigious consulting firm, identified the top 10 strategic technologies for 2018. A few of these technologies include the following:

- **Artificial Intelligence (AI) Foundation:** Creating systems that learn, adapt, and potentially act autonomously can enhance decision making and improve the customer experience.
- **Intelligent Things:** AI is driving advances for new intelligent things, including autonomous vehicles, robots, and drones as well as Internet of Things like thermostats, lights, and home appliances.
- **Cloud to the Edge:** Edge computing pushes data handling to the edge of the network, closer to the source of the data. Instead of sending data to the cloud server or central data center for processing, the device connects through a local gateway device, allowing faster analytics and reduced network pressure.
- **Immersive Experience:** Virtual, augmented, and mixed reality are changing the way that people perceive and interact with the digital world. “The virtual reality (VR) and augmented reality (AR) market is currently adolescent and fragmented. Interest is high, resulting in many novelty VR applications that deliver little real business value outside of advanced entertainment, such as video games and 360-degree spherical videos. To drive real tangible business benefit, enterprises must examine specific real-life scenarios where VR and AR can be applied to make employees more productive and enhance the design, training and visualization processes.”*

As you can see, a wide variety of projects use information technologies, and organizations rely on them for success.

Media Snapshot

One of Gartner’s top 10 strategic technologies for 2012 included application stores and marketplaces for smartphones and tablets. Gartner predicted that by 2014 there would be more than 70 billion mobile application downloads every year, but the actual number was almost double!* Facebook is by far the most downloaded app, and the most popular category of all apps continues to be games.

As of March 2017, Android users could download 2.8 million different apps, and Apple users could download 2.2 million. “In 2016, the global mobile internet user penetration has exceeded half the world’s population, while the average daily time spent accessing online content from a mobile device, such as a smartphone, a tablet computer or

wearable, has reached 185 minutes daily among Millennials, 110 minutes for Generation X and 43 daily minutes for Boomers.”*

Project Attributes

Projects come in all shapes and sizes. The following attributes help define a project further:

- *A project has a unique purpose.* Every project should have a well-defined objective. For example, Anne Roberts, the director of the Project Management Office in the chapter’s opening case, might sponsor an IT collaboration project to develop a list and initial analysis of potential IT projects that might improve operations for the company. The unique purpose of this project would be to create a collaborative report with ideas from people throughout the company. The results would provide the basis for further discussions and selecting projects to implement. As you can see from this example, projects result in a unique product, service, or result.
- *A project is temporary.* A project has a definite beginning and end. In the IT collaboration project, Anne might form a team of people to work immediately on the project, and then expect a report and an executive presentation of the results in one month.
- *A project drives change and enables value creation.* A project is initiated to bring about a change in order to meet a need or desire. Its purpose is to achieve a specific objective which changes the context (a living situation, in this house project example) from a current state to a more desired or valued future state.
- *A project is developed using progressive elaboration.* Projects are often defined broadly when they begin, and as time passes, the specific details of the project become clearer. Therefore, projects should be developed in increments. A project team should develop initial plans and then update them with more detail based on new information.
- *A project requires resources, often from various areas.* Resources include people, hardware, software, and other assets. Many projects cross departmental or other boundaries to achieve their unique purposes. For the IT collaboration project, people from IT, marketing, sales, distribution, and other areas of the company would need to work together to develop ideas.
- *A project should have a primary customer or sponsor.* Most projects have many interested parties or stakeholders, but for a project to succeed someone must take the primary role of sponsorship. The **project sponsor** usually provides the direction and funding for the project. Executive support is crucial to project success, as described in later chapters.

Anne Roberts would be the sponsor for the IT collaboration project.

- *A project involves uncertainty.* Because every project is unique, it is sometimes difficult to define its objectives clearly, estimate how long it will take to complete, or determine how much it will cost. External factors also cause uncertainty, such as a supplier going out of business or a project team member needing unplanned time off. This uncertainty is one of the main reasons project management is so challenging, especially on projects involving new technologies.

An effective [project manager](#) is crucial to a project's success. Project managers work with the project sponsors, team, and the other people involved to achieve project goals.

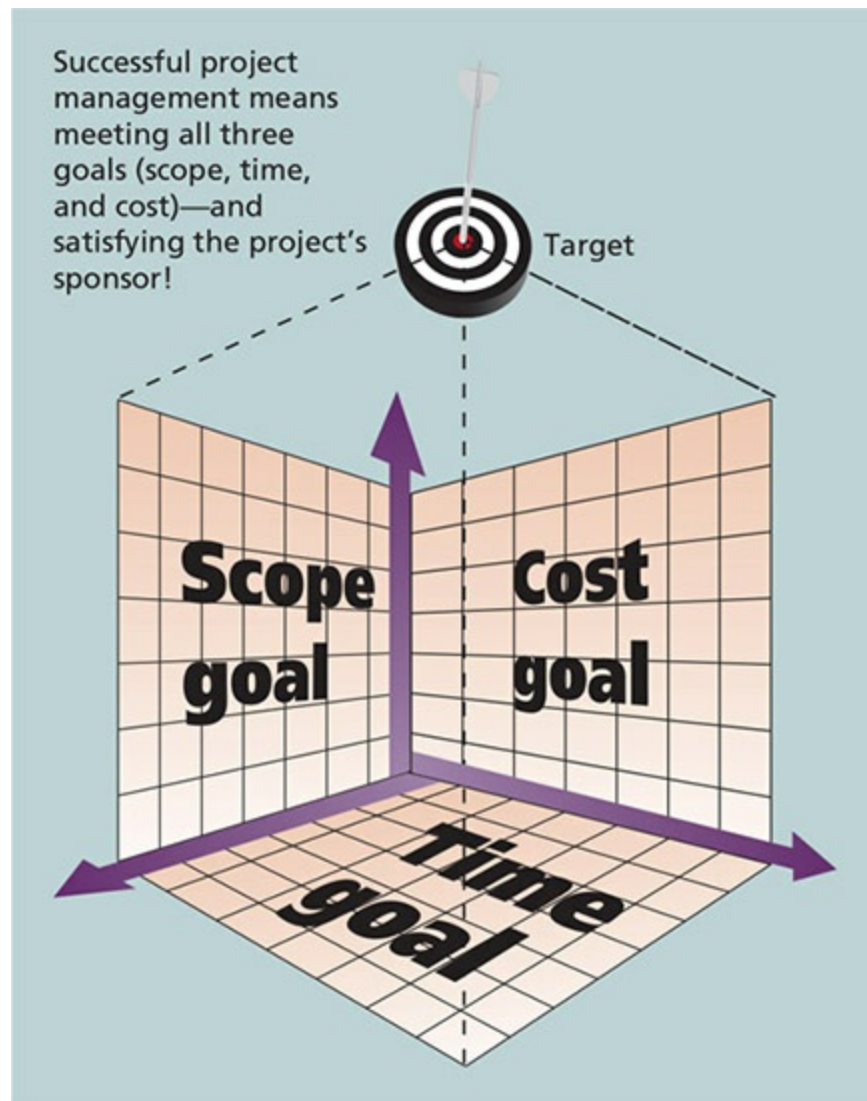
Project Constraints

Every project is constrained in different ways, often by its scope, time, and cost goals. These limitations are sometimes referred to in project management as the [triple constraint](#). To create a successful project, a project manager must consider scope, time, and cost and balance these three often-competing goals:

- *Scope:* What work will be done as part of the project? What unique product, service, or result does the customer or sponsor expect from the project? How will the scope be verified?
- *Time:* How long should it take to complete the project? What is the project's schedule? How will the team track actual schedule performance? Who can approve changes to the schedule?
- *Cost:* What should it cost to complete the project? What is the project's budget? How will costs be tracked? Who can authorize changes to the budget?

Figure 1-1 illustrates the three dimensions of the triple constraint. Each area—scope, time, and cost—has a target at the beginning of the project. For example, the IT collaboration project might have an initial scope of producing a 40- to 50-page report and a one-hour presentation on about 30 potential IT projects. The project manager might further define project scope to include providing a description of each potential project, an investigation of what other companies have implemented for similar projects, a rough time and cost estimate, and assessments of the risk and potential payoff as high, medium, or low. The initial time estimate for this project might be one month, and the cost estimate might be \$45,000–\$50,000. These expectations provide targets for the scope, time, and cost dimensions of the project.

Figure 1-1. Project constraints



Note that the scope and cost goals in this example include ranges—the report can be 40 to 50 pages long and the project can cost between \$45,000 and \$50,000. Because projects involve uncertainty and limited resources, projects rarely finish according to their original scope, time, and cost goals. Instead of discrete target goals, it is often more realistic to set a range for goals, such as spending between \$45,000 and \$50,000 and having a 40- to 50-page report. These goals might require hitting the target, but not the bull's eye.

Managing the triple constraint involves making trade-offs between scope, time, and cost goals for a project. For example, you might need to increase the budget for a project to meet scope and time goals. Alternatively, you might have to reduce the scope of a project to meet time and cost goals. Experienced project managers know that you must decide which aspect of the triple constraint is most important. If time is most important, you must often change the initial scope and cost goals to meet the schedule. If scope goals are most important, you may need to adjust time and cost goals.

To generate project ideas for the IT collaboration project, suppose that the project manager sent an e-mail survey to all employees, as planned. The initial time and cost estimate may have been one week and \$5,000 to collect ideas using this e-mail survey. Now, suppose that the e-mail survey generated only a few good project ideas, but the scope goal was to collect at least 30 good ideas. Should the project team use a different method like focus groups or interviews to collect ideas? Even though it was not in the initial scope, time, or cost estimates, it would really help the project. Because good ideas are crucial to project success, it would make sense to inform the project sponsor that adjustments are needed.

Although the triple constraint describes how the basic elements of a project interrelate, other elements can also play significant roles. Quality is often a key factor in projects, as is customer or sponsor satisfaction. Some people, in fact, refer to the *quadruple constraint* of project management, which includes quality as well as scope, time, and cost. A project team may meet scope, time, and cost goals but might fail to meet quality standards and satisfy the sponsor. For example, Anne Roberts may receive a 50-page report describing 30 potential IT projects and hear a presentation that summarizes the report. The project team may have completed the work on time and within the cost constraint, but the quality may have been unacceptable.

Other factors might also be crucial to a particular project. On some projects, resources are the main concern. For example, the entertainment industry often needs particular actors for movies or television shows. Project goals must be adjusted based on when particular people are available. Risk can also affect major project decisions. A company might wait to start a project until the risks are at an acceptable level. The project manager should be communicating with the sponsor throughout the project to make sure it is meeting expectations. **Chapter 10**, Project Communications Management, and Chapter 13, Project Stakeholder Management, address communicating with stakeholders and understanding their expectations in greater detail.

How can you avoid the problems that occur when you meet scope, time, and cost goals, but lose sight of customer satisfaction? The answer is *good project management, which includes more than managing project constraints*.

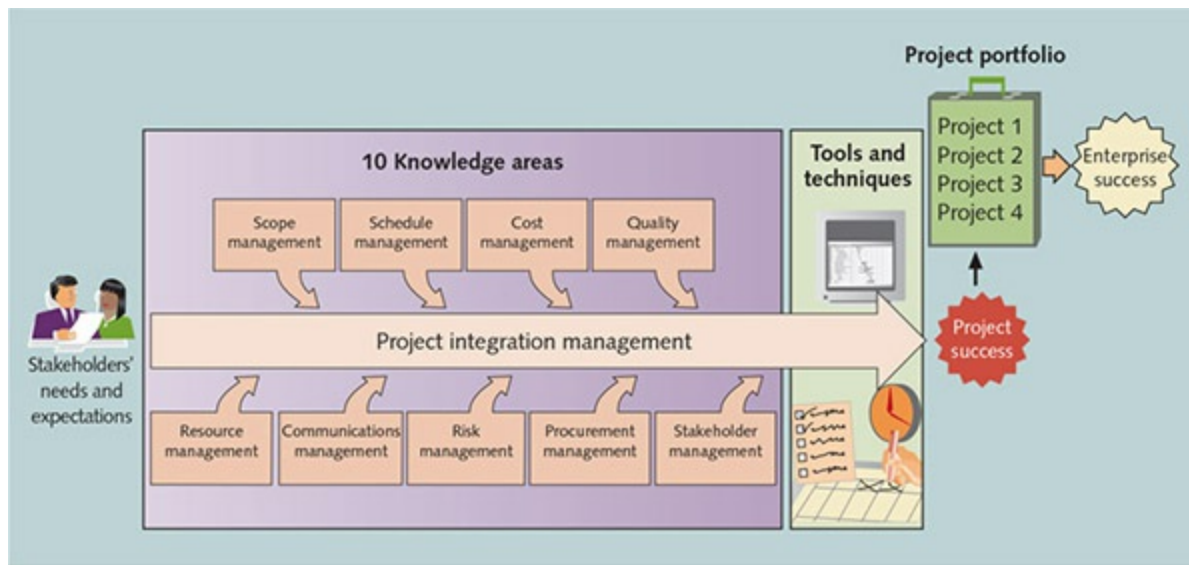
What Is Project Management?

Project management is “the application of knowledge, skills, tools, and techniques to project activities to meet project requirements.”* Project managers must not only strive to meet specific scope, time, cost, and quality goals of projects, but also facilitate the entire process to meet the needs and expectations of people involved in project activities or affected by them.

Figure 1-2 illustrates a framework to help you understand project management. Key elements of this framework include the project stakeholders, project management knowledge

areas, project management tools and techniques, and the contribution of successful projects to the enterprise.

Figure 1-2. Project management framework



Project Stakeholders

Stakeholders are the people involved in or affected by project activities, and include the project sponsor, project team, support staff, customers, users, suppliers, and even opponents of the project. These stakeholders often have very different needs and expectations. A familiar example of a project is building a new house. There are several stakeholders in a home construction project.

- The project sponsors would be the potential new homeowners who would be paying for the house. They could be on a very tight budget, so would expect the contractor to provide a realistic idea of what type of home they could afford given their budget constraints. They would also need a realistic idea of when they could move in. Regardless of budget, they would expect the contractor to provide accurate estimates for the building costs. The new homeowners would have to make important decisions to keep the costs of the house within their budget. Can they afford to finish the basement right away? If they can afford to finish the basement, will it affect the projected move-in date? In this example, the project sponsors are also the customers and users of the product, which is the house.
- The house may require financing by a bank or other financial institution like a credit union, which will secure a legal interest (lien) in the property and the finished home. This institution is an example of a legal stakeholder who must be informed of any changes to the plans or schedule because the project is part of a legal contract.

- The project manager in this example would normally be the general contractor responsible for building the house. The project manager needs to work with all the project stakeholders to meet their needs and expectations.
- The project team for building the house would include several construction workers, electricians, and carpenters. These stakeholders would need to know exactly what work they must do and when they need to do it. They would need to know if the required materials and equipment will be at the construction site or if they are expected to provide the materials and equipment. Their work would need to be coordinated because many interrelated factors are involved. For example, the carpenter cannot put in kitchen cabinets until the walls are completed.
- Support staff might include the buyers' employers, the general contractor's administrative assistant, and people who support other stakeholders. The buyers' employers might expect their employees to complete their work but allow some flexibility so they can visit the building site or take phone calls related to building the house. The contractor's administrative assistant would support the project by coordinating meetings between the buyers, the contractor, suppliers, and other parties.
- Building a house requires many suppliers. The suppliers would provide the wood, windows, flooring, appliances, and other materials. Suppliers would expect exact details on the items they need to provide, and where and when to deliver those items.
- A project might have opponents. In this example, a neighbor might oppose the project because the workers make so much noise that she cannot concentrate on her work at home, or the noise might wake her sleeping children. She might interrupt the workers to voice her complaints or even file a formal complaint. Or, the neighborhood might have association rules concerning new home design and construction. If the homeowners do not follow these rules, they might have to halt construction due to legal issues. Even without such complaints, the home must comply with certain building codes and other restrictions; these considerations may also result in changes to the project's requirements, making the local government a stakeholder in the project.

As you can see from this example, projects have many different stakeholders, and they often have different interests. Stakeholders' needs and expectations are important in the beginning and throughout the life of a project. Successful project managers develop good relationships with project stakeholders to understand and meet their needs and expectations.

Project Management Knowledge Areas

Project management knowledge areas describe the key competencies that project

managers must develop. The center of Figure 1-2 shows the 10 knowledge areas of project management.

1. Project scope management involves defining and managing all the work required to complete the project successfully.
2. Project schedule management (formerly called project time management) includes estimating how long it will take to complete the work, developing an acceptable project schedule, and ensuring timely completion of the project.
3. Project cost management consists of preparing and managing the budget for the project.
4. Project quality management ensures that the project will satisfy the stated or implied needs for which it was undertaken.
5. Project resource management is concerned with making effective use of the people and physical resources involved with the project.
6. Project communications management involves generating, collecting, disseminating, and storing project information.
7. Project risk management includes identifying, analyzing, and responding to risks related to the project.
8. Project procurement management involves acquiring or procuring goods and services for a project from outside the performing organization.
9. Project stakeholder management includes identifying and analyzing stakeholder needs while managing and controlling their engagement throughout the life of the project.
10. Project integration management is an overarching function that affects and is affected by all of the other knowledge areas.

Project managers must have knowledge and skills in all 10 of these areas. This text includes an entire chapter on each of these knowledge areas because all of them are crucial to project success.

Project Management Tools and Techniques

Thomas Carlyle, a famous historian and author, stated, “Man is a tool-using animal. Without tools he is nothing, with tools he is all.” As the world continues to become more complex, it is even more important for people to develop and use tools, especially for managing important

projects. [Project management tools and techniques](#) assist project managers and their teams in carrying out work in all 10 knowledge areas. For example, some popular time-management tools and techniques include Gantt charts, project network diagrams, and critical path analysis. Table 1-1 lists some commonly used tools and techniques by knowledge area. You will learn more about these and other tools and techniques throughout this text.

Table 1-1. Common project management tools and techniques by knowledge area

Knowledge Area/Category	Tools and Techniques	Super Tools
<i>Integration management</i>	Project selection methods	Project management software
	Project management methodologies	Change requests
	Stakeholder analyses	Lessons-learned reports
	Work requests	
	Project charters	
	Project management plans	
	Change control boards	
	Project review meetings	
<i>Scope management</i>	Statements of work	Scope statements
	Scope management plans	Work breakdown structures
	Scope verification techniques	Requirements analyses
	Scope change controls	
<i>Schedule management</i>	Project network diagrams	Gantt charts
	Critical path analysis	
	Crashing	
	Fast tracking	
	Schedule performance measurements	
<i>Cost management</i>	Project budgets	
	Net present value	
	Return on investment	
	Payback analysis	

	Earned value management	
	Project portfolio management	
	Cost estimates	
	Cost management plans	
	Cost baselines	

Quality management

	Quality metrics	
	Checklists	
	Quality control charts	
	Pareto diagrams	
	Fishbone diagrams	
	Maturity models	
	Statistical methods	
	Test plans	

Resource management

	Motivation techniques	
	Empathic listening	
	Responsibility assignment matrices	
	Project organizational charts	
	Resource histograms	
	Team building exercises	

Communications management

	Communications management plans	Kick-off meetings
	Conflict management	Progress reports
	Communications media selection	
	Status reports	
	Virtual communications	
	Templates	
	Project websites	

Risk management

	Risk management plans	
	Risk registers	
	Probability/impact matrices	

Procurement management

Make-or-buy analyses

Contracts

Requests for proposals or quotes

Source selections

Supplier evaluation matrices

A survey of 753 project and program managers was conducted to rate several project management tools. Respondents rated tools on a scale of 1–5 (low to high) based on the extent of their use and the potential of the tools to help improve project success. “Super tools” were defined as those that had high use and high potential for improving project success. These super tools included software for task scheduling (such as project management software), scope statements, requirement analyses, and lessons-learned reports. Tools that are already used extensively and have been found to improve project performance include progress reports, kick-off meetings, Gantt charts, and change requests.

These super tools appear in column 3 of **Table 1-1**.^{*} Note that project stakeholder management was not a separate knowledge area at the time of this survey.

The *PMBOK® Guide – Sixth Edition* now lists tools and techniques based on their purpose, as follows:

- *Data gathering*: benchmarking, brainstorming, check sheets, checklists, focus groups, interviews, market research, questionnaires and surveys, and statistical sampling
- *Data analysis*: alternatives analysis, assessment of other risk parameters, assumption and constraint analysis, cost of quality, cost-benefit analysis, decision tree analysis, document analysis, earned value analysis, and several other tools fit in this category
- *Data representation*: affinity diagrams, cause-and-effect diagrams, control charts, flow charts, hierarchical charts, histograms, logical data models, matrix diagrams, matrix-based charts, mind mapping, probability and impact matrix, scatter diagrams, stakeholder engagement assessment matrix, stakeholder mapping/representation, and text-oriented formats
- *Decision making*: multi-criteria decision analysis and voting
- *Communication*: feedback and presentations

- *Interpersonal and team skills*: active listening, communication styles assessment, conflict management, cultural awareness, decision making, emotional intelligence, facilitation, influencing, leadership, meeting management, motivation, negotiation, networking, nominal group, observation/conversation, political awareness, team building
- *Ungrouped*: several other tools fit in this category

These long lists of tools and techniques can be overwhelming. This text will focus on those used most often and with the most potential, providing the context and detailed examples for using them. It is crucial for project managers and their team members to determine which tools will be most useful for their particular projects. Selecting the appropriate tools and techniques (as well the processes, inputs, outputs, and life cycle phases, discussed later in this book) is part of project tailoring. Project management should be tailored to meet the unique needs of projects, organizations, and most importantly, people. After all, projects are done by, and for, people.

Despite its advantages, project management is not a silver bullet that guarantees success on all projects. Some projects, such as those involving new technologies, have a higher degree of uncertainty, so it is more difficult to meet their scope, schedule, and cost goals. Project management is a very broad, often complex discipline. What works on one project may not work on another, so it is essential for project managers to continue to develop their knowledge and skills in managing projects. It is also important to learn from the mistakes and successes of past projects.

What Went Right?

Follow-up studies by the Standish Group (see the previously quoted “CHAOS” study in the What Went Wrong? passage) showed improvement in the statistics for IT projects:

- The number of successful projects (those completed on time, on budget with a satisfactory result) was 29 percent in 2015 based on a sample of over 50,000 software development projects worldwide. The number of failed projects (those canceled or not used after implementation) was 19 percent. That leaves 52 percent of projects as challenged (over budget, late, and/or poorly implemented). These numbers include projects of all sizes and methodologies.
- The 2015 CHAOS study also summarized the success rates of projects by size, showing that 62 percent of small projects were successful from 2011 to 2015 compared to only 2 percent of grand, 6 percent of large, 9 percent of medium, and 21 percent of moderate size projects. Small projects are obviously easier to complete successfully.

- Agile approaches were also measured across all project sizes from 2011 to 2015, showing that 39 percent of all agile projects were successful compared to 11 percent of waterfall projects. For small projects, 58 percent of agile projects were successful compared to 44 percent of waterfall projects. About 10,000 projects were included for these statistics.*

According to the PMI research, across all industries, the average percentage of projects that are deemed failures is 14 percent; the average for IT projects deemed failures in 2016 also is 14 percent.*

Project Success

How do you define the success or failure of a project? The list that follows outlines a few common criteria for measuring the success of a project, illustrating each with an example of upgrading 500 desktop computers within three months for \$300,000:

1. *The project met scope, time, and cost goals.* If all 500 computers were upgraded and met other scope requirements, the work was completed in three months or less, and the cost was \$300,000 or less, you could consider the project successful. The Standish Group studies used this definition of success, but several people question this simple definition of project success and the methods used for collecting the data. (Search for articles by Robert L. Glass to read more about this debate.)
2. *The project satisfied the customer/sponsor.* Even if the project met initial scope, time, and cost goals, the users of the computers or their managers might not be satisfied. Perhaps the project manager or team members never returned calls or were rude. Perhaps users had their daily work disrupted during the upgrades or had to work extra hours due to the upgrades. If the customers were not happy with important aspects of the project, it would be deemed a failure. Conversely, a project might not meet initial scope, time, and cost goals, but the customer could still be very satisfied. Perhaps the project team took longer and spent more money than planned, but they were very polite and helped the users and managers solve several work-related problems. Many organizations implement a customer satisfaction rating system to measure project success instead of tracking only scope, time, and cost performance.
3. *The results of the project met its main objective, such as making or saving a certain amount of money, providing a good return on investment, or simply making the sponsors happy.* Even if the project cost more than estimated, it took longer to complete, and the project team was hard to work with, the project would be successful if users were happy with the

upgraded computers, based on this criterion. As another example, suppose that the sponsor approved the upgrade project to provide a good return on investment by speeding up work and therefore generating more profits. If those goals were met, the sponsor would deem the project a success, regardless of other factors involved.

Why do some IT projects succeed and others fail? **Table 1-2** summarizes the results of the 2015 CHAOS study. The factors that contribute most to the success of IT projects are listed in order of importance. Executive sponsorship is the most important factor, followed by emotional maturity of the organization. A few of the top success factors relate to good scope management, such as having clear business objectives and optimization. Project management expertise continues to be a key success factor. In fact, experienced project managers, who can often help influence all of these factors to improve the probability of project success, led 97 percent of successful projects, based on an earlier CHAOS study.

Table 1-2. **What helps projects succeed?**

Factors of Success	Points
Executive sponsorship	15
Emotional maturity	15
User involvement	15
Optimization	15
Skilled resources	10
Agile process	7
Modest execution	6
Project management expertise	5
Clear business objectives	4

Source: The Standish Group, “CHAOS Manifesto 2015” (2015).

A U.S. government report listed the top three reasons why federal technology projects succeed:

1. Adequate funding
2. Staff expertise
3. Engagement from all stakeholders

Notice that the CHAOS study list does not include adequate funding. Most nongovernment companies must either find adequate funds for important projects or cancel projects if they

cannot be funded or get an adequate return. Government projects often require that funds be allocated a year or more before they even start, and estimates often fall short. “The government has struggled when acquiring technology thanks to the convoluted nature of the federal contracting process and the shortage of qualified contracting officers and technical personnel. Critics argue that federal agencies get little return for the \$80 billion the government spends annually on IT. . . . ‘History has shown that government IT projects frequently face challenges of meeting cost, schedule or performance goals,’ said Sen. Susan Collins (R-Maine) in a statement.”*

It is interesting to compare success factors for IT projects in the United States with those in other countries. A survey of 247 information systems project practitioners in mainland found that relationship management is viewed as a top success factor for information systems in China, while it is not mentioned in U.S. studies. The study also suggested that having competent team members is less important in China than in the United States. The Chinese, like the Americans, included top management support, user involvement, and a competent project manager as vital to project success.*

It is also important to look beyond individual project success rates and focus on how organizations as a whole can improve project performance. Research comparing companies that excel in project delivery—the “winners”—with those that do not found four significant best practices:

1. *Use an integrated toolbox.* Companies that consistently succeed in managing projects clearly define what needs to be done in a project, by whom, when, and how. They use an integrated toolbox, including project management tools, methods, and techniques. They carefully select tools, align them with project and business goals, link them to metrics, and provide them to project managers to deliver positive results.
2. *Grow project leaders.* The winners know that strong project managers—referred to as project leaders—are crucial to project success. They also know that a good project leader needs to be a business leader as well, with strong interpersonal and intrapersonal skills. Companies that excel in project management often grow or develop their project leaders internally, providing them with career opportunities, training, and mentoring.
3. *Develop a streamlined project delivery process.* Winning companies have examined every step in the project delivery process, analyzed fluctuations in workloads, searched for ways to reduce variation, and eliminated bottlenecks to create a repeatable delivery process. All projects go through clear stages and clearly define key milestones. All project leaders use a shared road map, focusing on key business aspects of their projects while integrating goals across all parts of the organization.

4. *Measure project health using metrics.* Companies that excel in project delivery use performance metrics to quantify progress. They focus on a handful of important measurements and apply them to all projects. Metrics often include customer satisfaction, return on investment, and percentage of schedule buffer consumed.*

Project managers play an important role in making projects, and therefore organizations, successful. Project managers work with the project sponsors, the project team, and other stakeholders to meet project goals. They also work with sponsors to define success for particular projects. Good project managers do not assume that their definition of success is the same as the sponsors'. They take the time to understand their sponsors' expectations and then track project performance based on important success criteria.

Program and Project Portfolio Management

About one-quarter of the world's gross domestic product is spent on projects. Projects make up a significant portion of work in most business organizations or enterprises, and managing those projects successfully is crucial to enterprise success. Two important concepts that help projects meet enterprise goals are the use of programs and project portfolio management.

Programs

A **program** is "a group of related projects, subsidiary programs, and program activities managed in a coordinated manner to obtain benefits and control not available from managing them individually."* As you can imagine, it is often more economical to group projects together to help streamline management, staffing, purchasing, and other work. Programs are not large projects; a **megaproject** is a very large project that typically costs over US\$1 billion, affects over one million people, and lasts several years. For examples, the Panama Canal Expansion Project was a megaproject that took 11 years and \$5.25 billion to complete. The following are examples of common programs in the IT field.

- *Infrastructure:* An IT department often has a program for IT infrastructure projects. This program could encompass several projects, such as providing more wireless Internet access, upgrading hardware and software, enhancing computer security, and developing and maintaining corporate standards for IT.
- *Applications development:* This program could include several projects, such as updating an enterprise resource planning (ERP) system, purchasing a new off-the-shelf billing system, or developing a new capability for a customer relationship management system.
- *User support:* In addition to the many operational tasks related to user support, many IT departments have several projects to support users. For example, a project might

provide a better e-mail system or develop technical training for users.

A [program manager](#) provides leadership and direction for the project managers heading the projects within a program. Program managers also coordinate the efforts of project teams, functional groups, suppliers, and operations staff supporting the projects to ensure that products and processes are implemented to maximize benefits. Program managers are responsible for more than the delivery of project results; they are change agents responsible for the success of products and processes developed by those projects. For example, the NASA International Space Station Program is led by a program manager who oversees all U.S. projects involved with the station and is accountable for achieving their objectives, funding, and contribution to scientific knowledge.

Program managers often have review meetings with all their project managers to share important information and coordinate important aspects of each project. Many program managers worked as project managers earlier in their careers, and they enjoy sharing their wisdom and expertise with their project managers. Effective program managers recognize that managing a program is much more complex than managing a single project. They recognize that technical and project management skills are not enough—program managers must also possess strong business knowledge, leadership capabilities, and communication skills.

Project Portfolio Management

In many organizations, project managers also support an emerging business strategy of [project portfolio management](#) or [portfolio management](#), as called in this text, in which organizations group and manage projects and programs as a portfolio of investments that contribute to the entire enterprise's success. Portfolio managers help their organizations make wise investment decisions by helping to select and analyze projects from a strategic perspective. Portfolio managers may or may not have previous experience as project or program managers. It is most important that they have strong financial and analytical skills and understand how projects and programs can contribute to meeting strategic goals.

Figure 1-3 illustrates the differences between project management and project portfolio management. Notice that the main distinction is a focus on meeting tactical or strategic goals. Tactical goals are generally more specific and short term than strategic goals, which emphasize long-term goals for an organization. Individual projects often address tactical goals, whereas portfolio management addresses strategic goals. Project management addresses questions like “Are we carrying out projects well?”, “Are projects on time and on budget?”, and “Do project stakeholders know what they should be doing?”

Figure 1-3. Project management compared to project portfolio management



Portfolio management addresses questions like “Are we working on the right projects?”, “Are we investing in the right areas?”, and “Do we have the right resources to be competitive?” PMI defines a **portfolio** as “projects, programs, subsidiary portfolios, and operations managed as a group to achieve strategic objectives.”*

Many organizations use a more disciplined approach to portfolio management by developing guidelines and software tools to assist in it. The Project Management Institute (described later in this chapter) first published the *Organizational Project Management Maturity Model (OPM3®) Knowledge Foundation* in 2003.* OPM3® describes the importance of not only managing individual projects or programs well but also following organizational project management to align projects, programs, and portfolios with strategic goals. OPM3® is a standard that organizations can use to measure their organizational project management maturity against a comprehensive set of best practices.

Best Practice

A **best practice** is “an optimal way recognized by industry to achieve a stated goal or objective.”* Rosabeth Moss Kanter, a professor at Harvard Business School and well-known author and consultant, says that visionary leaders know “the best practice secret: Stretching to learn from the best of the best in any sector can make a big vision more

likely to succeed.”* Kanter also emphasizes the need to have measurable standards for best practices. An organization can measure performance against its own past, against peers, and, even better, against potential. Kanter suggests that organizations need to continue to reach for higher standards. She suggests the following exercise regimen for business leaders who want to adapt best practices in an intelligent way to help their own organizations:

- Reach high. Stretch. Raise standards and aspirations. Find the best of the best and then use it as inspiration for reaching full potential.
- Help everyone in your organization become a professional. Empower people to manage themselves through benchmarks and standards based on best practice exchange.
- Look everywhere. Go far afield. Think of the whole world as your laboratory for learning.

Robert Butrick, author of *The Project Workout*, wrote an article on best practices in project management for the *Ultimate Business Library’s Best Practice* book. He suggests that organizations need to follow basic principles of project management, including these two mentioned earlier in this chapter:

- Make sure your projects are driven by your strategy. Be able to demonstrate how each project you undertake fits your business strategy, and screen out unwanted projects as soon as possible.
- Engage your stakeholders. Ignoring stakeholders often leads to project failure. Be sure to engage stakeholders at all stages of a project, and encourage teamwork and commitment at all times.*

Table 1-3 provides a comparative overview of project, program, and portfolio management. **Organizational project management** is a “framework in which portfolio, program, and project management are integrated with organizational enablers in order to achieve strategic objectives.”*

Table 1-3. Comparative Overview of Portfolios, Programs, and Projects			
	Projects	Programs	Portfolios
Definition	A project is a temporary endeavor undertaken to create a unique product,	A program is a group of related projects, subsidiary programs, and program	A portfolio is a collection of projects, programs, subsidiary portfolios, and

service, or result.

activities that are managed in a coordinated way to obtain benefits not available from managing them individually.

operations managed as a group to achieve strategic objectives.

Management

Project managers manage the project team to meet the project objectives.

Programs are managed by program managers who ensure that program benefits are delivered as expected, by coordinating the activities of a program's components.

Portfolio managers may manage or coordinate portfolio management staff, or program and project staff that may have reporting responsibilities into the aggregate portfolio.

Monitoring

Project managers monitor and control the work of producing the products, services, or results that the project was undertaken to produce.

Program managers monitor the progress of program components to ensure that the overall goals, schedules, budget, and benefits of the program are met.

Portfolio managers monitor strategic changes and aggregate resource allocation, performance results, and risk of the portfolio.

Success

Success is measured by product and project quality, timeliness, budget compliance, and degree of customer satisfaction.

A program's success is measured by the program's ability to deliver its intended benefits to an organization, and by the program's efficiency and effectiveness in delivering those benefits.

Success is measured in terms of the aggregate investment performance and benefit realization of the portfolio.

Source: Project Management Institute, Inc., *A Guide to the Project Management Body of Knowledge (PMBOK® Guide) – Sixth Edition (2017)*.

Organizational Project Management

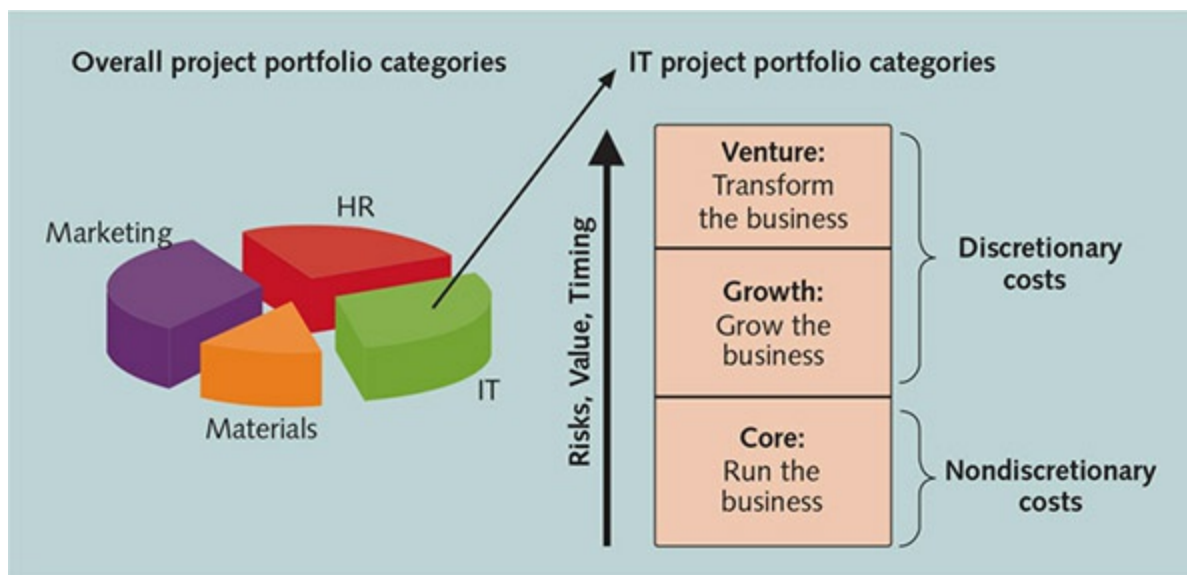
Organizations group projects into portfolios to help them make better investment decisions, such as increasing, decreasing, discontinuing, or changing specific projects or programs based on their financial performance, risks, resource utilization, and similar factors that affect business value. If a construction firm has much higher profit margins on apartment buildings than single-family homes, for example, it might choose to pursue more apartment building projects. The firm might also create a new project to investigate ways to increase profits for single-family home projects. On the other hand, if the company has too many projects focused on financial performance and not enough focused on improving its work force, the portfolio manager might suggest initiating more projects to support that strategic goal. Just like a personal financial portfolio, a business's portfolio should be diversified to account for risk.

By grouping projects into portfolios, organizations can better tie their projects to meeting strategic goals. Portfolio management can also help organizations do a better job of managing its human resources by hiring, training, and retaining workers to support the projects in the organization's portfolio. For example, if the construction firm needs more people with experience in building apartment buildings, they can make necessary adjustments by hiring or training current workers in the necessary skills.

As you can imagine, project portfolio management is not an easy task. **Figure 1-4** illustrates one approach for project portfolio management in which one large portfolio exists for the entire organization. This allows top management to view and manage all projects at an enterprise level. Sections of the portfolio are then broken down to improve the management of projects in each sector. For example, a company might have the main portfolio categories shown in the left part of Figure 1-4—marketing, materials, IT, and human resources (HR)—and divide each of those categories further to address its unique concerns. The right part of this figure shows how the IT projects could be categorized in more detail to assist in their management. In this example, there are three basic IT project portfolio categories:

- *Venture*: Projects in this category help transform the business. For example, the large retail chain described in the opening case might have an IT project to provide kiosks in stores and similar functionality on the Internet where customers and suppliers could quickly provide feedback on products or services. This project could help transform the business by developing closer partnerships with customers and suppliers.
- *Growth*: Projects in this category would help the company increase its revenues. For example, a company might have an IT project to provide information on its corporate website in a new language, such as Chinese or Japanese. This capability could help the company grow its business in those countries.
- *Core*: Projects in this category must be accomplished to run the business. For example, an IT project to provide computers for new employees would fall under this category.

Figure 1-4. Sample project portfolio approach



In **Figure 1-4**, the costs of Core IT projects are nondiscretionary, which means that the

company has no choice in whether to fund them. Core IT Projects must be funded for the company to stay in business. Projects in the Venture or Growth category are discretionary costs because the company can use its own discretion or judgment in deciding whether to fund them; these projects are not critical to the company fulfilling its mission. The arrow in the center of Figure 1-4 indicates that the risks and value of projects normally increase as you move from Core to Growth to Venture projects. In addition, timeliness becomes increasingly important; growth and venture projects, more than core projects, must be done within a certain time frame to be effective. However, some core projects can also be high risk, have high value, and require good timing. As you can see, many factors are involved in portfolio management.

The Role of the Project Manager

You have already read that project managers must work closely with the other stakeholders on a project, especially the sponsor and project team. They are also more effective if they are familiar with the 10 project management knowledge areas and the various tools and techniques related to project management. Experienced project managers help projects succeed. But what do project managers do, exactly? What skills do they really need to do a good job? What is PMI's talent triangle? The next section provides brief answers to these questions, and the rest of this book gives more insight into the role of the project manager. Even if you never become a project manager, you will probably be part of a project team, and it is important for team members to help their project managers.

Project Manager Job Description

A project manager can have many different job descriptions, which can vary tremendously based on the organization and the project. In fact, PMI includes a page on their website to answer the question, "Who are project managers?" In addition to saying that project managers are organized, passionate, and goal-oriented individuals who drive business results by leading projects, PMI emphasizes that they are also change agents who work well under pressure and enjoy challenging work environments.

Project management jobs can be found in every country and every industry. Sites like indeed.com listed hundreds of thousands of job openings in 2018. Monster.com has a job category for project management, and their site says that project managers "smoothly link management, clients and staff to keep projects rolling. To be successful in a project management job, you'll need people skills, business acumen and technical competence."* Here are a few edited postings:

- *Project manager for a consulting firm:* Plans, schedules, and controls activities to fulfill identified objectives applying technical, theoretical, and managerial skills to satisfy

project requirements. Coordinates and integrates team and individual efforts and builds positive professional relationships with clients and associates.

- *Project manager for a computer systems firm:* Works independently within established practices to assist in the development and implementation process of projects involving departmental, vendor relationships, and/or cross-functional teams. Coordinates with internal/external clients to gather business requirements and coordinate project plans. Monitor projects from initiation through delivery ensuring completion of the project on schedule.
- *IT project manager for a nonprofit consulting firm:* Responsibilities include business analysis, requirements gathering, project planning, budget estimating, development, testing, and implementation. Responsible for working with various resource providers to ensure that development is completed in a timely, high-quality, and cost-effective manner.

The job description for a project manager can vary by industry and by organization, but most project managers perform similar tasks regardless of these differences. In fact, project management is a skill needed in every major IT field, from database administrator to network specialist to technical writer. Because demand for project managers is high, some organizations have hired new college graduates to fill positions normally held by experienced professionals. For example, Boom Lab, a consulting company named by the Star Tribune as a top workplace in 2017, is growing quickly by finding, training, and placing talented people as project coordinators. As new project coordinators gain experience and credentials, they often continue their careers by managing larger projects, becoming program managers, or transitioning into other management positions.

Advice for Young Professionals

How do you know if you would be a good project manager? Being a project manager is a demanding yet rewarding profession, for the right person. Below are a few questions to ask yourself:

- Do you get frustrated by bad bosses? Do you think you could do a better job?
- Are you interested in understanding the big picture of how organizations work and how your individual work or your project fits in?
- Have you had other leadership roles, such as being a team captain, president of a club, or entrepreneur of a small business? Did you enjoy it? Did others think you did a good job?

- Are you good at mentoring others? Do people ask you for help in developing their skills or your advice on what to do?

If you answered “yes” to most of these questions, you should consider being a project manager. If you prefer just focusing on your own work and want to remain in a technical position, there’s absolutely nothing wrong with that. Just remember that it will help you in the long run if you at least understand what project managers do and assist them as needed doing things like creating estimates, identifying risks, etc.

What if you try it but no longer want to stay in a project management career path? You can often go back to your former, more technical role, and move along that career path. Or, many ex-project managers move into higher level management positions, such as director, vice president, or even CEO. Some become consultants, educators, or entrepreneurs. Their experience leading projects makes them marketable in several different professions.

Suggested Skills for Project Managers

Project managers need to have a wide variety of skills and be able to decide which skills are more important in different situations. *A Guide to the Project Management Body of Knowledge*—the *PMBOK® Guide*—now includes a chapter called “The Role of the Project Manager.” It uses the analogy of a project manager being like a conductor or a large orchestra. Conductors are not expected to be able to play every instrument in the orchestra, but they should possess musical knowledge, understanding, and experience required to conduct an orchestra. Similarly, a project manager in IT should possess project management and IT knowledge, understanding of the project and how it will fit into the organization, and experience in general management and human relations or soft skills needed to guide the project team.

This chapter introduced the 10 project management knowledge areas, as well as some general tools and techniques project managers use. The following section focuses on the IT application area, including skills required in the project environment, general management, and soft skills. Note that the *PMBOK® Guide – Sixth Edition* describes project management competencies based on the talent triangle, described in the following section.

The project environment differs from organization to organization and project to project, but some skills will help in almost all project environments. These skills include understanding change and understanding how organizations work within their social, political, and physical environments. Project managers must be comfortable leading and handling change, because most projects introduce changes in organizations and involve changes within the projects

themselves. Project managers need to understand the organization in which they work and how that organization develops products and provides services. The skills and behavior needed to manage a project for a Fortune 100 company in the United States may differ greatly from those needed to manage a government project in Poland. **Chapter 2**, The Project Management and Information Technology Context, provides detailed information on these topics.

Project managers should also possess general management knowledge and skills. They should understand important topics related to financial management, accounting, procurement, sales, marketing, contracts, manufacturing, distribution, logistics, the supply chain, strategic planning, tactical planning, operations management, organizational structures and behavior, personnel administration, compensation, benefits, career paths, and health and safety practices. On some projects, it will be critical for the project manager to have a lot of experience in one or several of these general management areas. On other projects, the project manager can delegate detailed responsibility for some of these areas to a team member, support staff, or even a supplier. Even so, the project manager must be intelligent and experienced enough to know which of these areas are most important and who is qualified to do the work. The project manager must make all key project decisions and take responsibility for them.

Achieving high performance on projects requires soft skills, otherwise called human relations skills. Some of these soft skills include effective communication, influencing the organization to get things done, leadership, motivation, negotiation, conflict management, and problem-solving. Why do project managers need good soft skills? One reason is that to understand, navigate, and meet stakeholders' needs and expectations, project managers need to lead, communicate, negotiate, solve problems, and influence the organization at large. They need to be able to listen actively to what others are saying, help develop new approaches for solving problems, and then persuade others to work toward achieving project goals. Project managers must lead their project teams by providing vision, delegating work, creating an energetic and positive environment, and setting an example of appropriate and effective behavior. Project managers must focus on teamwork skills to employ people effectively. They need to be able to motivate different types of people and develop *esprit de corps* within the project team and with other project stakeholders. Because most projects involve changes and trade-offs between competing goals, it is important for project managers to have strong coping skills as well. Project managers need to be able to cope with criticism and constant change. Project managers must be flexible, creative, and sometimes patient in working toward project goals; they must also be persistent in making project needs known.

Finally, project managers, especially those managing IT projects, must be able to make effective use of technology as it relates to the specific project. Making effective use of

technology often includes special product knowledge or experience with a particular industry.

Project managers must make many decisions and deal with people in a wide variety of disciplines, so it helps tremendously to have a project manager who is confident in using the special tools or technologies that are the most effective in particular settings. Project managers do not normally have to be experts on any specific technology, but they have to know enough to build a strong team and ask the right questions to keep things on track. For example, project managers for large IT projects do not have to be experts in the field of IT, but they must have working knowledge of various technologies and understand how the project would enhance the business. Many companies have found that good business managers can be very good IT project managers because they focus on meeting business needs and rely on key project members to handle the technical details.

A 2017 article published by CIO.com listed the six traits of highly effective project managers as follows:

1. Be a strategic business partner.
2. Encourage and recognize valuable contributions.
3. Respect and motivate stakeholders.
4. Be fully vested in success.
5. Stress integrity and accountability.
6. Work in the gray/Be able to deal with ambiguity.*

All project managers should continue to develop their knowledge and experience in project management, general management, soft skills, and the industries they support. IT project managers must be willing to develop more than their technical skills to be productive team members and successful project managers. Everyone, no matter how technical they are, should develop business and soft skills.

PMI Talent Triangle® and the Importance of Leadership Skills*

PMI developed a talent triangle to emphasize the types of skills project managers need to continuously develop. The talent triangle includes the following:

1. *Technical project management skills*: Understanding the knowledge areas, process groups, and project management tools and techniques fall into this category.

2. *Strategic and business management skills*: Topics include strategic planning and financial management (described in more detail in Chapter 4), accounting, marketing, and other topics listed in the previous section.
3. *Leadership skills*: Leadership and management are terms often used interchangeably, although there are differences. Generally, a **leader** focuses on long-term goals and big-picture objectives, while inspiring people to reach those goals. A **manager** often deals with the day-to-day details of meeting specific goals. Some people say that, “Managers do things right, and leaders do the right things.” “Leaders determine the vision, and managers achieve the vision.” “You lead people and manage things.”

Leadership is a soft skill, and there is no one best way to be a leader. Peter Northouse, author of a popular text called *Leadership: Theory and Practice*, says, “In the past 60 years, as many as 65 different classification systems have been developed to define the dimensions of leadership.”* Some classification systems focus on group processes, while others focus on personality traits or behaviors. For example, the *PMBOK® Guide – Sixth Edition* briefly describes the following leadership styles:

1. **Laissez-faire** : Meaning “let go,” this hands-off approach lets teams determine their own goals and how to achieve them.
2. **Transactional** : This management by exception approach focuses on achieving goals or compliance by offering team members appropriate rewards and punishments.
3. **Servant leader** : People using this approach focus on relationships and community first and leadership is secondary.
4. **Transformational** : By working with others to identify needed changes, these leaders empower others and guide changes through inspiration.
5. **Charismatic** : These people can inspire others based on their enthusiasm and confidence.
6. **Interactional** : This leadership style is a combination of transactional, transformational, and charismatic.

There are many different leadership styles in addition to the six listed above, and the one thing most experts agree on is that the best leaders are able to adapt their style to the needs of the situation. Also, while it may seem from the above list that such styles are mutually exclusive, the fact is that most leaders exhibit more than one leadership style as they adapt to their situations. Finally, the names of these styles should not be interpreted as either good or bad by their labels—they can each be appropriate and effective given the circumstances of a

situation.

Daniel Goleman, author of *Emotional Intelligence*, also wrote a book called *Primal Leadership*, which describes six different styles of leadership and situations where they are most appropriate:

1. *Visionary*: Needed when an organization needs a new direction, and the goal is to move people toward a new set of shared dreams. The leader articulates where the group is going, but lets them decide how to get there by being free to innovate, experiment, and take calculated risks.
2. *Coaching*: One-on-one style that focuses on developing individuals, showing them how to improve their performance. This approach works best with workers who show initiative and request assistance.
3. *Affiliative*: Emphasizes the importance of teamwork and creating harmony by connecting people to each other. This approach is effective when trying to increase morale, improve communication, or repair broken trust.
4. *Democratic*: Focuses on people's knowledge and skills and creates a commitment to reaching shared goals. This leadership style works best when the leader needs the collective wisdom of the group to decide on the best direction to take for the organization.
5. *Pacesetting*: Used to set high standards for performance. The leader wants work to be done better and faster and expects everyone to put forth their best effort.
6. *Commanding*: Most often used, also called autocratic or military style leadership. This style is most effective in a crisis or when a turnaround is needed.

"The goal for leaders should be to develop a solid understanding of the different styles of leadership and their implications, and reach the point where choosing the right one for the situation becomes second nature to them."*

Project managers often take on the role of both leader and manager. Good project managers know that people make or break projects, so they must set a good example to lead their team to success. They are aware of the greater needs of their stakeholders and organizations, so they are visionary in guiding their current projects and in suggesting future ones.

As mentioned earlier, program managers need the same skills as project managers. They often rely on their past experience as project managers, strong business knowledge, leadership capability, and communication skills to handle the responsibility of overseeing the

multiple projects that make up their programs. It is most important that portfolio managers have strong financial and analytical skills and understand how projects and programs can contribute to meeting strategic goals.

Companies that excel in project, program, and portfolio management grow project leaders, emphasizing development of business and leadership skills. Instead of thinking of leaders and managers as specific people, it is better to think of people as having leadership skills, such as being visionary and inspiring, and management skills, such as being organized and effective. Therefore, the best project, program, and portfolio managers have leadership and management characteristics; they are visionary yet focused on the bottom line. Above all else, they focus on achieving positive results!

Careers for IT Project Managers

As shown earlier, the IT industry continues to grow, and the need for people to lead IT projects has remained solid. Most IT workers will be involved in project work, so it's important to understand basic concepts of project management to help those projects succeed.

IT executives listed the “ten hot tech skills” they planned to hire for in 2017. Full-stack software development took first place, replacing programming and application development. Full-stack developers are familiar with all layers in computer software development and have experience in several programming languages and platforms. They also understand different roles in IT (DevOps, design, quality assurance, database administration, analytics, etc.) to effectively solve business problems. Project management again came in second, as shown in **Table 1-4**.

Table 1-4.	Ten hot tech skills for 2017
1.	Full-stack software development
2.	Project management
3.	Cyber-security
4.	Networking
5.	User experience/user interface (UX/UI) design
6.	Quality assurance (QA)/testing
7.	Cloud engineering
8.	Big data
9.	Machine learning/artificial intelligence

Source: Sharon Florentine, “10 IT skills that employers need in 2017,” CIO from IDG (February 1, 2017).

The Project Management Profession

The project management profession is growing at a very rapid pace. To understand this line of work, it is helpful to briefly review the history of project management, learn about the Project Management Institute and some of its services (such as certification), and examine the growth in project management software.

History of Project Management

Most people think that project management is a 20th century invention. But Mark Kozak-Holland, certified PMP® and author of books that mine history for insight about project management, says that’s wrong. He notes that major historical projects closely resemble today’s project management best practices. About his 2011 book *The History of Project Management*, he said, “The general perception of most people is that project management started in the mid-20th century, or started earlier with Henry Gantt and his charts. . . . Yet, how were all the great projects of the past delivered? Think about the Giza Pyramid, the Parthenon, the Coliseum, the Gothic Cathedrals of Medieval Europe, the great voyages of exploration, the Taj Mahal, and the mega projects of the industrial revolutions. Was project management used on these projects? Were the concepts of project management even understood? Can we connect modern and ancient project management?” Kozak-Holland’s answer to these questions is “yes.” You can see the PMBOK® process groups and techniques from the knowledge areas in all of these historical projects. Project management has been around since 2550 B.C.E.*

Although people have worked on projects for centuries, most agree that the modern concept of project management began with the Manhattan Project, which the U.S. military led to develop the atomic bomb in World War II. The Manhattan Project involved many people with different skills at several different locations. It also clearly separated the overall management of the project’s mission, schedule, and budget under General Leslie R. Groves and the technical management of the project under the lead scientist, Dr. Robert Oppenheimer. The Manhattan Project lasted about three years and cost almost \$2 billion in 1946.

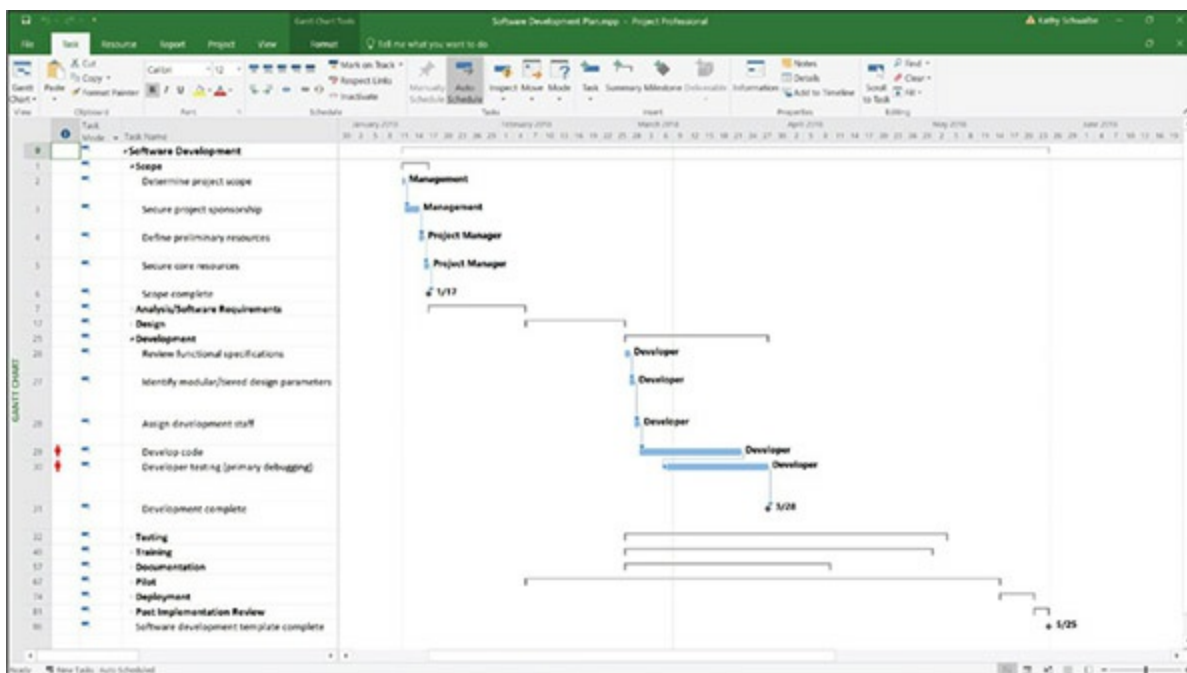
In developing the project, the military realized that scientists and other technical specialists often did not have the desire or the necessary skills to manage large projects. For example, after being asked several times for each team member’s responsibilities at the new Los

Alamos laboratory in 1943, Dr. Oppenheimer tossed the project organization chart at his director and said, “Here’s your damn organization chart.”* Project management was recognized as a distinct discipline requiring people with special skills and, more importantly, the desire to lead project teams.

In 1917, long before the Manhattan project, Henry Gantt developed the famous Gantt chart for scheduling work in factories. A **Gantt chart** is a standard format for displaying project schedule information by listing project activities and their corresponding start and finish dates in calendar form. Initially, managers drew Gantt charts by hand to show project tasks and schedule information. This tool provided a standard format for planning and reviewing all the work on early military projects.

Today’s project managers still use the Gantt chart as the primary tool to communicate project schedule information, but with the aid of computers, it is no longer necessary to draw the charts by hand, and they are easier to share and disseminate to project stakeholders. **Figure 1-5** displays a Gantt chart created with Project 2016, the most widely used project management software today. You will learn more about using Project 2016 in Appendix A, which is available on the Companion website for this text.

Figure 1-5. Sample Gantt chart from Project 2016 software development plan template



During the Cold War years of the 1950s and 1960s, the military continued to play an important role in refining several project management techniques. Members of the U.S. Navy Polaris missile/submarine project first used network diagrams in 1958. These diagrams helped managers model the relationships among project tasks, which allowed them to create schedules that were more realistic. You will learn more about Gantt charts, network diagrams,

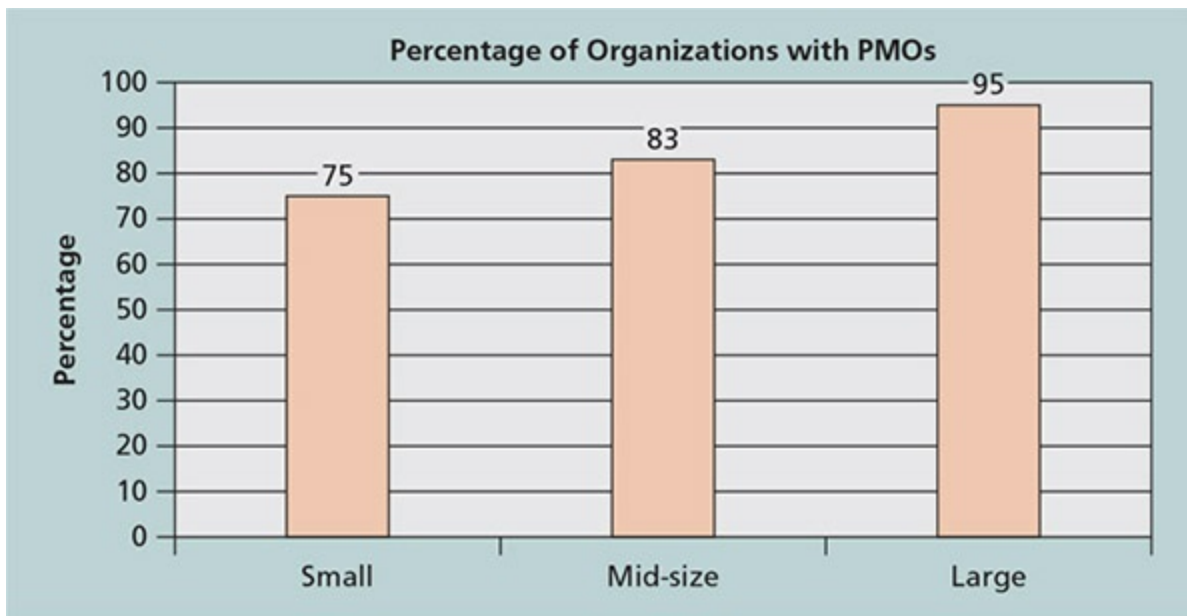
and other time management concepts in [Chapter 6](#), Project Schedule Management.

By the 1970s, the U.S. military and its civilian suppliers had developed software to assist in managing large projects. Early project management software was very expensive to purchase, and it ran exclusively on mainframe computers. For example, Artemis was an early project management software product that helped managers analyze complex schedules for designing aircraft. A full-time employee was often required to run the complicated software, and expensive pen plotters were used to draw network diagrams and Gantt charts.

As computer hardware became smaller and more affordable and software companies developed graphical, easy-to-use interfaces, project management software became less expensive and more widely used. This made it possible—and affordable—for many industries worldwide to use project management software on all types and sizes of projects. New software makes basic tools such as Gantt charts and network diagrams inexpensive, easy to create, and available for anyone to update. See the section later in this chapter on project management software for more information.

In the 1990s, many companies began creating Project Management Offices to help them handle the increasing number and complexity of projects. A [Project Management Office \(PMO\)](#) is an organizational group responsible for coordinating the project management function throughout an organization. A 2016 study found that 85 percent of U.S. organizations reported having PMOs. Figure 1-6 shows the percentage based on size. Small organizations were defined as those having revenues less than \$100 million; mid-size between \$100 million and \$1 billion, and large over \$1 billion. These percentages have grown from previous surveys, showing the growing importance of using standard project management processes in organizations of all sizes. The study also found that PMOs in high-performing organizations were twice as old, on average, as those in low performers (6 vs. 3 years old).*

Figure 1-6. Percentage of organizations with PMOs by size



Source: PM Solutions, "The State of the Project Management Office (PMO) 2016," 2016.

There are different ways to structure a PMO, and they can have various roles and responsibilities. Organizations continue to modify their PMOs to ensure they add value to their unique situations. For some organizations with very mature project management processes and experienced managers, a small PMO focusing on organizing all project data might be all that is needed. For an organization new to project management, a larger PMO might be needed focusing on training and standards. PM Solutions identified three key factors that are playing major roles in the growth of PMOs:

1. The growing strategic value of the PMO
2. The increased role of the PMO in training
3. The ever-present challenge of resource management

Below are possible goals of a PMO:

- Collect, organize, and integrate project data for the entire organization.
- Ensure that the organization's approaches for project management include accepted and validated best practices.
- Audit project documentation and offer feedback on project managers' approaches and compliance with standards.
- Develop and maintain templates, tools, and standards for project documents and project

methodologies to be used.

- Develop or coordinate training in various project management topics.
- Provide a formal career path for project managers.
- Provide project management consulting services.
- Provide a structure or department that project managers belong to while they are assigned to a project or are between projects.

By the end of the 20th century, people in virtually every industry around the globe began to investigate and apply different aspects of project management to their projects. The sophistication and effectiveness with which project management tools are being applied and used today is enabling companies to do business, use resources, and respond to market requirements with greater speed and accuracy.

Many colleges, universities, and companies around the world now offer courses related to different aspects of project management. You can even earn bachelor's, master's, and doctoral degrees in project management. PMI reported that formal education programs in project management continue to grow, especially in China and India, where many infrastructure projects are needed. "In China, for example, the 104 institutions offering project management programs receive more than 20,000 applications each year." As projects become more global and teams are no longer stationed in the same city or even country, students are learning a common project management language no matter where they seek their education.*

The problems organizations have in managing projects, increasing education in project management, and the belief that it can make a difference continue to contribute to the growth of this field.

The Project Management Institute

Although many professional societies suffer from declining membership, the [Project Management Institute \(PMI\)](#), an international professional society for project managers founded in 1969, has continued to attract and retain members, reporting more than 500,000 members worldwide by the end of 2017. Because so many people work on projects in different industries throughout the world, PMI has partnered with projectmanagement.com to create a global online community. The site provides over 14,000 articles by industry experts, over 1,000 templates to save you time and effort, and over 8,000 peer connections through a social networking system and discussion forums.

Global Issues

Based on a survey of more than 1,000 project management leaders across a variety of experience levels and industries, several global dynamics are forcing organizations to rethink their practices:

- Talent development for project and program managers is a top concern. Seventy percent of organizations have a career path for project and program management, but most are still informal and not documented.
- Basic project management techniques are core competencies. Seventy percent of organizations said that they always or often use basic practices like change management and risk management on their projects.
- Organizations want to use more agile approaches to project management. One-quarter of survey respondents said they now use agile techniques, and agile project management was the most requested article topic.
- Benefits realization of projects is a key metric. Organizations know that they need to align projects and programs with the organization's business strategy.*

PMI Student Membership

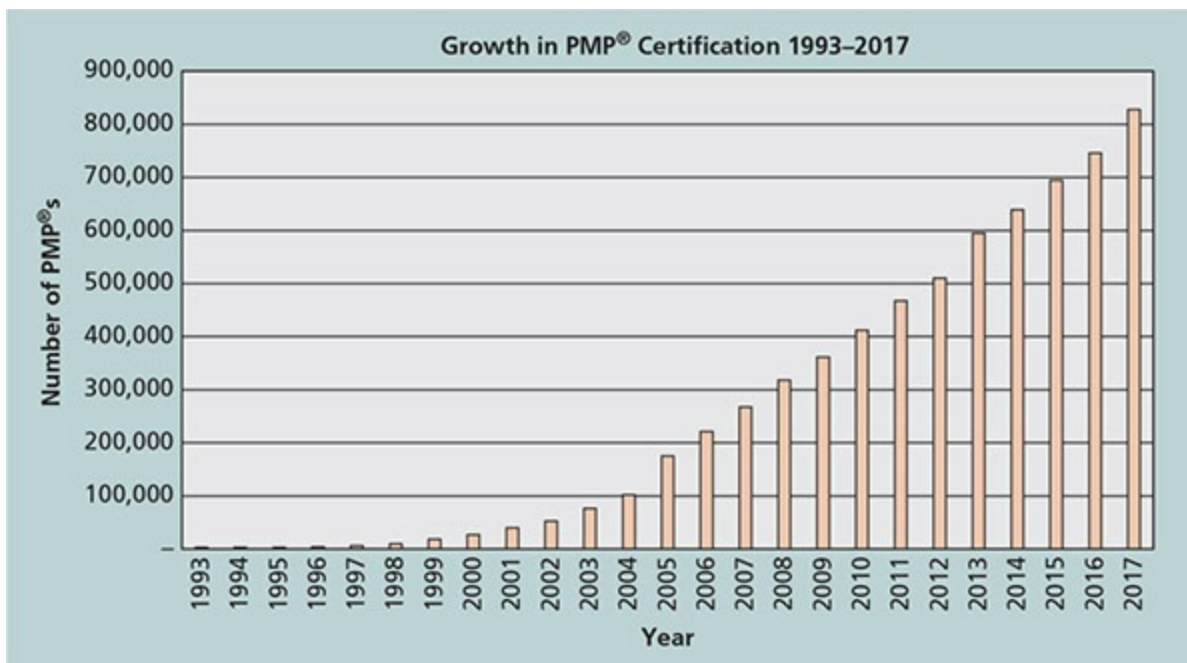
As a student, you can join PMI for a reduced fee (\$32 versus \$139 in 2018). Consult PMI's website (www.pmi.org) for more information. With student membership, you can network with other project management students by joining a local PMI chapter. Many welcome students to attend free events, including talks and job networking. You can volunteer your services to help develop your skills and serve your community. You can also qualify for the Certified Associate in Project Management (CAPM®) certification without work experience. However, if you do have enough experience, the PMP® is more marketable, as described in the next section. Even if you intend to eventually achieve a PMP® certification, achieving an initial CAPM® certification while you are gaining the needed additional experience indicates to many employers that you are serious about project management as a career and that you already know the fundamentals of the profession. This can often result in your being among their preferred choices for appointment to a PM role when it becomes available.

Project Management Certification

Professional certification is an important factor in recognizing and ensuring quality in a profession. PMI provides certification as a **Project Management Professional (PMP®)** — someone who has documented sufficient project experience and education, agreed to follow the PMI code of professional conduct, and demonstrated knowledge of project management by passing a comprehensive examination. Note that you do not need work experience to qualify for CompTIA's Project+ certification or PMI's CAPM® certification, so college graduates just entering the workforce can earn these certifications and become more marketable.

The number of people earning PMP® certification continues to increase. In 1993, there were about 1,000 certified project management professionals. At the end of December 2017, there were 827,960 active PMP®s.* Figure 1-7 shows the rapid growth in the number of people earning PMP® certification from 1993 to 2017.

Figure 1-7. Growth in PMP® certification, 1993–2017



Source: Data points from Project Management Institute, PMI Today® (annual issues).

Several studies show that organizations supporting technical certification programs tend to operate in more complex IT environments and are more efficient than organizations that do not support certification. Likewise, organizations that support PMP® certification see the value of investing in programs to improve their employees' knowledge in project management. Many employers today require specific certifications to ensure that their workers have current skills, and job seekers find that they often have an advantage when they earn and maintain marketable certifications. Global Knowledge listed PMP® certification as number 5 in their list of top-paying certifications for 2017.*

As IT projects become more complex and global in nature, the need for people with demonstrated knowledge and skills in project management will continue. Just as passing the CPA exam is a standard for accountants, passing the PMP® exam is becoming a standard for project managers. Some companies require that all project managers be PMP® certified. Project management certification is also enabling professionals in the field to share a common base of knowledge. For example, any person with PMP® certification can list, describe, and use the 10 project management knowledge areas. Sharing a common base of knowledge is important because it helps advance the theory and practice of project management. PMI also offers additional certifications, including agile techniques, scheduling, risk, program management, portfolio management, and business analysis.

Ethics in Project Management

Ethics, loosely defined, is a set of principles that guides decision making based on personal values of what is considered right and wrong. Making ethical decisions is an important part of project managers' personal and professional lives because it generates trust and respect with other people. Project managers often face ethical dilemmas. If project managers can make more money by taking bribes, should they? No! Should project managers accept subpar work to meet a deadline? No! Ethics guide us in making these types of decisions.

PMI approved a Code of Ethics and Professional Conduct that took effect in January 2007. This code applies not only to PMP®s but to all PMI members who hold a PMI certification, apply for a PMI certification, or serve PMI in a volunteer capacity.

It is vital for project management practitioners to conduct their work in an ethical manner. Even if you are not affiliated with PMI, these guidelines can help you conduct your work in an ethical manner, which helps the profession earn the confidence of the public, employers, employees, and all project stakeholders. The PMI Code of Ethics and Professional Conduct includes short chapters addressing vision and applicability, responsibility, respect, fairness, and honesty. A few excerpts from this document include the following:

“As practitioners in the global project management community:

- | | |
|-------|--|
| 2.2.1 | We make decisions and take actions based on the best interests of society, public safety, and the environment. |
| 2.2.2 | We accept only those assignments that are consistent with our background, experience, skills, and qualifications. |
| 2.2.3 | We fulfill the commitments that we undertake—we do what we say we will do. |
| 3.2.1 | We inform ourselves about the norms and customs of others and avoid engaging in behaviors they might consider disrespectful. |
| 3.2.2 | We listen to others' points of view, seeking to understand them. |

- | | |
|-------|--|
| 3.2.3 | We approach directly those persons with whom we have a conflict or disagreement. |
| 4.2.1 | We demonstrate transparency in our decision-making process. |
| 4.2.2 | We constantly reexamine our impartiality and objectivity, taking corrective action as appropriate. |
| 4.3.1 | We proactively and fully disclose any real or potential conflicts of interest to appropriate stakeholders. |
| 5.2.1 | We earnestly seek to understand the truth. |
| 5.2.2 | We are truthful in our communications and in our conduct.”* |

In addition, PMI added a new series of questions to the PMP® certification exam in March 2002 and continues to include this topic to emphasize the importance of ethics and professional responsibility.

Project Management Software*

The project management and software development communities have definitely responded to the need to provide more software to assist in managing projects. There are hundreds of tools available, ranging from free online or smartphone apps to enterprise tools costing thousands of dollars to implement and high monthly fees per user. Deciding which project management software to use has become a project in itself. Microsoft Project continues to lead the Project Portfolio Management (PPM) market with 35 percent of the \$874 million market, followed by Oracle (19 percent), ServiceNow, Inc. (7 percent), and SAP and Autodesk (5 percent each).*

See Appendix A (which is available on the Companion website for this text) for details on the various configurations available for Microsoft Project and detailed instructions for using Project Professional 2016. This section provides a summary of the basic types of project management software available and references for finding more information.

Free Trials and Information on Using Project 2016, MindView, Basecamp, and Other Software

A 30-day evaluation copy of “Project Online Professional” is available from Microsoft’s website. You can also access a 30-day trial version of MindView software for PCs, Macs, or an online version at www.matchware.com. Basecamp is a totally online project management tool. Educators can request a free Basecamp account without a time restriction from www.basecamp.com.

Note that *Appendix A: Brief Guide to Microsoft Project Professional 2016* is available on the Companion website for this text.

Many people still use basic productivity software such as Microsoft Word and Excel to perform many project management functions, including determining project scope, schedule, and cost, assigning resources, and preparing project documentation. People often use productivity software instead of specialized project management software because they already have it and know how to use it. However, there are hundreds of project management software tools that provide specific functionality for managing projects. These project management software tools can be divided into three general categories based on functionality and price:

- *Low-end tools*: These tools provide basic project management features and generally cost less than \$200 per user or a low monthly fee for online software. They are often recommended for small projects and single users. Most of these tools allow users to create Gantt charts, which cannot be done easily using current productivity software. Some of these tools are available online while others are stand-alone desktop applications. There are also several smartphone applications, and many online tools include smartphone integration. Examples of popular low-end tools include Basecamp, Smartsheet, and Zoho Projects.
- *Midrange tools*: A step up from low-end tools, midrange tools are designed to handle larger projects, multiple users, and multiple projects. All of these tools can produce Gantt charts and network diagrams, and can assist in critical path analysis, resource allocation, project tracking, status reporting, and other tasks. Prices range from about \$200 to \$600 per user or require a monthly fee per user. Microsoft Project (Professional, to be specific) is still the most widely used project management software today in this category and in general. There is also an enterprise or PPM version of Microsoft Project, as described briefly below and in more detail from Microsoft's website.
- *High-end tools*: Another category of project management software is high-end tools, sometimes referred to as PPM or enterprise project management software, as described earlier. These tools provide robust capabilities to handle very large projects, dispersed workgroups, and enterprise and portfolio management functions that summarize and combine individual project information to provide an enterprise view of all projects. These products are generally licensed on a per-user basis, integrate with enterprise database management software, and are accessible via the Internet and smartphones. In mid-2002, Microsoft introduced the first version of its Enterprise Project Management software, and in 2003, it introduced the Microsoft Enterprise Project Management solution, which was updated several times since then. In 2008, Oracle acquired Primavera Software, Inc., another popular tool for project-intensive industries.

Several free or open-source tools are also available. For example, ProjectLibre, LibrePlan, and OpenProject are all free open-source project management tools. Remember, however, that these tools are developed, managed, and maintained by volunteers and may not be well supported.

There are also several software tools that focus on managing agile projects. A January 2018 Google search found a list of 55 agile project management solutions from the site www.softwareadvice.com. Tools listed included Asana, Trello, Jira, Wrike, and VersionOne, to name a few. Of course Microsoft Project and other tools listed earlier can also be used to manage agile projects.

Remember that there is *much* more to managing successful projects than using software.

There are many reasons to study project management, particularly as it relates to IT projects. The number of IT projects continues to grow in almost every industry, the complexity of these projects continues to increase, and the profession of project management continues to expand and mature. As more people study and work in this important field, the success rate of IT projects should continue to improve.

Case Wrap-Up

In addition to overseeing their two high-priority projects (computer security and online collaboration) Anne Roberts worked with the VPs and the CEO to form teams to help identify other potential IT projects that would support their business strategies. They formed a project team to implement a portfolio project management software tool across the organization. They formed another team to develop project-based reward systems for all employees. They also authorized funds for a project to educate all employees in project management and agile, to help people earn PMP® and related certifications, and to develop a mentoring program. Anne had successfully convinced everyone that effectively managing projects was crucial to their company's future.

Chapter Summary

Many people and organizations have a new or renewed interest in project management as the number of projects continues to grow and their complexity continues to increase. The success rate of IT projects has more than doubled since 1995, but still only about one-third are successful in meeting scope, time, and cost goals. Using a more disciplined approach to managing projects can help projects and organizations succeed.

A project is a temporary endeavor undertaken to create a unique product, service, or result. An IT project involves the use of hardware, software, and networks. Projects are unique, temporary, and developed incrementally; they require resources, have a sponsor, and involve uncertainty. The triple constraint of project management refers to managing the scope, time, and cost dimensions of a project. It is important to address these dimensions as well as other constraints (such as quality, resources, and risks) and to satisfy the project sponsor.

Project management is the application of knowledge, skills, tools, and techniques to project activities to meet project requirements. Stakeholders are the people involved in or affected by project activities. A framework for project management includes the project stakeholders, project management knowledge areas, and project management tools and techniques. The 10 knowledge areas are project integration management, scope, schedule, cost, quality, resource, communications, risk, procurement, and stakeholder management. There are many tools and techniques in each knowledge area. There are different ways to define project success, and project managers must understand the criteria that define success for their unique projects.

A program is a group of related projects, subsidiary programs, and program activities managed in a coordinated way to obtain benefits and control that are not available from managing the projects individually. Project portfolio management involves organizing and managing projects and programs as a portfolio of investments that contribute to the entire enterprise's success. Portfolio management emphasizes meeting strategic goals, while project management focuses on tactical goals. Studies show that user involvement is crucial to project success, as are other factors like executive support and clear business objectives.

Project managers play a key role in helping projects and organizations succeed. They must perform various job duties, possess many skills, and continue to develop skills in project management, general management, and their application area, such as IT. Soft skills, especially leadership, are particularly important for project managers.

The profession of project management continues to grow and mature. In the United States, the military took the lead in project management and developed many tools such as Gantt charts and network diagrams, but today people use project management in virtually every

industry around the globe. The Project Management Institute (PMI) is an international professional society that provides several certifications and upholds a code of ethics. Today, hundreds of project management software products are available to assist people in managing projects.

Discussion Questions

1. Why is there a new or renewed interest in the field of project management?
2. What is a project, and what are its main attributes? How is a project different from what most people do in their day-to-day jobs? What is the triple constraint? What other factors affect a project?
3. What is project management? Briefly describe the project management framework, providing examples of stakeholders, knowledge areas, tools and techniques, and project success factors.
4. What is a program? What is a project portfolio? Discuss the relationship between projects, programs, and portfolio management and the contributions that each makes to enterprise success.
5. What is the role of the project manager? What are suggested skills for all project managers and for IT project managers? Why is leadership so important for project managers? How is the job market for IT project managers?
6. Briefly describe some key events in the history of project management. What role do the Project Management Institute and other professional societies play in helping the profession?
7. What functions can you perform with project management software? What are the main differences between low-end, midrange, and high-end project management tools?
8. Discuss ethical decisions that project managers often face. Do you think a professional code of ethics makes it easier to work in an ethical manner?

Quick Quiz

1. By 2027, employers will need over ____ million individuals working in project management-oriented roles.
 - a. 27
 - b. 47

c. 67

d. 87

2. Which of the following is not a potential advantage of using good project management?

a. Shorter development times

b. Higher worker morale

c. Lower cost of capital

d. Higher profit margins

3. A ____ is a temporary endeavor undertaken to create a unique product, service, or result.

a. program

b. process

c. project

d. portfolio

4. Which of the following is not an attribute of a project?

a. Projects are unique.

b. Projects are developed using progressive elaboration.

c. Projects have a primary customer or sponsor.

d. Projects involve little uncertainty.

5. Which of the following is not part of the triple constraint of project management?

a. Meeting scope goals

b. Meeting time goals

c. Meeting communications goals

d. Meeting cost goals

6. ____ is the application of knowledge, skills, tools, and techniques to project activities to

meet project requirements.

- a. Project management
- b. Program management
- c. Project portfolio management
- d. Requirements management

7. Project portfolio management addresses ____ goals of an organization, while project management addresses ____ goals.

- a. strategic, tactical
- b. tactical, strategic
- c. internal, external
- d. external, internal

8. Several application development projects done for the same functional group might best be managed as part of a ____ .

- a. portfolio
- b. program
- c. investment
- d. collaborative

9. Which of the following is not true?

- a. Most American companies have a project management office.
- b. You can earn an advanced degree in project management from hundreds of colleges and universities.
- c. Employers are looking for project management skills in new graduates.
- d. PMI's talent triangle includes leadership and information technology skills along with project management.

10. What is the name of one of the popular certifications provided by the Project Management Institute?

- a. Certified Project Manager (CPM)
- b. Project Management Professional (PMP®)
- c. Project Management Expert (PME)
- d. Project Management Mentor (PMM)

Exercises

1. Read at least two of the first five references cited in this chapter with statistics about the importance of IT and project management. Create a short paper or presentation summarizing which information is most interesting and intriguing to you and why.
2. Find someone who works as a project manager or someone who works on projects involving IT, such as a worker in your school's IT department or a project manager active in a professional group, like PMI. Prepare several interview questions to learn more about projects and project management, and then ask your questions in person, through e-mail, over the phone, or using other technology. Write a summary of your findings.
3. Write a paper summarizing key information available on the PMI's website (www.pmi.org). Also read and summarize two recent reports from PMI, including "Pulse of the Profession®: Success Rates Rise: Transforming the High Cost of Low Performance: (2017). Note: Instructors can break this into two exercises by specifying the second report.
4. Find any example of a real project with a real project manager. Feel free to use projects in the media (such as the Olympics, television shows, or movies) or a project from your work, if applicable. Write a paper describing the project in terms of its scope, time, and cost goals. Also describe other impacts on a project, such as quality, resources, and risks. Discuss what went right and wrong on the project and the role of the project manager and sponsor. Also describe whether the project was a success, and why. Include at least one reference and cite it on the last page.
5. Watch a free online video on the history of project management created by Mark Kozak-Holland (<https://www.youtube.com/watch?v=C1uxCBx2-UQ>). Summarize how the project management knowledge areas can be applied to building the Giza Pyramid Project.
6. Research articles and tools on project portfolio management. Summarize the advantages of performing project portfolio management as well as challenges.
7. Skim through Appendix A on Microsoft Project 2016 (available on the Companion website for this text). Review information about Project 2016 from the Microsoft website

(www.microsoft.com). Research three other project management software tools, including at least one smartphone app. Write a paper answering the following questions:

- a. What functions does project management software provide that you cannot do easily using other tools such as a spreadsheet or database?
 - b. How do the different tools you reviewed compare with Project 2016, based on cost of the tool, key features, and other relevant criteria?
 - c. How can organizations justify investing in enterprise or portfolio project management software?
8. Research information about PMP® and CAPM® certifications. Find at least two articles on this topic. What are benefits of certification in general? Do you think it is worthwhile for most project managers to get certified? Is it something you would consider? Write a paper summarizing your findings and opinions.
9. Review PMI's Code of Ethics and Professional Conduct. Find and summarize two articles related to ethics in project management.

Key Terms

[best practice](#) p. 20

[charismatic](#) p. 20

[DevOps](#) p. 21

[enterprise project management software](#)

[ethics](#) p. 34

[Gantt chart](#) p. 28

[interactional](#) p. 28

[laissez-faire](#) p. 28

[leader](#) p. 26

[manager](#) p. 26

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project portfolio management or portfolio management p. 18

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servant leader p. 7

stakeholders p. 10

transactional p. 10

transformational p. 10

triple constraint p. 7

Chapter 2. The Project Management and Information Technology Context

Learning Objectives

After reading this chapter, you will be able to:

- Define the systems view of project management and how it applies to information technology (IT) projects
- Summarize organizations, including the four frames, organizational structures, and organizational culture
- Explain why stakeholder management and top management commitment are critical for a project's success
- Distinguish between project and product life cycles
- Discuss the unique attributes and diverse nature of IT projects
- Summarize recent trends affecting IT project management, including globalization, outsourcing, virtual teams, and agile project management

Opening Case

Tom Walters was watching the 2018 Super Bowl with friends and noticed players, coaches, and referees using tablets to review plays. It reminded him of a bad experience he had over ten years ago when he tried to convince his college to require students to use tablets. Tom had accepted a new position as the Director of Information Technology at his small, private college after having been a respected faculty member for 15 years. The college offered a variety of programs in the liberal arts and professional areas. Enrollment included 1,500 full-time traditional students and about 1,000 working adults who attended evening programs. Like other institutions of higher learning, the use of IT at the college had grown tremendously, but only a few classrooms on campus had computers for the instructors and students, and most other classrooms had only instructor stations and projection systems. Tom knew that several colleges throughout the country require that all students lease or own laptops or tablets and that these colleges incorporate technology into most courses.

This idea fascinated him at the time. He and two other members of the IT department visited a local college that had required all students to lease laptops for the past three years, and they were very impressed with what they saw and heard. Because tablets were becoming more popular, they thought it would make more sense to require tablets instead of laptops. Tom had heard how easy it was for faculty members to create interactive course materials that would run on tablets; these materials also could help

reduce the cost of textbooks, a concern expressed by many students. Tom and his staff developed plans to start requiring students either to lease or purchase tablets at their college starting the next academic year.

Tom sent an e-mail to all faculty and staff that September briefly describing his plans. He did not get much response, however, until the February faculty meeting. As he described some of the details of his plan, the chairs of the History, English, Philosophy, and Economics departments all voiced opposition to the idea. They eloquently stated that the college was not a technical training school and that they did not have time to write their own course materials to run on tablets. They liked the books they used, and students could already buy books in an electronic format, but most preferred the print versions. Members of the Computer Science department voiced their concern that almost all of their students already had state-of-the-art laptops and would not want to pay a mandatory fee to lease less-powerful tablets. The director of the adult education program expressed her concern that many adult-education students would balk at an increase in fees or required technology. Tom was in shock to hear his colleagues' responses, especially after he and his staff had spent a lot of time planning how to implement tablets at their campus. He remembered being totally confused at the time due to his lack of understanding of organizational change. He wondered how the National Football League handled their implementation of tablets.

Many of the theories and concepts of project management are not difficult to understand. What *is* difficult is implementing them in various environments. Project managers must consider many different issues when managing projects. Just as each project is unique, so is its environment. This chapter discusses some of the concepts involved in understanding the project environment, such as using a systems approach, understanding organizations, managing stakeholders, matching product life cycles to the project environment, understanding the context of IT projects, and reviewing recent trends that affect IT project management.

A Systems View of Project Management

Even though projects are temporary and intended to provide a unique product or service, you cannot run projects in isolation. If project managers lead projects in isolation, it is unlikely that they will ever truly serve the needs of the organization. Therefore, projects must operate in a broad organizational environment, and project managers need to consider projects within the greater organizational context. To handle complex situations effectively, project managers need to take a holistic view of a project and understand how it relates to the larger organization. [Systems thinking](#) describes this holistic view of carrying out projects within

the context of the organization.

What Is a Systems Approach?

The term **systems approach** emerged in the 1950s to describe a holistic and analytical approach to solving complex problems that includes using a systems philosophy, systems analysis, and systems management. Systems are sets of interacting components that work within an environment to fulfill some purpose. For example, the human body is a system composed of many subsystems, including the nervous system, the skeletal system, the circulatory system, and the digestive system. Organizations are also systems, with people in various roles working together to design, develop, deliver, and sell various products and services. A **systems philosophy** is an overall model for thinking about things as systems.

Systems analysis is a problem-solving approach that requires defining the scope of the system, dividing it into components, and then identifying and evaluating its problems, opportunities, constraints, and needs. Once this is completed, the systems analyst then examines alternative solutions for improving the current situation; identifies an optimum, or at least satisfactory, solution or action plan; and examines that plan against the entire system. **Systems management** addresses the business, technological, and organizational issues associated with creating, maintaining, and modifying a system.

Using a systems approach is critical to successful project management. If top management and project managers are to understand how projects relate to the whole organization, they must follow a systems philosophy. They must use systems analysis to address needs with a problem-solving approach. They must use systems management to identify key issues in business, technological, and organizational spheres related to each project in order to identify and satisfy key stakeholders and do what is best for the entire organization.

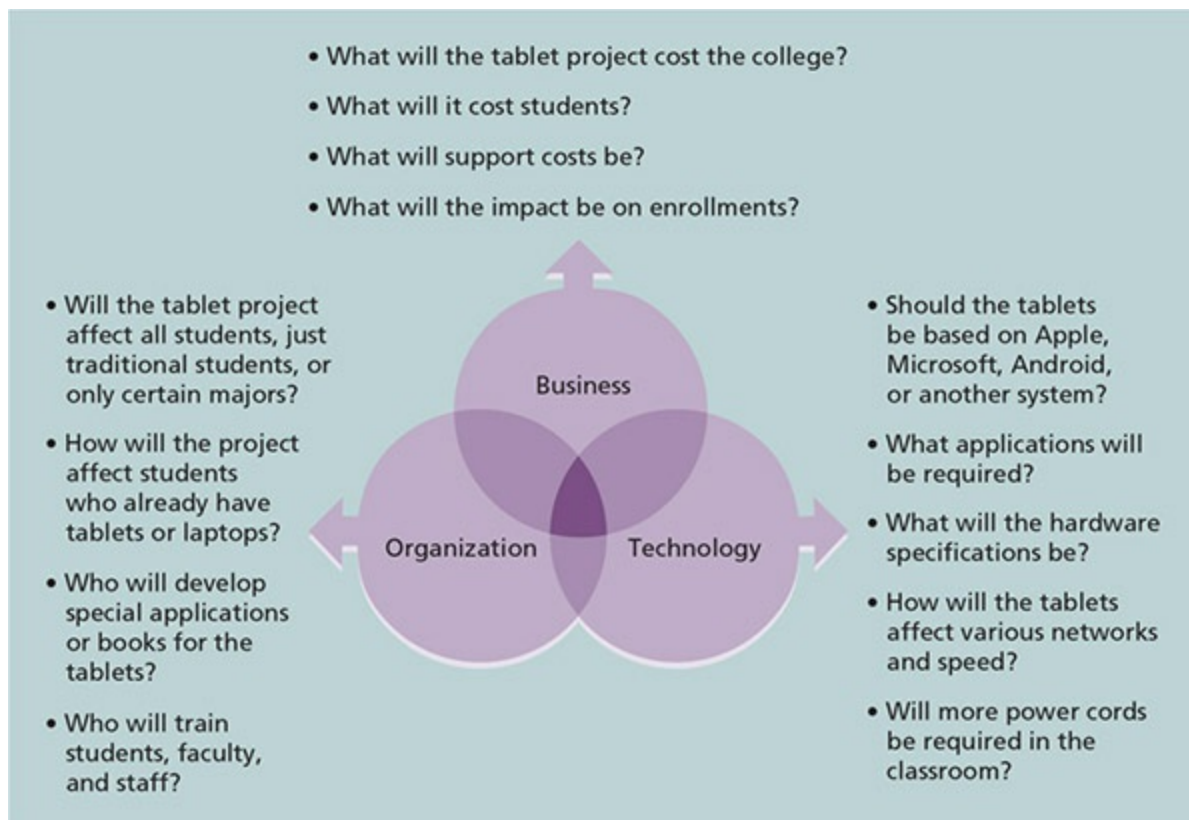
In the chapter's opening case, Tom Walters planned the tablet project without using a systems approach. Members of his IT department did all of the planning. Even though Tom sent an e-mail describing the tablet project to all faculty and staff, he did not address many of the organizational issues involved in such a complex project. Most faculty and staff are very busy at the beginning of the fall term, and many may not have read the entire message. Others may have been too busy to communicate their concerns to the IT department. Tom was unaware of the effects the tablet project would have on other parts of the college. He did not clearly define the business, technological, and organizational issues associated with the project. Tom and the IT department began work on the tablet project in isolation. If they had taken a systems approach, considering other dimensions of the project and involving key stakeholders, they could have identified and addressed many of the issues raised at the February faculty meeting *before* the meeting.

The Three-Sphere Model for Systems Management

Many business and IT students understand the concepts of systems and performing a systems analysis. At the same time, they often overlook systems management. However, addressing the three spheres of systems management—business, organization, and technology—can have a huge impact on selecting and managing projects successfully.

Figure 2-1 provides a sample of business, organizational, and technological issues that could be factors in the tablet project. In this case, technological issues, though not simple by any means, are probably the least difficult to identify and resolve. However, projects must address issues in all three spheres of the systems management model. Although it is easier to focus on the immediate and sometimes narrow concerns of a particular project, project managers and other staff must recognize the effects of any project on the interests and needs of the entire system or organization. The college president and senior administrators, in particular, will focus on whether the tablet project adds value to the college as a whole.

Figure 2-1. Three-Sphere model for systems management



Many IT professionals become captivated with the technology and day-to-day problem solving involved in working with information systems. They tend to become frustrated with many of the “people problems” or politics involved in most organizations. In addition, many IT professionals ignore important business questions, such as “Does it make financial sense to pursue this new technology?” or “Should the company develop this software in-house or

purchase it off the shelf?” Using a more holistic approach helps project managers integrate business and organizational issues into their planning. It also helps them look at projects as a series of interrelated phases. When you integrate business and organizational issues into project management planning and look at projects as a series of interrelated phases, you do a better job of ensuring project success.

Advice for Young Professionals

It’s difficult enough trying to understand the various technologies an organization uses. How can you begin to understand the business and organizational aspects? First of all, make it a priority. Don’t just focus on the technology, no matter how exciting it seems to you. Even if you take just a few minutes each day learning about other aspects of the organization, that’s a start. Second, tell your boss or other people you work with that you want to understand how the entire organization works. Ask important questions like how the company makes money, who key customers are, what the priorities are for the year, what meetings you can attend or documents you can read to gain more knowledge, etc. Third, fourth, and fifth: network, network, network! Find out which people inside or outside of your organization can help you in developing a systems approach. You might be surprised how quickly you can move up in your career once you understand the big picture.

Understanding Organizations

The systems approach requires that project managers always view their projects in the context of the larger organization. Organizational issues are often the most difficult part of working on and managing projects. In fact, many people believe that most projects fail because of organizational issues like company politics. Project managers often do not spend enough time identifying all the stakeholders involved in projects, especially the people opposed to the projects. Also, project managers often do not spend enough time considering the political context of a project or the culture of the organization. To improve the success rate of IT projects, it is important for project managers to develop a better understanding of people as well as organizations.

The Four Frames of Organizations

As shown in **Figure 2-2**, you can try to understand organizations better by focusing on different perspectives. Organizations can be viewed as having four different frames: structural, human resources, political, and symbolic.*

- The **structural frame** deals with how the organization is structured (usually depicted in an organizational chart) and focuses on different groups' roles and responsibilities to meet the goals and policies set by top management. This frame is very rational and focuses on coordination and control. For example, within the structural frame, a key IT issue is whether a company should centralize the IT personnel in one department or decentralize across several departments. You will learn more about organizational structures in the next section.
- The **human resources (HR) frame** focuses on producing harmony between the needs of the organization and the needs of people. It recognizes that mismatches can occur between the needs of the organization and those of individuals and groups, and works to resolve any potential problems. For example, many projects might be more efficient for the organization if employees worked 80 or more hours a week for several months. However, this work schedule would conflict with the personal lives and health of many employees. Important IT issues related to the human resources frame are the shortage of skilled IT workers within the organization and unrealistic schedules imposed on many projects.
- The **political frame** addresses organizational and personal politics. **Politics** in organizations take the form of competition among groups or individuals for power, resources, and leadership. The political frame emphasizes that organizations are coalitions composed of varied individuals and interest groups. Often, important decisions need to be made about the allocation of scarce resources. Competition for resources makes conflict a central issue in organizations, and power improves the ability to obtain those resources. Project managers must pay attention to politics and power if they are to be effective. It is important to know who opposes your projects as well as who supports them. Important IT issues related to the political frame are the differences in power between central functions and operating units or between functional managers and project managers.
- The **symbolic frame** focuses on symbols and meanings. In this frame, the most important aspect of any event in an organization is not what actually happened, but what it means. Was it a good sign that the CEO came to a kick-off meeting for a project, or was it a threat? The symbolic frame also relates to the company's culture. How do people dress? How many hours do they work? How do they run meetings? Many IT projects are international and include stakeholders from various cultures. Understanding those cultures is also a crucial part of the symbolic frame.

Figure 2-2. Perspectives on organizations*

Structural frame: Roles and responsibilities, coordination, and control. Organizational charts help describe this frame.	Human resources frame: Providing harmony between needs of the organization and needs of people.
Political frame: Coalitions composed of varied individuals and interest groups. Conflict and power are key issues.	Symbolic frame: Symbols and meanings related to events. Culture, language, traditions, and image are all parts of this frame.

Source: Bolman and Deal.

*Lee G. Bolman and Terrence E. Deal, *Reframing Organizations* (San Francisco: Jossey-Bass, 1991).

Project managers must learn to work within all four frames to function well in organizations. Organizational issues are discussed further in **Chapter 9**, Project Resource Management, Chapter 10, Project Communications Management, and Chapter 13, Project Stakeholder Management. The following sections on organizational structures, organizational culture, stakeholder management, and the need for top management commitment provide additional information related to the structural and political frames.

What Went Wrong?

In a paper titled “A Study in Project Failure,” two researchers examined the success and failure of 214 IT projects over an eight-year period in several European countries. The researchers found that only one in eight (12.5 percent) were considered successful in terms of meeting scope, time, and cost goals. The authors made the following conclusions about factors that contribute to a project’s failure:

“Our evidence suggests that the *culture* within many organisations is often such that leadership, stakeholder and risk management issues are not factored into projects early on and in many instances cannot formally be written down for *political* reasons and are rarely discussed openly at project board or steering group meetings although they may be discussed at length behind closed doors. . . . Despite attempts to make software development and project delivery more rigorous, a considerable proportion of delivery effort results in systems that do not meet user expectations and are subsequently cancelled. In our view this is attributed to the fact that very few organisations have the infrastructure, education, training, or management discipline to bring projects to successful completion.”*

Organizational Structures

Many discussions of organizations focus on their structure, which can take many forms. Three general classifications of organizational structures are functional, project or project-oriented, and matrix. **Figure 2-3** portrays these three structures, and Table 2-1 describes several more. A **functional organizational structure** is the hierarchy most people think of when picturing an organizational chart. Functional managers or vice presidents in specialties such as engineering, manufacturing, IT, and human resources report to the chief executive officer (CEO). Their staffs have specialized skills in their respective disciplines. For example, most colleges and universities have very strong functional organizations. Only faculty members in the business department teach business courses; faculty in the history department teach history; faculty in the art department teach art, and so on. If the college or university offers graduate and undergraduate programs, they might also have a divisional structure to further distinguish roles and responsibilities for those programs.

Figure 2-3. Functional, project, and matrix organizational structures

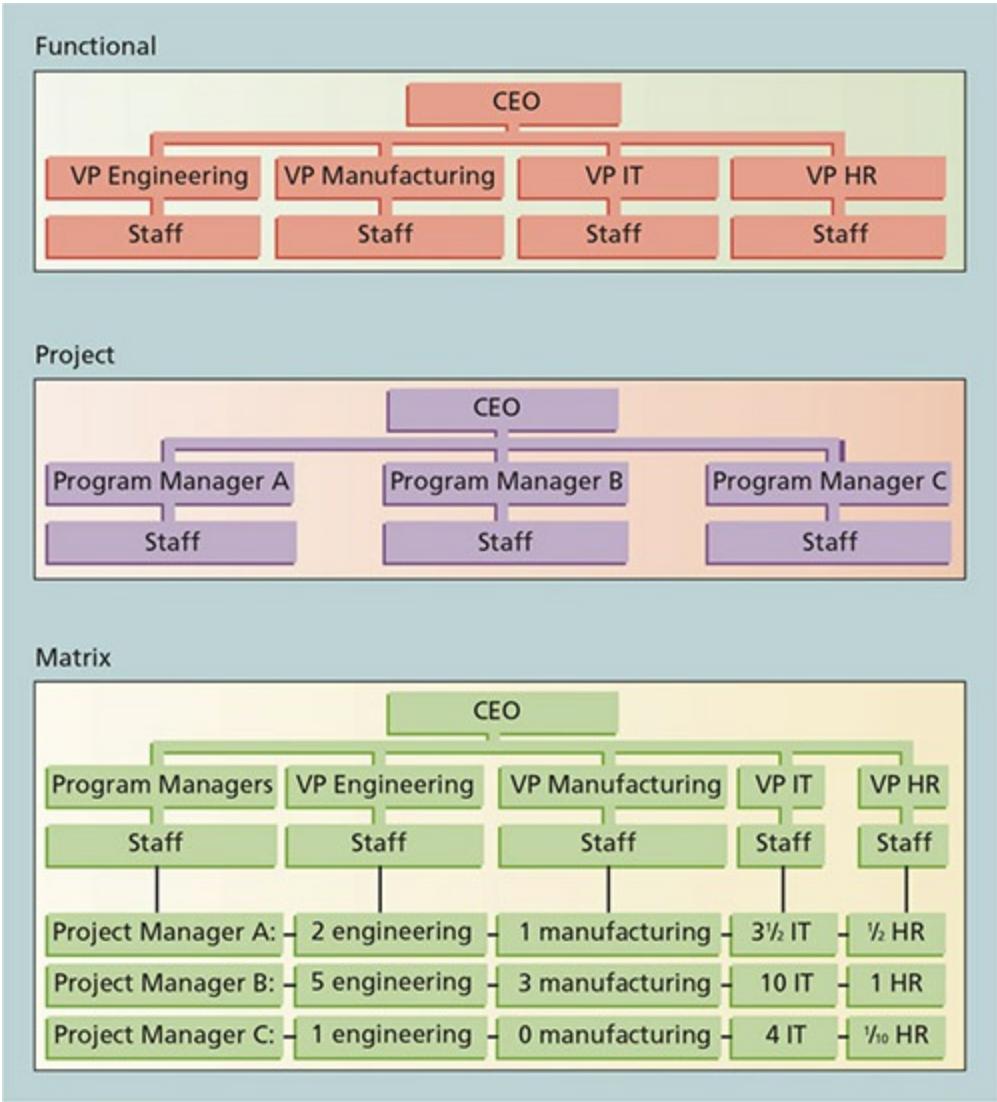


Table 2-1.

Influences of organizational structures on projects

Project Characteristics						
Organizational Structure Type	Work Groups Arranged by:	Project Manager's Authority	Project Manager's Role	Resource Availability	Who Manages the Project Budget?	Project Management Administrative Staff
Organic or Simple	Flexible; people working side-by-side	Little or none	Part-time; may or may not be a designated job role like coordinator	Little or none	Owner or operator	Little or none
Functional (centralized)	Job being done (e.g., engineering, manufacturing)	Little or none	Part-time; may or may not be a designated job role like coordinator	Little or none	Functional manager	Part-time
Multi-divisional (may replicate functions for each division with little centralization)	One of: product; production processes; portfolio; program; geographic region; customer type	Little or none	Part-time; may or may not be a designated job role like coordinator	Little or none	Functional manager	Part-time
Matrix - strong	By job function, with project manager as a function	Moderate to high	Full-time designated job role	Moderate to high	Project manager	Full-time
Matrix - weak	Job function	Low	Part-time; done as part of another job and not a designated job role like coordinator	Low	Functional manager	Part-time
Matrix - balanced	Job function	Low to moderate	Part-time; embedded in the functions as a skill and may not be a designated job role like coordinator	Low to moderate	Mixed	Part-time
Project-oriented (composite, hybrid)	Project	High to almost total	Full-time designated job role	High to almost total	Project manager	Full-time
Virtual	Network structure with nodes at points of	Low to moderate	Full-time or part-time	Low to moderate	Mixed	Could be full-time or part-time

	contact with other people					
Hybrid	Mix of other types	Mixed	Mixed	Mixed	Mixed	Mixed
PMO *	Mix of other types	High to almost total	Full-time designated job role	High to almost total	Project manager	Full-time

Source: Project Management Institute, Inc., *A Guide to the Project Management Body of Knowledge (PMBOK® Guide – Sixth Edition)* (2017).

A **project organizational structure** also is hierarchical, but instead of functional managers or vice presidents reporting to the CEO, program managers report to the CEO. Their staffs have a variety of skills needed to complete the projects within their programs. An organization that uses this structure earns its revenue primarily from performing projects for other groups under contract. For example, many defense, architectural, engineering, and consulting companies use a project organizational structure. These companies often hire people specifically to work on particular projects.

A **matrix organizational structure** represents the middle ground between functional and project structures. Personnel often report both to a functional manager and to one or more project managers. For example, IT personnel at many companies often split their time between two or more projects, but they report to their manager in the IT department. Project managers in matrix organizations have staff from various functional areas working on their projects, as shown in Figure 2-3. Matrix organizational structures can be strong, weak, or balanced, based on the amount of control exerted by the project managers. Problems can occur if project team members are assigned to several projects in a matrix structure and the project manager does not have adequate control of their time.

Table 2-1 summarizes how organizational structures influence projects and project managers, based on the *PMBOK® Guide – Sixth Edition*. Note that several additional organizational structures (besides functional, project oriented, and matrix) are listed, including organic or simple, multi-divisional, virtual, hybrid, and PMO. Project managers have the most authority in a pure project-oriented or PMO organizational structure and the least amount of authority in a pure functional, organic/simple, or multi-divisional organizational structure. It is important that project managers understand their current organizational structure. For example, if someone in a functional organization is asked to lead a project that requires strong support from several different functional areas, he or she should ask for top management sponsorship. This sponsor should solicit support from all relevant functional managers to ensure that they cooperate on the project and that qualified people are available to work as needed. The project manager might also ask for a separate budget to pay for

project-related trips, meetings, and training or to provide financial incentives to the people supporting the project.

Even though project managers have the most authority in the project and PMO organizational structures, this type of organization can be inefficient for the company as a whole. Assigning staff full time to the project often creates underutilization and misallocation of staff resources. For example, if a technical writer is assigned full time to a project, but has no project work on a particular day, the organization is wasting money by paying that person a full-time wage. Project organizations may also miss economies of scale that are available through pooling requests for materials with other projects. Most large organizations, therefore, use a variety of structures. For example, General Motors is first organized divisionally by Global Market (North America, Asia, Europe, etc.), then divisionally by consumer product (Chevy, Buick, Cadillac, etc.), and then within those by function (Marketing, Manufacturing, IT, Accounting, Design, Sales, Quality, Engineering, etc.), and finally within each of those by project. No wonder project managers struggle to understand their organizations!

Disadvantages such as these illustrate the benefit of using a systems approach to managing projects. For example, the project manager might suggest hiring an independent contractor to do the technical writing work instead of using a full-time employee. This approach would save the organization money while still meeting the needs of the project. When project managers use a systems approach, they are better able to make decisions that address the needs of the entire organization.

Organizational Culture

Just as an organization's structure affects its ability to manage projects, so does its culture. Organizational culture is a set of shared assumptions, values, and behaviors that characterize the functioning of an organization. It often includes elements of all four frames described previously. Organizational culture is very powerful, and many people believe that the underlying causes of many companies' problems are not in the organizational structure or staff; they are in the culture. It is also important to note that the same organization can have different subcultures. The IT department may have a different organizational culture than the finance department, for example. Some organizational cultures make it easier to manage projects.

According to Stephen P. Robbins and Timothy Judge, authors of a popular textbook on organizational behavior, there are 10 characteristics of organizational culture:

1. *Member identity*: The degree to which employees identify with the organization as a whole rather than with their type of job or profession. For example, project managers or

team members might feel more dedicated to their company or project team than to their job or profession, or they might not have any loyalty to a particular company or team. As you can guess, an organizational culture in which employees identify more with the whole organization are more conducive to a good project culture.

2. *Group emphasis*: The degree to which work activities are organized around groups or teams, rather than individuals. An organizational culture that emphasizes group work is best for managing projects.
3. *People focus*: The degree to which management's decisions take into account the effect of outcomes on people within the organization. A project manager might assign tasks to certain people without considering their individual needs, or the project manager might know each person very well and focus on individual needs when assigning work or making other decisions. Good project managers often balance the needs of individuals and the organization.
4. *Unit integration*: The degree to which units or departments within an organization are encouraged to coordinate with each other. Most project managers strive for strong unit integration to deliver a successful product, service, or result. An organizational culture with strong unit integration makes the project manager's job easier.
5. *Control*: The degree to which rules, policies, and direct supervision are used to oversee and control employee behavior. Experienced project managers know it is often best to balance the degree of control to get good project results.
6. *Risk tolerance*: The degree to which employees are encouraged to be aggressive, innovative, and risk seeking. An organizational culture with a higher risk tolerance is often best for project management because projects often involve new technologies, ideas, and processes.
7. *Reward criteria*: The degree to which rewards, such as promotions and salary increases, are allocated according to employee performance rather than seniority, favoritism, or other nonperformance factors. Project managers and their teams often perform best when rewards are based mostly on performance.
8. *Conflict tolerance*: The degree to which employees are encouraged to air conflicts and criticism openly. It is very important for all project stakeholders to have good communications, so it is best to work in an organization where people feel comfortable discussing differences openly.
9. *Means-ends orientation*: The degree to which management focuses on outcomes rather than on techniques and processes used to achieve results. An organization with a

balanced approach in this area is often best for project work.

10. *Open-systems focus*: The degree to which the organization monitors and responds to changes in the external environment. As you learned earlier in this chapter, projects are part of a larger organizational environment, so it is best to have a strong open-systems focus.*

As you can see, there is a definite relationship between organizational culture and successful project management. Project work is most successful in an organizational culture where employees identify more with the organization, where work activities emphasize groups, and where there is strong unit integration, high risk tolerance, performance-based rewards, high conflict tolerance, an open-systems focus, and a balanced focus on people, control, and means orientation.

Focusing on Stakeholder Needs

Recall from **Chapter 1** that project stakeholders are the people involved in or affected by project activities. Stakeholders can be internal or external to the organization, directly involved in the project, or simply affected by the project. Internal project stakeholders include the project sponsor, project team, support staff, and internal customers of the project. Other internal stakeholders include top management, other functional managers, and other project managers. Projects affect these additional internal stakeholders because they use the organization's limited resources. Thus, while additional internal stakeholders may not be directly involved in the project, they are still stakeholders because the project affects them in some way. External project stakeholders include the project's customers (if they are external to the organization), competitors, suppliers, and other external groups potentially involved in the project or affected by it, such as government officials or concerned citizens.

Because the purpose of project management is to meet project requirements and satisfy stakeholders, it is critical that project managers take adequate time to identify, understand, and manage relationships with all project stakeholders. Using the four frames of organizations to think about project stakeholders can help you meet their expectations. See Chapter 13, Project Stakeholder Management, for more information.

Consider again the tablet project from the opening case. Tom Walters seemed to focus on just a few internal project stakeholders. He viewed only part of the structural frame of the college. Because his department would do most of the work in administering the tablet project, he concentrated on those stakeholders. He did not even involve the main customers for this project—the students at the college. Even though Tom sent an e-mail to faculty and staff, he did not hold meetings with senior administrators or faculty at the college. Tom's view of the project stakeholders was very limited.

During the faculty meeting, it became evident that the tablet project had many stakeholders in addition to the IT department and students. If Tom had expanded his view of the structural frame of his organization by reviewing an organizational chart for the entire college, he could have identified other key stakeholders. He would have been able to see that the project would affect academic department heads and members of different administrative areas, especially if he wanted faculty members to develop customized course materials themselves. If Tom had focused on the human resources frame, he would have been able to tap into his knowledge of the school and identify people who would most support or oppose requiring tablets. By using the political frame, Tom could have considered the main interest groups that would be most affected by the project's outcome. Had he used the symbolic frame, Tom could have tried to address what moving to a tablet environment would really mean for the college. He then could have anticipated some of the opposition from people who were not in favor of increasing the use of technology on campus. He also could have solicited a strong endorsement from the college president or dean before talking at the faculty meeting.

Tom Walters, like many new project managers, learned the hard way that technical and analytical skills were not enough to guarantee success in project management. To be more effective, he had to identify and address the needs of different stakeholders and understand how his project related to the entire organization. And unlike the NFL, his college was not being paid to use tablets, as described in the following [Media Snapshot](#).

Media Snapshot

Prior to the 2014 football season, Microsoft paid the NFL \$400 million as part of a five-year deal to use their Surface as “the official tablet of the NFL.” The Microsoft logo is on the tablets as well as the replay monitors used by officials. The Surfaces replaced banks of printers used in the past to print out images of each play, a slow, wasteful process. The Surfaces run a special app that links with cameras to provide instant video of plays. Users can review plays over and over again and mark up images with a stylus. All 32 NFL teams were involved, and the deal was renewed for a sixth year in 2017. Yusuf Mehdi, corporate vice president at Microsoft, said, “We’re excited to help the NFL change the game with Surface devices being used by players and coaches on the sidelines to make more informed decisions, referees using Surface for instant replay to enhance the speed of the game, and teams using Microsoft products in their business and football operations.”*

Was it a smooth transition to using the tablets? Not quite. During week one of the season at least two television announcers mistakenly referred to the tablets as iPads, giving Apple unexpected exposure. Microsoft also had to defend the use of tablets after the New England Patriots stopped using them. Coach Bill Belichick “ranted that he was ‘done with the tablets’ during a five-minute-long expression of pure frustration at the Surface

tablets and the technology surrounding them. 'I'm going to stick with pictures, which several of our other coaches do, as well, because there just isn't enough consistency in the performance of the tablets. I just can't take it anymore,' said Belichick, weeks after smashing a Surface tablet on the sidelines."*

The Importance of Top Management Commitment

A very important factor in helping project managers successfully lead projects is the level of commitment and support they receive from top management. Without this commitment, many projects will fail. Some projects have a senior manager called a [champion](#) who acts as a key advocate for a project. The sponsor can serve as the champion, but often another manager can more successfully take on this role. As described earlier, projects are part of the larger organizational environment, and many factors that might affect a project are out of the project manager's control. Several studies cite executive support as one of the key factors associated with virtually all project success.

Top management commitment is crucial to project managers for the following reasons:

- Project managers need adequate resources. The best way to kill a project is to withhold the required money, human resources, and visibility. If project managers have top management commitment, they will also have adequate resources and not be distracted by events that do not affect their specific projects.
- Project managers often require approval for unique project needs in a timely manner. For example, on large IT projects, top management must understand that unexpected problems may result from the nature of the products being developed and the specific skills of people on the project team. The team might need additional hardware and software halfway through the project for proper testing, or the project manager might need to offer special pay and benefits to attract and retain key project personnel. With top management commitment, project managers can meet these needs.
- Project managers must have cooperation from people in other parts of the organization. Because most IT projects cut across functional areas, top management must help project managers deal with the political issues that often arise. If certain functional managers are not responding to project managers' requests for necessary information, top management must step in to encourage the functional managers to cooperate.
- Project managers often need someone to mentor and coach them on leadership issues. Many IT project managers come from technical positions and are inexperienced as managers. Senior managers should take the time to give advice on how to be good

leaders. They should encourage new project managers to take classes to develop leadership skills and allocate the time and funds for managers to do so.

IT project managers work best in an environment in which top management values IT. Working in an organization that values good project management and sets standards for its use also helps project managers succeed.

The Need for Organizational Commitment to Information Technology

Another factor that affects the success of IT projects is the organization's commitment to IT in general. It is very difficult for an IT project to be successful if the organization itself does not value IT. Many companies have realized that IT is integral to their business and have created a vice president or equivalent position for the head of IT, often called the Chief Information Officer (CIO). Some companies assign people from non-IT areas to work full time on large projects and increase involvement from end users of the systems. Some CEOs even take a strong leadership role in promoting the use of IT in their organizations and empower employees to use IT effectively.

Best Practice

A major element of good practice concerns [IT governance](#), which addresses the authority for and control of key IT activities in organizations, including IT infrastructure, IT use, and project management. (The term *project governance* can also be used to describe a uniform method of controlling all types of projects.) The IT Governance Institute (ITGI) was established in 1998 to advance international thinking and standards in directing and controlling an organization's use of technology. Effective IT governance helps ensure that IT supports business goals, maximizes investment in IT, and addresses IT-related risks and opportunities. A 2004 book by Peter Weill and Jeanne Ross titled *IT Governance: How Top Performers Manage IT Decision Rights for Superior Results* * includes research indicating that firms with superior IT governance systems have 20 percent higher profits than firms with poor governance.

A lack of IT governance can be dangerous, as evidenced by three well-publicized IT project failures in Australia: Sydney Water's customer relationship management system, the Royal Melbourne Institute of Technology's academic management system, and One. Tel's billing system. Researchers explained how these projects were catastrophic for their organizations, primarily due to a severe lack of IT governance, which the researchers dubbed *managerial IT unconsciousness* in a subsequent article:

“All three projects suffered from poor IT governance. Senior management in all three organizations had not ensured that prudent checks and balances were in place to enable them to monitor either the progress of the projects or the alignment and impact of the new systems on their business. Proper governance, particularly with respect to financial matters, auditing, and contract management, was not evident. Also, project-level planning and control were notably absent or inadequate—with the result that project status reports to management were unrealistic, inaccurate, and misleading.”*

The leadership style of the CIO plays a crucial role in gaining organizational commitment to IT as well as motivation and support for IT workers. A survey found that 76 percent of CIOs in companies in Europe, the Middle East, and Africa (EMEA) need to adapt their leadership style to fully embrace digital business. “Command-and-control leadership doesn’t suit this digital world,” said Dave Aron, vice president and Gartner Fellow. “In fact, it can be an obstacle. Vision and inspiration are typically the most powerful attributes of digital leaders. CIOs must accept to flip from ‘control first’ to vision first. In EMEA, 65 percent of CIOs said that they need to decrease their time on commanding IT, while 45 percent of them said they need to increase their visionary leadership.”*

Empowering employees at all levels to effectively use IT is also crucial. For example, Hilton Worldwide won a prestigious Customer Relationship Management (CRM) award by enabling its employees to create their own solution for improving customer service and loyalty. In addition to using the company’s Satisfaction and Loyalty Tracking (SALT) customer analytics software to deliver key information in a timely manner, team members created a more personal process to focus on using data to improve the guest experience called HEART: Hear the Guest; Empathize with the Guest; Apologize to the Guest; Resolve the Issue; and Thank the Guest. By following this process along with timely data, Hilton Worldwide dramatically increased its customer loyalty score, which leads to higher profits.*

The Need for Organizational Standards

Another problem in most organizations is a lack of standards or guidelines to follow when performing project management. These standards or guidelines might be as simple as providing standard forms or templates for common project documents, examples of good project management plans, or guidelines for how project managers should provide status information to top management. The content of a project management plan and instructions for providing status information might seem like common sense to senior managers, but many new IT project managers have never created plans or created a nontechnical status report. Top management must support the development of these standards and guidelines, and encourage or even enforce their use. For example, an organization might require all

potential project information to be reported in a standard format to make project portfolio management decisions. If a project manager does not submit a potential project in the proper format, it could be rejected.

As you saw in **Chapter 1**, some organizations invest heavily in project management by creating a project management office or center of excellence, which assists project managers in achieving project goals and maintaining project governance. Rachel Hollstadt, founder and retired CEO of a project management consulting firm, suggests that organizations consider adding a new position, a Chief Project Officer (CPO). Some organizations develop career paths for project managers; some require that all project managers have Project Management Professional (PMP®) certification and that all employees have some type of project management training. The implementation of such standards demonstrates an organization's commitment to project management.

Project and Product Life Cycles

Because projects operate as part of a system and involve uncertainty, it is good practice to divide projects into several phases. The same can be said for developing products. This section describes the various project and product life cycles.

Project Life Cycle

A **project life cycle** is a collection of project phases. Projects pass through these phases from their start to their completion. The *PMBOK® Guide – Sixth Edition* describes the generic life cycle to include the following four phases:

1. Starting the project
2. Organizing and preparing
3. Carrying out the work
4. Finishing the project

These phases should not be confused with the project management process groups of initiating, planning, executing, monitoring and controlling, and closing, as described in **Chapter 3**.

In general, project life cycles define what work will be performed in each phase, what deliverables will be produced and when, who is involved in each phase, and how management will control and approve work produced in each phase. A **deliverable** is a product or service, such as a technical report, a training session, a piece of hardware, or a segment of software

code, produced or provided as part of a project. (See Chapter 5, Project Scope Management, for detailed information on deliverables.)

In early phases of a project life cycle, resource needs are usually lowest and the level of uncertainty is highest. Project stakeholders have the greatest opportunity to influence the final characteristics of the project's products, services, or results during the early phases of a project life cycle. It is much more expensive to make major changes to a project during latter phases. During the middle phases of a project life cycle, the certainty of completing the project improves as it continues, and as more information is known about the project requirements and objectives. Also, more resources are usually needed than during the initial or final phase. The final phase of a project focuses on ensuring that project requirements were met and that the project sponsor approves completion of the project.

Product Life Cycles

Just as a *project* has a life cycle, so does a *product*. IT projects help develop products and services such as new software, hardware, networks, research reports, and training on new systems. A [product life cycle](#) is a process used to define, create, and deliver products. Developing a product, such as a new information system, a car, a building, and so on, often involves many projects.

All products follow some type of life cycle—cars, buildings, even amusement parks. The Walt Disney Company, for example, follows a rigorous process to design, build, and test new products. Disney assigns project managers to oversee the development of all new products, such as rides, parks, and cruise lines. Likewise, major automotive companies follow product life cycles to create new cars, trucks, and other products.

Most IT professionals are familiar with the concept of a product life cycle, especially for developing software. Software development projects are one subset of IT projects. Many IT projects involve researching, analyzing, and then purchasing and installing new hardware and software with little or no actual software development required. However, some projects involve minor modifications to enhance existing software or to integrate one application with another. Other projects involve a major amount of software development. A [systems development life cycle \(SDLC\)](#) is a framework for describing the phases of developing information systems.

The *PMBOK® Guide – Sixth Edition* briefly describes five product or development life cycles. Two factors are important in deciding which life cycle to use: the degree of change in requirements and the frequency of delivery of useful results. For example, for a product with a low degree of change in requirements and low frequency of delivery, a predictive life cycle would be appropriate.

1. *Predictive life cycle*: The scope, schedule, and cost are determined early, and changes to scope are carefully managed. PMI also refers to predictive life cycles as waterfall.
2. *Iterative life cycle*: The scope is determined early, but time and cost estimates are modified as the understanding of the product increases. Iterations are used to develop the product through a series of repeated cycles to add to the functionality of the product. This approach works best when there is a high degree of change and a low frequency of delivery.
3. *Incremental life cycle*: Deliverables are produced through a series of iterations that add functionality within a set time frame. The deliverable is not complete until after the final iteration. This approach works best when there is a low degree of change and a high frequency of delivery.
4. *Adaptive life cycle*: Stakeholders define and approve the detailed scope before the start of an iteration, producing a useable product at the end of each iteration. PMI also refers to adaptive life cycles as agile or change-driven. This approach works best when there is a high degree of change and a high frequency of delivery.
5. *Hybrid life cycle*: A combination of approaches is used based on the nature of the work. For example, some deliverables might have a low degree of change and low frequency of delivery such as weekly progress reports, a high degree of change and a high frequency of delivery such as certain software features, and so on.

Many organizations today use a hybrid approach to product development where a predictive set of steps is used as an overall means to coordinate more detailed steps that are adaptively managed. Predictive and adaptive approaches are not a mutually exclusive choice.

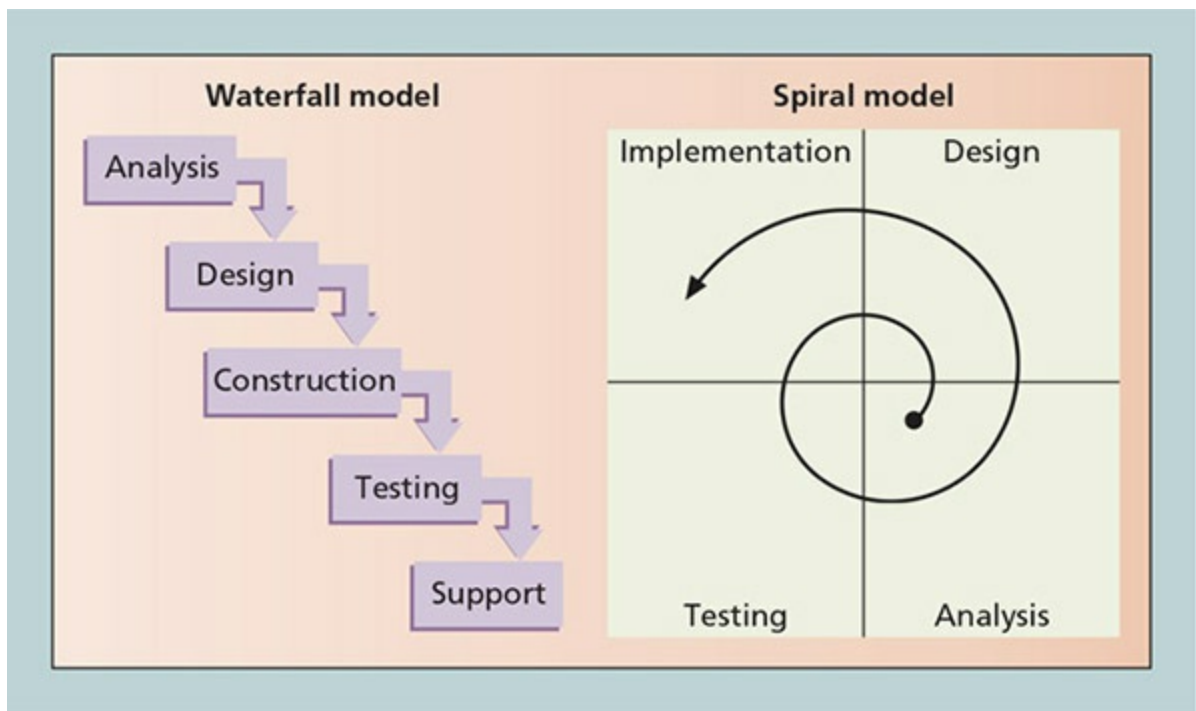
In addition to the waterfall model for the SDLC, other predictive life cycles include the spiral model, the prototyping model, and the Rapid Application Development (RAD) model. The project team spends a large portion of the project attempting to clarify the requirements of the entire system and then producing a design. Users are often unable to see any tangible results in terms of working software for an extended period. Below are brief descriptions of several predictive SDLC models.*

- The waterfall life cycle model has well-defined, linear stages of systems analysis, design, construction, testing, and support. This life cycle model assumes that requirements will remain stable after they are defined. The waterfall life cycle model is used when risk must be tightly controlled and when changes must be restricted after the requirements are defined. The waterfall approach is used in many large-scale systems projects where complexity and cost are so high that the more rigid steps of the approach help to ensure

careful completion of all deliverables. The waterfall approach also makes sense for IT projects that do not involve software development, such as upgrading all the routers and switches in a company to support VoIP phones.*

- The spiral life cycle model was developed based on refinements of the waterfall model as applied to large government software projects. It recognizes the fact that most software is developed using an iterative or spiral approach rather than a linear approach. The project team is open to changes and revisions later in the project life cycle, and returns to the requirements phase to more carefully clarify and design the revisions. This approach is suitable for projects in which changes can be incorporated with reasonable cost increases or with acceptable time delays. **Figure 2-4** illustrates the differences between the waterfall and spiral life cycle models.

Figure 2-4. Waterfall and spiral life cycle models



- The prototyping life cycle model is used for developing software prototypes to clarify user requirements for operational software. It requires heavy user involvement, and developers use a model to generate functional requirements and physical design specifications simultaneously. Developers can throw away or keep prototypes, depending on the project. This approach is often used in systems that involve a great deal of user interface design, such as website projects, in systems that automate previously manual functions, or in systems that change the nature of how something is done, such as mobile applications.
- The RAD life cycle model uses an approach in which developers work with an evolving

prototype. This life cycle model also requires heavy user involvement and helps produce systems quickly without sacrificing quality. Developers use RAD tools such as CASE (computer-aided software engineering), JRP (joint requirements planning), and JAD (joint application design) to facilitate rapid prototyping and code generation. These tools are often used in reporting systems in which programmers enter parameters into software to generate reports for user approval. When approved, the same parameters will generate the final production system without further modification by the programmer.

The Importance of Project Phases and Management Reviews

Due to the complexity and importance of many IT projects and their resulting products, it is important to take time to review the status of a project at each phase. A project should successfully pass through each of the main project phases before continuing to the next. Because the organization usually commits more money as a project continues, a management review should occur after each phase to evaluate progress, potential success, and continued compatibility with organizational goals.

Management reviews, called [phase exits](#) , [phase gate reviews](#) , or [kill points](#) , are very important for keeping projects on track and determining if they should be continued, redirected, or terminated. Recall that projects are just one part of the entire system of an organization. Changes in other parts of the organization might affect a project's status, and a project's status might likewise affect events in other parts of the organization. By breaking projects into phases, top management can make sure that the projects are still compatible with other needs of the organization.

Take another look at the opening case. Suppose Tom Walters' college conducted a study sponsored by the college president on increasing the use of technology. At the end of the concept phase, the project team could have presented information to the president, faculty, and other staff members that described different options for increasing the use of technology, an analysis of what competing colleges were doing, and results of a survey of local stakeholders' opinions on the subject. This presentation at the end of the concept phase represents one form of a management review. Suppose the study reported that 90 percent of surveyed students, faculty, and staff strongly opposed the idea of requiring all students to have tablets, and that many adult students said they would attend other colleges if they were required to pay for the additional technology. The college would probably decide not to pursue the idea any further. Had Tom taken a phased approach, he and his staff would not have wasted time and money developing detailed plans.

In addition to formal management reviews, it is important to have top management involvement throughout the life cycle of most projects. It is unwise to wait until the end of project or product phases to have management inputs. Many projects are reviewed by management on a regular basis, such as weekly or even daily, to make sure they are progressing well. Everyone wants to be successful in accomplishing goals at work, and having management involvement ensures that a company can accomplish its project and organizational goals.

What Went Right?

Having specific deliverables and kill points at the end of project or product phases helps managers make better decisions about whether to proceed, redefine, or kill a project. Improvement in IT project success rates reported by the Standish Group has been due in part to an increased ability to know when to cancel failing projects. Standish Group Chairman Jim Johnson made the following observation: “The real improvement that I saw was in our ability to—in the words of Thomas Edison—know when to stop beating a dead horse. . . . Edison’s key to success was that he failed fairly often; but as he said, he could recognize a dead horse before it started to smell. . . . In information technology we ride dead horses—failing projects—a long time before we give up. But what we are seeing now is that we are able to get off them; able to reduce cost overrun and time overrun. That’s where the major impact came on the success rate”*

Another example of the power of management oversight comes from Huntington Bancshares, Inc. This company, like many others, has an [executive steering committee](#), a group of senior executives from various parts of the organization who regularly review important corporate projects and issues. The Ohio-based, \$26 billion bank holding company completed a year-long website redesign using XML technology to give its online customers access to real-time account information as well as other banking services. The CIO, Joe Gottron, said there were “four or five very intense moments” when the whole project was almost stopped due to its complexity. The executive steering committee met weekly to review the project’s progress and discuss work planned for the following week. Gottron said the meetings ensured that “if we were missing a beat on the project, no matter which company [was responsible], we were on top of it and adding additional resources to make up for it.”*

Some projects still go on for a long time before being killed. Blizzard, producer of the popular massive multiplayer online (MMO) game World of Warcraft, decided to cancel the Titan game project after spending over seven years in development. According to Blizzard co-founder and CEO Mike Morhaime, “We set out to make the most ambitious thing that you could possibly imagine. And it didn’t come together. We didn’t find the fun.

We didn't find the passion. . . . We'd rather cut out a game we put a lot of time and resources into than put out something that might. . . ." Chris Metzen, Blizzard's senior vice president of story and franchise development, finished Morhaime's sentence: "Damage the relationship. Smash the trust."*

The Context of Information Technology Projects

The project context has a critical impact on which product development life cycle will be most effective for a particular software development project. Likewise, several issues unique to the IT industry have a critical impact on managing IT projects. These include the nature of projects, the characteristics of project team members, and the diverse nature of technologies involved.

The Nature of IT Projects

Unlike projects in many other industries, IT projects are diverse. Some involve a small number of people installing off-the-shelf hardware and associated software. Others involve hundreds of people analyzing several organizations' business processes and then developing new software in a collaborative effort with users to meet business needs. Even for small hardware-oriented projects, a wide diversity of hardware types can be involved—personal computers, mainframe computers, network equipment, kiosks, laptops, tablets, or smartphones. The network equipment might be wireless, cellular based, or cable based, or might require a satellite connection. The nature of software development projects is even more diverse than hardware-oriented projects. A software development project might include creating a simple, stand-alone Microsoft Excel or Access application or a sophisticated, global e-commerce system that uses state-of-the-art programming languages and runs on multiple platforms.

IT projects also support every possible industry and business function. Managing an IT project for a film company's animation department requires different knowledge and skills than a project to improve a federal tax collection system or to install a communication infrastructure in a third-world country. Because of the diversity of IT projects and the newness of the field, it is important to develop and follow best practices in managing these varied projects. Developing best practices gives IT project managers a common starting point and method to follow with every project.

Characteristics of IT Project Team Members

Because IT projects are diverse, the people involved come from diverse backgrounds and

possess different skills. The resulting diverse project teams provide a significant advantage because they can analyze project requirements from a more robust systems view. Many companies purposely hire graduates with degrees in other fields such as business, mathematics, or the liberal arts to provide different perspectives on IT projects. Even with these different educational backgrounds, however, there are common job titles for people working on most IT projects, such as business analyst, programmer, network specialist, database analyst, quality assurance expert, technical writer, security specialist, hardware engineer, software engineer, and system architect. Within the category of programmer, several other job titles describe the specific technologies used, such as Java programmer, PHP programmer, and C/C++/C# programmer.

Some IT projects require the skills of people in just a few job functions, but some require inputs from many or all of them. Occasionally, IT professionals move between these job functions, but more often people become technical experts in one area or they decide to move into a management position. It is also rare for technical specialists or project managers to remain with the same company for a long time. In fact, many IT projects include a large number of contract workers. Working with this “army of free agents,” as author Rob Thomsett calls them, creates special challenges.

Diverse Technologies

Many of the job titles for IT professionals reflect the different technologies required to hold those positions. Differences in technical knowledge can make communication between professionals challenging. Hardware specialists might not understand the language of database analysts, and vice versa. Security specialists may have a hard time communicating with business analysts. People within the same IT job function often do not understand each other because they use different technology. For example, someone with the title of programmer can often use several different programming languages. However, if programmers are limited in their ability to work in multiple languages, project managers might find it more difficult to form and lead more versatile project teams.

Another problem with diverse technologies is that many of them change rapidly. A project team might be close to finishing a project when it discovers a new technology that can greatly enhance the project and better meet long-term business needs. New technologies have also shortened the time frame many businesses have to develop, produce, and distribute new products and services. This fast-paced environment requires equally fast-paced processes to manage and produce IT projects and products.

Recent Trends Affecting Information Technology Project Management

Recent trends such as increased globalization, outsourcing, virtual teams, and agile project management are creating additional challenges and opportunities for IT project managers and their teams. Each of these trends and suggestions for addressing them are discussed in this section.

Globalization

In his popular book *The World Is Flat*, Thomas L. Friedman describes the effects of globalization, which has created a “flat” world where everyone is connected and the “playing field” is level for many more participants.* Lower trade and political barriers and the digital revolution have made it possible to interact almost instantaneously with billions of other people across the planet, and for individuals and small companies to compete with large corporations. Friedman also discusses the increase in “uploading,” in which people share information through blogging, podcasts, and open-source software.

IT is a key enabler of globalization, as shown in the following statistics. Note that total worldwide population was over 7.6 billion at the end of 2017.

- “The internet has 3.5 billion users
- There are 3.03 billion active social media users
- 91% of retail brands use 2 or more social media channels
- 81% of all small and medium businesses use some kind of social platform
- Internet users have an average of 7.6 social media accounts
- Social media users grew by 121 million between Q2 2017 and Q3 2017
- That works out at a new social media user every 15 seconds
- Facebook Messenger and Whatsapp handle 60 billion messages a day”*

Globalization has significantly affected the field of IT. Even though major IT companies such as Microsoft and IBM started in the United States, much of their business is global—indeed, companies and individuals throughout the world contribute to the growth of information technologies, and work and collaborate on various IT projects.

It is important for project managers to address several key issues when working on global projects:

- *Communications*: Because people work in different time zones, speak different languages,

have different cultural backgrounds, and celebrate different holidays, it is important to address how people will communicate in an efficient and timely manner. A communications management plan is vital. For details, see the plan described in Chapter 10, Project Communications Management.

- *Trust*: Trust is an important issue for all teams, especially when they are global teams. It is important to start building trust immediately by recognizing and respecting others' differences and the value they add to the project.
- *Common work practices*: It is important to align work processes and develop a modus operandi with which everyone agrees and is comfortable. Project managers must allow time for the team to develop these common work practices. Using special tools, as described next, can facilitate this process.
- *Tools*: IT plays a vital role in globalization, especially in enhancing communications and work practices. Many people use free tools such as Skype, Google Docs, or social media to communicate. Many project management software tools include their own communications and collaboration features in an integrated package. IBM continues to be the leader in providing collaboration tools to businesses in over 175 countries, followed by Oracle in 145 countries, SAP in 130 countries, and Microsoft in 113 countries.* Work groups must investigate options and decide which tools will work best for their projects. Security is often a key factor in deciding which tools to use.

After researching over 600 global organizations, KPMG International summarized several suggestions for managing global project teams:

- Employ greater project discipline for global projects; otherwise, weaknesses within the traditional project disciplines may be amplified by geographical differences.
- Think globally but act locally to align and integrate stakeholders at all project levels.
- Consider collaboration over standardization to help balance the goals and project approach.
- Keep momentum going for projects, which will typically have a long duration.
- Consider the use of newer, perhaps more innovative, tools and technology.*

Outsourcing

As described in detail in Chapter 12, Project Procurement Management, [outsourcing](#) is an organization's acquisition of goods and services from an outside source. The term [offshoring](#)

is sometimes used to describe outsourcing from another country. Offshoring is a natural outgrowth of globalization. IT projects continue to rely more and more on outsourcing, both within and outside their country boundaries.

Some organizations remain competitive by using outsourcing to their advantage. Many organizations have found ways to reduce costs by outsourcing, even though the practice can be unpopular in their home countries. For example, outsourcing was an important topic in the 2012 U.S. Republican presidential debates, as candidates discussed why Apple hires half a million low-paid workers in the Far East to assemble its products. A *New York Times* article explained that outsourcing is not just about low costs. “One former executive described how [Apple] relied upon a Chinese factory to revamp iPhone manufacturing just weeks before the device was due on shelves. Apple had redesigned the iPhone’s screen at the last minute, forcing an assembly line overhaul. New screens began arriving at the plant near midnight. A foreman immediately roused 8,000 workers inside the company’s dormitories, according to the executive. Each employee was given a biscuit and a cup of tea, guided to a workstation and within half an hour started a 12-hour shift fitting glass screens into beveled frames. Within 96 hours, the plant was producing over 10,000 iPhones a day. ‘The speed and flexibility is breathtaking,’ the executive said. ‘There’s no American plant that can match that.’”*

Because of the increased use of outsourcing for IT projects, project managers need to become more familiar with many global and procurement issues, including working on and managing virtual teams.

Virtual Teams

Several factors, such as the cost and time required for travel or employee relocation, the ability to communicate and work across vast distances, the advantages of hiring people in locations that have a lower cost of living, and worker preferences for flexible work hours, have contributed to a significant increase in virtual project teams. A [virtual team](#) is a group of people who work together despite time and space boundaries using communication technologies. Team members might all work for the same company in the same country, or they might include employees as well as independent consultants, suppliers, or even volunteers providing their expertise from around the globe.

The main advantages of virtual teams include the following:

- Lowering costs because many virtual workers do not require office space or support beyond their home offices
- Providing more expertise and flexibility or increasing competitiveness and responsiveness by having team members across the globe working any time of day or

night

- Improving the balance between work and life for team members by eliminating fixed office hours and the need to travel to work

Global Issues

Outsourcing also has disadvantages. For example, Apple benefits from manufacturing products in China, but it had big problems there after its iPhone 4S launch in January 2012 caused fighting between migrant workers who were hired by scalpers to stand in line to buy the phones. When Apple said it would not open its store in Beijing, riots resulted and people attacked security guards. The Beijing Apple Store has had problems before. In May 2011, four people were injured when a crowd waiting to buy the iPad 2 turned ugly. Market analysts blamed Apple for not marketing or distributing its products well in China.* In 2017, Apple continued to have problems with China as sales and market share there have been declining.

Disadvantages of virtual teams include the following:

- Isolating team members who may not adjust well to working in a virtual environment
- Increasing the potential for communications problems because team members cannot use body language or other nonverbal communications to understand each other and build relationships and trust
- Reducing the ability for team members to network and transfer information informally
- Increasing the dependence on technology to accomplish work

Like any team, a virtual team should focus on achieving a common goal. Research on virtual teams reveals a growing list of factors that influence their success:

- *Team processes*: It is important to define how the virtual team will operate. For example, teams must agree on how and when work will be done, what technologies will be used, how decisions will be made, and other important process issues.
- *Leadership style*: The project manager's leadership style affects all teams, especially virtual ones.
- *Trust and relationships*: Many virtual teams fail because of a lack of trust. It is difficult to

build relationships and trust from a distance. Some project managers like to have a face-to-face meeting so team members can get to know each other and build trust. If such a meeting is not possible, phone or video conferences can help.

- *Team member selection and role preferences:* Dr. Meredith Belbin defined a team role as “a tendency to behave, contribute and interrelate with others in a particular way.”* It is important to select team members carefully and to form a team in which all roles are covered. All virtual team members must also understand their roles on the team. (Visit www.belbin.com for more information on this topic.)
- *Task-technology fit:* IT is more likely to have a positive impact on individual performance if the capabilities of the technologies match the tasks that the user must perform.
- *Cultural differences:* It is important to address cultural differences, including how people with authority are viewed, how decisions are made, how requests or questions are communicated, and how workers prefer to operate (in collaboration or individually). These cultural differences vary from location to location and affect many aspects of the team.
- *Computer-mediated communication:* It is crucial to provide reliable and appropriate computer-mediated communication to virtual team members, including e-mail, instant messaging, text messaging, and chat rooms. If you rely on these technologies to bring the virtual team together, you need to ensure that they actually work, or you risk increasing the distance that can exist across virtual boundaries.
- *Team life cycles:* Just as projects and products have life cycles, so do teams. Project managers must address the team life cycle, especially when assigning team members and determining deliverable schedules.
- *Incentives:* Virtual teams may require different types of incentives to accomplish high-quality work on time. They do not have the benefit of physical contact with their project managers or other team members, so it is important to provide frequent positive incentives, such as a thank-you via e-mail or phone, or even a bonus on occasion. Negative incentives, such as fines or withholding payment, can also be effective if virtual team members are not being productive.
- *Conflict management:* Even though they might never physically meet, virtual teams still have conflicts. It is important to address conflict management, as described in more detail in Chapter 9, Project Resource Management.

Several studies have tried to determine factors that are correlated positively to the effectiveness of virtual teams. Research suggests that team processes, trusting relationships,

leadership style, and team member selection provide the strongest correlations to team performance and team member satisfaction.*

Agile

Earlier the adaptive or agile approach to product development was discussed. [Agile](#) means being able to move quickly and easily, but some people feel that project management, *as they have seen it used*, does not allow people to work quickly or easily. Early software development projects often used a waterfall approach, but as technology and businesses became more complex, the approach often became difficult to use because requirements were unknown or continuously changing. Agile today means using an approach where requirements and solutions evolve through collaboration. Agile can be used for software development or in any environment in which the requirements are unknown or change quickly.

Interest in agile continues to increase in the project management community. The *PMBOK® Guide – Sixth Edition* includes brief information describing considerations for agile/adaptive environments for each knowledge area. The Standard for Project Management includes an appendix called Agile, Iterative, Adaptive, and Hybrid Project Environments. PMI also partnered with the Agile Alliance® to publish the Agile Practice Guide in 2017.

It is interesting to note that PMI and many other sources now focus on organizational agility, not just specific agile techniques, like Scrum, described later in this section. “Organizational agility is not just about being fast, it also implies the capacity to remain in touch with customer needs. Transforming is much more than the technology going agile. It is a change in mindset across the organization to focus on becoming adaptable and driving business value to customers....People and processes are important for higher levels of agility, and culture is the key that unlocks agility’s value.”*

As emphasized in this chapter and hopefully throughout the text, projects are not done in isolation. Project managers and their teams must use a systems view to make their projects and their organizations successful.

The Manifesto for Agile Software Development

In the business world, the term *agile* was first applied to software development projects. In February 2001, a group of 17 people that called itself the Agile Alliance developed and agreed on the Manifesto for Agile Software Development, as follows:

We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value the following:

- Individuals and interactions over processes and tools
- Working software over comprehensive documentation
- Customer collaboration over contract negotiation
- Responding to change over following a plan*

The person or organization that implements Agile is responsible for interpreting and applying the preceding values.

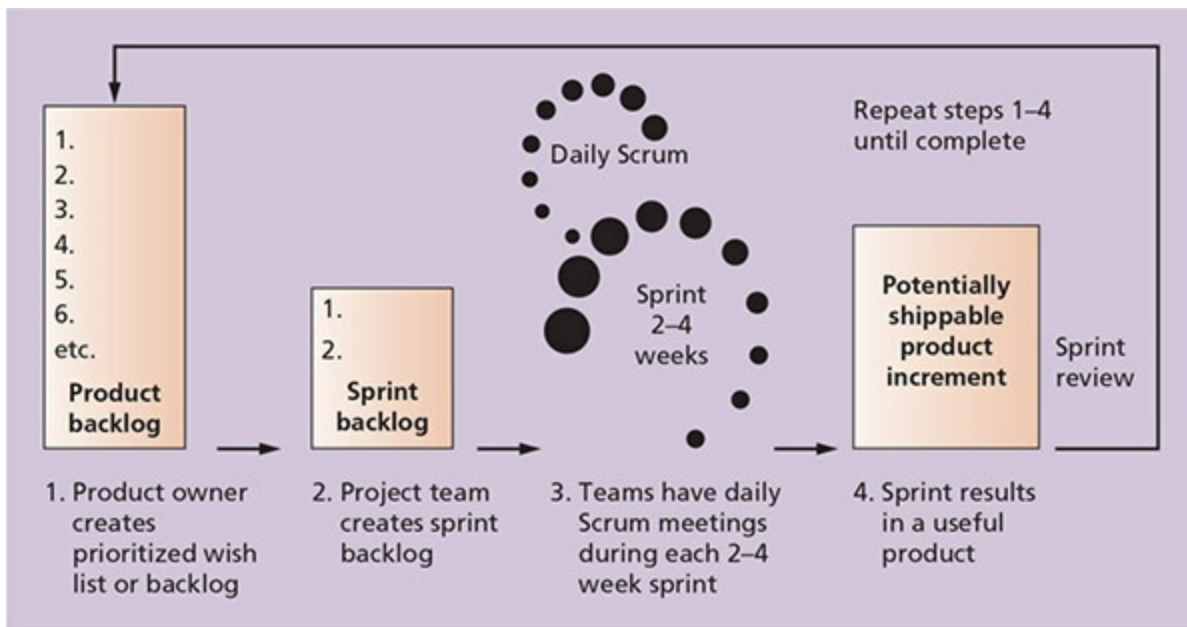
Some people associate Agile with specific techniques such as Scrum.

Scrum

According to the Scrum Alliance, [Scrum](#) is the leading agile development method for completing projects with a complex, innovative scope of work. The term was coined in 1986 in a Harvard Business Review study that compared high-performing, cross-functional teams to the Scrum formation used by rugby teams. The basic Scrum framework is summarized in the following list and illustrated in Figure 2-5:

- A *product owner* creates a prioritized wish list called a *product backlog*.
- During *sprint planning*, the team pulls a small chunk from the top of that wish list, a *sprint backlog*, and decides how to implement those pieces.
- The team has a certain amount of time, a *sprint*, to complete its work—usually two to four weeks—but meets each day to assess its progress (*daily Scrum*).
- Along the way, the *ScrumMaster* keeps the team focused on its goal.
- At the end of the sprint, the work should be *potentially shippable*, as in ready to hand to a customer, put on a store shelf, or show to a stakeholder.
- The sprint ends with a *sprint review* and *retrospective*.
- As the next sprint begins, the team chooses another chunk of the product backlog and begins working again.

Figure 2-5. Scrum framework



The cycle repeats until enough items in the product backlog have been completed, the budget is depleted, or a deadline arrives. Which of these milestones marks the end of the work is entirely specific to the project. No matter which of these reasons stops work, Scrum ensures that the most valuable work has been completed when the project ends.*

In the Axosoft video “Scrum in 10 Minutes,” Hamid Shojaee, an experienced software developer who has worked with several major corporations, briefly explains key concepts like product backlogs, team roles, sprints, and burndown charts.* Techniques from the just-in-time inventory control method [Kanban](#) can be used in conjunction with Scrum. Kanban was developed in Japan by Toyota Motor Corporation. It uses visual cues to guide workflow. For example, teams can place cards on boards to show the status of work in the backlog, such as new, in progress, and complete. Cards on the board are moved to the right to show progress in completing work. Kanban also helps limit work in progress by making a bottleneck visible so people can collaborate to solve problems that created the bottleneck. Kanban helps improve day-to-day workflow, while Scrum provides the structure for improving the organization of projects.* Scrum was initially applied to software development projects, but today other types of projects use this technique to help focus on teamwork, complete the most important work first, and add business value.

Agile, the *PMBOK® Guide*, and a New Certification

The *PMBOK® Guide* describes best practices for *what* should be done to manage projects. Agile is a methodology that describes *how* to manage projects. One could view Agile and the Scrum framework as methods that break down a big project into several smaller projects, defining the scope for each one. Project teams can have brief meetings each day to decide how

to get the most important work done first without calling the meetings “Scrums.”

As stated earlier in the chapter, several different methods are related to developing information systems and other products. Because projects are unique, someone must decide what processes are needed and how they should be performed. Project teams can follow one specific process, a hybrid of several, or their own customized approach. Organizations must also address the larger topic of organizational agility.

The Project Management Institute (PMI) recognized the increased interest in Agile, and introduced a new certification in 2011 called Agile Certified Practitioner (ACP). As stated on the PMI website in early 2018, “The PMI-ACP® is our fastest growing certification, and it’s no wonder. Organizations that are highly agile and responsive to market dynamics complete more of their projects successfully than their slower-moving counterparts—75 percent versus 56 percent—as shown in our 2015 Pulse of the Profession® report.”* At the end of November 2017, there were 18,329 people who earned the ACP Certification.*

One reason for increased interest in Agile is the hope that it will somehow make project management easier. Many books, courses, and consultants are capitalizing on this “new” approach. However, seasoned project managers understand that they have always had the option of customizing how they run projects. They also know that project management is not easy, even when using Agile. Organizations have also understood that they need to strive for a culture that allows them to move quickly to succeed in a rapidly changing world.

As you can see, working as an IT project manager or team member is an exciting and challenging job. The excitement and challenge come from the focus on successfully completing projects that will have a positive impact on the organization as a whole.

Case Wrap-Up

When the Super Bowl ended, Tom told his friends how the tablet idea for his college was killed. After several people voiced concerns about the idea at the faculty meeting, the president of the college directed that a committee be created to formally review the concept of requiring students to have tablets. Because the college was dealing with several important enrollment-related issues, the president named the vice president of enrollment to head the committee. Other people soon volunteered or were assigned to the committee, including Tom Walters as head of Information Technology, the director of the adult education program, the chair of the Computer Science department, and the chair of the History department. The president also insisted that the committee include at least two members of the student body. The president knew everyone was busy, and he questioned whether the tablet idea was a high-priority issue for the college. He directed the committee to present a proposal at next month’s faculty meeting, either to

recommend creating a formal project team to fully investigate requiring tablets or to recommend terminating the concept. At the next faculty meeting, few people were surprised to hear the recommendation to terminate the concept. Tom Walters learned that he had to pay much more attention to the needs of the entire college before proceeding with detailed IT plans. Tom and his friends found several articles online about the NFL's use of Microsoft Surface tablets. They laughed when they read about the Patriots refusing to use them after a while and their coach smashing one on the sidelines. Tom said, "If Coach Belichick had one at the end of this game, he would have smashed another one!"

Chapter Summary

Projects operate in an environment broader than the project itself. Project managers need to take a systems approach in order to successfully manage projects; they need to consider projects within the greater organizational context. To ensure project success, IT project managers need to integrate business and organizational issues as well as technology into project planning.

Organizations have four different frames: structural, human resources, political, and symbolic. Project managers need to understand all of these aspects of organizations to be successful. The structural frame focuses on different groups' roles and responsibilities to meet the goals and policies set by top management. The human resources frame focuses on producing harmony between the needs of the organization and the needs of people. The political frame addresses organizational and personal politics. The symbolic frame focuses on symbols and meanings.

The structure of an organization has strong implications for project managers, especially regarding the amount of authority they have. The three basic organizational structures are functional, matrix, and project. Project managers have the most authority in a pure project organization, an intermediate amount of authority in a matrix organization, and the least amount of authority in a pure functional organization.

Organizational culture also affects project management. Projects are more likely to succeed in a culture where employees have a strong identity with the organization, where work activities emphasize groups, and where there is strong unit integration, high risk tolerance, performance-based rewards, high conflict tolerance, an open-systems focus, and a balance among the dimensions of people focus, control, and means orientation.

Project stakeholders are individuals and organizations who are actively involved in the project or whose interests may be positively or negatively affected by the project's execution or successful completion. Project managers must identify and understand the different needs of all stakeholders involved with their projects.

Top management commitment is crucial for project success. Because projects often affect many areas in an organization, top management must assist project managers if they are to do a good job of project integration. Organizational commitment to IT is also important to the success of IT projects. Development standards and guidelines assist most organizations in managing projects.

A project life cycle is a collection of phases. Traditional project phases include concept,

development, implementation, and close-out. Projects often develop products, which follow product life cycles. Project managers must understand the specific life cycles of their products as well as the general project life cycle model.

A project should successfully pass through each project phase in order to continue to the next phase. A management review should occur at the end of each project phase, and more frequent management inputs are often needed. These management reviews and inputs are important for keeping projects on track and determining if projects should be continued, redirected, or terminated.

Project managers need to consider several factors affected by the unique context of IT projects. The diverse nature of these projects and the wide range of business areas and technologies involved make IT projects especially challenging to manage. Leading project team members with a variety of specialized skills and understanding rapidly changing technologies are also important considerations.

Several recent trends have affected IT project management. Increased globalization, outsourcing, virtual teams, and agile project management have changed the way many IT projects are staffed and managed. Project managers must stay abreast of these and other trends and discover ways to use them effectively.

Discussion Questions

1. What does it mean to take a systems view of a project? How does taking this view apply to project management?
2. Explain the four frames of organizations. How can they help project managers understand the organizational context for their projects?
3. Briefly explain the differences between functional, matrix, and project organizations. Describe how each structure affects the management of a project.
4. Describe how organizational culture is related to project management. What type of culture promotes a strong project environment?
5. Discuss the importance of top management commitment and the development of standards for successful project management. Provide examples to illustrate the importance of these items based on your experience on any type of project.
6. What are the phases in a traditional project life cycle? How does a project life cycle differ from a product life cycle? Why does a project manager need to understand both?
7. What makes IT projects different from other types of projects? How should project

managers adjust to these differences?

8. Define globalization, outsourcing, virtual teams, and agile project management, and describe how these trends are changing IT project management.

Quick Quiz

1. Which of the following is not part of the three-sphere model for systems management?
 - a. Business
 - b. Information
 - c. Technology
 - d. Organization
2. Which of the four frames of organizations addresses how meetings are run, employee dress codes, and expected work hours?
 - a. Structural
 - b. Human resources
 - c. Political
 - d. Symbolic
3. Personnel in a ____ organizational structure often report to two or more bosses.
 - a. functional
 - b. project
 - c. matrix
 - d. hybrid
4. Project work is most successful in an organizational culture where all of the following characteristics are important except ____ .
 - a. member identity
 - b. group emphasis
 - c. risk tolerance

- d. control
5. A ____ is a product or service, such as a technical report, a training session, or hardware, produced or provided as part of a project.
- a. deliverable
 - b. product
 - c. work package
 - d. tangible goal
6. Which of the following statements is false?
- a. An analysis project life cycle is a collection of project phases.
 - b. A product life cycle is the same as a project life cycle.
 - c. The waterfall approach is an example of a predictive life cycle model.
 - d. Agile is an example of an adaptive life cycle model.
7. Which of the following terms describes a framework of the phases involved in developing information systems?
- a. Systems development life cycle
 - b. Rapid application development
 - c. Predictive life cycle
 - d. Extreme programming
8. The nature of IT projects is different from the nature of projects in many other industries because they are very ____ .
- a. expensive
 - b. technical
 - c. diverse
 - d. challenging
9. What term describes an organization's acquisition of goods and services from an outside

source in another country?

- a. Globalization
- b. Offshoring
- c. Exporting
- d. Global sourcing

10. ____ is the leading agile development method.

- a. Extreme programming
- b. Sprint
- c. Kanban
- d. Scrum

Exercises

1. Summarize the three-sphere model for systems management in your own words. Then use your own project experience or interview someone who recently completed an IT project, and list several business, technology, and organizational issues addressed during the project. Which issues were most important to the project, and why? Summarize your answers in a short paper or presentation.
2. Apply the four frames of organizations to a possible project that involves the development of a new technology like mobile banking, online retail, or social media. Work with two other class members in a virtual environment on this exercise. Write a short paper or presentation that summarizes your analysis and opinions of how working virtually helped or hindered your results.
3. Search the Internet for two interesting articles about software development life cycles, including agile development. Review the website www.agilealliance.org. What do these sources say about project management? Write a summary of your findings and opinions on this topic, and cite your references.
4. Search the Internet and scan IT industry magazines or websites to find an example of an IT project that had problems due to organizational issues. Write a short paper summarizing the key stakeholders for the project and how they influenced the outcome.
5. Write a short summary of an article that discusses the importance of top management support for successful IT projects. Your summary should include your opinion on this

topic.

6. Research the trend of using virtual teams. Review the information on team role theory from www.belbin.com and other related sources. Write a summary of your findings, and cite at least three references. Also include any personal experience and your opinion on the topic. For example, what role(s) would you prefer to play on a team? Do you like working on virtual teams? If you have not yet worked on one, how do you think it would be different from working on a face-to-face team?
7. Research the agile movement as it applies to software development and other types of projects. See how many books and websites are available on the topic. Is there evidence to show that agile projects are more successful than others? What agile principles can be applied to all types of projects? Write a short paper that summarizes your findings, and cite at least three references. Also include any personal experience and your opinion on the topic.
8. Watch videos about Scrum and Kanban by Axosoft or other organizations. Summarize what you learned and any questions you have in a short paper. Try to find answers to your questions and cite your sources.
9. Review the Opening and Closing case about Tom Walters tablet project. Also find at least two articles about the NFL's adoption of tablets. Using the four frames of organizations presented in this chapter, contrast the structural, human resource, political, and symbolic frames as they relate to adopting tablets for Tom's college versus the NFL. What do you think the main factors were in canceling Tom's project early versus continuing the NFL tablet project?

Key Terms

[agile](#) p. 69

[champion](#) p. 54

[deliverable](#) p. 57

[executive steering committee](#) p. 63

[functional organizational structure](#) p. 49

[human resources \(HR\) frame](#) p. 47

[IT governance](#) p. 55

[Kanban](#) p. 71

[kill point](#) p. 62

[matrix organizational structure](#) p. 50

[offshoring](#) p. 66

[organizational culture](#) p. 51

[outsourcing](#) p. 66

[phase gate review phase exit](#) p. 62

[political frame](#) p. 48

[politics](#) p. 48

[predictive life cycle](#)

[product life cycle](#) p. 57

[project life cycle](#) p. 57

[Project organizational structure](#) p. 50

[Scrum](#) p. 70

[structural frame](#) p. 47

[symbolic frame](#) p. 48

[systems analysis](#) p. 45

[systems approach](#) p. 45

[systems development life cycle \(SDLC\)](#) p. 60

[systems management](#) p. 45

[systems philosophy](#) p. 45

[systems thinking](#) p. 45

[virtual team](#) p. 67

Chapter 4. Project Integration Management

Learning Objectives

After reading this chapter, you will be able to:

- Describe an overall framework for project integration management as it relates to the other project management knowledge areas and the project life cycle
- Discuss the strategic planning process and apply different project selection methods
- Explain the importance of creating a project charter to formally initiate projects
- Describe project management plan development, understand the content of these plans, and describe approaches for creating them
- Explain project execution, its relationship to project planning, the factors related to successful results, and tools and techniques to assist in directing and managing project work
- Apply the principles of knowledge management to the various aspects of project integration
- Describe the process of monitoring and controlling a project
- Define the integrated change control process, relate this to the steps for planning for and managing changes on information technology (IT) projects, and create an appropriate change control system for a project that incorporates both
- Explain the importance of developing and following good procedures for closing projects
- Describe how software can assist in project integration management
- Discuss considerations for agile/adaptive environments

Opening Case

Nick Carson recently became project manager of a critical biotech enterprise at his Silicon Valley company. This project involved creating the hardware and software for a next generation (next-gen) DNA-sequencing instrument used in assembling and analyzing the human genome. Several companies were competing to build smaller, faster sequencing instruments that would reduce the costs and improve the quality of data analysis in this rapidly changing field. The biotech project was the company's largest endeavor, and it had tremendous potential for future growth and revenue.

Unfortunately, there were problems managing this large project. It had been under way

for three years and had already gone through three different project managers. Nick had been the lead software developer on the project before top management made him the project manager. The CEO told him to do whatever it took to deliver the first version of the product in four months and a production version in nine months. Negotiations for a potential corporate buyout with a larger company influenced top management's sense of urgency to complete the project.

Highly energetic and intelligent, Nick had the technical background to make the project a success. He delved into the technical problems and found some critical flaws that kept the next-gen DNA-sequencing instrument from working. Nevertheless, he was having difficulty in his new role as project manager. Although Nick and his team got the product out on time, top management was upset because Nick did not focus on managing all aspects of the project. He never provided them with accurate schedules or detailed plans of what was happening on the project. Instead of performing the work of project manager, Nick had taken on the role of software integrator and troubleshooter. Nick, however, did not understand top management's complaints—he delivered the product, didn't he? Didn't they realize how valuable he was?

What is Project Integration Management?

Project integration management involves coordinating all of the other project management knowledge areas throughout a project's life cycle. This integration ensures that all the elements of a project come together at the right times to complete a project successfully. According to the *PMBOK® Guide – Sixth Edition*, six main processes are involved in project integration management:

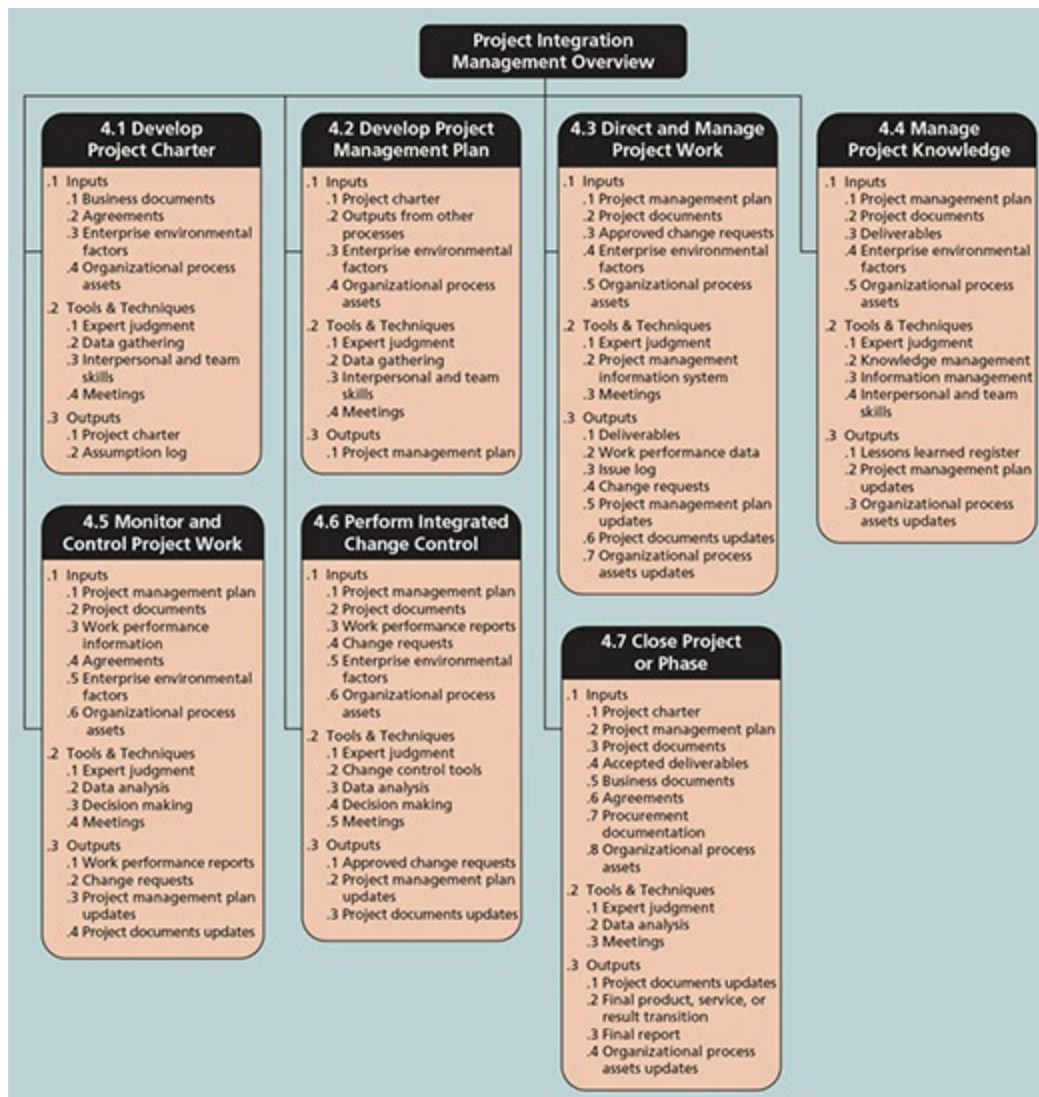
1. *Developing the project charter* involves working with stakeholders to create the document that formally authorizes a project—the charter.
2. *Developing the project management plan* involves coordinating all planning efforts to create a consistent, coherent document—the project management plan.
3. *Directing and managing project work* involves carrying out the project management plan by performing the activities included in it.
4. *Managing project knowledge* involves using existing knowledge and creating new knowledge to achieve project objectives while also contributing to organizational learning.
5. *Monitoring and controlling project work* involves overseeing activities to meet the

performance objectives of the project.

6. *Performing integrated change control* involves identifying, evaluating, and managing changes throughout the project life cycle.
7. *Closing the project or phase* involves finalizing all activities to formally close the project or phase.

Figure 4-1 summarizes the inputs, tools and techniques, and outputs of project integration management.

Figure 4-1. Project integration management summary



Source: Project Management Institute, Inc. *A Guide to the Project Management Body of Knowledge, (PMBOK® Guide) – Sixth Edition* (2017), Figure 4-1, Page 71.

Many people consider project integration management the key to overall project success. Someone must take responsibility for coordinating all of the people, plans, and work required to complete a project. Someone must focus on the big picture of the project and steer the

project team toward successful completion. Someone must make the final decisions when conflicts occur among project goals or people. Someone must communicate key project information to top management. These responsibilities belong to the project manager, whose chief means for accomplishing all these tasks is project integration management.

What Went Wrong?

Everyone knows that making sure people are paid is crucial to employee satisfaction, so when a payroll system project doesn't work, it is a disaster. Tens of thousands of Canadian federal government workers faced a third year of payroll system torment in 2018. The Phoenix system project, started in 2009 and budgeted to cost C\$309.5, is estimated to cost over C\$1 billion and take three additional years to complete properly. The main goal of the system was to reduce payroll processing overhead and staffing costs.

"In what is quickly moving into contention as one of the worst government-managed IT implementations ever, over half the 290,000 plus civil servants paid through the IBM-developed Phoenix pay system have been underpaid, overpaid, or not paid at all since its rollout began in February of 2016. . . . According to Canada's Auditor General, nearly 50,000 government workers have had to wait over a year to get their pay straightened out."*

Particularly on large projects, the project manager's main job is project integration management. To perform well in that role, it is important to understand the organization's needs and staff your project with skilled workers capable of making the project a success.

Good project integration management is critical to providing stakeholder satisfaction. Project integration management includes [interface management](#), which involves identifying and managing the points of interaction between various elements of a project. The primary tools for interface management are communication and relationships. The number of interfaces can increase exponentially as the number of people involved in a project increases. Thus, one of the most important jobs of a project manager is to establish and maintain good communication and relationships across organizational interfaces. The project manager must communicate well with all project stakeholders, including customers, the project team, top management, other project managers, and opponents of the project.

What happens when a project manager does not communicate well with all stakeholders? In the chapter's opening case, Nick Carson seemed to ignore a key stakeholder for the next-gen DNA-sequencing instrument project—his top management. Nick was comfortable working

with other members of the project team, but he was not familiar with his new job as project manager or the needs of the company's top management. Nick continued to do his old job of software developer and took on the added role of software integrator. He mistakenly thought project integration management meant software integration management and focused on the project's technical problems. He totally ignored what project integration management is really about—integrating the work of all of the people involved in the project by focusing on good communication and relationship management. Recall that project management is applying knowledge, skills, tools, and techniques to meet project requirements, while also meeting or exceeding stakeholder needs and expectations. Nick did not take the time to find out what top management expected from him as the project manager; he assumed that completing the project on time and within budget was sufficient to make them happy. Yes, top management should have made its expectations more clear, but Nick should have taken the initiative to get the guidance he needed.

In addition to not understanding project integration management, Nick did not use holistic or systems thinking (see [Chapter 2](#)). He burrowed into the technical details of his particular project. He did not stop to think about what it meant to be the project manager, how this project related to other projects in the organization, or top management's expectations of him and his team.

Project integration management must occur within the context of the entire organization, not just within a particular project. The project manager must integrate the work of the project with the ongoing operations of the organization. In the opening case, Nick's company was negotiating a potential buyout with a larger company. Consequently, top management needed to know when the next-gen DNA-sequencing instrument would be ready, how big the market was for the product, and if the company had enough in-house staff to continue to manage projects like this one in the future. Top management wanted to see a project management plan and a schedule to help monitor the project's progress and show the potential buyer what was happening. When top managers tried to talk to Nick about these issues, he soon returned to discussing the technical details of the project. Even though Nick was very bright, he had no experience or real interest in many of the company's business aspects. Project managers must always view their projects in the context of the changing needs of their organizations and respond to requests from top management. Likewise, top management must keep project managers informed of major issues that could affect their projects and strive to make processes consistent throughout their organization.

Following a standard process for managing projects can help prevent some of the typical problems new and experienced project managers face, including communicating with and managing stakeholders. Before organizations begin projects, however, they should go through a formal process to decide what projects to pursue.

Strategic Planning and Project Selection

Successful leaders look at the big picture or strategic plan of the organization to determine what types of projects will provide the most value. Some may argue that project managers should not be involved in strategic planning and project selection because top management is usually responsible for these types of business decisions. But successful organizations know that project managers can provide valuable insight into the project selection process.

Strategic Planning

Strategic planning involves determining long-term objectives by analyzing the strengths and weaknesses of an organization, studying opportunities and threats in the business environment, predicting future trends, and projecting the need for new products and services. Strategic planning provides important information to help organizations identify and then select potential projects.

Many people are familiar with **SWOT analysis**—analyzing **S**trengths, **W**eaknesses, **O**pportunities, and **T**hreats—which is one tool used in strategic planning. For example, a group of four people who want to start a new business in the film industry could perform a SWOT analysis to help identify potential projects. They might determine the following based on a SWOT analysis:

Strengths:

- As experienced professionals, we have numerous contacts in the film industry.
- Two of us have strong sales and interpersonal skills.
- Two of us have strong technical skills and are familiar with several filmmaking software tools.
- We all have impressive samples of completed projects.

Weaknesses:

- None of us have accounting or financial experience.
- We have no clear marketing strategy for products and services.
- We have little money to invest in new projects.
- We have no company website and limited use of technology to run the business.

Opportunities:

- A potential client has mentioned a large project she would like us to bid on.
- The film industry continues to grow.
- There are two major conferences this year where we could promote our company.

Threats:

- Other individuals or companies can provide the services we can.
- Customers might prefer working with more established individuals and organizations.
- There is high risk in the film business.

Based on their SWOT analysis, the four entrepreneurs outline potential projects as follows:

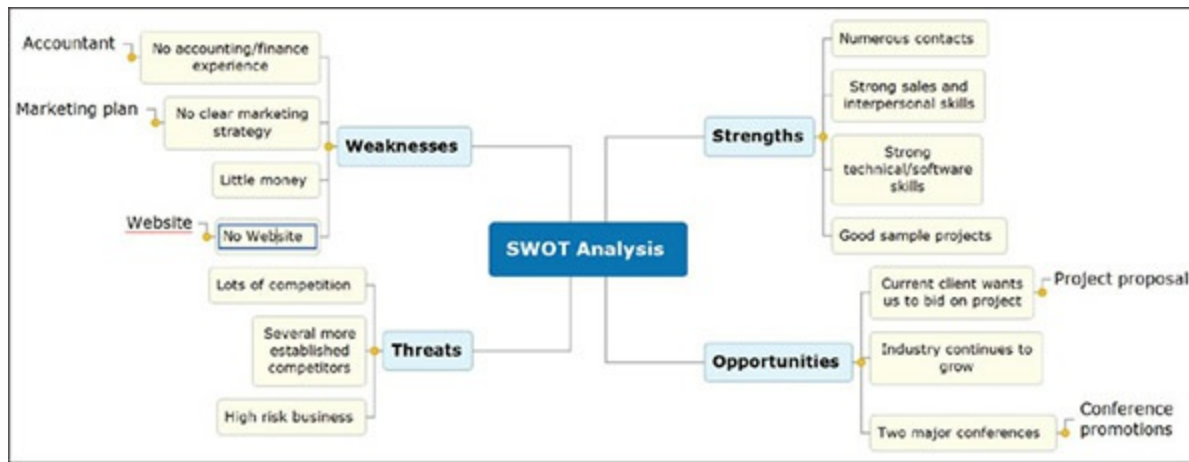
- Find an external accountant or firm to help run the business.
- Hire someone to develop a company website, focusing on our experience and past projects.
- Develop a marketing plan.
- Develop a strong proposal to get the large project the potential client mentioned.
- Plan to promote the company at two major conferences this year.

Some people like to perform a SWOT analysis by using [mind mapping](#), a technique that uses branches radiating from a core idea to structure thoughts and ideas. The human brain does not work in a linear fashion. People come up with many unrelated ideas. By capturing those ideas in a visual mind map format, you can often generate more ideas than by creating lists. You can create mind maps by hand, using sticky notes, using presentation software such as Microsoft PowerPoint, or by using mind mapping software.

Figure 4-2 shows a sample mind map for the new film industry business SWOT analysis. This diagram was created using MindView Business Edition software by MatchWare. Notice that this map has four main branches representing strengths, weaknesses, opportunities, and threats. Ideas in each category are added to the appropriate branch, and sub-branches are shown for some ideas under those categories. Notice that several branches end with a project

idea, such as Project proposal, Conference promotions, Accountant, Marketing plan, and Website. You can easily format the project ideas to make them stand out, as shown in Figure 4-2. From this example, you can see that no project ideas are identified to address strengths or threats, so these areas should be discussed further.

Figure 4-2. Mind map of a SWOT analysis to help identify potential projects

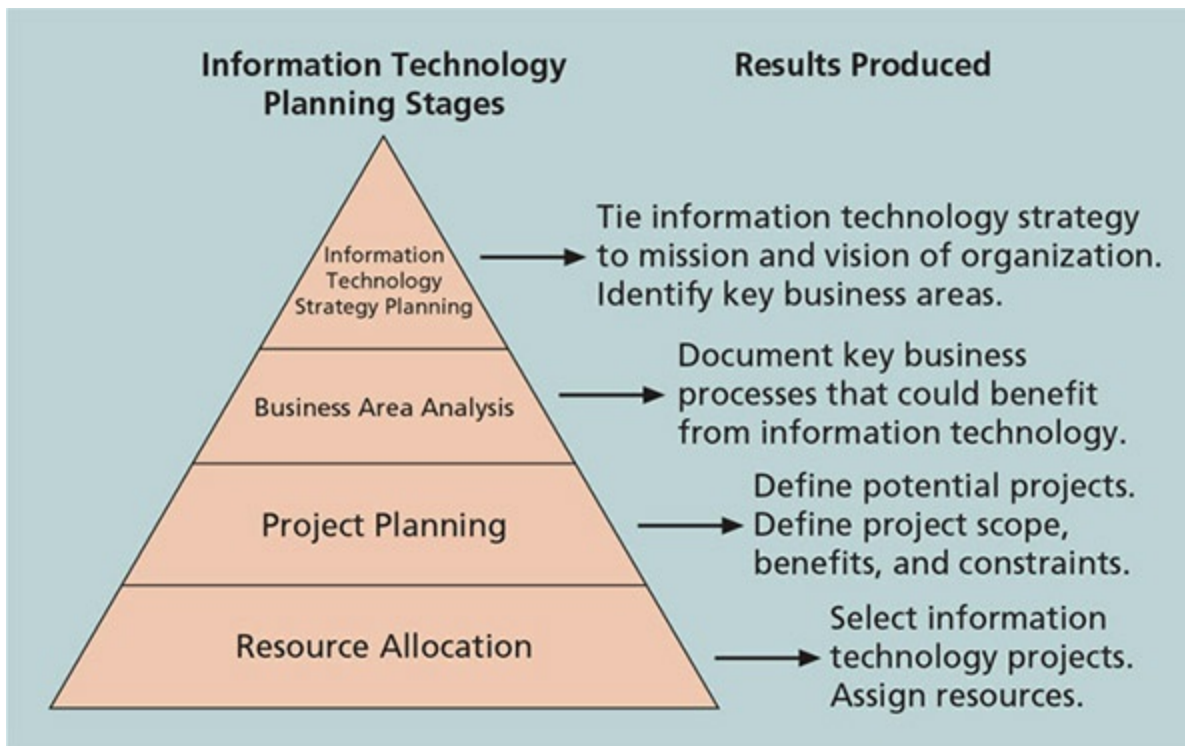


Identifying Potential Projects

The first step in project management is deciding what projects to do in the first place. Therefore, project initiation starts with identifying potential projects, using realistic methods to select which projects to work on, and then formalizing their initiation by issuing some sort of project charter.

In addition to using a SWOT analysis, organizations often follow a detailed process for project selection. **Figure 4-3** shows a four-stage process for selecting IT projects. Note the hierarchical structure of this model and the results produced from each stage.

Figure 4-3. Planning process for selecting IT projects



In the first stage of the selection process, starting at the top of the hierarchy, a steering committee develops an IT strategic plan that is tied to the organization's overall strategic plan. In many organizations, this steering committee consists of managers from departments throughout the company to ensure that all projects are selected in the best interests of the entire organization. The head of the Project Management Office (PMO) would be part of this committee because the PMO acts as the central location for keeping track of all project activities. It is very important to have managers from outside the IT department assist in developing the IT strategic plan, as they can help IT personnel understand organizational strategies and identify the business areas that support them. At the end of this stage, the organization should have a well-defined list of IT strategic goals.

After identifying strategic goals, the next stage in the process for selecting IT projects is to perform a business area analysis. This analysis outlines business processes that are central to achieving strategic goals and helps determine which processes could most benefit from IT. In the next stage, the organization starts defining the scope, benefits, and constraints of potential IT projects. The last stage in the process for selecting IT projects is choosing which projects to do and assigning resources for working on them.

Aligning IT with Business Strategy

Notice that aligning IT projects with business strategy is at the heart of selecting IT projects. This is consistently a top concern for CIOs. It is often difficult to educate line managers on

technology's possibilities and limitations and keep IT professionals in tune with changing business needs. Most organizations face thousands of problems and opportunities for improvement. To get the most value from technology, an organization's strategic plan should guide the IT project selection process. Recall from **Chapter 2**'s Best Practice feature that IT governance is an important part of ensuring that IT supports business goals. IT governance helps organizations maximize their investments in IT and address IT-related risks and opportunities.

An organization must develop a strategy for using IT to define how it will support the organization's objectives. This IT strategy must align with the organization's strategic plans. In fact, research shows that supporting explicit business objectives is the top reason cited for why organizations invest in IT projects. Other top criteria for investing in IT projects include supporting implicit business objectives and providing financial incentives, such as a good internal rate of return (IRR) or net present value (NPV).^{*} Using a balanced scorecard approach, as described later in this chapter also helps align projects with business strategy. Gartner estimated worldwide IT spending at \$3.6 trillion in 2018, an increase of 3.5 percent from 2017. The majority of survey respondents felt that the CIO role was becoming more strategic, reporting that 62 percent sit on the executive board, an all-time high from past surveys.^{*}

Information systems often are central to business strategy. Author Michael Porter, who developed the concepts of the strategic value of competitive advantage and value chain, and many other experts have emphasized the importance of using IT to support strategic plans and provide a competitive advantage. Many information systems are classified as strategic because they directly support key business strategies. For example, information systems can support an organizational strategy of being a low-cost producer. As one of the largest retailers in the United States, Walmart's inventory control system is a classic example of such a strategic system. Information systems can support a strategy of providing specialized products or services that set a company apart from others in the industry. Consider the classic example of Federal Express's introduction of online package tracking systems. FedEx was the first company to provide this type of service, which gave it a competitive advantage until others developed similar systems. Information systems can also support a strategy of selling to a particular market or occupying a specific product niche. Owens Corning developed a strategic information system that boosted the sales of its home-insulation products by providing its customers with a system for evaluating the energy efficiency of building designs. The editor in chief of *CIO* magazine stated, "The smart use of technology is always a key component in driving innovation and creating a competitive edge. We see IT value proving itself across the board in many organizations."^{*}

In 2017 Fortune released their Business by Design list, highlighting the 24 companies they believe are using technology and globalization to find their competitive advantage. A few excerpts from their analysis include the following:

- “No company tops Apple (aapl, -0.24%) for demonstrating the strategic power of great design and learning to ‘think different.’ While there is a raging debate about whether or not Apple has lost some of its design mojo in recent years, as the story ‘Has Apple Lost Its Design Mojo’ explores, the world’s most valuable company continues to push boundaries. Meanwhile, a host of other leading companies, including Alphabet, Amazon, and Nike, have achieved success by expanding design capabilities.
- Dyson is the U.K.’s biggest investor in robotics and artificial intelligence research. In September, the company launched the Dyson Institute of Engineering and Technology, a university within its office grounds, to feed its growing headcount of engineers and scientists, which Dyson predicts will double to 6,000 by 2020.
- It wasn’t so long ago that Samsung found itself in a courtroom defending its creativity against Apple. But the company’s decades-long bid to move beyond its reputation as a budget brand has paid off. Today Samsung is tech’s largest spender on R&D. And its TVs, phones, appliances, services, and offices? Covetable.
- Capital One has adopted design thinking as a mantra to reinvent itself as a software company and innovation incubator, rather than a traditional bank. After acquiring design firms Adaptive Path and Monsoon, Capital One has recently rolled out fresh digital features, from an emoji-enabled SMS chatbot to GPS-tracked transaction histories. In early 2018 it will unveil its 1717 Innovation Center in Richmond, a 42,000-square-foot facility housing an experience design research lab and, through a partnership with an incubator program, some 50 startups.”*

Methods for Selecting Projects

Organizations identify many potential projects as part of their strategic planning processes, and they need to narrow down the list of potential projects to the ones that will be of most benefit. They often rely on experienced project managers to help them make project selection decisions. Selecting projects is not an exact science, and many methods exist for selecting projects. Five common techniques are as follows:

- Focusing on broad organizational needs

- Categorizing IT projects
- Performing net present value or other financial analyses
- Using a weighted scoring model
- Implementing a balanced scorecard

In practice, many organizations use a combination of these approaches to select projects. Each approach has advantages and disadvantages, and it is up to management to determine the best approach for selecting projects based on their particular organization.

Focusing on Broad Organizational Needs

Top managers must focus on meeting their organizations' many needs when deciding what projects to undertake, when to undertake them, and to what level. Projects that address broad organizational needs are much more likely to be successful because they will be important to the organization. For example, a broad organizational need might be to improve safety, increase morale, provide better communications, or improve customer service. However, it is often difficult to provide a strong justification for many IT projects related to these broad organizational needs. For example, estimating the financial value of such projects is often impossible, even though everyone agrees that they have a high value. As the old proverb says, "It is better to measure gold roughly than to count pennies precisely."

One method for selecting projects based on broad organizational needs is to determine whether they first meet three important criteria: *need*, *funding*, and *will*. Do people in the organization agree that the project needs to be done? Does the organization have the desire and capacity to provide adequate funds to perform the project? Is there a strong will to make the project succeed? For example, many visionary CEOs can describe a broad need to improve certain aspects of their organizations, such as communications. Although they cannot specifically describe how to improve communications, they might allocate funds to projects that address this need. As projects progress, the organization must reevaluate the need, funding, and will for each project to determine if it should be continued, redefined, or terminated.

Categorizing IT Projects

Another method for selecting projects is based on various categorizations, such as the project's impetus, time window, and general priority. The impetus for a project is often to respond to a problem, an opportunity, or a directive.

- **Problems** are undesirable situations that prevent an organization from achieving its goals. These problems can be current or anticipated. For example, users of an information system may be having trouble logging on to the system or getting information in a timely manner because the system has reached its capacity. In response, the company could initiate a project to enhance the current system by adding more access lines or upgrading the hardware with a faster processor, more memory, or more storage space.
- **Opportunities** are chances to improve the organization. For example, the project described in the chapter's opening case involves creating a new product that can make or break the entire company.
- **Directives** are new requirements imposed by management, government, or some external influence. For example, many projects that involve medical technologies must meet rigorous government requirements.

Organizations select projects for any of these reasons. It is often easier to get approval and funding for projects that address problems or directives because the organization must respond to these categories to avoid hurting their business. Many problems and directives must be resolved quickly, but managers must also apply systems thinking and seek opportunities for improving the organization through IT projects.

Another categorization for IT projects is based on timing: How long will it take to complete a project and what is the deadline for completing it? For example, some potential projects must be finished within a specific time window; otherwise, they are no longer valid projects. Some projects can be completed very quickly—within a few weeks, days, or even minutes. Many organizations have an end-user support function to handle very small projects that can be completed quickly. However, even though many IT projects can be completed quickly, it is still important to prioritize them.

Organizations can also categorize IT projects as having high, medium, or low priority based on the current business environment. For example, if it is crucial to cut operating costs quickly, projects that have the most potential to do so would be given a high priority. An organization should always complete high-priority projects first, even if a low- or medium-priority project could be finished in less time. Usually an organization has many more potential IT projects than it can undertake at one time, so it is crucial to work on the most important projects first.

Performing Financial Analyses

Financial considerations are an important aspect of the project selection process, whether

economic times are tough or the economy is growing. As authors Dennis Cohen and Robert Graham put it, “Projects are never ends in themselves. Financially they are always a means to an end, cash.”* Many organizations require an approved business case before pursuing projects, and financial projections are a critical component of the business case. (See Chapter 3 for a sample business case.) Three primary methods for projecting the financial value of projects include net present value analysis, return on investment, and payback analysis. Because project managers often deal with business executives, they must understand how to speak business language, which often boils down to the following important financial concepts.

Net Present Value Analysis

Most people know that a dollar earned today is worth more than a dollar earned five years from now—a principle called the time value of money. Many projects have financial implications that extend into the future. In order to evaluate potential projects equally, you need to consider their net present value.

Net present value (NPV) analysis is a method of calculating the expected net monetary gain or loss from a project by calculating the value of all expected future cash inflows and outflows at the present time. An organization should consider only projects with a positive NPV if financial value is a key criterion for project selection. A positive NPV means that the return from a project exceeds the **cost of capital**—the return available from investing the capital elsewhere. In other words, the cost of capital is the rate of return that could have been earned by putting the same money into a different investment with equal risk. Projects with higher NPVs are preferred to projects with lower NPVs, if all other factors are equal.

To calculate NPV, you must assume a certain discount rate. The **discount rate** is the interest rate used to discount cash flows. It takes into account not just the time value of money but also the risk or uncertainty of future cash flows. The greater the uncertainty of future cash flows, the higher the discount rate. It is also called the **capitalization rate** or the **opportunity cost of capital**.

Figure 4-4 illustrates this concept in Microsoft Excel for two different projects. Note that this example starts discounting immediately in Year 1 and uses a 10 percent discount rate. You can use the NPV function in Excel to calculate the NPV quickly. Detailed steps for performing this calculation manually are presented later.

Figure 4-4. Net present value example

	A	B	C	D	E	F	G
1	Discount rate	10%					
2							
3	PROJECT 1	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	TOTAL
4	Benefits	\$0	\$2,000	\$3,000	\$4,000	\$5,000	\$14,000
5	Costs	\$5,000	\$1,000	\$1,000	\$1,000	\$1,000	\$9,000
6	Cash flow	(\$5,000)	\$1,000	\$2,000	\$3,000	\$4,000	\$5,000
7	NPV	\$2,316					
8		Formula =npv(b1,b6:f6)					
9							
10	PROJECT 2	YEAR 1	YEAR 2	YEAR 3	YEAR 4	YEAR 5	TOTAL
11	Benefits	\$1,000	\$2,000	\$4,000	\$4,000	\$4,000	\$15,000
12	Costs	\$2,000	\$2,000	\$2,000	\$2,000	\$2,000	\$10,000
13	Cash flow	(\$1,000)	\$0	\$2,000	\$2,000	\$2,000	\$5,000
14	NPV	\$3,201					
15		Formula =npv(b1,b13:f13)					
16							
17							

Note that totals are equal, but NPVs are not because of the time value of money

Figure 4-4 lists the projected benefits first, followed by the costs, and then the calculated cash flow amount. Note that the sum of the **cash flow**—benefits minus costs or income minus expenses—is the same for both projects at \$5,000. The net present values are different, however, because they account for the time value of money. Project 1 has a negative cash flow of \$5,000 in the first year, while Project 2 has a negative cash flow of only \$1,000 in the first year. Although both projects have the same total cash flows without discounting, they are not of comparable financial value. Project 2's NPV of \$3,201 is better than Project 1's NPV of \$2,316. NPV analysis, therefore, is a method for making equal comparisons between cash flows for multiyear projects.

To determine NPV, follow these steps:

1. Determine the estimated costs and benefits for the life of the project and the products it creates. For example, JWD Consulting assumed that its project would produce a system in about six months that would be used for three years, so costs are included in Year 0, when the system is developed. Ongoing system costs and projected benefits are included for Years 1, 2, and 3.
2. Determine the discount rate. In **Figure 4-4**, the discount rate is 10 percent per year.
3. Calculate the net present value. Most spreadsheet software has a built-in function to calculate NPV. For example, **Figure 4-4** shows the formula that Microsoft Excel uses: =npv(discount rate, range of cash flows), where the discount rate is in cell B1 and the range of cash flows for Project 1 are in cells B6 through F6. (See Chapter 7, Project Cost

Management, for more information on cash flow and other cost-related terms.) To use the NPV function, you must have a row or column in the spreadsheet for the cash flow each year, which is the benefit amount for that year minus the cost amount.

The result of the formula yields an NPV of \$2,316 for Project 1 and \$3,201 for Project 2. Because both projects have positive NPVs, they are good candidates for selection. However, because Project 2 has an NPV that is 38 percent higher than Project 1, it would be the better choice. If the two numbers are close, then other methods should be used to help decide which project to select.

The mathematical formula for calculating NPV is

$$\text{NPV} = \sum_{t=0 \dots n} A_t / (1 + r)^t$$

where t equals the year of the cash flow, n is the last year of the cash flow, A is the amount of cash flow each year, and r is the discount rate.

If you cannot enter the data into spreadsheet software, you can perform the calculations by hand or with a calculator. First, determine the annual [discount factor](#)—a multiplier for each year based on the discount rate and year—and then apply it to the costs and benefits for each year. The formula for the discount factor is $1/(1 + r)^t$, where r is the discount rate, such as 8 percent, and t is the year. For example, the discount factors used in Figure 4-5 are calculated as follows:

$$\text{Year 0 : discount factor} = 1/(1 + 0.08)^0 = 1$$

$$\text{Year 1 : discount factor} = 1/(1 + 0.08)^1 = 0.93$$

$$\text{Year 2 : discount factor} = 1/(1 + 0.08)^2 = 0.86$$

$$\text{Year 3 : discount factor} = 1/(1 + 0.08)^3 = 0.79$$

Figure 4-5. JWD Consulting net present value and return on investment example

Discount rate	8%					
Assume the project is completed in Year 0			Year			
	0	1	2	3	Total	
Costs	140,000	40,000	40,000	40,000		
Discount factor	1	0.93	0.86	0.79		
Discounted costs	140,000	37,200	34,400	31,600	243,200	
Benefits	0	200,000	200,000	200,000		
Discount factor	1	0.93	0.86	0.79		
Discounted benefits	0	186,000	172,000	158,000	516,000	
Discounted benefits - costs	(140,000)	148,800	137,600	126,400	272,800	← NPV
Cumulative benefits - costs	(140,000)	8,800	146,400	272,800		
ROI	112%					
			Payback in Year 1			

After determining the discount factor for each year, multiply the costs and benefits each year by the appropriate discount factor. (Note that the discount factor in this case is rounded to two decimal places.) For example, in **Figure 4-5**, the discounted cost for Year 1 is $\$40,000 \times 0.93 = \$37,200$. Next, sum all of the discounted costs and benefits each year to get a total. The total discounted costs in Figure 4-5 are \$243,200. To calculate the NPV, subtract the total discounted costs from the total discounted benefits. In this example, the NPV is $\$516,000 - \$243,200 = \$272,800$.

When calculating NPV, some organizations refer to the investment year or years for project costs as Year 0 and do not discount costs in Year 0. Other organizations start discounting immediately based on their financial procedures; it's simply a matter of preference for the organization.

The discount rate can also vary, often based on the prime rate and other economic considerations. Some people consider it to be the rate at which the organization could borrow money for the project. Financial experts in your organization can tell you what discount rate to use.

When calculating NPV, you can enter costs as negative numbers instead of positive numbers, and you can list costs first and then benefits. For example, **Figure 4-5** shows the financial calculations used in the JWD Consulting business case for the project management intranet site project described in Chapter 3. Note that the discount rate is 8 percent, costs are not discounted in Year 0, the discount factors are rounded to two decimal places, costs are listed

first, and costs are entered as positive numbers. Also note that costs and benefits are discounted before they are summed. The NPV and other calculations are the same; only the format is different. A project manager needs to check with the organization to learn its guidelines for when discounting starts, what discount rate to use, and what format the organization prefers.

Return on Investment

Another important financial consideration is return on investment. **Return on investment (ROI)** is the result of subtracting the project costs from the benefits and then dividing by the costs. For example, if you invest \$100 today and next year it is worth \$110, your ROI is $(\$110 - 100)/100$ or 0.10 (10 percent). Note that the ROI is always a percentage. It can be positive or negative. For multiyear projects, it is best to use discounted costs and benefits when calculating ROI. Figure 4-5 shows an ROI of 112 percent, which you calculate as follows:

$$\text{ROI} = (\text{total discounted benefits} - \text{total discounted costs}) / \text{discounted costs}$$

$$\text{ROI} = (516,000 - 243,200) / 243,200 = 112\%$$

The higher the ROI is, the better. An ROI of 112 percent is outstanding.

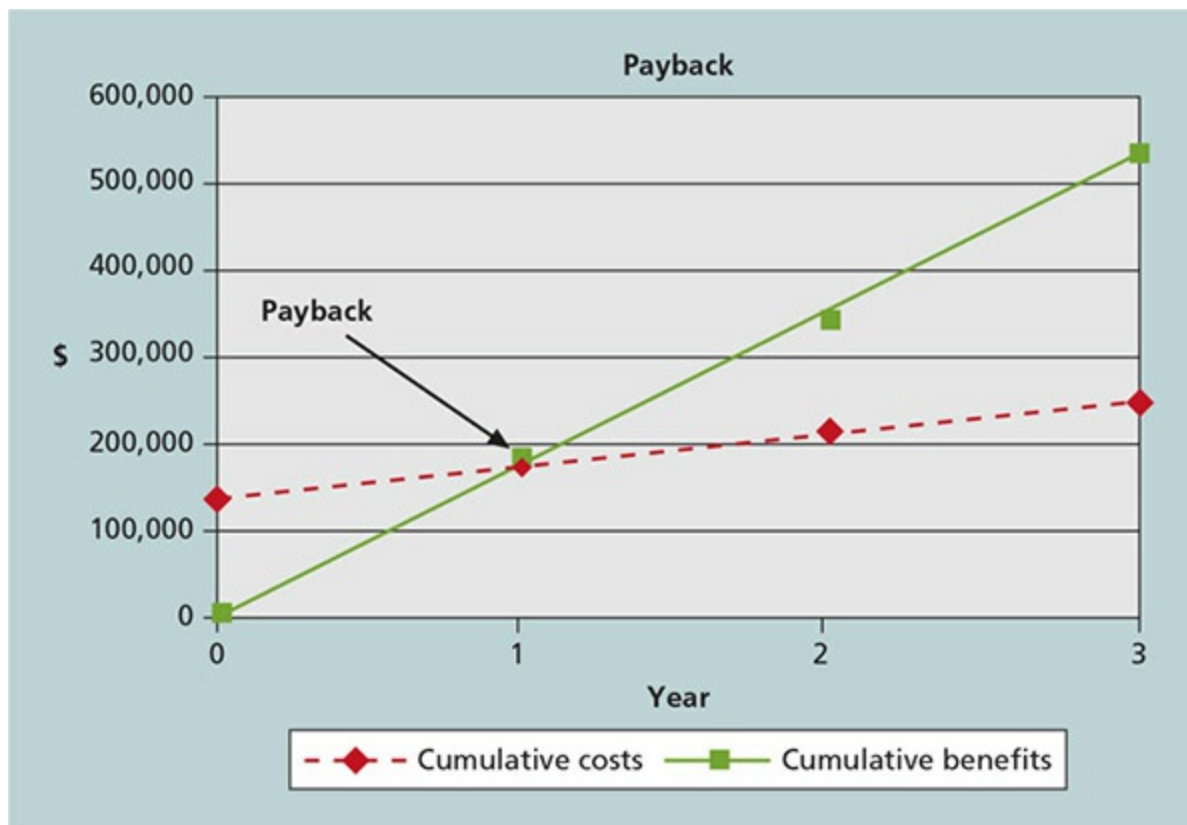
Many organizations have a required rate of return for projects. The **required rate of return** is the minimum acceptable rate of return on an investment. For example, an organization might have a required rate of return of at least 10 percent for projects. The organization bases the required rate of return on what it could expect to receive elsewhere for an investment of comparable risk. You can also determine a project's **internal rate of return (IRR)** by finding what discount rate results in an NPV of zero for the project. You can use the Goal Seek function in Excel to set the cell that contains the formula for NPV to the value 0 by changing the cell that contains the discount rate. The resulting discount rate is the IRR. For example, in Figure 4-4, you could set cell b7 to zero while changing cell b1 to find that the IRR for Project 1 is 27 percent.

Payback Analysis

Payback analysis is another important financial tool when selecting projects. **Payback period** is the amount of time it will take to recoup the total dollars invested in a project, in terms of net cash inflows. In other words, payback analysis determines how much time will elapse before accrued benefits overtake accrued and continuing costs. Payback occurs when the net cumulative benefits equal the net cumulative costs or when the net cumulative benefits minus costs equal zero. Figure 4-5 shows how to find the payback period. The cumulative benefits minus costs for Year 0 is \$140,000. Adding that number to the discounted benefits minus costs for Year 1 results in \$8,800. Because that number is positive, the payback occurs in Year 1.

Creating a chart helps illustrate more precisely when the payback period occurs. **Figure 4-6** charts the cumulative discounted costs and cumulative discounted benefits each year using the numbers from Figure 4-5. Note that the lines cross around Year 1. This is the point where the cumulative discounted benefits equal the cumulative discounted costs, so that the cumulative discounted benefits minus costs is zero. Beyond this point, discounted benefits exceed discounted costs and the project shows a profit. An early payback period, such as in the first or second year, is considered very good.

Figure 4-6. Charting the payback period for the JWD consulting project



Many organizations have requirements for the length of the payback period of an investment. They might require all IT projects to have a payback period of less than two years or even one year, regardless of the estimated NPV or ROI. Dan Hoover, vice president and area director of Ciber Inc., an international systems integration consultancy, suggests that organizations, especially small firms, should focus on payback period when making IT investment decisions. "If your costs are recovered in the first year," Hoover said, "the project is worthy of serious consideration, especially if the benefits are high. If the payback period is more than a year, it may be best to look elsewhere."* However, organizations must also consider long-range goals when making technology investments. Many crucial projects cannot achieve a payback so quickly or be completed in such a short time period.

To aid in project selection, project managers must understand the organization's financial expectations for projects. Top management must also understand the limitations of financial

estimates, particularly for IT projects. For example, it is very difficult to develop good estimates of projected costs and benefits for IT projects. You will learn more about estimating costs and benefits in **Chapter 7**, Project Cost Management.

Using a Weighted Scoring Model

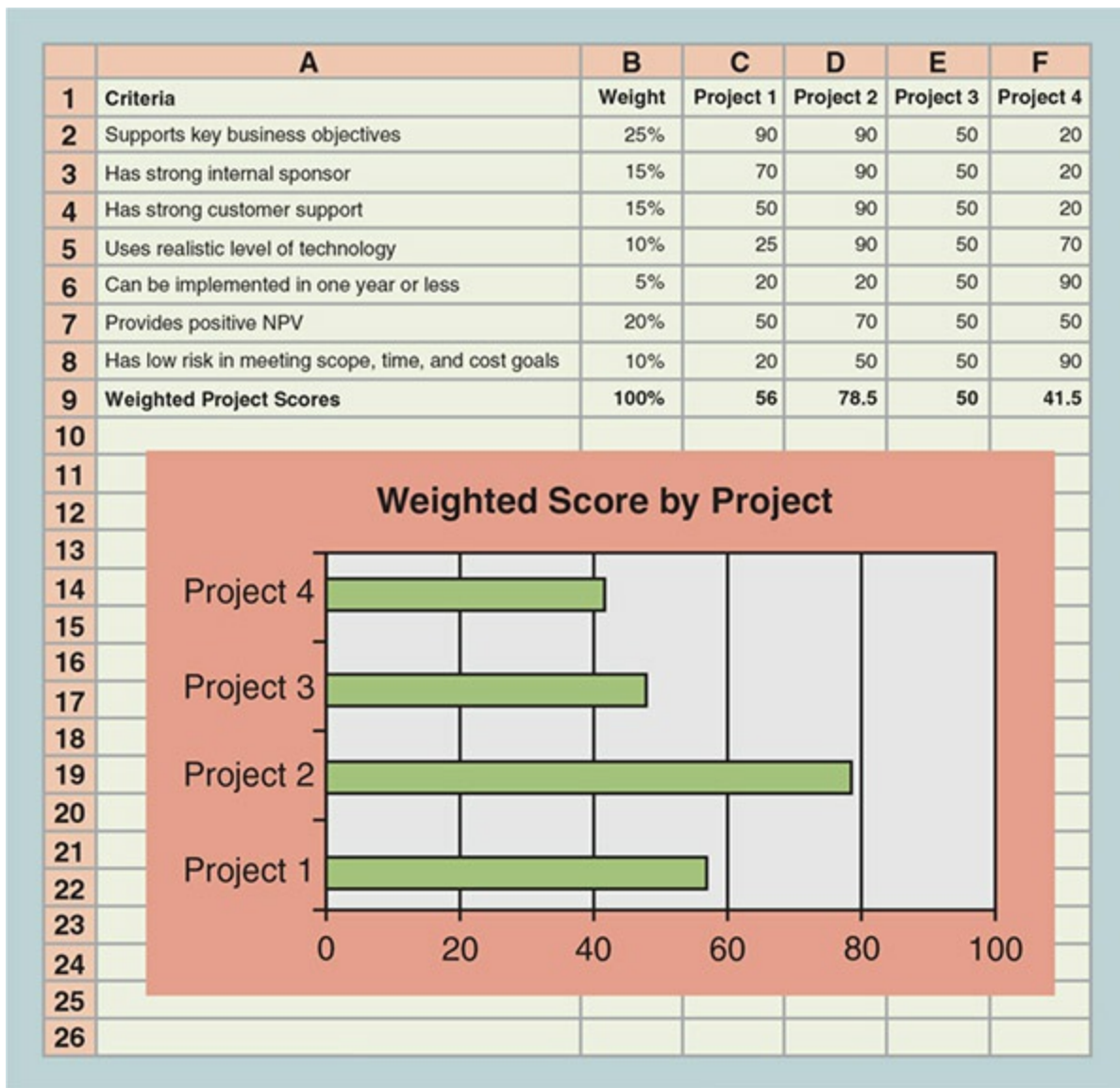
A **weighted scoring model** is a tool that provides a systematic process for selecting projects based on many criteria. These criteria can include factors such as meeting broad organizational needs; addressing problems, opportunities, or directives; the amount of time needed to complete the project; the overall priority of the project; and projected financial performance of the project.

The first step in creating a weighted scoring model is to identify criteria that are important to the project selection process. It often takes time to develop and reach agreement on these criteria. Holding facilitated brainstorming sessions or using groupware to exchange ideas can aid in developing these criteria. Possible criteria for IT projects include the following:

- Supports key business objectives or strategies
- Has strong internal sponsor
- Has strong customer support
- Uses realistic level of technology
- Can be implemented in one year or less
- Provides positive NPV
- Has low risk in meeting scope, time, and cost goals

Next, you assign a weight to each criterion based on its importance. Once again, determining weights requires consultation and final agreement. You can assign weights based on percentages; the weights of the criteria must total 100 percent. You then assign numerical scores to each criterion (e.g., 0 to 100) for each project. The scores indicate how much each project meets each criterion. At this point, you can use a spreadsheet application to create a matrix of projects, criteria, weights, and scores. **Figure 4-7** provides an example of a weighted scoring model to evaluate four different projects. After assigning weights for the criteria and scores for each project, you calculate a weighted score for each project by multiplying the weight for each criterion by its score and adding the resulting values.

Figure 4-7. Sample weighted scoring model for project selection



For example, you calculate the weighted score for Project 1 in [Figure 4-7](#) as

$$25\% * 90 + 15\% * 70 + 15\% * 50 + 10\% * 25 + 5 * 20 + 20\% * 50 + 10\% * 20 = 56$$

Note that in this example, Project 2 would be the obvious choice for selection because it has the highest weighted score. Creating a bar chart to graph the weighted scores for each project allows you to see the results at a glance.

If you create the weighted scoring model in a spreadsheet, you can enter the data, create and copy formulas, and perform a “what-if” analysis. For example, suppose that you want to change the weights for the criteria. By having the weighted scoring model in a spreadsheet, you can easily change the weights and update the weighted scores and charts automatically. This capability allows you to investigate various options for different stakeholders quickly. Ideally, the result should reflect the group’s consensus, and any major disagreements should be documented.

You can also establish how well projects meet criteria by assigning points. For example, a project might receive 10 points if it definitely supports key business objectives, 5 points if it somewhat supports them, and 0 points if it is totally unrelated to key business objectives. With a point model, you can simply add all the points to determine the best projects for selection, without having to multiply weights and scores and sum the results.

You can also determine minimum scores or thresholds for specific criteria in a weighted scoring model. For example, suppose that an organization should not consider a project if it does not score at least 50 out of 100 on every criterion. You can build this type of threshold into the weighted scoring model to reject projects that do not meet these minimum standards. As you can see, weighted scoring models can aid in project selection decisions.

Implementing a Balanced Scorecard

Drs. Robert Kaplan and David Norton developed another approach to help select and manage projects that align with business strategy. A **balanced scorecard** is a strategic planning and management system that helps organizations align business activities to strategy, improve communications, and monitor performance against strategic goals. The Gartner Group estimates that over half of large U.S. organizations use this approach. The balanced scorecard has evolved over time. “The ‘new’ balanced scorecard transforms an organization’s strategic plan from an attractive but passive document into the ‘marching orders’ for the organization on a daily basis. It provides a framework that not only provides performance measurements but helps planners identify what should be done and measured.”* You can find several examples of balanced scorecards from manufacturing companies like Shat-R-Shield to non profits like the Kenya Red Cross at www.balancedscorecard.org.

As you can see, organizations can use many approaches to select projects. Many project managers have some say in which projects their organizations select for implementation. Even if they do not, they need to understand the motives and overall business strategies for the projects they are managing. Project managers and team members are often called upon to explain the importance of their projects, and understanding project selection methods can help them represent the project effectively.

Developing a Project Charter

After top management decides which projects to pursue, it is important to let the rest of the organization know about these projects. Management needs to create and distribute documentation to authorize project initiation. This documentation can take many different forms, but one common form is a project charter. A **project charter** is a document that formally recognizes the existence of a project and provides direction on the project’s objectives and management. It authorizes the project manager to use organizational

resources to complete the project. Ideally, the project manager plays a major role in developing the project charter.

Instead of project charters, some organizations initiate projects using a simple letter of agreement, while others use much longer documents or formal contracts. Key project stakeholders should sign a project charter to acknowledge agreement on the need for and intent of the project. A project charter is a key output of the initiation process, as described in **Chapter 3**.

The *PMBOK® Guide – Sixth Edition* lists inputs, tools, techniques, and outputs of the six project integration management processes. The following inputs are helpful in developing a project charter:

- *A business case*: As explained in Chapter 3, many projects require a business case to justify their investment. Information in the business case, such as the project objective, high-level requirements, and time and cost goals, is included in the project charter.
- *Benefits management plan*: This document describes how and when the projected benefits of the project will be delivered and how they will be managed. Topics covered can include the target benefits, strategic alignment, timeframe for realizing benefits, benefits owner, metrics, assumptions, and risks.
- *Agreements*: If you are working on a project under contract for an external customer, the contract or agreement should include much of the information needed for creating a good project charter. Some people might use a contract or agreement in place of a charter. However, many contracts are difficult to read and can often change, so it is still a good idea to create a project charter.
- *Enterprise environmental factors*: These factors include relevant government or industry standards, the organization's infrastructure, and marketplace conditions. Managers should review these factors when developing a project charter.
- *Organizational process assets*: **Organizational process assets** include formal and informal plans, policies, procedures, guidelines, information systems, financial systems, management systems, lessons learned, and historical information that can influence a project's success.

The main tools and techniques for developing a project charter are expert judgment, data gathering, interpersonal and team skills, and meetings. Experts from inside and outside the organization should be consulted when creating a project charter to make sure it is useful and realistic. Facilitators often make it easier for experts to collaborate and provide useful information.

The two outputs of the process to develop a project charter are the charter itself and an assumption log. Although the format of project charters can vary tremendously, they should include at least the following basic information:

- The project's title and date of authorization
- The project manager's name and contact information
- A summary schedule, including the planned start and finish dates; if a summary milestone schedule is available, it should also be included or referenced
- A summary of the project's budget or reference to budgetary documents
- A brief description of the project objectives, including the business need or other justification for authorizing the project
- Project success criteria, including project approval requirements and who signs off on the project
- A summary of the planned approach for managing the project, which should describe stakeholder needs and expectations, important assumptions, and constraints, and should refer to related documents, such as a communications management plan, as available
- A roles and responsibilities matrix
- A sign-off section for signatures of key project stakeholders
- A comments section in which stakeholders can provide important comments related to the project

Unfortunately, many internal projects, like the one described in the opening case of this chapter, do not have project charters. They often have a budget and general guidelines, but no formal, signed documentation. If Nick had a project charter to refer to—especially if it included information for managing the project—top management would have received the business information it needed, and managing the project might have been easier. Project charters are usually not difficult to write. The difficult part is getting people with the proper knowledge and authority to write and sign the project charter. Top management should have reviewed the charter with Nick because he was the project manager. In their initial meeting, they should have discussed roles and responsibilities, as well as their expectations of how Nick should work with them. If there is no project charter, the project manager should work with key stakeholders, including top management, to create one. **Table 4-1** shows a possible

charter that Nick could have created for completing the next-gen DNA-sequencing instrument project.

Table 4-1. Project charter for the next-gen DNA-sequencing instrument completion project

Project Title: Next-gen DNA-Sequencing Instrument Completion Project

Date of Authorization: February 1

Project Start Date: February 1 **Projected Finish Date:** November 1

Key Schedule Milestones:

- Complete first version of the software by June 1
- Complete production version of the software by November 1

Budget Information: The firm has allocated \$1.5 million for this project, and more funds are available if needed. The majority of costs for this project will be internal labor. All hardware will be outsourced.

Project Manager: Nick Carson, (650) 949-0707, ncarson@dnaconsulting.com

Project Objectives: The Next-gen DNA-sequencing instrument project has been under way for three years. It is a crucial project for our company. This is the first charter for the project; the objective is to complete the first version of the instrument software in four months and a production version in nine months.

Main Project Success Criteria: The software must meet all written specifications, be thoroughly tested, and be completed on time. The CEO will formally approve the project with advice from other key stakeholders.

Approach:

- Hire a technical replacement for Nick Carson and a part-time assistant as soon as possible.
- Within one month, develop a clear work breakdown structure, scope statement, and Gantt chart detailing the work required to complete the Next-gen DNA-sequencing instrument.
- Purchase all required hardware upgrades within two months.
- Hold weekly progress review meetings with the core project team and the sponsor.
- Conduct thorough software testing per the approved test plans.

ROLES AND RESPONSIBILITIES

Name	Role	Position	Contact Information
Ahmed Abrams	Sponsor	CEO	aabrams@dnaconsulting.com
Nick Carson	Project Manager	Manager	ncarson@dnaconsulting.com
Susan Johnson	Team Member	DNA Expert	sjohnson@dnaconsulting.com
Renyong Chi	Team Member	Testing Expert	rchi@dnaconsulting.com
Erik Haus	Team Member	Programmer	ehaus@dnaconsulting.com
Bill Strom	Team Member	Programmer	bstrom@dnaconsulting.com
Maggie Elliot	Team Member	Programmer	melliot@dnaconsulting.com

Sign-off: (signatures of all the above stakeholders)

**Ahmed
Abrams**

Nick Carson

**Susan
Johnson**

Renyong Chi

Erik Haus

Bill Strom

Maggie Elliot

Comments: (handwritten or typed comments from above stakeholders, if applicable)

“I want to be heavily involved in this project. It is crucial to our company’s success, and I expect everyone to help make it succeed.”—ahmed abrams

“The software test plans are complete and well documented. If anyone has questions, do not hesitate to contact me.”—Renyong Chi

Many projects fail because of unclear requirements and expectations, so starting with a project charter makes a lot of sense. If project managers are having difficulty obtaining support from project stakeholders, for example, they can refer to the agreements listed in the project charter. Note that the sample project charter in **Table 4-1** includes several items under the Approach section to help Nick in managing the project and the sponsor in overseeing it. To help Nick make the transition to project manager, the charter said that the company would hire a technical replacement and part-time assistant for Nick as soon as possible. To help Ahmed, the project sponsor, feel more comfortable with how the project was

being managed, items were included to ensure proper planning and communications. Recall from Chapter 2 that executive support contributes the most to successful IT projects. Because Nick was the fourth project manager on this project, top management at his company obviously had problems choosing and working with project managers.

Taking the time to discuss, develop, and sign off on a simple project charter could have prevented several problems in this case.

In addition to a project charter, it is a good idea to create an assumption log. Assumptions are usually identified in the business case, but additional assumptions should be documented throughout the project. Assumptions can be related to technology, estimates, schedules, risks, etc. A simple document or spreadsheet can be used to log assumptions. Information could include an assumption ID, date, source, category, description, and status.

After creating a project charter and assumption log, the next step in project integration management is preparing a project management plan.

Developing a Project Management Plan

To coordinate and integrate information across project management knowledge areas and across the organization, there must be a good project management plan. A [project management plan](#) is a document used to coordinate all project planning documents and help guide a project's execution and control. Plans created in the other knowledge areas are considered subsidiary parts of the overall project management plan. Project management plans also document project planning assumptions and decisions regarding choices, facilitate communication among stakeholders, define the content, extent, and timing of key management reviews, and provide a baseline for progress measurement and project control. Project management plans should be dynamic, flexible, and subject to change when the environment or project changes. These plans should greatly assist the project manager in leading the project team and assessing project status.

To create and assemble a good project management plan, the project manager must practice the art of project integration management, because information is required from all of the project management knowledge areas. Working with the project team and other stakeholders to create a project management plan will help the project manager guide the project's execution and understand the overall project.

The main inputs for developing a project management plan include the project charter, outputs from other processes, enterprise environment factors, and organizational process assets. The main tool and technique is expert judgments are expert judgment, data gathering, interpersonal and team skills, and meetings, and the output is a project management plan.

Project Management Plan Contents

The project management plan briefly describes the overall scope, schedule, and cost baselines for the project. Specific plans in each of those knowledge areas provide more detailed baseline information. For example, the project management plan might provide a high-level budget baseline for the entire project, whereas the cost baseline prepared as part of the project cost management knowledge area provides detailed cost projections by WBS by month. The project management plan can also include a project life cycle description and development approach, as shown in **Table 4-2**. Also notice the many other project documents that project teams can create, as needed, for their projects.

Table 4-2. Project management plan and project documents

Project Management Plan	Project Documents	
1. Scope management plan	1. Activity attributes	14. Project communications
2. Requirements management plan	2. Activity list	15. Project schedule
3. Schedule management plan	3. Assumption log	16. Project schedule network diagram
4. Cost management plan	4. Basis of estimates	17. Project scope statement
5. Quality management plan	5. Change log	18. Project team assignments
6. Resource management plan	6. Cost estimates	19. Quality control measurements
7. Communications management plan	7. Cost forecasts	20. Quality metrics
8. Risk management plan	8. Duration estimates	21. Quality report
9. Procurement management plan	9. Issue log	22. Requirements documentation
10. Stakeholder engagement plan	10. Lessons learned register	33. Test and evaluation documents
11. Change management plan	11. Milestone list	
12. Configuration management plan	12. Physical resource assignments	
13. Scope baseline	13. Project calendars	
14. Schedule baseline	23. Requirements traceability matrix	
15. Cost baseline	24. Resource breakdown structure	
16. Performance measurement baseline	25. Resource calendars	
17. Project life cycle description	26. Resource requirements	
18. Development approach	27. Risk register	
	28. Risk report	
	29. Schedule data	
	30. Schedule forecasts	
	31. Stakeholder register	
	32. Team charter	

Project management plans should be dynamic, flexible, and receptive to change when the environment or project changes. These plans should greatly assist the project manager in leading the project team and assessing project status. Just as projects are unique, so are project plans. For a small project involving a few people over a couple of months, a project charter, scope statement, and Gantt chart might be the only formal project planning documents needed; there would not be a need for a separate project management plan. A large project involving 100 people over three years would benefit from having a detailed project management plan and separate plans for each knowledge area. Because all project plans should help guide the completion of the particular project, they should be tailored as needed for each project.

There are, however, common elements to most project management plans, as follows:

- Introduction/overview of the project
- Project organization
- Management and technical processes (including project lifecycle description and development approach, as applicable)
- Work to be performed (scope)
- Schedule and budget information
- References to other project planning documents

Using Guidelines to Create Project Management Plans

Many organizations use guidelines to create project management plans. Microsoft Project 2016 and other project management software packages come with several template files to use as guidelines. However, do not confuse a project management plan with a Gantt chart. The project management plan is much more than a Gantt chart, as described earlier.

Many government agencies also provide guidelines for creating project management plans. For example, the U.S. Department of Defense (DOD) Standard 2167, Software Development Plan, describes the format for contractors to use when creating a software development plan for DOD projects. The Institute of Electrical and Electronics Engineers (IEEE) Standard 1058–1998 describes the contents of its Software Project Management Plan (SPMP). **Table 4-3**

provides some of the categories for the IEEE SPMP. Companies that work on software development projects for the Department of Defense must follow this standard or a similar standard.

Table 4-3. Sample contents for the IEEE software project management plan (SPMP)	
Major Section Headings	Section Topics
Overview	Purpose, scope, and objectives; assumptions and constraints; project deliverables; schedule and budget summary; evolution of the plan
Project Organization	External interfaces; internal structure; roles and responsibilities
Managerial Process Plan	Start-up plans (estimation, staffing, resource acquisition, and project staff training plans); work plan (work activities, schedule, resource, and budget allocation); control plan; risk management plan; closeout plan
Technical Process Plans	Process model; methods, tools, and techniques; infrastructure plan; product acceptance plan
Supporting Process Plans	Configuration management plan; verification and validation plan; documentation plan; quality assurance plan; reviews and audits; problem resolution plan; subcontractor management plan; process improvement plan

Source: IEEE Standard 1058–1998.

In many private organizations, specific documentation standards are not as rigorous; however, there are usually guidelines for developing project management plans. It is good practice to follow the organization’s standards or guidelines for developing project management plans to facilitate their execution. The organization can work more efficiently if all project management plans follow a similar format. Recall from **Chapter 1** that companies that excel in project management develop and deploy standardized project delivery systems.

The winners clearly spell out what needs to be done in a project, by whom, when, and how. For this they use an integrated toolbox, including PM tools, methods, and techniques. . . . If a scheduling template is developed and used over and over, it becomes a repeatable action that leads to higher productivity and lower uncertainty. Sure, using scheduling templates is neither a breakthrough nor a feat. But laggards exhibited almost no use of the templates. Rather, in constructing schedules their project managers started with a clean sheet, a clear waste of time.*

In the chapter’s opening case, Nick Carson’s top managers were disappointed because he did not provide the project planning information they needed to make important business decisions. They wanted to see detailed project management plans, including schedules and a means for tracking progress. Nick had never created a project management plan or even a simple progress report before, and the organization did not provide templates or examples to

follow. If it had, Nick might have been able to deliver the information top management was expecting.

Directing and Managing Project Work

Directing and managing project work involves managing and performing the work described in the project management plan, one of the main inputs for this process. Other inputs include approved change requests, enterprise environmental factors, and organizational process assets. The majority of time on a project is usually spent on execution, as is most of the project's budget.

The application area of the project directly affects project execution because products are created during the execution phase. For example, the next-gen DNA-sequencing instrument from the opening case and all associated software and documentation would be produced during project execution. The project team would need to use its expertise in biology, hardware and software development, and testing to create the product successfully.

The project manager needs to focus on leading the project team and managing stakeholder relationships to execute the project management plan successfully. Project resource management, communications management, and stakeholder management are crucial to a project's success. See Chapters 9, 10, and 13, respectively, for more information on these knowledge areas. If the project involves a significant amount of risk or outside resources, the project manager also needs to be well versed in project risk management and project procurement management. See Chapters 11 and 12 for details on those knowledge areas. Many unique situations occur during project execution, so project managers must be flexible and creative in dealing with them. Review the situation that Erica Bell faced during project execution in [Chapter 3](#). Also review the ResNet case study (available on the Companion website for this text) to understand the execution challenges that project manager Peeter Kivestu and his project team faced.

Coordinating Planning and Execution

In project integration management, project planning and execution are intertwined and inseparable activities. The main function of creating a project management plan is to guide project execution. A good plan should help produce good products or work results and should document what constitutes good work results. Updates to plans should reflect knowledge gained from completing work earlier in the project. Anyone who has tried to write a computer program from poor specifications appreciates the importance of a good plan. Anyone who has had to document a poorly programmed system appreciates the importance of good execution.

A common sense approach to improving the coordination between project plan development and execution is to follow this simple rule: Those who will do the work should plan the work. All project personnel need to develop both planning and executing skills, and they need experience in these areas. In IT projects, programmers who have to write detailed specifications and then create the code from them become better at writing specifications. Likewise, most systems analysts begin their careers as programmers, so they understand what type of analysis and documentation they need to write good code. Although project managers are responsible for developing the overall project management plan, they must solicit input from project team members who are developing plans in each knowledge area.

Providing Strong Leadership and a Supportive Culture

Strong leadership and a supportive organizational culture are crucial during project execution. Project managers must lead by example to demonstrate the importance of creating good project plans and then following them in project execution. Project managers often create plans for things they need to do themselves. If project managers follow through on their own plans, their team members are more likely to do the same.

What Went Right

In *Pulse of the Profession®: Capturing the Value of Project Management 2015*, PMI found that organizations that excel in project management are indeed capturing business value. They complete about 80 percent of their projects successfully (defined by meeting the scope on time and on budget) and waste 13 times less money than low-performing counterparts. “Their rigorous approach to project, program and portfolio management **improves their ability to execute strategy** and creates a competitive advantage.”*

The surprising news in this survey of 2,800 global professionals was that only 12 percent of organizations were considered to be high performers, a percentage that has remained unchanged since 2012. The main industry group surveyed was in IT (19 percent), followed by financial services (11 percent).

In order to improve, organizations must make some major *cultural changes*. They need to make sure that everyone fully understands the value of project management, require that executive sponsors are fully engaged on projects and programs, and align their projects to the organization’s strategy.

Good project execution also requires a supportive organizational culture. For example, organizational procedures can help or hinder project execution. If an organization has useful

guidelines and templates for project management that everyone in the organization follows, it will be easier for project managers and their teams to plan and do their work. If the organization uses the project plans as the basis for performing work and monitoring progress during execution, the culture will promote the relationship between good planning and execution. On the other hand, if organizations have confusing or bureaucratic project management guidelines that hinder getting work done or measuring progress against plans, project managers and their teams will be frustrated.

Even with a supportive organizational culture, project managers may sometimes find it necessary to break the rules to produce project results in a timely manner. When project managers break the rules, politics will play a role in the results. For example, if a particular project requires use of nonstandard software, the project manager must use political skills to convince concerned stakeholders that using standard software would be inadequate. Breaking organizational rules—and getting away with it—requires excellent leadership, communication, and political skills.

Capitalizing on Product, Business, and Application Area Knowledge

In addition to strong leadership, communication, and political skills, project managers need to possess product, business, and application area knowledge to execute projects successfully. It is often helpful for IT project managers to have prior technical experience or at least a working knowledge of IT products. For example, if the project manager were leading a team to help define user requirements, it would be helpful to understand the language of the business and technical experts on the team. See [Chapter 5](#), Project Scope Management, for more information on collecting requirements.

Many IT projects are small, so project managers may be required to perform some technical work or mentor team members to complete the project. For example, a three-month project to develop a Web-based application with only three team members would benefit most from a project manager who can complete some of the technical work. On larger projects, however, the project manager's primary responsibility is to lead the team and communicate with key project stakeholders. The project manager does not have time to do the technical work. In this case, it is usually best that the project manager understand the business and application area of the project more than the technology involved.

On very large projects the project manager *must* understand the business and application area of the project. For example, Northwest Airlines completed a series of projects in recent years to develop and upgrade its reservation systems. The company spent millions of dollars and had more than 70 full-time people working on the projects at peak periods. The project

manager, Peeter Kivestu, had never worked in an IT department, but he had extensive knowledge of the airline industry and the reservations process. He carefully picked his team leaders, making sure they had the required technical and product knowledge. ResNet was the first large IT project at Northwest Airlines led by a business manager instead of a technical expert, and it was a roaring success. Many organizations have found that large IT projects require experienced general managers who understand the business and application area of the technology, not the technology itself.

Project Execution Tools and Techniques

Directing and managing project work requires specialized tools and techniques, some of which are unique to project management. Project managers can use specific tools and techniques to perform activities that are part of execution processes. These include the following:

- *Expert judgment:* Anyone who has worked on a large, complex project appreciates the importance of expert judgment in making good decisions. Project managers should not hesitate to consult experts on different topics, such as what methodology to follow, what programming language to use, and what training approach to follow.
- *Meetings:* Meetings are crucial during project execution. Face-to-face meetings with individuals or groups of people are important, as are phone and virtual meetings. Meetings allow people to develop relationships, pick up on important body language or tone of voice, and have a dialogue to help resolve problems. It is often helpful to establish set meeting times for various stakeholders. For example, Nick could have scheduled a short meeting once a week with senior managers. He could have also scheduled 10-minute standup meetings every morning for the project team.
- *Project management information systems:* As described in Chapter 1, hundreds of project management software products are on the market today. Many large organizations use powerful enterprise project management systems that are accessible via the Internet and tie into other systems, such as financial systems. Even in smaller organizations, project managers or other team members can create Gantt charts that include links to other planning documents on an internal network. For example, Nick or his assistant could have created a detailed Gantt chart for their project in Project 2016 and created links to other key planning documents created in Word, Excel, or PowerPoint. Nick could have shown the summary tasks during the progress review meetings, and if top management had questions, Nick could have shown them supporting details. Nick's team could also have set baselines for completing the project and tracked their progress toward achieving those goals. See Appendix A (available on the Companion website for this text) for details on using Project 2016 to perform these functions, and for samples of Gantt

charts and other useful outputs from project management software.

Although project management information systems can aid in project execution, project managers must remember that positive leadership and strong teamwork are critical to successful project management. Project managers should delegate the detailed work involved in using these tools to other team members and focus on providing leadership for the whole project to ensure project success. Stakeholders often focus on the most important output of execution from their perspective: the deliverables. For example, a production version of the next-gen DNA-sequencing instrument was the main deliverable for the project in the opening case. Of course, many other deliverables were created along the way, such as software modules, tests, and reports. Other outputs of project execution include work performance information, change requests, and updates to the project management plan and project documents.

Project managers and their teams are most often remembered for how well they executed a project and handled difficult situations. Likewise, sports teams around the world know that the key to winning is good execution. Team coaches can be viewed as project managers, with each game a separate project. Coaches are often judged on their win-loss record, not on how well they planned for each game. On a humorous note, when one losing coach was asked what he thought about his team's execution, he responded, "I'm all for it!"

Managing Project Knowledge

PMI added this new process to the *PMBOK® Guide* in 2017 to highlight the importance of managing project knowledge. There are two basic types of knowledge:

1. **Explicit knowledge** : This type of knowledge can be easily explained using words, pictures, or numbers and is easy to communicate, store, and distribute. Examples include information found in textbooks and encyclopedias as well as project documents and plans.
2. **Tacit knowledge** : Unlike explicit knowledge, tacit knowledge, sometimes called informal knowledge, is difficult to express and is highly personal. Examples include beliefs, insight, and experience. It is often shared through conversations and interactions between people. Many organizations set up programs like mentorships, communities of practice, or workshops to assist in passing on tacit knowledge.

Knowledge management should be done before, during, and after projects are completed. It is often very difficult to accomplish. Organizations can provide explicit knowledge in writing or other formats, including audio, pictures, and videos. They can also build a culture of trust and sharing so people work together to pass on tacit knowledge. One of the main outputs of

managing project knowledge is a lessons-learned register.

A lessons-learned register should document challenges, problems, realized risks and opportunities, and other content to assist in knowledge management on current and future projects. Contents can include a lessons learned ID, date identified, owner, name, category, situation, and recommendation. For example, if Nick from the opening case documented lessons learned, he might highlight one important lesson of understanding sponsor expectations, listing himself as the owner, communications as the category, being offered a severance package as the situation that prompted the lesson, and a recommendation to ask the project sponsor about expectations on a regular basis.

It is best if project team members document lessons learned throughout the life of the project. At the end of the project, all lessons learned should be both discussed with the team and archived in a lessons learned repository, a key organizational process asset.

Advice for Young Professionals

Hopefully you are interested in finding and using existing knowledge as well as creating new knowledge. Many college students excel in this area based on their experiences doing rigorous coursework. To stand out in your job, consider volunteering to be in charge of creating your project team's lessons-learned register. You could create a shared online space where team members and other stakeholders can share information. Be sure to read through and organize and edit inputs, if needed. Consider pairing up with a more experienced person to develop something that is of real value to your team and organization.

Monitoring and Controlling Project Work

On large projects, many project managers say that 90 percent of the job is communicating and managing changes. Changes are inevitable on most projects, so it's important to develop and follow a process to monitor and control changes.

Monitoring project work includes collecting, measuring, and disseminating performance information. It also involves assessing measurements and analyzing trends to determine what process improvements can be made. The project team should continuously monitor project performance to assess the overall health of the project and identify areas that require special attention.

The project management plan, project documents, work performance information,

agreements, enterprise environmental factors, and organizational process assets are all important inputs for monitoring and controlling project work.

The project management plan provides the baseline for identifying and controlling project changes. A [baseline](#) is a starting point, a measurement, or an observation that is documented so that it can be used for future comparison. For example, the project management plan includes a section that describes the work to perform on a project. This section of the plan describes the key deliverables for the project, the products of the project, and quality requirements. The schedule section of the project management plan lists the planned dates for completing key deliverables, and the budget section of the plan provides the planned cost of these deliverables. The project team must focus on delivering the work as planned. If the project team or someone else causes changes during project execution, the team must revise the project management plan and have it approved by the project sponsor. Many people refer to different types of baselines, such as a cost baseline or schedule baseline, to describe different project goals more clearly and performance toward meeting them.

Schedule and cost forecasts, validated changes, and work performance information provide details on how project execution is going. The main purpose of this information is to alert the project manager and project team about issues that are causing problems or might cause problems in the future. The project manager and project team must continuously monitor and control project work to decide if corrective or preventive actions are needed, what the best course of action is, and when to act.

Media Snapshot

Few events get more media attention than the Olympic Games. Imagine all the work involved in planning and executing an event that involves thousands of athletes from around the world with millions of spectators. The 2002 Olympic Winter Games and Paralympics took five years to plan and cost more than \$1.9 billion. PMI presented the Salt Lake Organizing Committee (SLOC) with the Project of the Year award for delivering world-class games that, according to the International Olympic Committee, “made a profound impact upon the people of the world.”*

Four years before the Games began, the SLOC used a Primavera software-based system with a cascading color-coded WBS to integrate planning. A year before the Games, the team added a Venue Integrated Planning Schedule to help integrate resource needs, budgets, and plans. For example, this software helped the team coordinate different areas involved in controlling access into and around a venue, such as roads, pedestrian pathways, seating and safety provisions, and hospitality areas, saving nearly \$10 million.

When the team experienced a budget deficit three years before the Games, it separated

“must-have” items from “nice-to-have” items and implemented a rigorous expense approval process. According to Matthew Lehman, SLOC managing director, using classic project management tools turned a \$400 million deficit into a \$100 million surplus.

The SLOC also used an Executive Roadmap, a one-page list of the top 100 activities during the Games, to keep executives apprised of progress. Activities were tied to detailed project information within each department’s schedule. A 90-day highlighter showed which managers were accountable for each integrated activity. Fraser Bullock, SLOC Chief Operating Officer and Chief, said, “We knew when we were on and off schedule and where we had to apply additional resources. The interrelation of the functions meant they could not run in isolation—it was a smoothly running machine.”*

Important outputs of monitoring and controlling project work include change requests and work performance reports. Change requests include recommended corrective and preventive actions and defect repairs. Corrective actions should result in improvements in project performance. Preventive actions reduce the probability of negative consequences associated with project risks. Defect repairs involve bringing defective deliverables into conformance with requirements. For example, if project team members have not been reporting hours that they worked, a corrective action would show them how to enter the information and let them know that they need to do it. An example of a preventive action might be modifying a time-tracking system screen to avoid common errors people have made in the past. A defect repair might be having someone redo an incorrect entry. Many organizations use a formal change request process and forms to keep track of project changes, as described in the next section. Work performance reports include status reports, progress reports, memos, and other documents used to communicate performance.

Performing Integrated Change Control

Integrated change control involves identifying, evaluating, and managing changes throughout the project life cycle. The three main objectives of integrated change control are as follows:

- *Influencing the factors that create changes to ensure that changes are beneficial:* To ensure that changes are beneficial and that a project is successful, project managers and their teams must make trade-offs among key project dimensions, such as scope, time, cost, and quality.
- *Determining that a change has occurred:* To determine that a change has occurred, the project manager must know the status of key project areas at all times. In addition, the

project manager must communicate significant changes to top management and key stakeholders. Top management and other key stakeholders do not like surprises, especially ones that mean the project might produce less, take longer to complete, cost more than planned, or create products of lower quality.

- *Managing actual changes as they occur:* Managing change is a key role of project managers and their teams. It is important that project managers exercise discipline in managing the project to help minimize the number of changes that occur.

Important inputs to the integrated change control process include the project management plan, project documents, work performance information, change requests, enterprise environmental factors, and organizational process assets. Important outputs include approved change requests and updates to the project management plan and project documents.

Change requests are common on projects and occur in many different forms. They can be oral or written, formal or informal. For example, a project team member responsible for installing a server might ask the project manager if it is acceptable to order a server with a faster processor than planned. The server is from the same manufacturer and has the same approximate cost. Because this change is positive and should have no negative effects on the project, the project manager might give a verbal approval at the progress review meeting. Nevertheless, it is still important that the project manager document this change to avoid any potential problems. The appropriate team member should update the scope statement to include the new server specifications.

Still, keep in mind that many change requests can have a major impact on a project. For example, customers who change their minds about the number of hardware components they want as part of a project will have a definite impact on its scope and cost. Such a change might also affect the project's schedule. The project team must present such significant changes in written form to the project sponsor, and there should be a formal review process for analyzing and deciding whether to approve these changes.

Change is unavoidable and often expected on most IT projects. Technologies change, personnel change, organizational priorities change, and so on. A good change control system is also important for project success.

Change Control on IT Projects

From the 1950s to the 1980s, IT was often referred to as data automation or data processing. At that time, a widely held view of project management was that the project team should strive to do exactly what it planned, on time and within budget. The problem with this view

was that project teams could rarely meet original project goals, especially for projects that used new technologies. Stakeholders rarely agreed up front on the scope of the project or what the finished product should look like. Time and cost estimates created early in a project were rarely accurate.

Beginning in the 1990s, most project managers and top management realized that project management is a process of constant communication and negotiation about project objectives and stakeholder expectations. This view assumes that changes happen throughout the project life cycle and recognizes that changes are often beneficial to some projects. For example, if a project team member discovers a new hardware or software technology that could satisfy customers' needs for less time and money, the project team and key stakeholders should be open to making major changes in the project.

All projects will have some changes, and managing them is a key issue in project management, especially for IT projects. Many IT projects involve the use of hardware and software that is updated frequently. To continue the example from earlier in this section, the initial plan for ordering the server might have identified a model that used cutting-edge technology at the time. If the actual server order occurred six months later, it is quite possible that a more powerful server could be available at the same cost. This example illustrates a positive change. On the other hand, the server manufacturer specified in the project plan could go out of business, which would result in a negative change. IT project managers should be accustomed to such changes and build some flexibility into their project plans and execution. Customers for IT projects should also be open to meeting project objectives in different ways.

Some changes might make sense but be too large to fit into a current project. Remember that projects have scope, time, cost, and other goals, and changes often affect those goals. If the organization wants to meet time and cost goals, for example, it must control changes to the project's scope. Organizations often decide to document some change requests and include them in an upgrade to the current project.

Even if project managers, project teams, and customers are flexible, it is important that projects have a formal change control system. This formal system is necessary to plan for managing change.

Change Control System

A [change control system](#) is a formal, documented process that describes when and how official project documents may be changed. It also describes the people authorized to make changes, the paperwork required for these changes, and any automated or manual tracking systems the project will use. A change control system often includes a change control board, configuration management, and a process for communicating changes.

A **change control board (CCB)** is a formal group of people responsible for approving or rejecting changes to a project. The primary functions of a CCB are to provide guidelines for preparing change requests, evaluating change requests, and managing the implementation of approved changes. An organization could have key stakeholders for the entire organization on this board, and a few members could rotate based on the unique needs of each project. By creating a formal board and a process for managing changes, overall change control should improve.

However, CCBs can have some drawbacks, such as the time it takes to make decisions on proposed changes. CCBs often meet only once a week or once a month and may not make decisions in one meeting. Some organizations have streamlined processes for making quick decisions on smaller project changes. One company created a “48-hour policy,” in which task leaders on a large IT project would reach agreements on key decisions or changes within their expertise and authority. The person in the area most affected by this decision or change then had 48 hours to go to top management and seek approval. If the project team’s decision could not be implemented for some reason, the top manager consulted would have 48 hours to reverse the decision; otherwise, the project team’s decision was approved. This type of process is an effective way to deal with the many time-sensitive decisions or changes that project teams must make on IT projects.

Configuration management is another important part of integrated change control.

Configuration management ensures that the descriptions of the project’s products are correct and complete. It involves identifying and controlling the functional and physical design characteristics of products and their support documentation. Members of the project team, frequently called configuration management specialists, are often assigned to perform configuration management for large projects. Their job is to identify and document the functional and physical characteristics of the project’s products, control any changes to such characteristics, record and report the changes, and audit the products to verify conformance to requirements. Visit the Institute of Configuration Management’s website (www.icmhq.com) for more information on this topic.

Global Issues

Rapid changes in technology, such as the increased use of mobile roaming for communications, often cause governments around the world to take action.

Incompatible hardware, software, and networks can make communications difficult in some regions, and a lack of competition can cause prices to soar. Fortunately, a group called the Organisation for Economic Co-operation and Development (OECD) promotes policies that will improve the economic and social well-being of people around the world. In February 2012, the OECD called upon its members’ governments to boost competition

in international mobile roaming markets. “The OECD has detailed a series of measures that, if implemented would, it says: ‘encourage effective competition, raise consumer awareness and protection and ensure fairer prices.’ If these fail to produce results it says: ‘Governments should consider price regulation for roaming services,’ and that ‘Wholesale roaming services could be regulated by means of bilateral or multilateral wholesale agreements with mutually established price caps.’”*

OECD also encourages expansion of other technologies. By the end of 2013, wireless broadband penetration grew to 72.4 percent in the 34-country OECD area. Strong demand for smartphones and tablets helped wireless broadband subscriptions grow by 14.6 percent.* Wireless broadband penetration continues to grow globally, thanks in part to these agreements.

Another critical factor in change control is communication. Project managers should use written and oral performance reports to help identify and manage project changes. For example, on software development projects, most programmers must make their edits to a copy of the master file in a database; to ensure version control, programmers must “check out” the file to edit it. If two programmers are allowed to check out the same file, they must coordinate to merge their changes. In addition to written or formal communication methods, oral and informal communications are also important. Some project managers have stand-up meetings once a week or even every morning, depending on the nature of the project. The goal of a stand-up meeting is to quickly communicate what is most important for the project. For example, the project manager might have a stand-up meeting every morning with all of the team leaders. There might be a stand-up meeting every Monday morning with all interested stakeholders. Requiring participants to stand keeps meetings short and forces everyone to focus on the most important project events.

Why is good communication so critical to success? One of the most frustrating aspects of project change is not having everyone coordinated and informed about the latest project information. Again, it is the project manager’s responsibility to integrate all project changes so that the project stays on track. The project manager and staff members must develop a system for notifying everyone affected by a change in a timely manner. E-mail, real-time databases, cell phones, and the Web make it easy to disseminate the most current project information. You will learn more about good communication in **Chapter 10**, Project Communications Management.

Table 4-4 lists suggestions for performing integrated change control. As described earlier, project management is a process of constant communication and negotiation. Project managers should plan for changes and use appropriate tools and techniques, such as a change

control board, configuration management, and good communication. It is helpful to define procedures for making timely decisions about small changes, use written and oral performance reports to help identify and manage changes, and use software to assist in planning, updating, and controlling projects. An IT steering committee can also be used to represent the entire IT Portfolio—a key consideration in change management, especially when resources must shift to accommodate approved changes. In addition, shifts in strategic direction by the IT steering committee can reach down to influence existing projects, actually creating changes.

Table 4-4.	Suggestions for performing integrated change control
View project management as a process of constant communication and negotiation.	
Plan for change.	
Establish a formal change control system, including a change control board (CCB) and IT steering committee.	
Use effective configuration management.	
Define procedures for making timely decisions about smaller changes.	
Use written and oral performance reports to help identify and manage change.	
Use project management software and other software to help manage and communicate changes.	
Focus on leading the project team and meeting overall project goals and expectations.	

Project managers must also provide strong leadership to steer the project to successful completion. They must not get too involved in managing project changes. Project managers should delegate much of the detailed work to project team members and focus on providing overall leadership for the project in general. Remember, project managers must focus on the big picture and perform project integration management well to lead their team and organization to success.

Closing Projects or Phases

The last process in project integration management is closing the project or phase, which requires that you finalize all activities and transfer the completed or cancelled work to the appropriate people. The main inputs to this process are the project charter, project management plan, project documents, accepted deliverables, business documents, agreements, procurement documentation, and organizational process assets. The main tools and techniques are expert judgment, data analysis, and meetings. The outputs of closing projects are as follows:

- *Project documents updates*: All project documents should be reviewed and marked as final versions. Others may need to refer to these documents in the future, and it's important that they be accurate.
- *Final product, service, or result transition*: Project sponsors are usually most interested in making sure they receive delivery of the final products, services, or results they expected when they authorized the project. For items produced under contract, formal acceptance or handover includes a written statement that the terms of the contract were met. Internal projects can also include some type of project completion form.
- *Final report*: A final project report and presentation are also commonly used during project closing. The *PMBOK® Guide – Sixth Edition* suggests the final report include the following topics:
 - Summary level description of the project or phase
 - Scope objectives, the criteria used to evaluate the scope, and evidence that the completion criteria were met
 - Quality objectives, the criteria used to evaluate the project and product quality, and the verification and validation information
 - Schedule objectives, including planned and actual milestone delivery dates and reasons for variances
 - Cost objectives, including the acceptable cost range, actual costs, and reasons for variances
 - Summary of how the final project, service, or result achieved the benefits that the project was undertaken to address.
 - Summary of how the final project, service, or result achieved the business needs identified in the business plan.

- Summary of any risks or issues encountered on the project and how they were addressed
- *Organizational process asset updates:* The project team should provide a list of project documentation, project closure documents, and historical information produced by the project in a useful format. This information is considered a process asset. Project teams normally produce a final project report, which often includes a transition plan describing work to be done as part of operations after the project is completed. Teams also often write a lessons-learned report at the end of a project, and this information can be a tremendous asset for future projects. (See Chapter 10, Project Communications Management, for more information on creating final project reports, lessons-learned reports, and other project communications.) Several organizations also conduct a post-implementation review to analyze whether the project achieved what it set out to do. Information from this type of review also becomes an organizational process asset for future projects.

Using Software to Assist in Project Integration Management

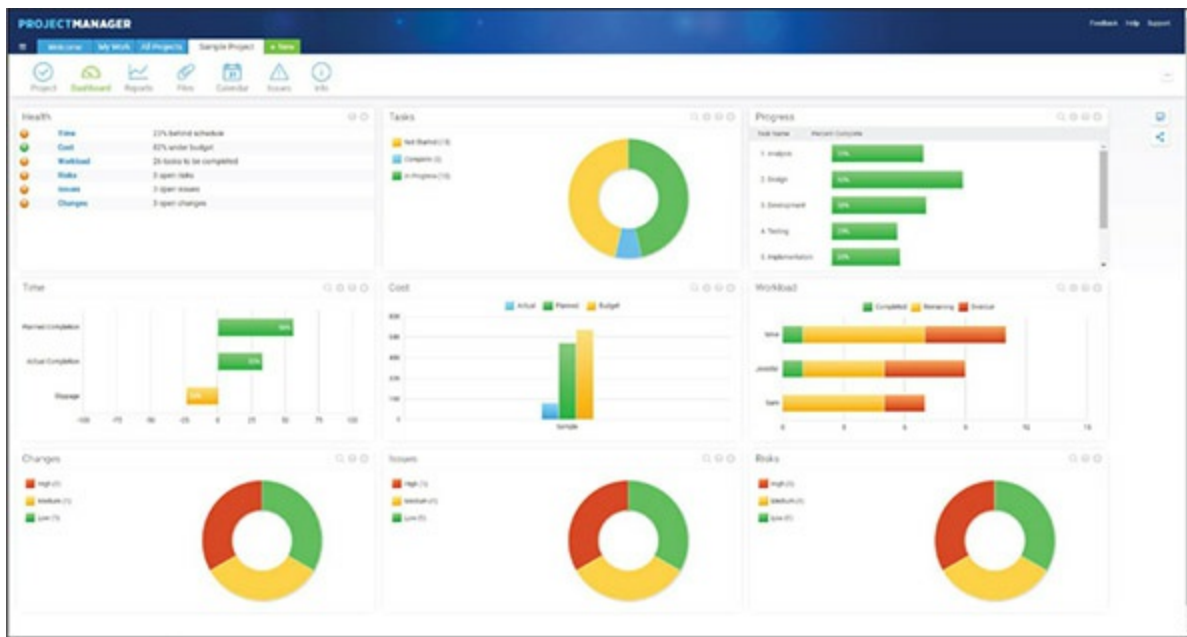
As described throughout this chapter, project teams can use various types of software to assist in project integration management. Project teams can create documents with word-processing software, give presentations with presentation software, track information with spreadsheets, databases, or customized software, and transmit information using various types of communication software.

Project management software is also an important tool for developing and integrating project planning documents, executing the project management plan and related project plans, monitoring and controlling project activities, and performing integrated change control. Small project teams can use low-end or midrange project management software to coordinate their work. For large projects, such as the Olympic Games described in the Media Snapshot earlier in this chapter, organizations may benefit most from high-end tools that provide enterprise project management capabilities and integrate all aspects of project management.

As you learned in **Chapter 1**, organizations can also use software to assist in project portfolio management and optimization. Portfolio management software often provides many types of charts or dashboards to help managers see the big picture in managing portfolios of projects. For example, Figure 4-8 shows a project dashboard from the software ProjectManager. Students can get a free 120-day trial of this software from www.projectmanager.com. All projects can benefit from using some type of project management information system to coordinate and communicate project information.

Figure 4-8.

Sample portfolio management software screens



Source: www.projectmanager.com

In recent years, the growth of cloud computing has transformed how, when, and where people work. Many project management software tools are now available in the cloud, as are other tools and services. Most business professionals and students now store their files using some type of cloud storage (Google Drive, Microsoft OneDrive, DropBox, etc.). Many cloud tools are accessible via smartphones and tablets as well as laptops and desktops. Cloud computing enables users to easily access and share information from any location at any time. Project integration management is not easy, but the cloud has definitely helped provide easier access to important information and applications.

Considerations for Agile/Adaptive Environments

The *PMBOK® Guide – Sixth Edition* includes a short section with this title for each knowledge area. The following information is provided for project integration management:

Iterative and agile approaches promote the engagement of team members as local domain experts in integration management. The team members determine how plans and components should integrate.

The expectations of the project manager as noted in the Key Concepts for Integration Management do not change in an adaptive environment, but control of the detailed product planning and delivery is delegated to the team. The project manager's focus is on building a collaborative decision-making environment and ensuring that the team has the ability to respond to changes. This collaborative approach can be further enhanced when team members possess a broad skill base rather than a narrow

specialization.*

For many large, traditional projects, several team members are not directly involved in integration management and do have specialized skills. It is up to the project managers and their team leads to focus on integration while team members focus on completing often complex, technical tasks. This does not mean that they should not contribute to project planning. As emphasized in this chapter, those who do the work should help plan the work.

Project managers using *any* product life cycle should focus on creating a collaborative decision-making environment and providing opportunities for team members to develop additional skills. Recall that the Agile Manifesto values the following:

- Individuals and interactions over processes and tools
- Working software over comprehensive documentation
- Customer collaboration over contract negotiation
- Responding to change over following a plan

As you can see, a lot of work is involved in project integration management. Project managers and their teams must focus on pulling all the elements of a project together to successfully complete it.

Case Wrap-Up

Without consulting Nick Carson or his team, Nick's CEO hired a new person, Jim Lansing, to act as a middle manager between himself and the people in Nick's department. The CEO and other top managers really liked Jim; he met with them often, shared ideas, and had a great sense of humor. He started developing standards that the company could use to help manage projects in the future. For example, he developed templates for creating plans and progress reports and put them on the company's intranet. However, Jim and Nick did not get along, especially after Nick heard that Jim told the CEO that he was hard to work with and preoccupied with the birth of his son.

Nick was very frustrated after hearing this and stormed into the CEO's office. The CEO suggested that Nick move to another department, but Nick did not like that so the CEO offered Nick a severance package to leave the company. Because of the planned corporate buyout, the CEO knew the company might have to let some people go anyway. Nick talked the CEO into giving him a two-month sabbatical he had earned plus a higher percentage on his stock options. After discussing the situation with his wife and realizing that he would get over \$70,000 if he resigned, Nick took the severance package. The CEO

called a meeting with Nick and Jim to give them both advice. He told Nick that he was one of the best technical people he ever worked with and that he would thrive in another organization in a technical role. He told Jim that he was a great organizer, but he had to learn to respect technical people more. They all left the meeting with a feeling of mutual respect.

Chapter Summary

Project integration management is usually the most important project management knowledge area, because it ties together all the other areas of project management. A project manager's primary focus should be on project integration management.

Before selecting projects to pursue, it is important for organizations to follow a strategic planning process. Many organizations perform a SWOT analysis to help identify potential projects based on their strengths, weaknesses, opportunities, and threats. IT projects should support the organization's overall business strategy. Common techniques for selecting projects include focusing on broad organizational needs, categorizing projects, performing financial analyses, developing weighted scoring models, and using balanced scorecards.

Project integration management includes the following processes:

- Developing the project charter, which involves working with stakeholders to create the document that formally authorizes a project. Project charters can have different formats, but they should include basic project information and signatures of key stakeholders.
- Creating an assumption log to document and track assumptions throughout the life of the project.
- Developing the project management plan, which involves coordinating all planning efforts to create a consistent, coherent document. The main purpose of project plans is to facilitate action.
- Directing and managing project work, which involves carrying out the project plans by performing the activities included in it. Project plan execution usually requires the majority of a project's budget.
- Managing project knowledge, which involves using existing knowledge and creating new knowledge to achieve project objectives while also contributing to organizational learning.
- Monitoring and controlling project work, which is necessary to meet the performance objectives of the project. The project team should continuously monitor project performance to assess the overall health of the project.
- Performing integrated change control, which involves identifying, evaluating, and managing changes throughout the project life cycle. A change control system often

includes a change control board (CCB), configuration management, and a process for communicating changes.

- Closing the project or phase involves finalizing all project activities. It is important to follow good procedures to ensure that all project activities are completed and that the project sponsor accepts delivery of the final products, services, or results.

Several types of software products are available to assist in project integration management.

Be sure to consider how project integration management can differ in agile/adaptive environments.

Discussion Questions

1. Describe project integration management. How does it relate to the project life cycle, stakeholders, and the other project management knowledge areas?
2. Describe options that organizations have for selecting projects that align with their mission or strategy, and describe how each might work differently in the selection of IT projects.
3. Summarize key work involved in each of the six processes of project integration management.
4. Either from your own experience or by searching the Internet, describe a well-planned and executed project. Describe a failed project. What elements of project integration might have contributed to the success or failure of each?
5. Discuss the importance of following a well-integrated change control process on IT projects. What consequences can result from not following these best practices? What types of change control would be appropriate for small IT projects? What types of change control would be appropriate for large ones?

Quick Quiz

1. Which of the following processes is not part of project integration management?
 - a. Developing the project business case
 - b. Developing the project charter
 - c. Developing the project management plan
 - d. Closing the project or phase

2. What is the last step in the four-stage planning process for selecting IT projects?
 - a. IT strategy planning
 - b. Business area analysis
 - c. Mind mapping
 - d. Resource allocation
3. Which of the following is not a best practice for new product development projects?
 - a. Aligning projects and resources with business strategy
 - b. Selecting projects that will take less than two years to provide payback
 - c. Focusing on customer needs in identifying projects
 - d. Assigning project managers to lead projects
4. A new government law requires an organization to report data in a new way. Which of the following categories would include a new information system project to provide this data?
 - a. Problem
 - b. Opportunity
 - c. Directive
 - d. Regulation
5. If estimates for total discounted benefits for a project are \$120,000 and total discounted costs are \$100,000, what is the estimated return on investment (ROI)?
 - a. \$20,000
 - b. \$120,000
 - c. 20 percent
 - d. 120 percent
6. A ____ is a document that formally recognizes the existence of a project and provides direction on the project's objectives and management.

- a. project charter
- b. contract
- c. business case
- d. project management plan

7. Which of the following items is not normally included in a project charter?

- a. The name of the project manager
- b. Budget information
- c. Stakeholder signatures
- d. A Gantt chart

8. ____ ensures that the descriptions of the project's products are correct and complete.

- a. Configuration management
- b. Integrated change control
- c. Integration management
- d. A change control board

9. Which of the following is not a suggestion for performing integrated change control?

- a. Use good configuration management.
- b. Minimize change.
- c. Establish a formal change control system.
- d. View project management as a process of constant communication and negotiation.

10. What tool and technique is used for all processes of project integration management?

- a. Project management software
- b. Templates
- c. Expert judgment
- d. All of the above

Exercises

1. Write a short paper based on the chapter's opening case. Answer the following questions:
 - a. What do you think the real problem was in this case?
 - b. Does the case present a realistic scenario? Why or why not?
 - c. Was Nick Carson a good project manager? Why or why not?
 - d. What should top management have done to help Nick?
 - e. What could Nick have done to be a better project manager?
2. Download a free trial of MindView mind mapping software from www.matchware.com or similar software and create a mind map of a SWOT analysis for your organization. Include at least two strengths, weaknesses, opportunities, and threats, and then provide ideas for at least three potential projects. Or, you can use your college or university for the SWOT analysis, focusing on what it can do to improve services for students.
3. Use spreadsheet software to create **Figures 4-4**, Figure 4-5, Figure 4-6 and Figure 4-7 in this text. Make sure your formulas work correctly.
4. Perform a financial analysis for a project using the format provided in **Figure 4-5**. Assume that the projected costs and benefits for this project are spread over four years as follows: Estimated costs are \$300,000 in Year 1 and \$40,000 each year in Years 2, 3, and 4. Estimated benefits are \$0 in Year 1 and \$120,000 each year in Years 2, 3, and 4. Use a 7 percent discount rate, and round the discount factors to two decimal places. Create a spreadsheet or use the business case financials template on the Companion website to calculate and clearly display the NPV, ROI, and year in which payback occurs. In addition, write a paragraph explaining whether you would recommend investing in this project, based on your financial analysis.
5. Create a weighted scoring model to determine grades for a course. Final grades are based on three exams worth 15 percent, 15 percent, and 25 percent, respectively; homework is worth 15 percent; and a group project is worth 30 percent. Enter scores for three students. Assume that Student 1 earns 100 percent (or 100) on every item. Assume that Student 2 earns 70 percent on each of the exams, 80 percent on the homework, and 95 percent on the group project. Assume that Student 3 earns 90 percent on Exam 1, 80 percent on Exam 2, 75 percent on Exam 3, 80 percent on the homework, and 70 percent on the group project. You can use the weighted scoring model template, create your own spreadsheet, or make the matrix by hand.
6. Develop an outline (major headings and subheadings only) for a project management

plan to create a mobile-friendly website for your class, and then fill in the details for the introduction or overview section. Assume that this website would include a home page with links to a syllabus for the class, lecture notes or other instructional information, links to the website for this textbook, links to other websites with project management information, and links to personal pages for each member of your class and future classes. Also, include a bulletin board and chat room feature where students and the instructor can exchange information. Assume that your instructor is the project sponsor, you are the project manager, your classmates are your project team, and you have three months to complete the project.

7. Research software mentioned in this chapter, such as software for assisting in project selection, enterprise project management software, or cloud-based applications. Find at least two references and summarize your findings in a short memo to senior management.
8. Watch videos and read articles to research how two different organizations did a good job in directing and managing project work. You can search “PMI Project of the Year award” to find examples. Find at least four references and summarize your findings in a short paper.
9. Read the report “2015 Pulse of the Profession®: Capturing the Value of Project Management” by PMI, mentioned in the What Went Right feature. Summarize key points in this study and your opinion of it. Describe some specific steps organizations can take to provide a supportive culture for project management.
10. Find two recent articles related to managing project knowledge. Summarize your findings in a short paper.

Running Case

Note: Additional running cases are provided on the Companion website, including the Manage Your Health, Inc. case from the seventh edition of this text, now with updated tasks based on this edition. Template files are also available on the Companion website. This running case starts here and continues through Chapter 13. Tasks based on this case are explained in the following Tasks section; throughout the book, these tasks will build on work done in previous chapters and scenarios.

Economic inequality is a huge issue. A recent study found that the world’s 80 wealthiest individuals own as much as the entire world’s poorest 3.5 billion people. The richest 1 percent of the world’s population control half of the world’s total wealth.* Many individuals, corporations, charities, and government agencies have projects and programs in place to attempt to tackle this and other important global issues such as sustainability, but there are many opportunities to do more.

A grassroots group of college students has decided to work together to do their part in making the world a better place. The students are from many different countries, and several of them met at global conferences, through study abroad experiences, or on various Internet groups. Strategic goals of this group include developing skills for both college students and for needy populations, sharing information on existing products and services that promote economic growth and sustainability, and promoting entrepreneurship. Leaders of this group were reviewing some ideas for projects (all with a significant IT component) that they could do to support their strategic goals:

1. *Global Treps* : Many people are familiar with the television show called *Shark Tank* where entrepreneurs (sometimes called “treps”) present their business ideas to a group of investors or sharks. Several colleges, high schools, and even elementary schools throughout the world hold unique versions of a shark tank like event. You believe that creating a non-profit organization with one central mobile-friendly website/application to assist groups in organizing these types of events would spark even more entrepreneurs throughout the world. You would plan to hold several shark tank like events during the term of the project and create a site and applications to help continue developing more global treps. This site/application would include the following capabilities:
 - Provide guidelines and templates for running a shark tank type of event.
 - Accept donations from potential investors targeted toward specific schools or organizations wishing to host an event (similar to the popular www.donorschoose.org site where people can fund teacher’s requests).
 - Accept ideas for needed new products or services.
 - Provide the ability for organizations to create their own custom site to solicit local participants and sharks, accept applications, and promote the winners as well as losers.
 - Research ideas for a mechanism where a certain percentage of all donations and profits earned by contestants are donated back to the Global Treps organization.
 - Provide an online version of the events by showing videos of contestants and live reactions of the sharks while also getting live inputs and donations from viewers.
2. *Change the Laws Campaign* : Launch a global campaign to change laws to reduce further income inequality and promote social responsibility. This project would also involve creating a mobile-friendly website/application that would include information about current and proposed laws, allow discussions of potential ideas to change laws, organize

people to contact appropriate lawmakers, etc.

3. *Wealthy Unite* : Develop a system to enable the richest people in the world to provide their input on how they can make the world a better place. Provide information on what several people are currently doing (i.e., Bill Gates, Warren Buffet, famous celebrities, etc.) to promote philanthropy. Allow others to donate to suggested causes and recommend other ways to reduce economic inequality.
4. *Global Smart Shoppers* : Develop a mobile app and website that recommends products and services produced by organizations that promote social responsibility. Customize the app so it works in any country in the user's desired language. Work with large companies that do not currently sell products or services in certain countries to expand to regions in need. Allow small companies to easily add their products and services to the shopping network.

Tasks

1. Summarize each of the proposed projects using a simple table format suitable for presentation to top management. Include the name of each project, identify how each one supports business strategies, assess the potential financial benefits and other benefits of each project, and provide your initial assessment of the value of each project. Write your results in a one-to two-page memo to top management, including appropriate back-up information and calculations.
2. Evaluate the four projects by preparing a weighted scoring model using the template provided on the Companion website for this text. Develop at least four criteria, assign weights to each criterion, assign scores, and then calculate the weighted scores. Print the spreadsheet and bar chart with the results. Also write a one-page paper that describes this weighted scoring model and the results.
3. Prepare a business case for the Global Treps project. Assume that the project will take six months to complete, use many volunteer hours, and cost about \$130,000 for hardware, software, travel, and labor. Use the business case template provided on the Companion website for this text. Be sure to research information on the television show and events held by colleges and other groups, which have been sparked by the need for more successful entrepreneurs. Also visit [DonorsChoose.org](https://www.donorschoose.org) to see how that site operates and look into steps for forming a non profit organization.
4. Prepare a draft project charter and an assumption log for the Global Treps project. Assume that the project will take six months to complete and have a

budget of \$130,000. Use the project charter template provided in this text and the sample project charter provided in **Table 4-1** as guides. For the assumption log, document at least two assumptions including an assumption ID, date, source, category, description, and status. You will be the project manager, and Dr. K. will be the project sponsor. Other team members will include Bobby, Ashok, Kim, and Alfreda. You plan to hold four shark tank like events plus develop the Global Treps site and application.

5. Start a lessons learned register to document knowledge learned on this project. Include a lessons learned ID, date, source, category, description, impact, recommendations, and proposed actions. Be creative in then making two entries.
6. Prepare a change request for the Global Treps project, using the template provided on the Companion website for this text. Assume that you have decided not to provide an online version of the show as it would be too much work for the initial project. Be creative when making up information.

Key Terms

[balanced scorecard](#) p.154

[baseline](#) p.166

[capitalization rate](#) p.148

[cash flow](#) p.148

[change control board \(CCB\)](#) p.169

[change control system](#) p.169

[configuration management](#) p.170

[cost of capital](#) p.148

[directives](#) p.147

[discount factor](#) p.149

[discount rate](#) p.148

[explicit knowledge](#)

[integrated change control](#) p.168

[interface management](#) p.140

[internal rate of return \(IRR\)](#) p.151

[mind mapping](#) p.142

[net present value \(NPV\) analysis](#) p.148

[opportunities](#) p.147

[opportunity cost of capital](#) p.148

[organizational process assets](#) p.155

[payback period](#) p.151

[problems](#) p.146

[project charter](#) p.155

[project integration management](#) p.138

[project management plan](#) p.158

[required rate of return](#) p.151

[return on investment \(ROI\)](#) p.150

[strategic planning](#) p.141

[SWOT analysis](#) p.142

[tacit knowledge](#)

[weighted scoring model](#) p.152

Chapter 5. Project Scope Management

Learning Objectives

After reading this chapter, you will be able to:

- List key reasons why good project scope management is important
- Describe the process of planning scope management
- Discuss methods for collecting and documenting requirements to meet stakeholder needs and expectations
- Explain the scope definition process and describe the contents of a project scope statement
- Discuss the process for creating a work breakdown structure using the analogy, top-down, bottom-up, and mind-mapping approaches
- Explain the importance of validating scope and how it relates to defining and controlling scope
- Given an information technology (IT) project situation, show how recommended approaches for controlling scope can improve the potential for project success
- Describe how software can assist in project scope management
- Discuss considerations for agile/adaptive environments

Opening Case

Kim Nguyen was leading a meeting to create the work breakdown structure (WBS) for her company's IT upgrade project. This project was necessary because of several high-priority, Internet-based applications the company was developing. The IT upgrade project involved creating and implementing a plan to make all employees' IT assets meet new corporate standards within nine months. These standards specified the minimum requirements for each desktop or laptop computer, including the type of processor, amount of memory, hard disk size, type of network connection, security features, and software. Kim knew that to perform the upgrades, the project team would first have to create a detailed inventory of all the current hardware, networks, and software in the entire company of 2,000 employees.

Kim had worked with other stakeholders to develop a project charter and initial scope statement. The project charter included rough cost and schedule estimates for the project and signatures of key stakeholders; the initial scope statement provided a start in defining the hardware, software, and network requirements as well as other information related to the project scope. Kim called a meeting with her project team and other stakeholders to further define the scope of the project. She wanted to get everyone's

ideas on what the project involved, who would do what, and how they could avoid scope creep. The company's new CEO, Walter Schmidt, was known for keeping a close eye on major projects. The company had started using a new project management information system that let everyone know the status of projects at a detailed and high level. Kim knew that a good WBS was the foundation for scope, time, and cost performance, but she had never led a team in creating one or allocating costs based on a WBS. Where should she begin?

What is Project Scope Management?

Many of the factors associated with project success, such as user involvement, clear business objectives, and optimized scope, are elements of project scope management.

A critically important and difficult aspect of project management is defining the scope of a project. **Scope** refers to *all* the work involved in creating the products of the project and the processes used to create them. Recall from Chapter 2 that a **deliverable** is a product created as part of a project. Deliverables can be product related, such as a piece of hardware or software, or process related, such as a planning document or meeting minutes. Project stakeholders must agree what the products of the project are and, to some extent, how they should be produced to define all of the deliverables.

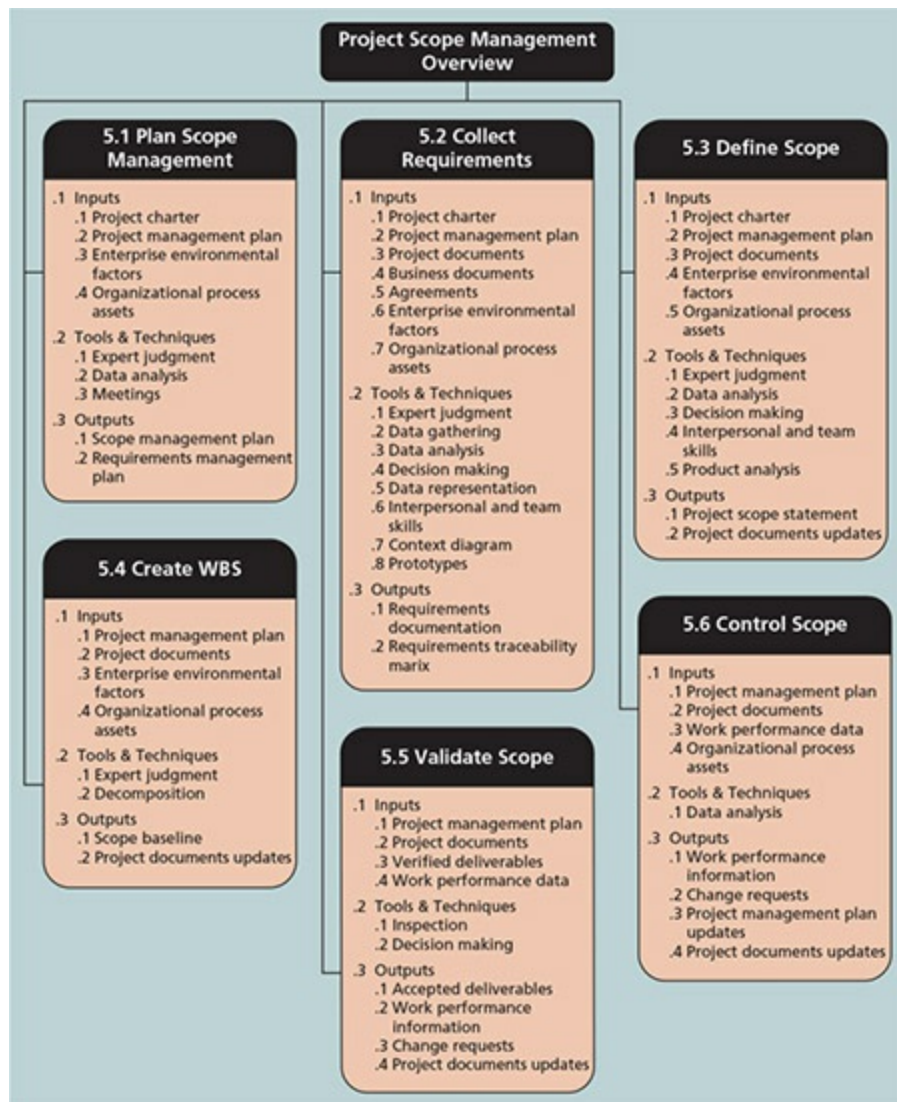
Project scope management includes the processes involved in defining and controlling what work is or is not included in a project. It ensures that the project team and stakeholders have the same understanding of what products the project will produce and what processes the project team will use to produce them. Six main processes are involved in project scope management:

1. *Planning scope management* involves determining how the project's scope and requirements will be managed.
2. *Collecting requirements* involves defining and documenting the features and functions of the products as well as the processes used for creating them.
3. *Defining scope* involves reviewing the scope management plan, project charter, requirements documents, and organizational process assets to create a scope statement, adding more information as requirements are developed and change requests are approved.
4. *Creating the WBS* involves subdividing the major project deliverables into smaller, more manageable components.

5. *Validating scope* involves formalizing acceptance of the project deliverables. Key project stakeholders, such as the customer and sponsor for the project, inspect and then formally accept the deliverables during this process. If the deliverables are not acceptable, the customer or sponsor usually requests changes.
6. *Controlling scope* involves controlling changes to project scope throughout the life of the project—a challenge on many IT projects. Scope changes often influence the team’s ability to meet project time and cost goals, so project managers must carefully weigh the costs and benefits of scope changes.

Figure 5-1 summarizes the inputs, tools and techniques, and outputs of project scope management.

Figure 5-1. Project scope management overview



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Planning Scope Management

The first step in project scope management is planning how the scope will be managed throughout the life of the project. After reviewing the project management plan, project charter, enterprise environmental factors, and organizational process assets, the project team uses expert judgment, data analysis, and meetings to develop two important outputs: the scope management plan and the requirements management plan.

The scope management plan is a subsidiary part of the project management plan, as described in [Chapter 4](#), Project Integration Management. It can be informal and broad or formal and detailed, based on the needs of the project. In fact, small projects may not need a written scope management plan, but large projects or highly technical projects often benefit from one. In general, a scope management plan includes the following information:

- *How to prepare a detailed project scope statement*: For example, are there templates or guidelines to follow? How much detail is needed to describe each deliverable?
- *How to create a WBS*: It is often difficult to create a good WBS. This section of the scope management plan would provide suggestions, samples, and resources for creating a WBS.
- *How to maintain and approve the WBS*: The initial WBS often changes, and project team members disagree on what should be included. The scope management plan describes guidelines for maintaining the WBS and getting approval for it.
- *How to obtain formal acceptance of the completed project deliverables*: It is extremely important to understand the process for obtaining formal acceptance of completed deliverables, especially for projects in which payments are based on formal acceptance.
- *How to control requests for changes to the project scope*: This process is related to performing integrated change control, as described in Chapter 4. Organizations often have guidelines for submitting, evaluating, and approving changes to scope, and this section of the scope management plan would specify how to handle change requests for the project.

Another important output of planning scope management is the requirements management plan. Before you learn about the contents of this document, it is important to understand what requirements are. The 1990 IEEE Standard Glossary of Software Engineering Terminology defines a requirement as follows:

1. A condition or capability needed by a user to solve a problem or achieve an objective.
2. A condition or capability that must be met or possessed by a system or system component to satisfy a contract, standard, specification, or other formally imposed document.
3. A documented representation of a condition or capability as in 1 or 2.*

The *PMBOK® Guide – Sixth Edition*, defines a [requirement](#) as “a condition or capability that is necessary to be present in a product, service, or result to satisfy a business need.” It further explains that requirements “include the quantified and documented needs and expectations of the sponsor, customer, and other stakeholders. These requirements need to be elicited, analyzed, and recorded in enough detail to be included in the scope baseline and be measured once project execution begins.”*

For example, the chapter’s opening case describes a project for upgrading IT assets to meet corporate standards. These standards specify the minimum requirements for each laptop, such as the type of processor, amount of memory, and hard disk size. The documented requirements for this project, therefore, might state that all laptops include a specific type of processor, a minimum amount of memory, and a minimum hard drive size.

For software projects, it is helpful to divide requirements development into the software engineering steps called *elicitation*, *analysis*, *specification*, and *validation*. These steps include all the activities involved in gathering, evaluating, and documenting requirements for a software or software-containing product. It is also important to use an iterative approach to defining requirements because they are often unclear early in a project.

The [requirements management plan](#) documents how project requirements will be analyzed, documented, and managed. A requirements management plan can include the following information:

- How to plan, track, and report requirements activities
- How to perform configuration management activities
- How to prioritize requirements
- How to use product metrics
- How to trace and capture attributes of requirements

What Went Right?

Several studies cite how difficult it is to manage requirements. Finding qualified people—business analysts—to do the job is equally difficult. The U.S. Bureau of Labor Statistics has projected the number of jobs for business analysts to increase 19 percent by 2022.* A PMI survey found that only 49 percent of respondents had the resources in place to do requirements management properly and 53 percent failed to use a formal process to validate requirements.*

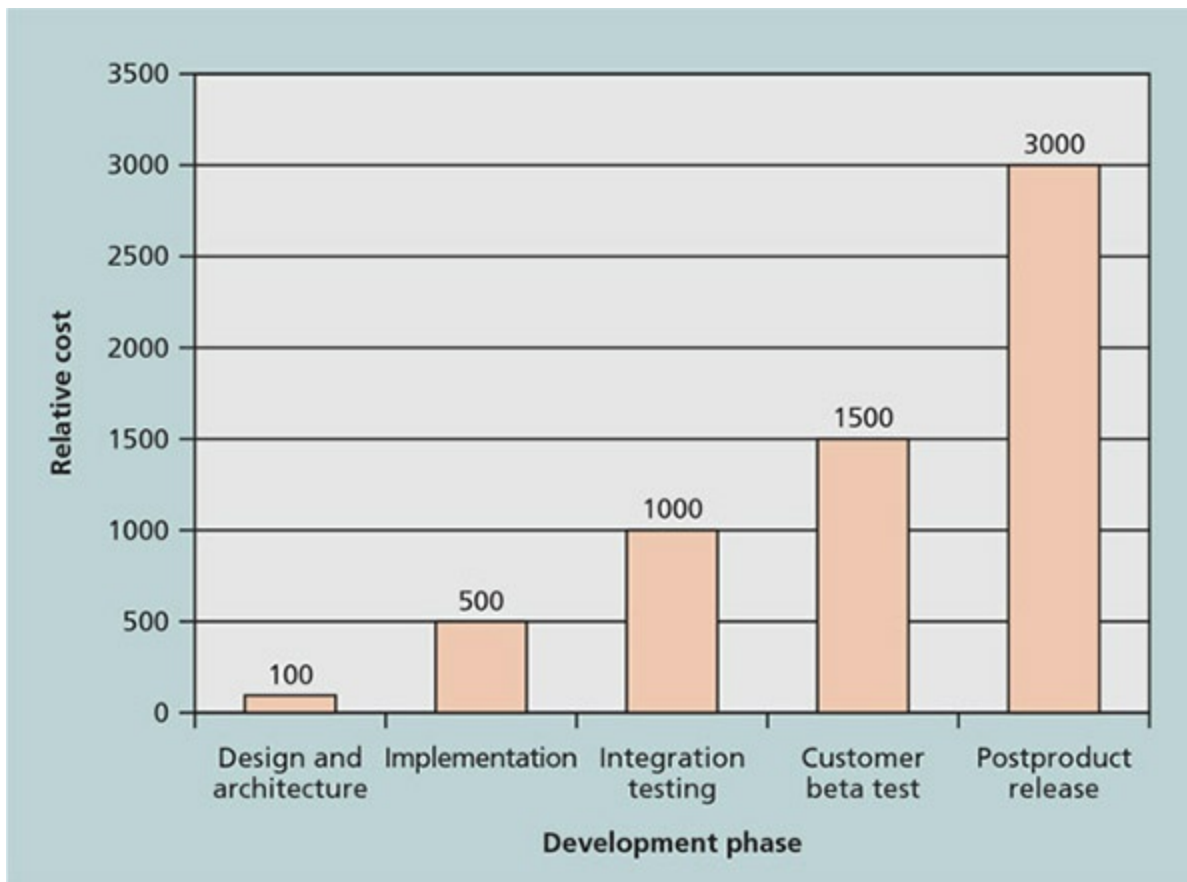
Fortunately, several organizations have recognized this need and have developed training and certification programs for business analysts.

- The International Institute of Business Analysis (IIBA®) issues both the Certified Business Analysis Professional (CBAP®) and Certification of Competency in Business Analysis (CCBA®) certifications. This organization had over 29,000 members in 2018 and published the third edition of the *Guide to Business Analysis Body of Knowledge (BABOK® Guide)*.
- The International Requirements Engineering Board (IREB) provides certification as a Certified Professional for Requirements Engineering (CPRE). Over 37,000 people in 73 countries held this certification in 2018.
- PMI began offering a new certification as a PMI Professional in Business Analysis (PMI-PBA®) in 2014.
- Several colleges and universities are offering majors and minors in business or data analytics at the undergraduate and graduate levels.

Collecting Requirements

The second step in project scope management is often the most difficult: collecting requirements. A major consequence of not defining requirements well is rework, which can consume up to half of project costs, especially for software development projects. As illustrated in **Figure 5-2**, it costs much more (up to 30 times more) to correct a software defect in later development phases than to fix it in the requirements phase. Everyone can cite examples in all types of industries of how important it is to understand requirements as early as possible. For example, if you are designing a house, it is much cheaper to decide where windows and walls will be on paper or a computer screen than after the entire house is framed. New processes and technologies are making it easier to define and implement requirements, but it is still one of the most challenging aspects of project scope management.

Figure 5-2. Relative cost to correct a software defect



Source: IBM Software Group, "Minimizing code defects to improve software quality and lower development costs," Rational Software (October 2008).

Best Practice

Google continues to be among the most admired companies in the world, well-known for being a great place to work and for being innovative in developing new products. James Whittaker, former engineering director at Google, responsible for testing Chrome, maps, and Google Web apps, wrote a book with co-authors Jason Arbon and Jeff Carollo (also former Google employees) called *How Google Tests Software*. Whittaker also worked for Microsoft and as a professor and is one of the best-known names in testing software.

In *How Google Tests Software*, the authors explain that Google used to be like other large companies: testing was not part of the mainstream, and people who did it were underappreciated and overworked. It took Google a long time to develop the people, processes, and technologies it uses today to develop software. A key part of success at Google was changing the culture. Google has fewer dedicated testers than most of its competitors have on a single product team. How is that possible? The authors explain that quality rests on the shoulders of those writing the code. The employees at Google do things the best, fastest way possible for them, knowing that coders are responsible for the quality of their own work. They don't rely on testers to ensure quality.

It is also interesting to note that Google does not believe in fads or buzzwords. In an interview, InfoQ asked the authors how Google's approach fits into the wider Agile community. Their response: "Google doesn't try to be part of the Agile community. We don't use the terminology of Scrums or bother with scrum masters and the like. We have crafted our own process of moving fast. It's a very Agile process that doesn't get bogged down with someone else's idea of what it means to be Agile. When you have to stop and define what it means to be Agile and argue what flavor of Agile you are, you just stopped being Agile."*

Part of the difficulty is that people often do not have a good process for collecting and documenting project requirements.

There are several ways to collect requirements. Interviewing stakeholders one on one is often very effective, although it can be expensive and time consuming. Holding focus groups and facilitated workshops, and using group creativity and decision-making techniques to collect requirements are normally faster and less expensive than one-on-one interviews.

Questionnaires and surveys can be efficient ways to collect requirements as long as key stakeholders provide honest and thorough information. Observation can also be a good technique for collecting requirements, especially for projects that involve improving work processes and procedures. For software development projects, prototyping and document analysis are common techniques for collecting requirements, as are context diagrams, which help to clarify the interfaces and boundaries of a project or process. On agile software development projects, the product owner creates the prioritized product backlog for each sprint, as shown in Chapter 3. [Benchmarking](#), or generating ideas by comparing specific project practices or product characteristics to those of other projects or products inside or outside the performing organization, can also be used to collect requirements.

Even though there are many ways to collect requirements, people who work on software projects in particular have considerable difficulty defining and managing requirements. One study revealed some interesting statistics:

- Eighty-eight percent of the software projects involved enhancing existing products instead of creating new ones.
- Eighty-six percent of respondents said that customer satisfaction was the most important metric for measuring the success of development projects, 82 percent said that feedback from customers and partners was the main source of product ideas and requirements, and 73 percent said that the most important challenge for their teams was gaining a clear understanding of what customers wanted, followed by documenting and

managing requirements.

- Seventy-five percent of respondents were managing projects with at least 100 requirements; 20 percent were managing projects with over 1,000 requirements.
- Seventy percent of respondents spent at least 10 percent of their time managing changes to requirements; 30 percent spent more than 25 percent of their time on such changes.
- The majority of software development teams used a hybrid methodology, 26 percent used waterfall or modified waterfall techniques, and 19 percent used agile techniques.
- Eighty-three percent of software development teams still use Microsoft Office applications such as Word and Excel as their main tools to communicate requirements.
- The respondents listed “requirements collaboration and management software” and “requirements modeling and visualization” as the top two software tools on their wish list, followed by test management and project management.*

The project’s size, complexity, importance, and other factors affect how much effort is spent on collecting requirements. For example, a team working on a project to upgrade the entire corporate accounting system for a multibillion-dollar company with more than 50 locations should spend a fair amount of time collecting requirements. On the other hand, a project to upgrade the hardware and software for a small accounting firm with only five employees would need a much smaller effort. In any case, it is important for a project team to decide how it will collect and manage requirements. It is crucial to gather inputs from key stakeholders and align the scope with business strategy, as described in Chapter 4.

Just as a project team can collect requirements in several ways, there are several ways to document the requirements. Project teams should first review the project charter because it includes high-level requirements for the project, and they should refer to the scope and requirements management plans. They should also review the stakeholder register and stakeholder management plan to ensure that all key stakeholders have a say in determining requirements. The format for documenting stakeholder requirements can range from a listing of all requirements on a single piece of paper to a room full of notebooks. People who have worked on complex projects, such as building a new airplane, know that the requirements documentation for a plane can weigh more than the plane itself! Requirements documents are often generated by software and include text, images, diagrams, videos, and other media. Requirements are often broken down into different categories such as functional requirements, service requirements, performance requirements, quality requirements, and training requirements.

In addition to preparing requirements documentation as an output of collecting

requirements, project teams often create a requirements traceability matrix. A [requirements traceability matrix \(RTM\)](#) is a table that lists requirements, their various attributes, and the status of the requirements to ensure that all are addressed. Table 5-1 provides an example of an RTM entry for the IT upgrade project described in the chapter’s opening case. An RTM can have many variations. For example, software requirements are often documented in an RTM that cross-references each requirement with related ones and lists specific tests to verify that they are met. Remember that the main purpose of an RTM is to maintain the linkage from the source of each requirement through its decomposition to implementation and validation. Search the Internet for more detailed examples of an RTM.

Table 5-1. Sample entry in a requirements traceability matrix				
Requirement No.	Name	Category	Source	Status
R32	Laptop memory	Hardware	Project charter and corporate laptop specifications	Complete. Laptops ordered meet memory requirement.

Defining Scope

The next step in project scope management is to provide a detailed definition of the work required for the project. Good scope definition is very important to project success because it helps improve the accuracy of time, cost, and resource estimates, it defines a baseline for performance measurement and project control, and it aids in communicating clear work responsibilities. The main tools and techniques used in defining scope include expert judgment, data analysis, decision making, interpersonal and team skills, and product analysis. The main outputs of scope definition are the project scope statement and project documents updates.

Table 5-2 shows the project charter for the IT upgrade project described in the opening case. Notice how information from the project charter provides a basis for further defining the project scope. The charter describes the high-level scope, time, and cost goals for the project objectives and success criteria, a general approach to accomplishing the project’s goals, and the main roles and responsibilities of important project stakeholders.

Table 5-2. Sample project charter	
Project Title: Information Technology (IT) Upgrade Project	
Project Start Date: March 4	Projected Finish Date: December 4
Key Schedule Milestones:	

- Inventory update completed April 15
- Hardware and software acquired August 1
- Installation completed October 1
- Testing completed November 15

Budget Information: Budgeted \$1,000,000 for hardware and software costs and \$500,000 for labor costs.

Project Manager: Kim Nguyen, (310) 555–2784, knguyen@course.com

Project Objectives: Upgrade hardware and software for all employees (approximately 2,000) within nine months based on new corporate standards. See attached sheet describing the new standards. Upgrades may affect servers as well as associated network hardware and software.

Main Project Success Criteria: The hardware, software, and network upgrades must meet all written specifications, be thoroughly tested, and be completed in nine months. Employee work disruptions will be minimal.

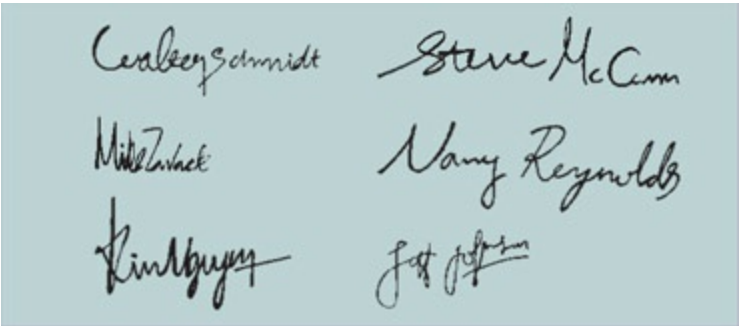
Approach:

- Update the IT inventory database to determine upgrade needs.
- Develop detailed cost estimate for project and report to CIO.
- Issue a request for quote to obtain hardware and software.
- Use internal staff as much as possible for planning, analysis, and installation.

ROLES AND RESPONSIBILITIES

Name	Role	Responsibility
Walter Schmidt	CEO	Project sponsor, monitor project
Mike Zwack	CIO	Monitor project, provide staff
Kim Nguyen	Project Manager	Plan and execute project
Jeff Johnson	Director of IT Operations	Mentor Kim
Nancy Reynolds	VP, Human Resources	Provide staff, issue memo to all employees about project
Steve McCann	Director of Purchasing	Assist in purchasing hardware and software

Sign-off: (Signatures of all the above stakeholders)

A collection of handwritten signatures on a light blue background. The signatures are arranged in two columns. The left column contains three signatures: 'Walter Schmidt', 'Mike Zwack', and 'Kim Nguyen'. The right column contains three signatures: 'Steve McCann', 'Nancy Reynolds', and 'Jeff Johnson'.

Comments: (Handwritten or typed comments from above stakeholders, if applicable)

“This project must be done within 10 months at the absolute latest.”—Mike Zwack, CIO

“We are assuming that adequate staff will be available and committed to supporting this project. Some work must be done after hours to avoid work disruptions, and overtime will be provided.”—Jeff Johnson and Kim Nguyen, IT department

Although contents vary, [project scope statements](#) should include at least a product scope description, product user acceptance criteria, and detailed information on all project deliverables. It is also helpful to document other scope-related information, such as the project boundaries, constraints, and assumptions. The project scope statement should also reference supporting documents, such as product specifications that will affect what products are created or purchased, or corporate policies, which might affect how products or services are produced. Many IT projects require detailed functional and design specifications for developing software, which also should be referenced in the detailed scope statement.

As time progresses, the scope of a project should become more clear and specific. For example, the project charter for the IT upgrade project in [Table 5-2](#) includes a short statement about the servers and other computers and software that the project may affect. Table 5-3 provides an example of how the scope becomes progressively more detailed in scope statements labeled Version 1 and Version 2.

Table 5-3. Further defining project scope

Project Charter:

Upgrades may affect servers . . . (listed under Project Objectives)

Project Scope Statement, Version 1:

Servers: If additional servers are required to support this project, they must be compatible with existing servers. If it is more economical to enhance existing servers, a detailed description of enhancements must be submitted to the CIO for approval. See current server specifications provided in Attachment 6. The CEO must approve a detailed plan describing the servers and their location at least two weeks before installation.

Project Scope Statement, Version 2:

Servers: This project will require purchasing 10 new servers to support Web, network, database, application, and printing functions. Virtualization will be used to maximize efficiency. Detailed descriptions of the servers are provided in a product brochure in Attachment 8, along with a plan describing where they will be located.

Notice in [Table 5-3](#) that the project scope statements often refer to related documents, which

can be product specifications, product brochures, or other plans. As more information becomes available and decisions are made related to project scope, such as specific products that will be purchased or changes that have been approved, the project team should update the project scope statement. The team might name different iterations of the scope statement Version 1, Version 2, and so on. These updates may also require changes to other project documents. For example, if the company must purchase servers from a supplier it has never worked with before, the procurement management plan should include information on working with that new supplier.

An up-to-date project scope statement is an important document for developing and confirming a common understanding of the project scope. It describes in detail the work to be accomplished on the project and is an important tool for ensuring customer satisfaction and preventing scope creep, as described later in this chapter.

Recall from **Chapter 1** the importance of addressing the triple constraint of project management—meeting scope, time, and cost goals for a project. Time and cost goals are normally straightforward. For example, the time goal for the IT upgrade project is nine months, and the cost goal is \$1.5 million. It is much more difficult to describe, agree upon, and meet the scope goal of many projects.

Media Snapshot

Inaccurate requirements gathering continues to be one of the main causes of project failure. Organizations lack the resources and maturity in requirements management, which comes at a price. The authors of the 2014 PMI in-depth report on requirements management state that “For every dollar spent on projects and programs, 5.1 percent is wasted due to poor requirements management. Put in more striking terms, this amounts to US\$51 million wasted for every US\$1 billion spent. That’s a lot of potential value falling through the cracks in the project-driven world.”*

The report provides the following survey results:

- Forty-seven percent of organizations report that inaccurate requirements management is the primary cause of projects not meeting their original goals and business objectives.
- Over half of organizations report not having adequate resources to perform requirements management well.
- Only 24 percent of organizations report doing well in recognizing and developing skills needed for effective management of requirements.

- Only 46 percent of organizations report using a formal process to ensure unbiased requirements validation for projects.

The report concludes by suggesting that organizations need to develop people, processes, and culture to improve requirements management.

Creating the Work Breakdown Structure

After collecting requirements and defining scope, the next step in project scope management is to create a work breakdown structure. A [work breakdown structure \(WBS\)](#) is a deliverable-oriented grouping of the work involved in a project that defines its total scope. Because most projects involve many people and many different deliverables, it is important to organize and divide the work into logical parts based on how the work will be performed. The WBS is a foundation document in project management because it provides the basis for planning and managing project schedules, costs, resources, and changes. Because the WBS defines the total scope of the project, some project management experts believe that work should not be done on a project if it is not included in the WBS. Therefore, it is crucial to develop a complete WBS.

The project management plan, project documents, enterprise environmental factors, and organizational process assets are the primary inputs for creating a WBS. In addition to expert judgment, the main tool or technique is [decomposition](#)—that is, subdividing project deliverables into smaller pieces. The outputs of the process of creating the WBS are the scope baseline and project documents updates. The [scope baseline](#) includes the approved project scope statement and its associated WBS and WBS dictionary.

What does a WBS look like? A WBS is often depicted as a task-oriented tree of activities, similar to an organizational chart. Many people like to create a WBS in chart form first to help them visualize the whole project and all of its main parts. **Figure 5-3** shows a WBS for an intranet project. The top box labeled Intranet represents the entire project, and it is called Level 1 of the WBS. In this example, the main products to be delivered are shown in Level 2, called website design, home page design, marketing pages, and sales pages. Below them are Level 3 items, such as site map, graphic design, and so on.

Figure 5-3. Sample intranet WBS organized by product

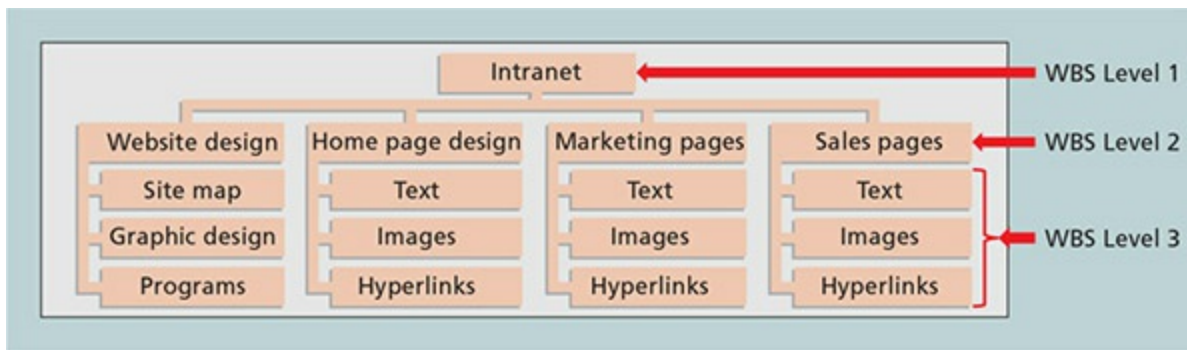
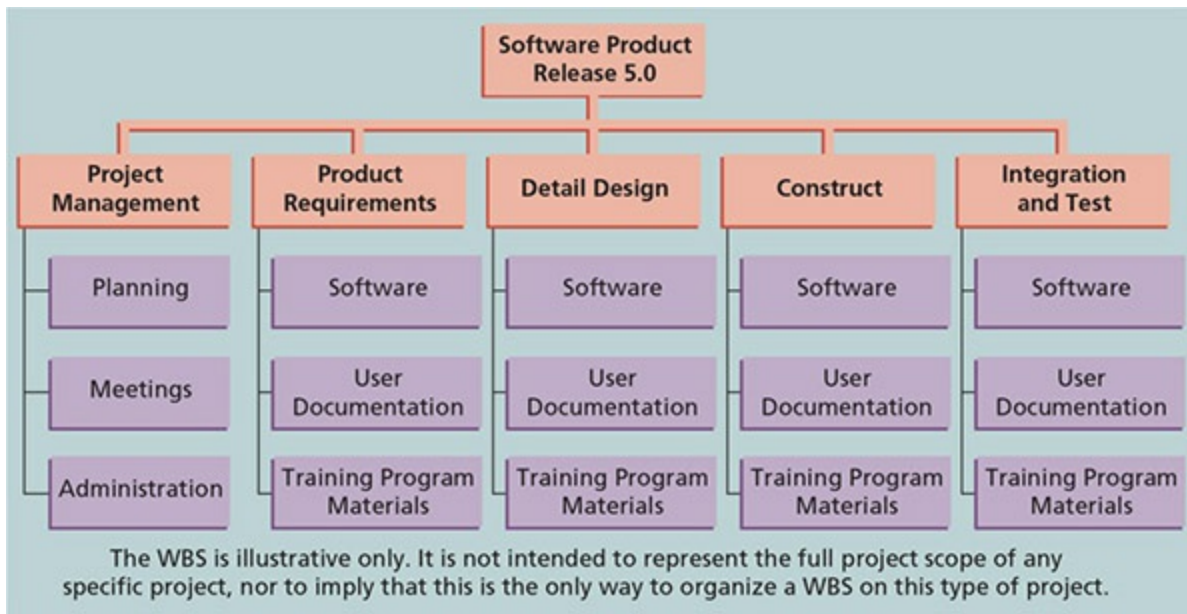


Figure 5-4 shows a software product release project created by PMI. Notice that instead of showing product deliverables in Level 2, it is organized around deliverables for each project phase (product requirements, detail design, construct, and integration and test) as well as another Level 2 item for project management. Table 5-4 shows this same WBS in a tabular or list format. This level numbering is based on the Project Management Institute’s *Practice Standard for Work Breakdown Structures, Second Edition* (2006).

Figure 5-4. Sample intranet WBS organized by phase in chart and tabular form



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Table 5-4. Tabular form of WBS

1.0	Software Product Release 5.0
1.1	Project Management
1.1.1	Planning
1.1.2	Meetings
1.1.3	Administration
1.2	Product Requirements

- 1.2.1 Software
- 1.2.2 User Documentation
- 1.2.3 Training Program Materials
- 1.3 Detail Design
 - 1.3.1 Software
 - 1.3.2 User Documentation
 - 1.3.3 Training Program Materials
- 1.4 Construct
 - 1.4.1 Software
 - 1.4.2 User Documentation
 - 1.4.3 Training Program Materials
- 1.5 Integration and Test
 - 1.5.1 Software
 - 1.5.2 User Documentation
 - 1.5.3 Training Program Materials

Many documents, such as contracts, use the tabular format, but the numbering scheme may vary. Project management software also uses the tabular format. The WBS provides the basis for creating the activities that are entered into the Task Name column in Microsoft Project, and the hierarchy or level of tasks is shown by indenting and numbering tasks within the software. Be sure to check with your organization to see which numbering scheme it prefers to use for WBSs. To avoid confusion, it is important to decide on a numbering scheme and then use it when referring to WBS items.

Also determine how you will name WBS items. Some organizations use only nouns to focus on describing deliverables, as PMI recommends. For example, instead of using the words “define requirements” on a WBS, use “requirements definition.” Activities on a schedule below requirements definition would include verb phrases, as discussed in [Chapter 6](#), Project Schedule Management. Remember that the main purpose of the WBS is to define all of the work required to complete a project, so focus on that while following your organization’s structural guidelines.

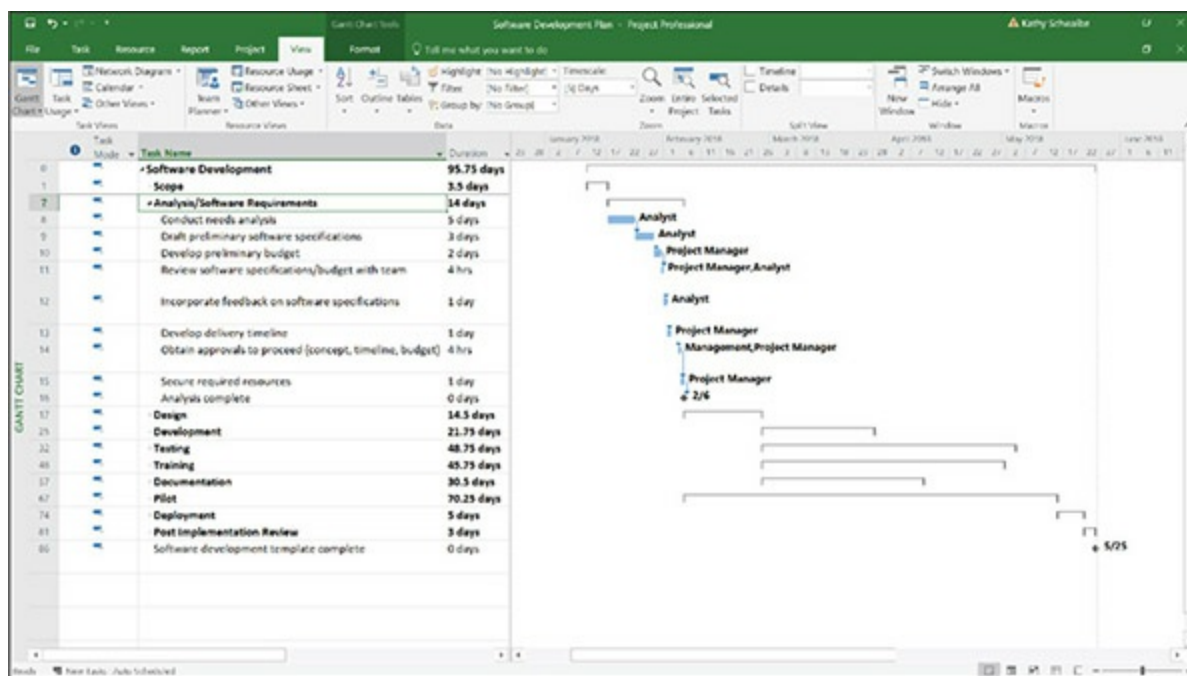
A [work package](#) is a task at the lowest level of the WBS. A work package also represents the level of work that the project manager monitors and controls. You can think of work packages in terms of accountability and reporting. If a project has a relatively short time frame and requires weekly progress reports, a work package might represent work completed in one week or less. If a project has a very long time frame and requires quarterly progress reports, a work package might represent work completed in one month or more. A work package might also be the procurement of a specific product or products, such as an item or items purchased from an outside source. A work package should be defined at the proper level so the project manager can clearly establish an estimate of the effort needed to complete it, estimate the

cost of all required resources, and evaluate the quality of the results when the work package is finished.

When using project management software, estimates of work time should be entered only at the work package level. The rest of the WBS items are just groupings or summary tasks for the work packages. The software automatically calculates duration estimates for various WBS levels based on data entered for each work package and the WBS hierarchy.

Figure 5-5 shows a template file from Microsoft Project 2016 for a Software Development project. The Level 2 WBS categories include scope, analysis/software requirements, design, development, testing training, documentation, pilot, deployment, and post implementation review. Activities are shown for the analysis/software requirements. Notice that the WBS is the basis for creating the Gantt chart, as described in more detail in Chapter 6, Project Schedule Management. You will learn more about Gantt charts in Chapter 6, Project schedule management.

Figure 5-5. Software development project template from Microsoft Project 2016



The example WBSs shown here are simplified so that they are somewhat easy to understand and construct. *Nevertheless, it is very difficult to create a good WBS.* To create a good WBS, you must understand the project and its scope and incorporate the needs and knowledge of the stakeholders. The project manager and the project team must decide as a group how to organize the work and how to include many levels in the WBS.

While many project managers have found that they should focus on doing the top levels well before becoming bogged down in more detailed levels, it is also true that more accurate

estimates of scope, time, and cost are obtained when the project is defined appropriately and in sufficient detail. Operating at too high a level increases project risk. The act of defining the WBS is meant to offset that risk by considering the project's details in advance of their execution.

Many people confuse tasks on a WBS with specifications. Tasks on a WBS represent work that needs to be done to complete the project. For example, if you are creating a WBS to redesign a kitchen, you might have Level 2 categories called design, purchasing, flooring, walls, cabinets, and appliances. Under flooring, you might have tasks to remove the old flooring, install the new flooring, and install the trim. You would not have tasks like "12 ft. by 14 ft. of light oak" or "flooring must be durable"; these are specifications.

Another concern when creating a WBS is how to organize it to provide the basis for the project schedule. You should focus on what work needs to be done and how it will be done, not when it will be done. In other words, the tasks do not have to be developed as a sequential list of steps. If you want some time-based flow for the work, you can create a WBS using the project management process groups of initiating, planning, executing, monitoring and controlling, and closing as Level 2 in the WBS. By doing this, not only does the project team follow good project management practice, the WBS tasks can also be mapped more easily against time. For example, **Figure 5-6** shows a WBS and Gantt chart for a website project, organized by the five project management process groups. Activities under initiating include selecting a project manager, forming the project team, and developing the project charter. Activities under planning include developing a scope statement, creating a WBS, and developing and refining other plans, which would be broken down in more detail for a real project. Concept, website design, website development, and rollout are WBS Level 3 items under executing. The executing tasks vary the most from project to project, but many of the activities under the other project management process groups would be similar for all projects. If you do not use the project management process groups in the WBS, you can have a Level 2 category called "project management" to make sure you account for all activities related to managing the project, as was done in Figure 5-4 from PMI. Remember that all work should be included in the WBS, including project management.

Figure 5-6.
groups

Website project Gantt chart organized by project management process

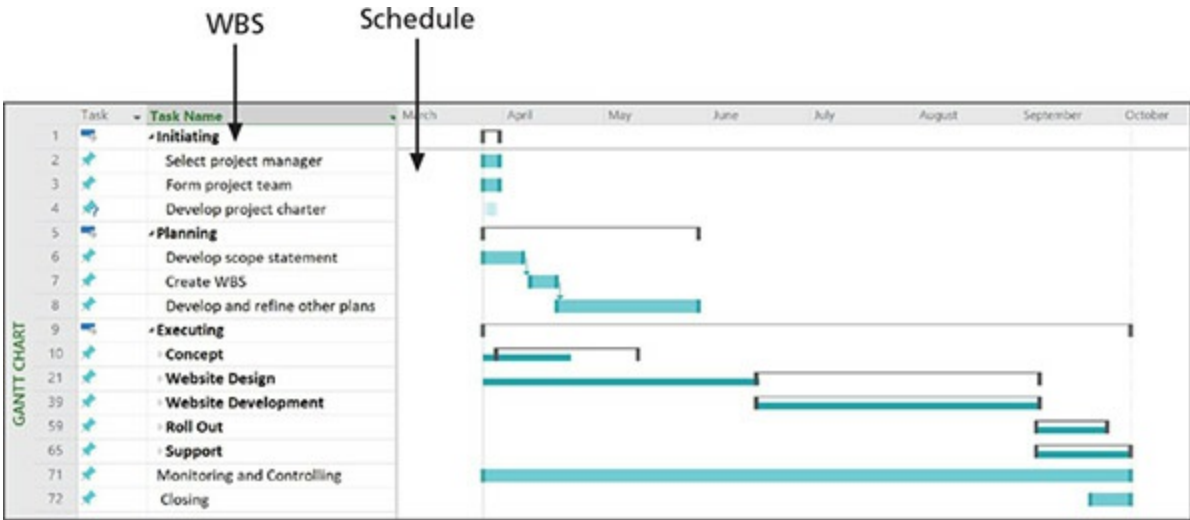


Table 5-5.

Executing tasks for JWD Consulting's WBS

- 3.0 Executing
 - 3.1 Survey
 - 3.2 User inputs
 - 3.3 Intranet site content
 - 3.3.1 Templates and tools
 - 3.3.2 Articles
 - 3.3.3 Links
 - 3.3.4 Ask the Expert
 - 3.3.5 User Requests feature
 - 3.4 Intranet site design
 - 3.5 Intranet site construction
 - 3.6 Intranet site testing
 - 3.7 Intranet site roll-out
 - 3.8 Site rollout
 - 3.9 Intranet site roll-out

Some project teams like to list every deliverable they need to produce and then use those as the basis for creating all or part of their WBS. In **Chapter 3**, JWD Consulting used the project management process groups for the Level 2 items in its WBS for the project management intranet site project. Then in breaking down the executing section, the project team focused on the product deliverables it had to produce. Table 5-5 shows the categories the team used for that part of the WBS. Recall that the scope statement should list and describe all of the deliverables required for the project. It is very important to ensure consistency between the project charter, scope statement, WBS, and Gantt chart to define the scope of the project accurately.

It is also very important to involve the entire project team and the customer in creating and reviewing the WBS. *People who will do the work should help to plan the work* by creating the WBS. Having group meetings to develop a WBS helps everyone understand *what* work must be done for the entire project and *how* it should be done, given the people involved. It also helps to identify where coordination between different work packages will be required.

Approaches to Developing Work Breakdown Structures

You can use several approaches to develop a WBS:

- Using guidelines
- The analogy approach
- The top-down approach
- The bottom-up approach
- The mind-mapping approach

Using Guidelines

If guidelines exist for developing a WBS, it is very important to follow them. Some organizations—the U.S. Department of Defense (DOD), for example—prescribe the form and content for WBSs for particular projects. Many DOD projects require contractors to prepare their proposals based on the DOD-provided WBS. These proposals must include cost estimates for each task in the WBS at a detailed and summary level. The cost for the entire project must be calculated by summing the costs of all of the lower-level WBS tasks. When DOD personnel evaluate cost proposals, they must compare the contractors' costs with the DOD's estimates. A large variation in costs for a certain WBS task often indicates confusion as to what work must be done.

Consider a large automation project for the U.S. Air Force. In the mid-1980s, the Air Force developed a request for proposals for the Local On-Line Network System (LONS) to automate 15 Air Force Systems Command bases. This \$250 million project involved providing the hardware and developing software for sharing documents such as contracts, specifications, and requests for proposals. The Air Force proposal guidelines included a WBS that contractors were required to follow in preparing their cost proposals. Level 2 WBS items included hardware, software development, training, and project management. The hardware item was composed of several Level 3 items, such as servers, workstations, printers, and network hardware. Air Force personnel reviewed the contractors' cost proposals against their internal cost estimate, which was also based on this WBS. Having a prescribed WBS helped

contractors to prepare their cost proposals and helped the Air Force to evaluate them.

Many organizations provide guidelines and templates for developing WBSs, as well as examples of WBSs from past projects. Microsoft Project 2016 comes with several templates, and more are available on Microsoft's website and other sites. At the request of many of its members, PMI developed a WBS *Practice Standard* to provide guidance for developing and applying the WBS to project management. The *Practice Standard* includes sample WBSs for a wide variety of projects in various industries, including projects for Web design, telecom, service industry outsourcing, and software implementation.

Project managers and their teams should review appropriate information to develop their unique project WBSs more efficiently. For example, Kim Nguyen and key team members from the opening case should review their company's WBS guidelines, templates, and other related information before and during the team meetings to create their WBS.

The Analogy Approach

Another method for constructing a WBS is the analogy approach. In the [analogy approach](#), you use a similar project's WBS as a starting point. For example, Kim Nguyen from the opening case might learn that one of her organization's suppliers did a similar IT upgrade project last year. She could ask them to share their WBS for that project to provide a starting point for her own project.

McDonnell Aircraft Company, now part of Boeing, provides an example of using an analogy approach when creating WBSs. McDonnell Aircraft Company designed and manufactured several different fighter aircraft. When creating a WBS for a new aircraft design, it started by using 74 predefined subsystems for building fighter aircraft based on past experience. There was a Level 2 WBS item for the airframe that was composed of Level 3 items such as a forward fuselage, center fuselage, aft fuselage, and wings. This generic, product-oriented WBS provided a starting point for defining the scope of new aircraft projects and developing cost estimates for new aircraft designs.

Some organizations keep a repository of WBSs and other project documentation on file to assist people working on projects. Project 2016 and many other software tools include sample files to assist users in creating a WBS and Gantt chart. Viewing examples of WBSs from similar projects allows you to understand different ways to create a WBS.

The Top-Down and Bottom-Up Approaches

Two other methods of creating WBSs are the top-down and bottom-up approaches. Most project managers consider the top-down approach of WBS construction to be conventional.

To use the **top-down approach**, start with the largest items of the project and break them into subordinate items. This process involves refining the work into greater and greater levels of detail. After finishing the process, all resources should be assigned at the work package level. The top-down approach is best suited to project managers who have vast technical insight and a big-picture perspective.

In the **bottom-up approach**, team members first identify as many specific tasks related to the project as possible. They then aggregate the specific tasks and organize them into summary activities, or higher levels in the WBS. For example, a group of people might be responsible for creating a WBS to develop an e-commerce application. Instead of looking for guidelines on how to create a WBS or viewing WBSs from similar projects, they could begin by listing detailed tasks they think they would need to perform in order to create the application. After listing these detailed tasks, they would group the tasks into categories. Then they would group these categories into higher-level categories. Some people have found that writing all possible tasks as notes and then placing them on a wall helps the team see all the work required for the project and develop logical groupings for performing the work. For example, a business analyst on the project team might know that it had to define user requirements and content requirements for the e-commerce application. These tasks might be part of the requirements documents the team would have to create as one of the project deliverables. A hardware specialist might know that the team had to define system requirements and server requirements, which would also be part of a requirements document. As a group, they might decide to put all four of these tasks under a higher-level item called “requirements definition” that would result in the delivery of a requirements document. Later, they might realize that requirements definition should fall under a broader category of concept design for the e-commerce application, along with other groups of tasks related to the concept design. The bottom-up approach can be very time consuming, but it can also be a very effective way to create a WBS. Project managers often use the bottom-up approach for projects that represent entirely new systems or approaches to doing a job, or to help create buy-in and synergy with a project team.

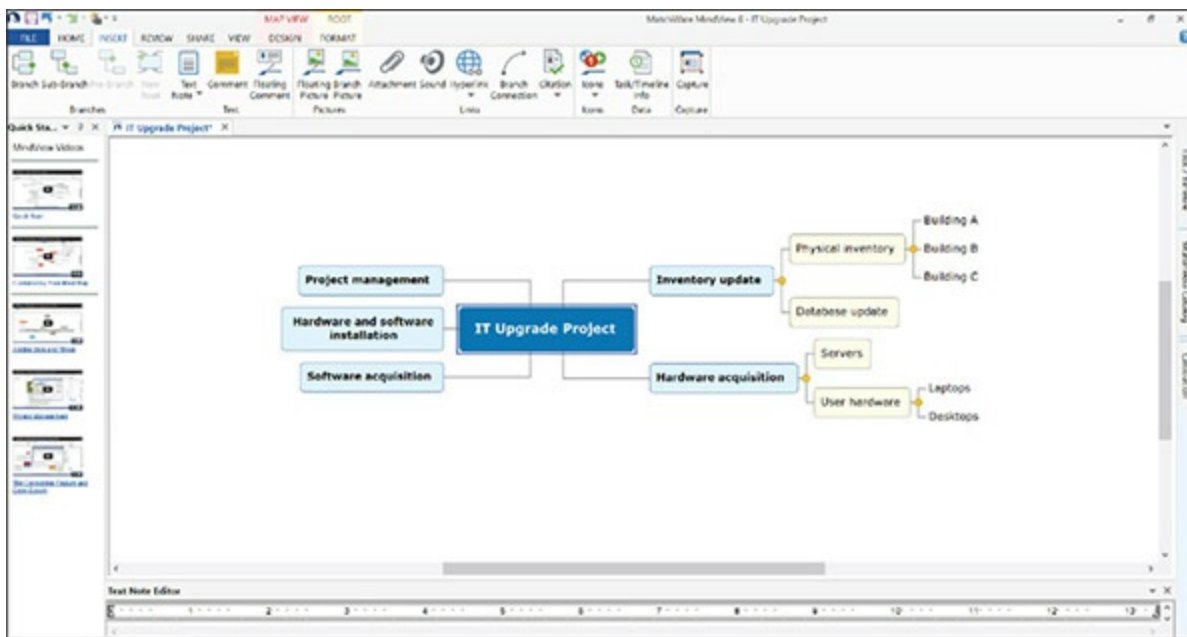
Mind Mapping

Some project managers like to use mind mapping to help develop WBSs. As described in **Chapter 4** during the discussion of SWOT analysis, mind mapping is a technique that uses branches radiating from a core idea to structure thoughts and ideas. Instead of writing down tasks in a list or immediately trying to create a structure for tasks, mind mapping allows people to write and even draw pictures of ideas in a nonlinear format. This more visual, less structured approach to defining and then grouping tasks can unlock creativity among individuals and increase participation and morale among teams.*

Figure 5-7 shows a diagram created with MatchWare’s MindView 6.0 software that uses mind

mapping to create a WBS for the IT upgrade project from the opening case. The rectangle near the center represents the entire project. Each of the four main branches radiating from the center represents the main tasks or Level 2 items for the WBS. Different people at the meeting who are creating this mind map might have different roles in the project, which could help in deciding the tasks and WBS structure. For example, Kim would want to focus on all of the project management tasks, and she might also know that they will be tracked in a separate budget category. People who are familiar with acquiring or installing hardware and software might focus on that work, and so on. The team would continue to add branches and items until they exhausted ideas for what work needs to be performed.

Figure 5-7. Sample mind-mapping technique for creating a WBS

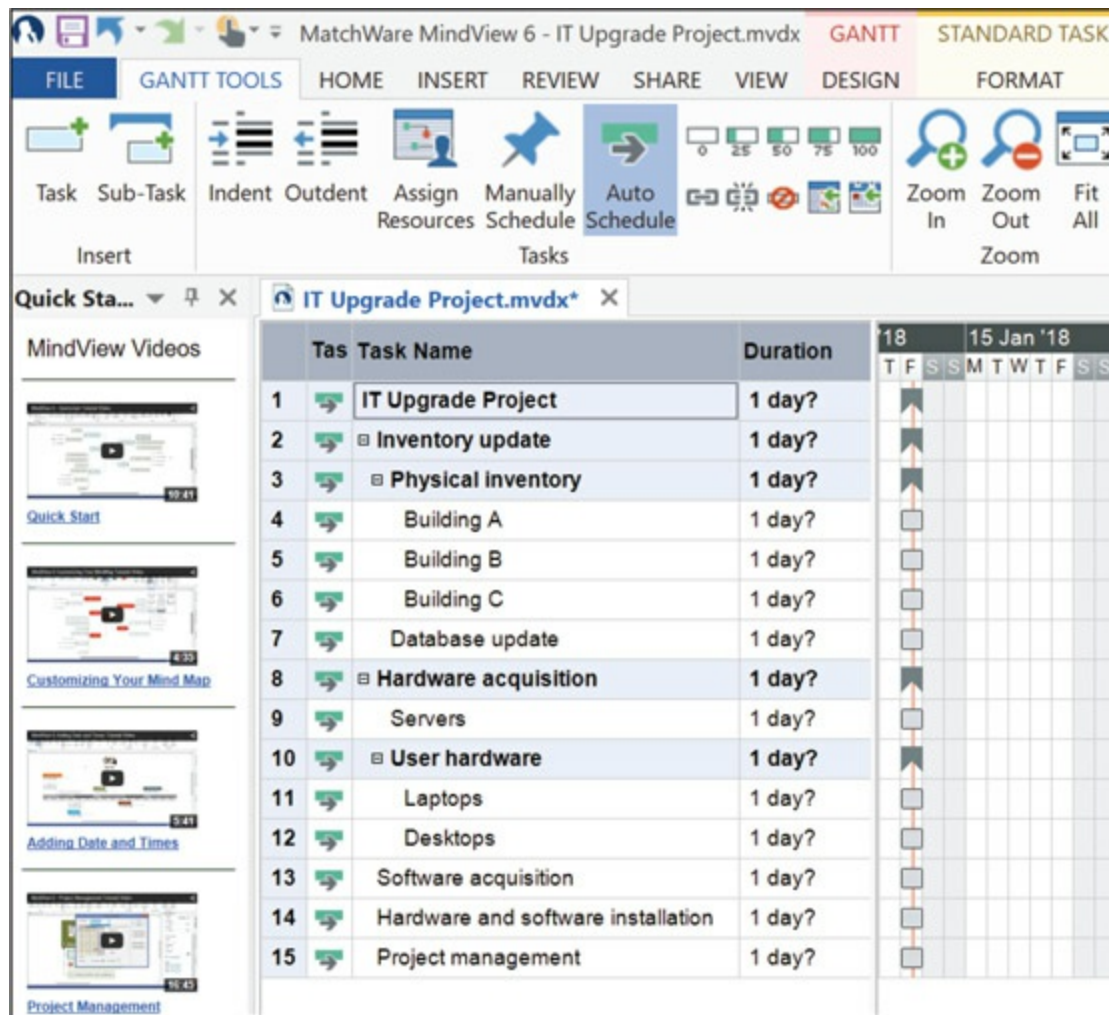


Source: MatchWare MindView 6.0

After discovering WBS items and their structure using the mind-mapping technique, you could then translate the information into chart or tabular form, as described earlier. A feature of MatchWare's MindView 6.0 software is that you can click a single icon to convert a mind map into a Gantt chart. The mind map provides the task list based on the WBS. MindView also lets you enter information about tasks, such as dependencies and durations, to generate a complete Gantt chart. You can also export your mind map into Microsoft Project. The WBS is entered in the Task List column, and the structure is created automatically based on the mind map. Figure 5-8 shows the resulting Gantt chart for the IT upgrade project in MindView 6.0.

Figure 5-8.

Gantt chart with WBS generated from a mind map



Source: MatchWare MindView 6.0

Mind mapping can be used for developing WBSs using the top-down or bottom-up approach. For example, you could conduct mind mapping for an entire project by listing the project in the center of a document, adding the main categories on branches radiating from the center, and then adding branches for appropriate subcategories. You could also develop a separate mind-mapping diagram for each deliverable and then merge them to create one large diagram for the entire project. You can also add items anywhere on a mind-mapping document without following a strict top-down or bottom-up approach. After the mind-mapping documents are complete, you can convert them into a chart or tabular WBS form.

Advice for Young Professionals

As mentioned in this chapter, it is very difficult to create a good WBS. Try to attend meetings in your organization where teams work together to create a WBS. If that's not possible, ask to see WBSs for projects that have been completed or are in process where you work. You can also do your own research to find examples of different WBSs. There are many template files available for Microsoft Project and other project management

software. Don't be afraid to ask questions. There is no one way to organize the work that needs to be done on a project. Remember to not leave anything essential out of a WBS, and do not include anything that is superfluous.

The WBS Dictionary

Many of the items listed on the sample WBSs are rather vague. What exactly does “Database update” mean, for example? The person responsible for this task might think that it does not need to be broken down any further, which could be fine. However, the task should be described in more detail so everyone has the same understanding of what it involves. What if someone else has to perform the task? What would you tell that team member to do? What will it cost to complete the task? More detailed information is needed to answer these and other questions.

A **WBS dictionary** is a document that provides detailed information about each WBS item. The term *dictionary* should not be confused with defining terms or acronyms in this case; such definitions belong in a glossary that would be included elsewhere in the project documentation. Instead, the WBS dictionary is a definition of the work involved in the task—a clarification that makes the summary description in the WBS easier to understand in terms of the approach taken to complete the work.

The format of the WBS dictionary can vary based on project needs. It might be appropriate to have a short paragraph describing each work package. For a more complex project, an entire page or more might be needed for each of the work package descriptions. Some projects might require that each WBS item describe the responsible organization, resource requirements, estimated costs, dependencies with other activities, and other information. Project teams often review WBS dictionary entries from similar tasks to get a better idea of how to create their entries.

In the IT upgrade project described in the opening case, Kim should work with her team and sponsor to determine the level of detail needed in the WBS dictionary. They should also decide where this information will be entered and how it will be updated. Kim and her team decided to enter all of the WBS dictionary information into their enterprise project management system, following departmental guidelines. **Table 5-6** is an example of one entry.

Table 5-6. Sample WBS dictionary entry

WBS Dictionary Entry March 20

Project Title: Information Technology (IT) Upgrade Project

WBS Item Number: 2.2

WBS Item Name: Database Update

Description: The IT department maintains an online database of hardware and software on the corporate intranet. We need to make sure that we know exactly what hardware and software employees are currently using and if they have any unique needs before we decide what to order for the upgrade. This task will involve reviewing information from the current database, producing reports that list each department's employees

and location, and updating the data after performing the physical inventory and receiving inputs from department managers. Our project sponsor will send a notice to all department managers to communicate the importance of this project and this particular task. In addition to general hardware and software upgrades, the project sponsors will ask the department managers to provide information for any unique requirements they might have that could affect the upgrades. This task also includes updating the inventory data for network hardware and software. After updating the inventory database, we will send an e-mail to each department manager to verify the information and make changes online as needed. Department managers will be responsible for ensuring that their people are available and cooperative during the physical inventory. Completing this task is dependent on WBS Item Number 2.1, Physical Inventory, and must precede WBS Item Number 3.0, Hardware and Software Acquisition.

The approved project scope statement and its associated WBS and WBS dictionary form the scope baseline. Performance in meeting project scope goals is based on this scope baseline.

Advice for Creating a WBS and a WBS Dictionary

As stated previously, creating a good WBS is no easy task and usually requires several iterations. Often, it is best to use a combination of approaches to create a project's WBS. Some basic principles, however, apply to creating any good WBS and its WBS dictionary.

- A unit of work should appear in only one place in the WBS.
- The work content of a WBS item is the sum of the WBS items below it.
- A WBS item is the responsibility of only one person, even though many people might be working on it.
- The WBS must be consistent with the way work actually will be performed; it should serve the project team first, and serve other purposes only if practical.
- Project team members should be involved in developing the WBS to ensure consistency and buy-in.
- Each WBS item must be documented in a WBS dictionary to ensure accurate understanding of the scope of work included and not included in that item.
- The WBS must be a flexible tool to accommodate inevitable changes while properly maintaining control of the work content in the project according to the scope statement.*

Validating Scope

It is difficult to create a good project scope statement and WBS for a project. It is even more difficult, especially on IT projects, to verify the project scope and minimize scope changes. Some project teams know from the start that the scope is very unclear and that they must work closely with the project customer to design and produce various deliverables. In this case, the project team must develop a process for scope validation that meets unique project needs. Careful procedures must be developed to ensure that customers are getting what they want and that the project team has enough time and money to produce the desired products and services.

Even when the project scope is fairly well defined, many IT projects suffer from [scope creep](#)—the tendency for project scope to keep getting bigger and bigger. There are many horror stories about IT projects failing due to problems such as scope creep, including a few classic examples in the following What Went Wrong? feature. For this reason, it is very important to verify the project scope with users throughout the life of the project and develop a process for controlling scope changes.

Scope creep can also be a good thing, if managed well. Later in this chapter, you can see how Northwest Airlines encouraged scope changes on its ResNet project and managed them well.

What Went Wrong?

A project scope that is too broad and grandiose can cause severe problems. Scope creep and an overemphasis on technology for technology's sake resulted in the bankruptcy of a large pharmaceutical firm, Texas-based FoxMeyer Drug. In 1994, the CIO was pushing for a \$65 million system to manage the company's critical operations. He did not believe in keeping things simple, however. The company spent nearly \$10 million on state-of-the-art hardware and software and contracted the management of the project to a prestigious (and expensive) consulting firm. The project included building an \$18 million robotic warehouse, which looked like something out of a science fiction movie, according to insiders. The scope of the project kept getting bigger and more impractical. The elaborate warehouse was not ready on time, and the new system generated erroneous orders that cost FoxMeyer Drug more than \$15 million in unrecovered excess shipments. In July 1996, the company took a \$34 million charge for its fourth fiscal quarter, and by August of that year, FoxMeyer Drug filed for bankruptcy.*

Another classic example of scope creep comes from McDonald's Restaurants. In 2001, the fast-food chain initiated a project to create an intranet that would connect its headquarters with all of its restaurants and provide detailed operational information in

real time. For example, headquarters would know if sales were slowing or if the grill temperature was correct in every single store—all 30,000 of them in more than 120 countries. McDonald's would not divulge detailed information, but they admitted that the project was too large in scale and scope. After spending \$170 million on consultants and initial implementation planning, McDonald's realized that the project was too much to handle and terminated it.*

Another major scope problem on IT projects is a lack of user involvement. A prime example occurred in the late 1980s at Northrop Grumman, which specializes in defense electronics, IT, advanced aircraft, shipbuilding, and space technology. An IT project team there became convinced that it should automate the review and approval process of government proposals. The team implemented a powerful workflow system to manage the whole process. Unfortunately, the end users of the system were aerospace engineers who preferred to work in a more casual, ad hoc fashion. They dubbed the system "Naziware" and refused to use it. This example illustrates an IT project that wasted millions of dollars developing a system that was not in touch with the way end users did their work.*

Failing to follow good project management processes and use off-the-shelf software also results in scope problems. 21st Century Insurance Group in Woodland Hills, California, paid Computer Sciences Corporation \$100 million to develop a system for managing business applications, including managing insurance policies, billing, claims, and customer service. After five years, the system was still in development and supported less than 2 percent of the company's business. Joshua Greenbaum, an analyst at Enterprise Applications Consulting, called the project a "huge disaster" and questioned the insurance company's ability "to manage a process that is pretty well known these days. . . . I'm surprised that there wasn't some way to build what they needed using off-the-shelf components and lower their risk."*

Scope validation involves formal acceptance of the completed project deliverables. This acceptance is often achieved by a customer inspection and then sign-off on key deliverables. To receive formal acceptance of the project scope, the project team must develop clear documentation of the project's products and procedures to evaluate whether they were completed correctly and satisfactorily. Recall from Chapter 4 that configuration management specialists identify and document the functional and physical characteristics of the project's products, record and report the changes, and audit the products to verify conformance to requirements. To minimize scope changes, it is crucial to do a good job of configuration management and validating project scope.

The project management plan, project documents, verified deliverables, and work performance data are the main inputs for scope validation. The main tools for performing scope validation are inspection and decision-making techniques. The customer, sponsor, or user inspects the work after it is delivered and decides if it meets requirements. The main outputs of scope validation are accepted deliverables, change requests, work performance information, and project documents updates. For example, suppose that Kim's team members deliver upgraded computers to users as part of the IT upgrade project. Several users might complain because the computers did not include special keyboards they need for medical reasons. Appropriate people would review this change request and take appropriate corrective action, such as getting sponsor approval for purchasing the special keyboards.

Global Issues

Many countries have had difficulties controlling the scope of large projects, especially those that involve advanced technologies and many different users. For example, the state government of Victoria, Australia, introduced a public transportation smart card, called myki, in 2010. Public Transport Victoria's description of the card included the following information: "Many cities around the world have public transport smart cards. myki has been designed to fit our State's unique needs. myki users enjoy an integrated ticketing system that works across the state on trains, trams and buses."*

Unfortunately, there were many problems in developing and implementing the smart cards. The \$1.35 billion system became valid on all forms of Melbourne public transportation in July 2010, three years and five months after it was meant to replace the Metcard. Users' initial reactions to the myki smart card were mixed, with several reports of myki readers not working on trams. Many skeptics said they would wait until problems were fixed before trying the new system.* Many articles described problems with the myki card, revealing obvious difficulty in validating the scope of this high-visibility project. The Public Transport Users Association (PTUA) compiled a long list of problems with the new card and suggested that people stick to using the old card for a while. Clearly, the new system did not meet user requirements and had major flaws. In January 2012, over 18 months after the myki rollout, 70 percent of users still used the old Metcard. The government decided to keep myki in June 2011 after estimating that it would cost taxpayers more than \$1 billion to scrap the troubled system. The Metcard was planned to be phased out for good by the end of 2012. PTUA president Daniel Bowen was not certain that the transition would go smoothly. "To a certain extent it makes sense to bring people across (to myki) because you can't have two systems running at the same time forever. . . . Once you get the majority of people using myki though, they'd better make sure it's humming, otherwise there could be chaos."* Myki continued to make headlines in 2015, but not in a good way. "Two years after myki became the only ticket in

town, frustrated commuters say the system is still riddled with issues.”*

Controlling Scope

As you learned in [Chapter 4](#) when you studied integrated change control, change is inevitable on projects, especially changes to the scope of IT projects. Scope control involves managing changes to the project scope while keeping project goals and business strategy in mind. Users often are not sure how they want screens to look or what functionality they will need to improve business performance. Developers are not exactly sure how to interpret user requirements, and they also have to deal with constantly changing technologies.

The goal of scope control is to influence the factors that cause scope changes, to ensure that changes are processed according to procedures developed as part of integrated change control, and to manage changes when they occur. You cannot do a good job of controlling scope if you do not first do a good job of collecting requirements, defining scope, and validating scope. How can you prevent scope creep when you have not agreed on the work to be performed and your sponsor has not validated that the proposed work is acceptable? You also need to develop a process for soliciting and monitoring changes to project scope. Stakeholders should be encouraged to suggest changes that will benefit the overall project and discouraged from suggesting unnecessary changes.

The project management plan, project documents, work performance data, and organizational process assets are the main inputs to scope control. An important tool for performing scope control is data analysis, including variance analysis. [Variance](#) is the difference between planned and actual performance. For example, if a supplier was supposed to deliver five special keyboards and you received only four, the variance would be one keyboard. The outputs of scope control include work performance information, change requests, project management plan updates, and project documents updates.

Table 1-2 in Chapter 1 lists the top 10 factors that help IT projects succeed. Four of these ten factors are related to scope validation and control: executive sponsorship, user involvement, optimization, and clear business objectives. To avoid project failures, therefore, it is crucial for IT project managers and their teams to improve user input and executive support and reduce incomplete and changing requirements.

The following sections provide more suggestions for improving scope management on IT projects.

Suggestions for Improving User Input

Lack of user input leads to problems with managing scope creep and controlling change. How can you manage this important issue? The following suggestions can help a project team improve user input:

- Develop a good project selection process for IT projects. Insist that all projects have a sponsor from the user organization. The sponsor should not work in the IT department, nor should the sponsor be the project manager. Project information, including the project charter, project management plan, project scope statement, WBS, and WBS dictionary, should be easily available in the organization. Making basic project information available will help avoid duplication of effort and ensure that the most important projects are the ones on which people are working.
- Have users on the project team. Some organizations require project managers to come from the business area of the project instead of the IT group. Some organizations assign co-project managers to IT projects, one from IT and one from the main business group. Users should be assigned full time to large IT projects and part-time to smaller projects. A key success factor in Northwest Airlines' ResNet project was training reservation agents—the users—how to write programming code for their new reservation system. (See the Companion website for this text to read the entire case study for the ResNet project.) Because the sales agents had intimate knowledge of the business, they provided excellent input and actually created most of the software.
- Have regular meetings with defined agendas. The idea of meeting regularly sounds obvious, but many IT projects fail because the project team members do not have regular interaction with users. They assume that they understand what users need without getting direct feedback. To encourage this interaction, users should sign off on key deliverables presented at meetings.
- Deliver something to project users and sponsors on a regular basis. If the delivered product is hardware or software, make sure it works first.
- Do not promise to deliver what the team cannot deliver in a particular time frame. Make sure the project schedule allows enough time to produce the deliverables.
- Locate users with the developers. People often get to know each other better by being in close proximity. If the users cannot be physically moved to be near developers during the entire project, they should set aside certain days to work in the same location.

Suggestions for Reducing Incomplete and Changing Requirements

Some requirement changes are expected on IT projects, but many projects have too many changes to their requirements, especially during later stages of the project life cycle when it is more difficult to implement changes. The following suggestions can help improve the requirements process:

- Develop and follow a requirements management process that includes procedures for determining initial requirements.
- Employ techniques such as prototyping, use case modeling, and Joint Application Design to understand user requirements thoroughly. [Prototyping](#) involves developing a working replica of the system or some aspect of the system. These working replicas may be throwaways or an incremental component of the deliverable system. Prototyping is an effective tool for gaining an understanding of requirements, determining the feasibility of requirements, and resolving user interface uncertainties. [Use case modeling](#) is a process for identifying and modeling business events, who initiated them, and how the system should respond to them. It is an effective tool for understanding requirements of information systems. [Joint Application Design \(JAD\)](#) uses highly organized and intensive workshops to bring together project stakeholders—the sponsor, users, business analysts, programmers, and so on—to jointly define and design information systems. These techniques also help users become more active in defining system requirements. Consult a systems analysis and design text for details on these techniques.
- Put all requirements in writing and keep them current and readily available. Several tools are available to automate this function. For example, a type of software called a requirements management tool aids in capturing and maintaining requirements information, provides immediate access to the information, and assists in establishing necessary relationships between requirements and information created by other tools.
- Create a requirements management database for documenting and controlling requirements. Computer Aided Software Engineering (CASE) tools or other technologies can assist in maintaining a repository for project data. A CASE tool's database can also be used to document and control requirements.
- Provide adequate testing to verify that the project's products perform as expected. Conduct testing throughout the project life cycle. Chapter 8, Project Quality Management, includes more information on testing.
- Use a process for reviewing requested requirements changes from a systems perspective. For example, ensure that project scope changes include associated cost and schedule changes. Require approval via signatures of appropriate stakeholders. It is crucial for the project manager to lead the team in its focus on achieving approved scope

goals and not getting sidetracked into doing additional work. For example, in his book *Alpha Project Managers*, Andy Crowe tried to uncover what the best or “alpha” project managers do differently from other project managers. One of these alpha project managers explained how he learned an important lesson about scope control:

Toward the end of some projects I’ve worked on, the managers made their teams work these really long hours. After the second or third time this happened, I just assumed that this was the way things worked. Then I got to work with a manager who planned everything out really well and ran the team at a good pace the whole time, and we kept on schedule. When the customer found out that things were on schedule, he kept trying to increase the scope, but we had a good manager this time, and she wouldn’t let him do it without adjusting the baselines. That was the first time I was on a project that finished everything on time and on budget, and I was amazed at how easy she made it look.*

- Emphasize completion dates. For example, a project manager at Farmland Industries, Inc. in Kansas City, Missouri, kept her 15-month, \$7 million integrated supply-chain project on track by setting the project deadline. She said, “May 1 was the drop-dead date, and everything else was backed into it. Users would come to us and say they wanted something, and we’d ask them what they wanted to give up to get it. Sticking to the date is how we managed scope creep.”*
- Allocate resources specifically for handling change requests. For example, Peeter Kivestu and his ResNet team at Northwest Airlines knew that users would request enhancements to the reservations system they were developing. They provided a special function key on the ResNet screen for users to submit their requests, and the project included three full-time programmers to handle these requests. Users made over 11,000 enhancement requests. The managers who sponsored the four main software applications had to prioritize the software enhancement requests and decide as a group what changes to approve. The three programmers then implemented as many items as they could, in priority order, given the time they had. Although they only implemented 38 percent of the requested enhancements, they were the most important ones, and the users were very satisfied with the system and process.

Using Software to Assist in Project Scope Management

Project managers and their teams can use several types of software to assist in project scope management. As shown in several of the figures and tables in this chapter, you can use word-processing software to create scope-related documents, and most people use spreadsheet or presentation software to develop various charts, graphs, and matrixes related to scope management. Mind-mapping software can be useful in developing a WBS. Project

stakeholders also transmit project scope management information using various types of communication software such as e-mail and assorted web-based applications.

Project management software helps you develop a WBS, which serves as a basis for creating Gantt charts, assigning resources, allocating costs, and performing other tasks. You can also use the templates that come with various project management software products to help you create a WBS for your project.

You can also use many types of specialized software to assist in project scope management. Many IT projects use special software for requirements management, prototyping, modeling, and other scope-related work. Because scope is such a crucial part of project management, many software products are available to assist in managing project scope. For example, Gartner estimates that the market for requirements definition and management tools was \$280 million in 2014 and growing over 3 percent annually. New tools focus on improving collaboration and speed to better address mass market needs.*

Considerations for Agile/Adaptive Environments

The *PMBOK® Guide – Sixth Edition* provides the following information for project scope management:

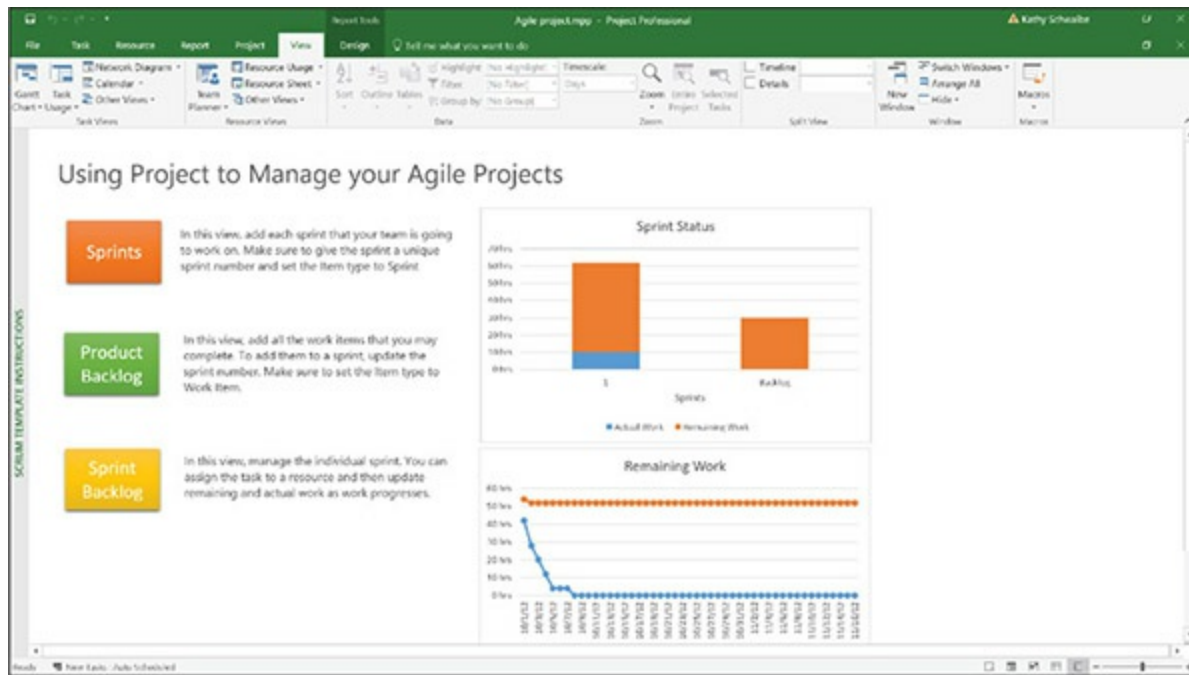
In projects with evolving requirements, high risk, or significant uncertainty, the scope is often not understood at the beginning of the project or it evolves during the project. Agile methods deliberately spend less time trying to define and agree on scope in the early stage of the project and spend more time establishing the process for its ongoing discovery and refinement. Many environments with emerging requirements find that there is often a gap between the real business requirements and the business requirements that were originally stated. Therefore, agile methods purposefully build and review prototypes and release versions in order to refine the requirements. As a result, scope is defined and redefined throughout the project. In agile approaches, the requirements constitute the backlog.*

As discussed in **Chapter 2**, stakeholders define and approve the detailed scope before the start of an iteration with an adaptive or agile product life cycle, producing a usable product at the end of each iteration. Does this mean that scope does not change for other product life cycles? As described in this chapter, the high-level requirements should always be defined early (in the project charter and initial scope statement), but the detailed scope develops over time. For example, in a large, traditional project like building a house, the owners set the high-level requirements (number of square feet, location, basic blueprint, etc.), but detailed requirements are defined later (which particular flooring, what model number for appliances, etc.). Also, remember a key difference between an agile approach and others is that it

provides *several* usable products during the project, while other approaches only deliver the final product at the end of the project.

You can still use project management software for agile projects. For example, **Figure 5-9** shows a screenshot of a Microsoft Project 2016 template called “Agile Project Management.” It includes links and brief instructions for creating sprints, the product backlog, and the sprint backlog. You can also create other charts, as shown in the figure.

Figure 5-9. Using Microsoft Project 2016 for agile projects



Project scope management is very important, especially on IT projects. After selecting projects, organizations must plan scope management, collect the requirements and define the scope of the work, break down the work into manageable pieces, validate the scope with project stakeholders, and manage changes to project scope. Using the basic project management concepts, tools, and techniques discussed in this chapter can help you manage project scope successfully.

Case Wrap-Up

Kim Nguyen reviewed guidelines for creating WBSs that were provided by her company and other sources. She had a meeting with the three team leaders for her project to get their input on how to proceed. They reviewed several sample documents and decided to have major groupings for their project based on updating the inventory database, acquiring the necessary hardware and software, installing the hardware and software, and performing project management. After they decided on a basic approach, Kim led a meeting with the entire project team of 12 people, with some attending virtually. She

reviewed the project charter and stakeholder register, described the basic approach they would use to collect requirements and define the project scope, and reviewed sample WBSs. Kim opened the floor for questions, which she answered confidently. She then let each team leader work with his or her people to start writing the detailed scope statement and their sections of the WBS and WBS dictionary. Everyone participated in the meeting, sharing their expertise and openly asking questions. Kim could see that the project was off to a good start.

Chapter Summary

Project scope management includes the processes to ensure that the project addresses all the work required to complete the project successfully. The main processes include planning scope management, collecting requirements, defining scope, creating the WBS, validating scope, and controlling scope.

The first step in project scope management is planning scope management. The project team reviews information and uses expert judgment and meetings to help create a scope management plan and requirements management plan.

The next step is collecting requirements, a crucial part of many IT projects. It is important to review the project charter and meet with key stakeholders listed in the stakeholder register when collecting requirements. The main outputs of this process are requirements documentation and a requirements traceability matrix.

A project scope statement is created in the scope definition process. This document often includes a product scope description, product user acceptance criteria, detailed information on all project deliverables, and information on project boundaries, constraints, and assumptions. There are often several versions of the project scope statement to keep scope information detailed and up to date.

A work breakdown structure (WBS) is a deliverable-oriented grouping of the work involved in a project that defines its total scope. The WBS forms the basis for planning and managing project schedules, costs, resources, and changes. You cannot use project management software without first creating a good WBS. A WBS dictionary is a document that provides detailed information about each WBS item. A good WBS is often difficult to create because of the complexity of the project. There are several approaches for developing a WBS, including using guidelines, the analogy approach, the top-down approach, the bottom-up approach, and mind mapping.

Validating scope involves formal acceptance of the completed project deliverables. Controlling scope involves controlling changes to the project scope.

Poor project scope management is one of the key reasons projects fail. For IT projects, it is important for good project scope management to have strong user involvement, executive support, a clear statement of requirements, and a process for managing scope changes.

Many software products are available to assist in project scope management. The WBS is a key concept in properly using project management software because it provides the basis for entering tasks.

Be sure to consider how project scope management can differ in agile/adaptive environments.

Discussion Questions

1. What is involved in project scope management, and why is good project scope management so important on IT projects?
2. What is involved in collecting requirements for a project? Why is it often difficult to do?
3. Discuss the process of defining project scope in more detail as a project progresses, going from information in a project charter to a project scope statement, WBS, and WBS dictionary.
4. Describe different ways to develop a WBS and explain why it is often so difficult to do.
5. What is the main technique used for validating scope? Give an example of scope validation on a project.
6. Using examples in this book or online, describe a project that suffered from scope creep. Could it have been avoided? How? Can scope creep be a good thing? When? What can organizations do to successfully manage inevitable changes in scope that are good for business?
7. Why do you need a good WBS to use project management software? What other types of software can you use to assist in project scope management?
8. How is scope management different on agile projects?

Quick Quiz

1. ____ refer(s) to all the work involved in creating the products of the project and the processes used to create them.
 - a. Deliverables
 - b. Milestones
 - c. Scope
 - d. Product development
2. Which tool or technique for collecting requirements is often the most expensive and time consuming?

- a. Interviews
 - b. Focus groups
 - c. Surveys
 - d. Observation
3. A ____ is a deliverable-oriented grouping of the work involved in a project that defines its total scope.
- a. scope statement
 - b. WBS
 - c. WBS dictionary
 - d. work package
4. What approach to developing a WBS involves writing down or drawing ideas in a nonlinear format?
- a. Top-down
 - b. Bottom-up
 - c. Analogy
 - d. Mind mapping
5. Assume that you have a project with major categories called planning, analysis, design, and testing. What level of the WBS would these items fall under?
- a. 0
 - b. 1
 - c. 2
 - d. 3
6. Which of the following is not a best practice that can help in avoiding scope problems on IT projects?
- a. Keep the scope realistic.

- b. Use off-the-shelf hardware and software whenever possible.
- c. Follow good project management processes.
- d. Don't involve too many users in scope management.

7. What constitutes requirements for agile projects?

- a. The Scrum
- b. The backlog
- c. The sprints
- d. The requirements list

8. Scope ____ is often achieved by a customer inspection and then sign-off on key deliverables.

- a. acceptance
- b. validation
- c. completion
- d. close-out

9. Which of the following is not a suggestion for improving user input?

- a. Develop a good project selection process for IT projects.
- b. Have users on the project team.
- c. Co-locate users with developers.
- d. Only have meetings as needed, not on a regular basis.

10. Project management software helps you develop a ____, which serves as a basis for creating Gantt charts, assigning resources, and allocating costs.

- a. project plan
- b. schedule
- c. WBS

d. deliverable

Exercises

1. You are working on a project to develop a new or enhanced system to help people at your college, university, or organization to find jobs. The system must be tailored to your student or work population and be very easy to use. Write a short paper describing how you would collect requirements for this system, and include at least five requirements in a requirements traceability matrix.
2. Read the report cited in the chapter called PMI's *Pulse of the Profession: Requirements Management—A Core Competency for Project and Program Success*. Find one or two other reports on requirements management written in the past year. Summarize the findings of the reports and your opinion on the topic.
3. Use MindView or similar software to develop a WBS for a project of your choice. You can use one of the examples in this chapter, or create your own. Take screenshots and print out the WBS in various formats (mind map, chart, and tabular).
4. Create the same WBS from **Exercise 3** using Microsoft Project or similar software and indenting categories appropriately. Use the outline numbering feature to display the outline numbers, or enter the numbers manually. Do not enter any durations or dependencies. See Appendix A or Project 2016's Help for instructions on creating a WBS. Print the resulting Gantt chart on one page, and be sure to display the entire Task Name column.
5. Using the file you created in MindView or similar software for **Exercise 3**, export the WBS into Project 2016 or similar software. Make adjustments as needed and then print the Gantt chart from Project 2016 or similar software. Also write a paragraph describing your opinion of using a mind map versus just entering information in the Task Name column.
6. Create a WBS for one of the following projects:

Introducing self-checkout registers at your school's bookstore

Providing a new Internet cafe onsite at your organization

Developing an app that you and a friend will create and sell online within three months

Earning your CAPM® or PMP® certification

Decide on all of the Level 2 categories for the WBS. Use your choice of software to create a WBS in chart form and as tasks in a Gantt chart. Do not enter any durations or

dependencies. Be sure to include all of the deliverables and work required for the project. Make notes of questions you had while completing this exercise.

7. Review three different template files from Microsoft Project 2016, MindView, or any other project management software tool. What do you think about the WBSs? Write a short paper summarizing your analysis, and provide at least three suggestions for improving one of the WBSs.
8. Research the benefits of and requirements for earning a certification in business analysis, as described in the What Went Right feature in this chapter. Summarize your findings in a short paper.

Running Case

You have been selected as the project manager for the Global Treps project. You helped to run a local shark tank like event at your college last year as part of a class project, so you have a general idea of what is involved. The schedule goal is six months, and the budget is \$120,000. Your favorite professor, Dr. K., and a few of her associates have agreed to fund the project. Your strengths are your organizational and leadership skills.

You are a senior, live on-campus, and get free room and board by being a resident assistant in your dorm. Bobby, a computer whiz who funded a lot of his college expenses by building websites, will be your main technical guy on the project. He goes to your college and lives off-campus. Three other people will form your core project team: Kim, a new college grad now working for a non profit group in Vietnam; Ashok, a business student in India; and Alfreda, a student in the United States planning to visit her home town in Ethiopia for two months in a few months. You will hold most meetings virtually, but you can meet face-to-face with Bobby and Dr. K. as needed. You have all known each other for at least a year and are excited to make this project a success.

You and your team members will do the work part-time while you finish school or work at other jobs, but you can use up to \$50,000 total to pay yourselves. You estimate that you will need another \$30,000 for travel expenses, \$20,000 for hardware and software, and the other \$20,000 will go toward organizing events, consultants, legal/business fees, etc.

Your goal is to develop a fully functioning website and test it by holding four events in four different countries. You'll make improvements to the site after those events, develop plans to scale it up, and recommend how to transition the project results into a successful business.

Recall from **Chapter 4** that this system would include the following capabilities:

- Provide guidelines and templates for running a shark tank type event

- Accept donations from potential investors targeted toward specific schools or organizations wishing to host an event (similar to the popular www.donorschoose.org site, where people can fund teachers' requests)
- Accept ideas for needed new products or services
- Provide the ability for organizations to create their own custom site to solicit local participants and sharks, accept applications, and promote the winners as well as losers.

Note that you decided *not* to include the idea of providing an online version of the event as part of the initial project as your sponsor and team decided that physical events would be most effective. You have also decided to limit the scope of this first project to provide the ability for 20 organizations to create their own custom websites. Your team members will screen the organizations and assist people in using the site to plan their events.

You plan to hold four shark tank like events within four months, using your team members abroad to help organize and run those events, plus one at your college. Your semester has just started, so you plan to hold your event at the end of the term. The project will fund refreshments for the events and prizes for the winners, with a budget of \$1,000 for each event. You don't think you'll get any donations via the new website before these events, but you'll try to have it set up to accept donations by the last month.

You will create some short videos to show people how to use the site and provide suggestions for holding the events.

After testing the site and getting customer feedback, you will make some changes and document recommendations for a follow-on project. You will also create a business plan recommending how to transition this project into a real business that can make a profit after two years.

Assume that you would pay for a new website and account through an online provider. Bobby would do most of the customization/programming for the site, but you would consider outsourcing or purchasing services to provide some of the capabilities like accepting donations and developing the short videos on the site. You would also buy a new laptop and Internet access for your three team members abroad so that they could share information with their contacts in those countries.

You and Dr. K. want to attend all of the events as part of the project, and you might include a full face-to-face meeting with the whole team if possible.

Tasks

1. Document your approach for collecting requirements for the project. Include at least five technical requirements and five non technical requirements in a requirements traceability matrix. Also develop a list of at least five questions you would like to ask the project sponsor. Let your instructor answer them for you, if applicable.
2. Develop a first version of a project scope statement for the project. Use the template provided on the Companion website for this text and the example in **Chapter 3** as guides. Be as specific as possible in describing product characteristics and requirements, as well as all of the project's deliverables. Be sure to include the Global Treps website, four shark tank like events, a business plan, and outsourced items (laptops, donation acceptance, video creation, etc.) as part of the project scope.
3. Develop a work breakdown structure for the project. Break down the work to Level 3 or Level 4, as appropriate. Use the template on the Companion website and samples in this text as guides. Print the WBS in list form. Be sure the WBS is based on the project charter (created for the **Chapter 4** Running Case), the project scope statement created in Task 2 above, and other relevant information.
4. Use the WBS you developed in **Task 3** to begin creating a Gantt chart using your choice of software. Do not enter any durations or dependencies. Print the resulting Gantt chart on one page, and be sure to display the entire Task Name column.
5. Develop a strategy for scope validation and change control for this project. Write a short paper summarizing key points of the strategy.

Key Terms

[analogy approach](#) p.200

[benchmarking](#) p.189

[bottom-up approach](#) p.201

[decomposition](#) p.194

[deliverable](#) p.184

[Joint Application Design \(JAD\)](#) p.209

[project scope management](#) p.184

[project scope statement](#) p.191

[prototyping](#) p.209

[requirement](#) p.186

[requirements management plan](#) p.187

[requirements traceability matrix \(RTM\)](#) p.191

[scope](#) p.184

[scope baseline](#) p.194

[scope creep](#) p.205

[scope validation](#) p.206

[top-down approach](#) p.201

[use case modeling](#) p.209

[variance](#) p.208

[WBS dictionary](#) p.203

[work breakdown structure \(WBS\)](#) p.194

[work package](#) p.196

Chapter 6. Project Schedule Management

Learning Objectives

After reading this chapter, you will be able to:

- Illustrate the importance that project schedules and good project schedule management can have in helping to make projects successful
- Discuss the process of planning schedule management
- Define activities as the basis for developing project schedules
- Describe how project managers use network diagrams and dependencies to assist in activity sequencing
- Explain how various tools and techniques help project managers perform activity duration estimates
- Use a Gantt chart for planning and tracking schedule information, find the critical path for a project, and describe how critical chain scheduling and the Program Evaluation and Review Technique (PERT) affect schedule development
- Compare how schedule management is addressed using Agile vs. more predictive project approaches
- Discuss how reality checks and discipline are involved in controlling and managing changes to the project schedule
- Describe how project management software can assist in project schedule management and review words of caution before using this software
- Discuss considerations for agile/adaptive environments

Opening Case

Sue Johnson was the project manager for a consulting company contracted to provide a new online registration system at a local college in nine months or less. This system had to be operational by May 1 so students could use it to register for the fall semester. Her company's contract had a stiff penalty clause if the system was not ready by then, and Sue and her team would get nice bonuses for doing a good job on this project and meeting the schedule. Sue knew that it was her responsibility to meet the schedule and manage scope, cost, and quality expectations. She and her team developed a detailed schedule and network diagram to help organize the project.

Developing the schedule turned out to be the easy part; keeping the project on track was more difficult. Managing personnel issues and resolving schedule conflicts were two of the bigger challenges. Many of the college's employees took unplanned vacations and

missed or rescheduled project review meetings. These changes made it difficult for the project team to follow the planned schedule for the system because the team had to have customer sign-off at various stages of the systems development life cycle. One senior programmer on Sue's project team quit, and she knew it would take extra time for a new person to get up to speed, especially since the exiting programmer did a poor job documenting how his code linked to other systems at the college. It was still early in the project, but Sue knew they were falling behind. What could she do to meet the operational date of May 1?

The Importance of Project Schedules

Managers often cite the need to deliver projects on time as one of their biggest challenges and the main cause of conflict. Perhaps part of the reason that schedule problems are so common is that time is easily measured and remembered. You can debate scope and cost overruns and make actual numbers appear closer to estimates, but once a project schedule is set, people remember the projected completion date, and anyone can quickly estimate schedule performance by subtracting the original time estimate from how long it really took to complete the project. People often compare planned and actual project completion times without taking into account the approved changes in the project. Time is the variable that has the least amount of flexibility. Time passes no matter what happens on a project.

Individual work styles and cultural differences may also cause schedule conflicts. For example, you will learn in Chapter 9, Project Resource Management, about the Myers-Briggs Type Indicator. One dimension of this team-building tool deals with attitudes toward structure and deadlines. Some people prefer detailed schedules and emphasize task completion. Others prefer to keep things open and flexible. Different cultures and even entire countries have different attitudes about schedules. For example, in some countries businesses close for several hours every afternoon to have siestas. Countries may have different holidays, which means not much work will be done at certain times of the year. Cultures may also have different perceptions of work ethic—some may value hard work and strict schedules while others may value the ability to remain relaxed and flexible.

Media Snapshot

In contrast to the 2002 Salt Lake City Winter Olympic Games (see [Chapter 4's](#) Media Snapshot), planning and scheduling were not well implemented for the 2004 Athens Summer Olympic Games or the 2014 Sochi Winter Olympic Games.

Many articles written before the Athens opening ceremonies predicted that the facilities

would not be ready in time. “With just 162 days to go to the opening of the Athens Olympics, the Greek capital is still not ready for the expected onslaught. . . . By now 22 of the 30 Olympic projects were supposed to be finished. This week the Athens Olympic Committee proudly announced 19 venues would be finished by the end of next month. That’s a long way off target.”* However, many people were pleasantly surprised by the amazing opening ceremonies, beautiful new buildings, and state-of-the-art security and transportation systems in Athens. For example, traffic flow, considered a major pre-Games hurdle, was superb. One spectator at the games commented on the prediction that the facilities would not be ready in time: “Athens proved them all wrong. . . . It has never looked better.”* The Greeks even made fun of critics by having construction workers pretend that they were still working as the ceremonies began. Unfortunately, the Greek government suffered a huge financial deficit because the games cost more than twice the planned budget.

The 2014 Winter Olympic Games in Sochi, Russia, suffered even greater financial losses. Originally budgeted at US\$12 billion, final costs reached over US\$51 billion, making it the most expensive games in history. Unlike Greece’s humorous response to challenges at the Athens games, Russian citizens were much more serious. Although in 2006, 86 percent of Sochi residents supported the games, by 2013 only 40 percent supported them. Russian citizens, especially those in Sochi, understood the negative impacts of hosting the games when they heard about the poor planning and huge cost overruns. “. . . the Sochi Olympics will continue to be a burden for the Russian state, with expenses for operation, maintenance and foregone interest and tax revenue in the order of USD 1.2 billion per year. The event also did not manage to improve the image of Russia in the world and among the domestic population support dropped over the seven years of its implementation, most notably among the local population.”*

With all the possibilities for schedule conflicts, it is important for project managers to use good project schedule management. [Project schedule management](#), simply defined, involves the processes required to ensure timely completion of a project. Six main processes are involved in project schedule management:

1. *Planning schedule management* involves determining the policies, procedures, and documentation that will be used for planning, executing, and controlling the project schedule.
2. *Defining activities* involves identifying the specific activities that the project team members and stakeholders must perform to produce the project deliverables. An [activity](#) or [task](#) is an element of work normally found on the work breakdown structure (WBS)

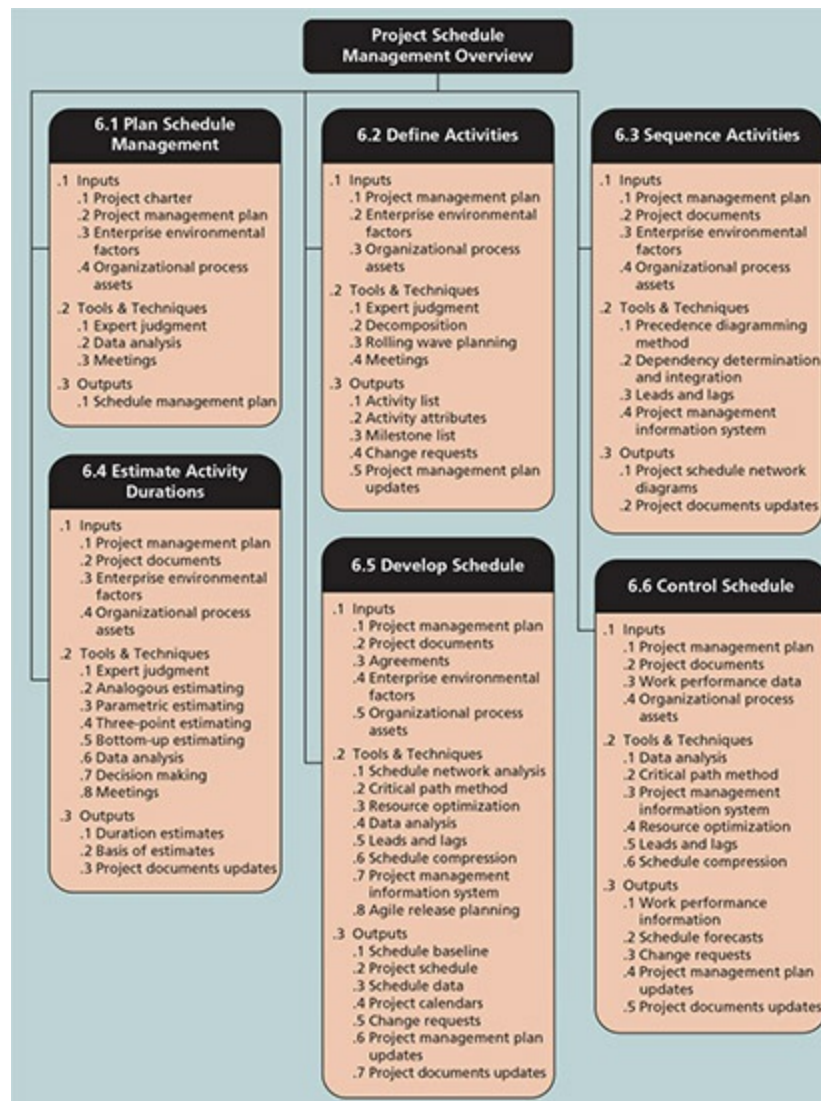
that has expected duration, cost, and resource requirements.

3. *Sequencing activities* involves identifying and documenting the relationships between project activities. Requirements, a resource breakdown structure, and project documents updates.
4. *Estimating activity durations* involves estimating the number of work periods that are needed to complete individual activities.
5. *Developing the schedule* involves analyzing activity sequences, resource requirements, and activity duration estimates to create the project schedule.
6. *Controlling the schedule* involves controlling and managing changes to the project schedule.

Figure 6-1 summarizes the inputs, tools and techniques, and outputs of project schedule management.

Figure 6-1.

Project schedule management overview



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You can improve project schedule management by performing these processes and by using some basic project management tools and techniques. Every manager is familiar with some form of scheduling, but most managers have not used several of the tools and techniques that are unique to project schedule management, such as Gantt charts, network diagrams, and critical path analysis.

Planning Schedule Management

The first step in project schedule management is planning how the schedule will be managed throughout the life of the project. Project schedules grow out of the basic documents that initiate a project. The project charter often mentions planned project start and end dates, which serve as the starting points for a more detailed schedule. After reviewing the project management plan, project charter, enterprise environmental factors, and organizational process assets, the project team uses expert judgment, analytical techniques, and meetings to

develop the schedule management plan.

The schedule management plan, like the scope management plan, can be informal and broad or formal and detailed, based on the needs of the project. In general, a schedule management plan includes the following information:

- *Project schedule model development*: Many projects include a schedule model, which contains project activities with estimated durations, dependencies, and other planning information that can be used to produce a project schedule. See Appendix A (available on the Companion website for this text) for information on using Microsoft Project 2016 to create a schedule model.
- *Level of accuracy and units of measure*: This section discusses how accurate schedule estimates should be and determines whether time is measured in hours, days, or another unit.
- *Control thresholds*: Variance thresholds, such as ± 10 percent, are established for monitoring schedule performance.
- *Rules of performance measurement*: For example, if team members are expected to track the percentage of work completed, this section specifies how to determine the percentages.
- *Reporting formats*: This section describes the format and frequency of schedule reports required for the project.
- *Process descriptions*: The schedule management plan also describes how all of the schedule management processes will be performed.

Defining Activities

You might think that all project work has been defined in enough detail after planning scope management, but it is often necessary to describe activities in more detail as part of schedule management. Defining activities involves identifying the specific actions that will produce the project deliverables in enough detail to determine resource and schedule estimates. The project team reviews the project management plan, enterprise environmental factors, and organizational process assets to begin defining activities. Outputs of this process include an activity list, activity attributes, a milestone list, change requests, and project management plan updates.

The [activity list](#) is a tabulation of activities to be included on a project schedule. The list should include the activity name, an activity identifier or number, and a brief description of

the activity. The [activity attributes](#) provide schedule-related information about each activity, such as predecessors, successors, logical relationships, leads and lags, resource requirements, constraints, imposed dates, and assumptions related to the activity. The activity list and activity attributes should agree with the WBS and WBS dictionary. Information is added to the activity attributes as it becomes available; this information includes logical relationships and resource requirements that are determined in later processes. Many project teams use an automated system to keep track of activity-related information.

A [milestone](#) on a project is a significant event that normally has no duration. It often takes several activities and a lot of work to complete a milestone, but the milestone itself is a marker to help in identifying necessary activities. Milestones are also useful tools for setting schedule goals and monitoring progress. For example, milestones on a project like the one in the chapter's opening case might include completion and customer sign-off documents, such as design documents and test plans; completion of specific products, such as software modules or installation of new hardware; and completion of important process-related work, such as project review meetings and tests. Not every deliverable or output created for a project is really a milestone. Milestones are the most important and visible events. For example, in the context of child development, parents and doctors check for *milestones*, such as a child first rolling over, sitting, crawling, walking, and talking. You will learn more about milestones later in this chapter.

Activity information is a required input to the other schedule management processes. You cannot determine activity sequencing, durations, develop the schedule, or control the schedule until you have a good understanding of project activities.

Recall the triple constraint of project management—balancing scope, time, and cost goals—and note the order of these items. Ideally, the project team and key stakeholders first define the project scope, then the time or schedule for the project, and then the project's cost. The order of these three items reflects the basic order of the processes in project schedule management: defining activities (further defining the scope), sequencing activities (further defining the time), and estimating activity resources and activity durations (further defining the time and cost). These processes are the basis for creating a project schedule.

The goal of defining activities is to ensure that the project team completely understands all the work it must do as part of the project scope so they can start scheduling the work. For example, a WBS item might be “Study report.” The project team must understand what the item means before team members can make schedule-related decisions. How long should the report be? Does it require a survey or extensive research to produce? What skill level does the report writer need to have? Further defining the task will help the project team determine how long it will take to do and who should do it.

The WBS is often dissected further as the project team members continue to define the activities required for performing the work. For example, the task “Study report” might be broken down into several subtasks describing smaller deliverables required to produce the report, such as survey development, survey administration, draft report, report edits, and final report. This process of progressive elaboration, one of the project attributes listed in **Chapter 1**, is sometimes called “rolling wave planning.”

As stated earlier, activities (or tasks, the term used in most project management software) are elements of work performed during the course of a project; they have expected durations, costs, and resource requirements. Defining activities also results in supporting detail to document important product information as well as assumptions and constraints related to specific activities. The project team should review the activity list and activity attributes with project stakeholders before moving on to the next step in project schedule management. If the team does not review these items with project stakeholders, it could produce an unrealistic schedule and deliver unacceptable results. For example, if a project manager simply estimated that the “time required to complete the Study report” deliverable would be one day and then had an intern or trainee write a 10-page report to complete that task, the result could be a furious customer who expected extensive research, surveys, and a 100-page report. Clearly defining the work is crucial to all projects. If there are misunderstandings about activities, requested changes may be required.

In the opening case of this chapter, Sue Johnson and her project team had a contract and detailed specifications for the college’s new online registration system. They also had to focus on meeting the May 1 date for delivering an operational system so the college could start using the new system for the new semester’s registration. To develop a project schedule, Sue and her team had to review the contract, detailed specifications, and desired operational date, then create an activity list, activity attributes, and milestone list. After developing more detailed definitions of project activities, Sue and her team would review them with the customers to ensure that the team was on the right track.

What Went Wrong?

At the U.S. Federal Bureau of Investigation (FBI), poor time management was one of the reasons behind the failure of Trilogy, a “disastrous, unbelievably expensive piece of vaporware, which was more than four years in the (un)making. The system was supposed to enable FBI agents to integrate intelligence from isolated information silos within the Bureau.”* In May 2006, the Government Accounting Agency said that the Trilogy project failed at its core mission of improving the FBI’s investigative abilities and was plagued with missed milestones and escalating costs.

The FBI rushed to develop the new system beginning in 2001, in response to the attacks on September 11. The need for new software was obvious to former FBI agent David J. Williams, who recalls joining a roomful of agents shortly after 9/11 to help with intelligence. Agents were wearing out the casters on their chairs by sliding back and forth among 20 old computer terminals, many of which were attached to various databases that contained different information.

Congressional hearings revealed numerous problems with the project. Its requirements were very loosely defined, there were many leadership changes throughout the project, and several contracts had no scheduled milestones for completion or penalties if work was late. The new system was finally completed in 2006 and cost more than \$537 million—more than a year late and \$200 million over budget.

The FBI replaced Trilogy with a new system called Sentinel and began training employees to use it in May 2007.* Unfortunately, history repeated itself as troubles still loomed with Sentinel in 2012. According to the FBI's original plan, Sentinel was to be completed by December 2009 at an estimated cost of \$425 million. The FBI later increased the estimated cost to \$451 million and extended the project completion date twice, so that the project was now two years behind schedule. During a test exercise in 2011, the system experienced two outages, and the FBI determined that its current hardware structure was inadequate. In 2014, several people complained that the system still wasn't working well. "You can't find what you need most of the time, or you get junk you don't want, but other than that, the FBI's long troubled, half-billion-dollar Sentinel computerized file system is coming along just fine."*

Sequencing Activities

After defining project activities, the next step in project schedule management is sequencing them or determining their dependencies. Inputs to the activity sequencing process include the project management plan, project documents (like the activity attributes, activity list, assumption log, and milestone list), enterprise environmental factors, and organizational process assets. The sequencing process involves evaluating the reasons for dependencies and the different types of dependencies.

Dependencies

A [dependency](#) or [relationship](#) pertains to the sequencing of project activities or tasks. For example, does a certain activity have to be finished before another can start? Can the project team do several activities in parallel? Can some overlap? Determining these relationships or

dependencies among activities is crucial for developing and managing a project schedule.

There are several types of dependencies among project activities:

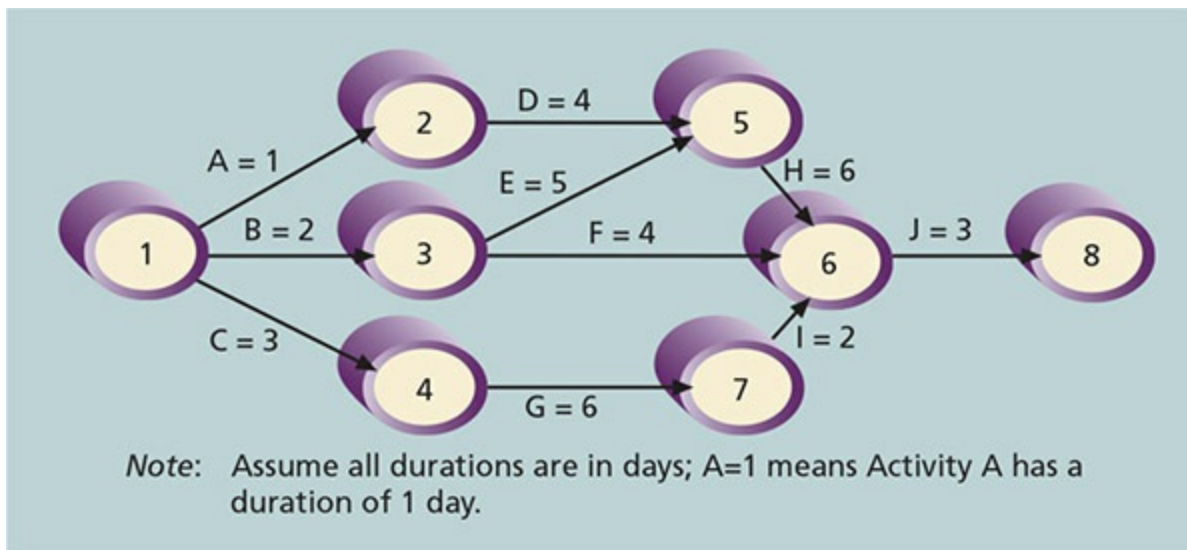
- **Mandatory dependencies** are inherent in the nature of the work being performed on a project. They are sometimes referred to as hard logic. For example, you cannot test code until after the code is written.
- **Discretionary dependencies** are defined by the project team. For example, a project team might follow good practice and not start the detailed design of a new information system until the users sign off on all of the analysis work. Discretionary dependencies are sometimes referred to as soft logic and should be used with care because they may limit later scheduling options.
- **External dependencies** involve relationships between project and non-project activities. For example, the installation of a new operating system and other software may depend on delivery of new hardware from an external supplier. Even though delivery of the hardware may not be included in the scope of the project, you should add an external dependency to it because late delivery will affect the project schedule.
- **Internal dependencies** involve relationships between project activities that are generally inside the project team's control. For example, if software is developed by the team, they can create dependencies such as performing unit testing before system testing.

Note that dependencies can be internal and mandatory, external and discretionary, and so on. As with activity definition, it is important that project stakeholders work together to define the activity dependencies in their project. If you do not define the sequence of activities, you cannot use some of the most powerful scheduling tools available to project managers: network diagrams and critical path analysis.

Network Diagrams

Network diagrams are the preferred technique for showing activity sequencing. A **network diagram** is a schematic display of the logical relationships among project activities and their sequencing. Some people refer to network diagrams as project schedule network diagrams or PERT charts. PERT is described later in this chapter. Figure 6-2 shows a sample network diagram for Project X.

Figure 6-2. Network diagram for project X



Note the main elements on this network diagram. The letters A through J represent activities with dependencies that are required to complete the project. These activities come from the WBS and activity definition process described earlier. The arrows represent the activity sequencing or relationships between tasks. For example, Activity A must be done before Activity D, and Activity D must be done before Activity H.

The format of this network diagram uses the [activity-on-arrow \(AOA\)](#) approach or the [arrow diagramming method \(ADM\)](#) —a network diagramming technique in which activities are represented by arrows and connected at points called nodes to illustrate the sequence of activities. A [node](#) is simply the starting and ending point of an activity. The first node signifies the start of a project and the last node represents the end.

Keep in mind that the network diagram represents activities that must be done to complete the project. It is not a race to get from the first node to the last node. *Every* activity on the network diagram must be completed in order to finish the project. Note also that not every item on the WBS needs to be shown on the network diagram; only activities with dependencies need to be shown. However, some people like to have start and end milestones and to list every activity. It is a matter of preference. For large projects with hundreds of activities, it might be simpler to include only activities with dependencies on a network diagram. Sometimes it is enough to put summary tasks on a network diagram or to break down the project into several smaller network diagrams.

Assuming that you have a list of the project activities and their start and finish nodes, follow these steps to create an AOA network diagram:

1. Find all of the activities that start at Node 1. Draw their finish nodes, and draw arrows between Node 1 and each of the finish nodes. Put the activity letter or name on the

associated arrow. If you have a duration estimate, write it next to the activity letter or name, as shown in **Figure 6-2**. For example, $A = 1$ means that the duration of Activity A is one day, week, or other standard unit of time. Be sure to put arrowheads on all arrows to signify the direction of the relationships.

2. Continue drawing the network diagram, working from left to right. Look for bursts and merges. **Bursts** occur when two or more activities follow a single node. A **merge** occurs when two or more nodes precede a single node. For example, in Figure 6-2, Node 1 is a burst because it goes into Nodes 2, 3, and 4. Node 5 is a merge preceded by Nodes 2 and 3.
3. Continue drawing the AOA network diagram until all activities are included.
4. As a rule of thumb, all arrowheads should face toward the right, and no arrows should cross on an AOA network diagram. You may need to redraw the diagram to make it look presentable.

Even though AOA or ADM network diagrams are generally easy to understand and create, a different method is more commonly used: the precedence diagramming method. The **precedence diagramming method (PDM)** is a network diagramming technique in which boxes represent activities. It is particularly useful for visualizing certain types of time relationships.

Figure 6-3 illustrates the types of dependencies that can occur among project activities based on a Microsoft Project help screen. After you determine the reason for a dependency between activities (mandatory, discretionary, or external), you must determine the type of dependency. Note that the terms *activity* and *task* are used interchangeably, as are *relationship* and *dependency*. The four types of dependencies or relationships between activities include the following:

- **Finish-to-start dependency** : A relationship in which the “from” activity or predecessor must finish before the “to” activity or successor can start. For example, you cannot provide user training until after software or a new system has been installed. Finish-to-start is the most common type of relationship or dependency, and AOA network diagrams use only finish-to-start dependencies.
- **Start-to-start dependency** : A relationship in which the “from” activity cannot start until the “to” activity or successor is started. For example, on IT projects, a group of activities might start simultaneously, such as the many tasks that occur when a new system goes live.
- **Finish-to-finish dependency** : A relationship in which the “from” activity must be

finished before the “to” activity can be finished. One task cannot finish before another finishes. For example, quality control efforts cannot finish before production finishes, although the two activities can be performed at the same time.

- **Start-to-finish dependency** : A relationship in which the “from” activity must start before the “to” activity can be finished. This type of relationship is rarely used, but it is appropriate in some cases. For example, an organization might strive to stock raw materials just in time for the manufacturing process to begin. A delay in starting the manufacturing process should delay completion of stocking the raw materials. Another example would be a babysitter who wants to finish watching a young child but is dependent on the parent’s arrival. The parent must show up or “start” before the babysitter can finish the task.

Figure 6-3. Task Dependency Types

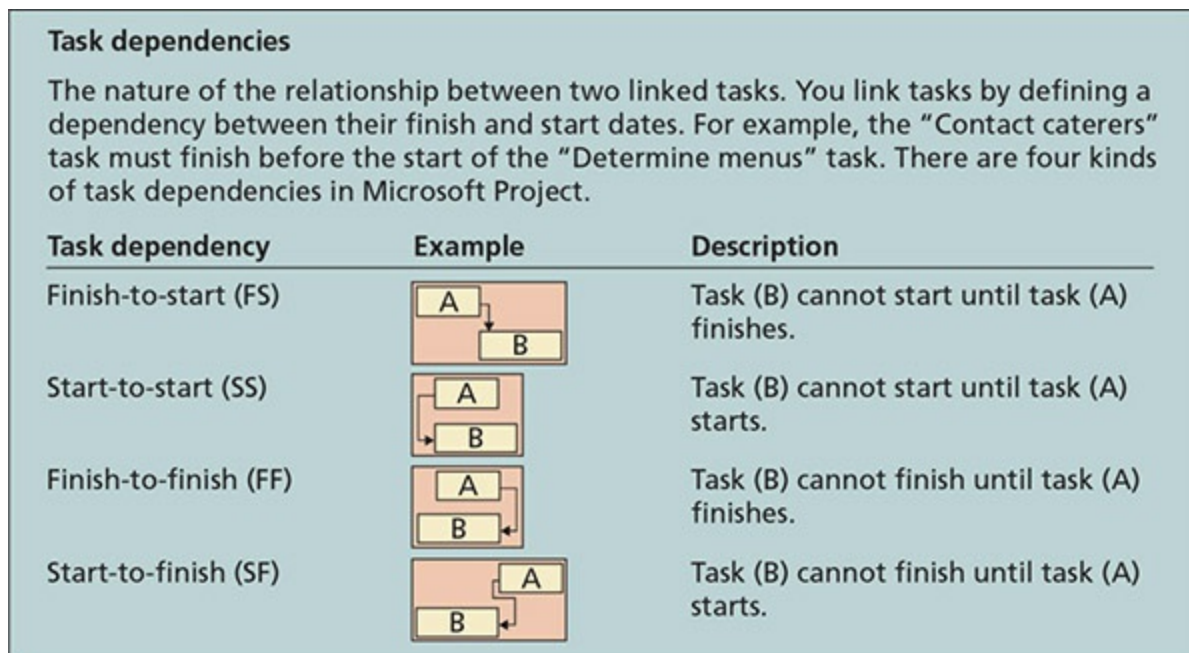


Figure 6-4 illustrates Project X using the precedence diagramming method. Notice that the activities are placed inside boxes, which represent the nodes on this diagram. Arrows show the relationships between activities. This figure was created using Microsoft Project, which automatically places additional information inside each node. Each task box includes the start and finish dates, which are labeled Start and Finish; the task ID number, labeled ID; the task’s duration, labeled Dur; and the names of resources, if any, that are assigned to the task. These resources are labeled Res. The border of boxes for tasks on the critical path appears automatically in red in the Microsoft Project network diagram view.

Figure 6-4. Precedence diagramming methods (PDM) network diagram for project X



The precedence diagramming method is used more often than AOA network diagrams and offers a number of advantages over the AOA technique. First, most project management software uses the precedence diagramming method. Second, using this method avoids the need to use dummy activities. [Dummy activities](#) have no duration and no resources, but are occasionally needed on AOA network diagrams to show logical relationships between activities. These activities are represented with dashed arrow lines and have zeros for their duration estimates. Third, the precedence diagramming method shows different dependencies among tasks, whereas AOA network diagrams use only finish-to-start dependencies. You will learn more about activity sequencing using Project 2016 in Appendix A (available on the Companion website for this text).

Estimating Activity Durations

After working with key stakeholders to define activities and determine their dependencies, the next process in project schedule management is to estimate the duration of activities. It is important to note that [duration](#) includes the actual amount of time worked on an activity *plus* elapsed time. For example, even though it might take one workweek or five workdays to do the actual work, the duration estimate might be two weeks to allow extra time needed to obtain outside information. The people or resources assigned to a task will also affect the task duration estimate. As another example, if someone asked for an estimate of when you plan to finish reading a particular book, you might give an answer of two months. Two months would be the duration estimate, even if you only plan to spend 20 hours actually reading the book.

Do not confuse duration with [effort](#), which is the number of workdays or work hours required to complete a task. (Duration is normally entered in the Duration column in software such as Microsoft Project 2016, while effort is entered in a Work column. See Appendix A—available on the Companion website for this text—for more information.) A duration estimate of one day could be based on eight hours of work or 80 hours of work, assuming that multiple people are working on a task that day. Duration relates to the time estimate on a calendar, not the effort estimate. In the previous example, you might plan to spend 20 hours

reading a book—the effort estimate—and spread that time out over two months—the duration.

Of course, duration and effort are related, so project team members must document their assumptions when creating duration estimates. The people who will actually do the work, in particular, should have a lot of say in duration estimates because their performances will be evaluated based on their ability to meet the estimates. It is also helpful to review similar projects and seek the advice of experts in estimating activity durations.

Project team members must also update the estimates as the project progresses. If scope changes occur on the project, the duration estimates should be updated to reflect those changes.

There are several inputs to activity duration estimates, including the project management plan, project documents, enterprise environmental factors, and organizational process assets. In addition to reviewing past project information, the team should review the accuracy of the duration estimates thus far on the project. For example, if team members find that all of their estimates have been much too long or short, they should update the estimates to reflect what they have learned.

One of the most important considerations in making activity duration estimates is the availability of resources, especially human resources. What specific skills do people need to do the work? What are the skill levels of the people assigned to the project? How many people are expected to be available to work on the project at any one time? Estimating activity resources is described in Chapter 9, Project Resource Management.

The outputs of activity duration estimates include the estimates themselves, the basis of estimates, and project documents updates. Duration estimates are often provided as a discrete number, such as four weeks; as a range, such as three to five weeks; or as a three-point estimate. A [three-point estimate](#) includes an optimistic, a most likely, and a pessimistic estimate, such as three weeks for the optimistic scenario, four weeks for the most likely scenario, and five weeks for the pessimistic scenario. The optimistic estimate is based on a best-case scenario, while the pessimistic estimate is based on a worst-case scenario. The most likely estimate, as you might expect, is based on a most likely or expected scenario. A three-point estimate is required for performing PERT estimates, as described later in this chapter, and for performing Monte Carlo simulations, as described in Chapter 11, Project Risk Management. Other duration estimating techniques include analogous and parametric estimating and reserve analysis, as described in Chapter 7, Project Cost Management. Expert judgment is also an important tool for developing good activity duration estimates.

Advice for Young Professionals

Some people find estimating to be challenging, especially for their own work. However, it is very important to develop this skill. You don't want to be stressing yourself out when you gave an overly optimistic estimate and need to work around the clock to get something done, nor do you want to lose out on an opportunity because you made a very high estimate that was unacceptable to a potential client or sponsor. Practice estimating how long it takes you to do different activities, and then take actual measurements. Defining the activity in detail, as described earlier in this chapter, will help you to make better estimates about each part of the activity. Also, don't be afraid to ask for help in making estimations. We often think that things take less time than they actually do, and people with experience can help us be more realistic with estimates. You could start with activities like writing reports, arranging meetings, or creating a presentation. You might think you can develop a presentation in four hours, for example, but you could find that you actually take twice or three times that much because you have to think it through, revise, get opinions, format graphics, or other similar parts of the activity. If you realize that an activity estimate might not be a good one, let your team know as soon as possible so that adjustments can be made early in the project.

Developing the Schedule

Schedule development uses the results of all the preceding project schedule management processes to determine the start and end dates of the project and its activities. Project schedule management processes often go through several iterations before a project schedule is finalized. The ultimate goal of developing a realistic project schedule is to provide a basis for monitoring project progress for the time dimension of the project. The main outputs of this process are a schedule baseline, project schedule, schedule data, project calendars, change requests, project management plan updates, and project documents updates. Some project teams create a computerized model to create a network diagram, enter resource requirements and availability by time period, and adjust other information to quickly generate alternative schedules. See Appendix A (available on the Companion website for this text) for information on using Project 2016 to assist in schedule development.

A few of the tools and techniques for schedule development include the following:

- A Gantt chart is a common tool for displaying project schedule information.
- Critical path analysis is a very important tool for developing and controlling project schedules.

- Critical chain scheduling is a technique that focuses on limited resources when creating a project schedule.
- PERT analysis is a means for considering schedule risk on projects.

The following sections provide samples of each of these tools and techniques and discuss their advantages and disadvantages.

Gantt Charts

Gantt charts provide a standard format for displaying project schedule information by listing project activities and their corresponding start and finish dates in calendar form. Gantt charts are sometimes referred to as bar charts because the activities' start and end dates are shown as horizontal bars. Figure 6-5 shows a simple Gantt chart for Project X created with Microsoft Project. Figure 6-6 shows a Gantt chart that is more sophisticated and is based on a software launch project from a template provided by Microsoft. The activities on the Gantt chart are driven by the deliverables on the WBS, and should coincide in turn with the activity list and milestone list. Notice that the Gantt chart for the software launch project contains milestones, summary tasks, individual task durations, and arrows showing task dependencies.

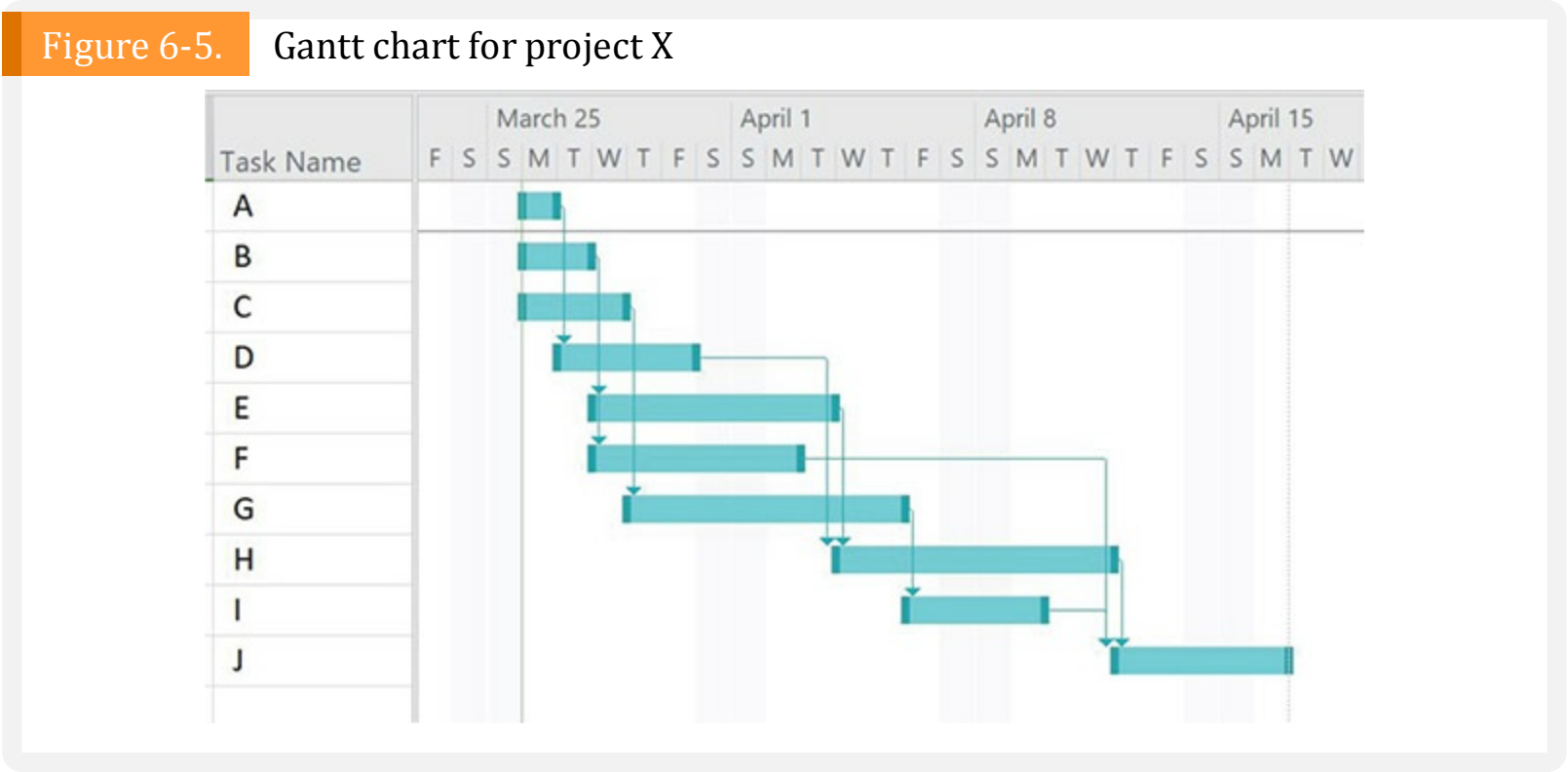
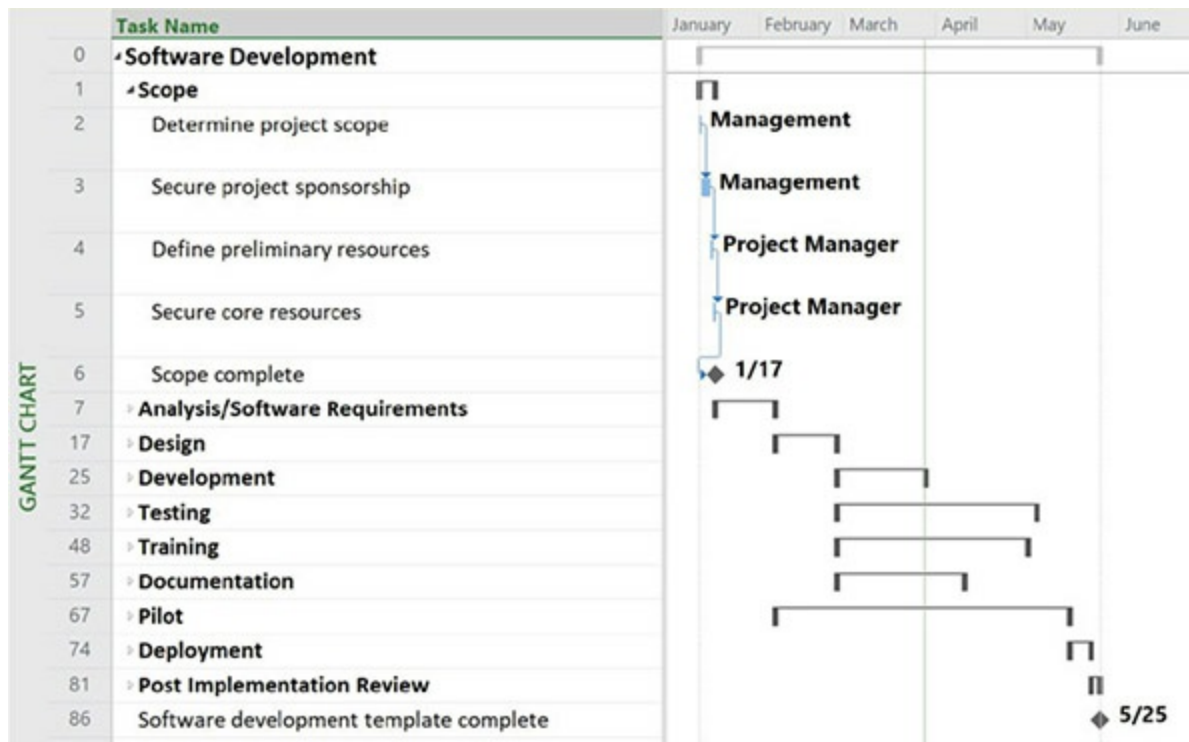


Figure 6-6. Gantt chart for software launch project



Notice the different symbols on the Gantt chart for the software launch project (see [Figure 6-6](#)):

- The black diamond represents a milestone. In [Figure 6-6](#), Task 6, “Scope complete,” is a milestone that occurs on January 17. For very large projects, top managers might want to see only milestones on a Gantt chart. Microsoft Project allows you to filter information displayed on a Gantt chart so you can easily show specific tasks, such as milestones.
- The thick black bars with rectangles at the beginning and end represent summary tasks. For example, the activities listed in rows 2-5 are all subtasks of the summary task called “Scope,” Task 1. WBS activities are referred to as tasks and subtasks in most project management software.
- The light blue horizontal bars for Tasks 2, 3, 4, and 5 represent the duration of each individual task.
- Arrows connecting these symbols show relationships or dependencies between tasks. Gantt charts often do not show dependencies, which is their major disadvantage. If dependencies have been established in Microsoft Project, they are automatically displayed on the Gantt chart.

Adding Milestones to Gantt Charts

Milestones can be a particularly important part of schedules, especially for large projects. Many people like to focus on meeting milestones, so you can create them to emphasize important events or accomplishments on projects. Normally, you create milestones by entering tasks with zero duration. In Microsoft Project, you can mark any task as a milestone by checking the appropriate box in the Advanced tab of the Task Information dialog box. The duration of the task will not change to zero, but the Gantt chart will show the milestone symbol to represent that task based on its start date. See Appendix A (available on the Companion website for this text) for more information.

To make milestones meaningful, some people use the SMART criteria to help define them. The [SMART criteria](#) are guidelines suggesting that milestones should be:

- Specific
- Measurable
- Assignable
- Realistic
- Time-framed

For example, distributing a marketing plan is specific, measurable, and assignable if everyone knows what should be in the marketing plan, how it should be distributed, how many copies should be distributed and to whom, and who is responsible for the actual delivery. Distributing the marketing plan is realistic and able to be time-framed if it is an achievable event and scheduled at an appropriate time.

Best Practice

In his book, *The Happiness Advantage*, Shawn Achor shares principles of positive psychology that can fuel success at work. One problem many people have at work is feeling overwhelmed. They complain that they are working all the time, but they cannot get their work done. Achor suggests that the 20-second rule can help people improve their focus by minimizing barriers to change.

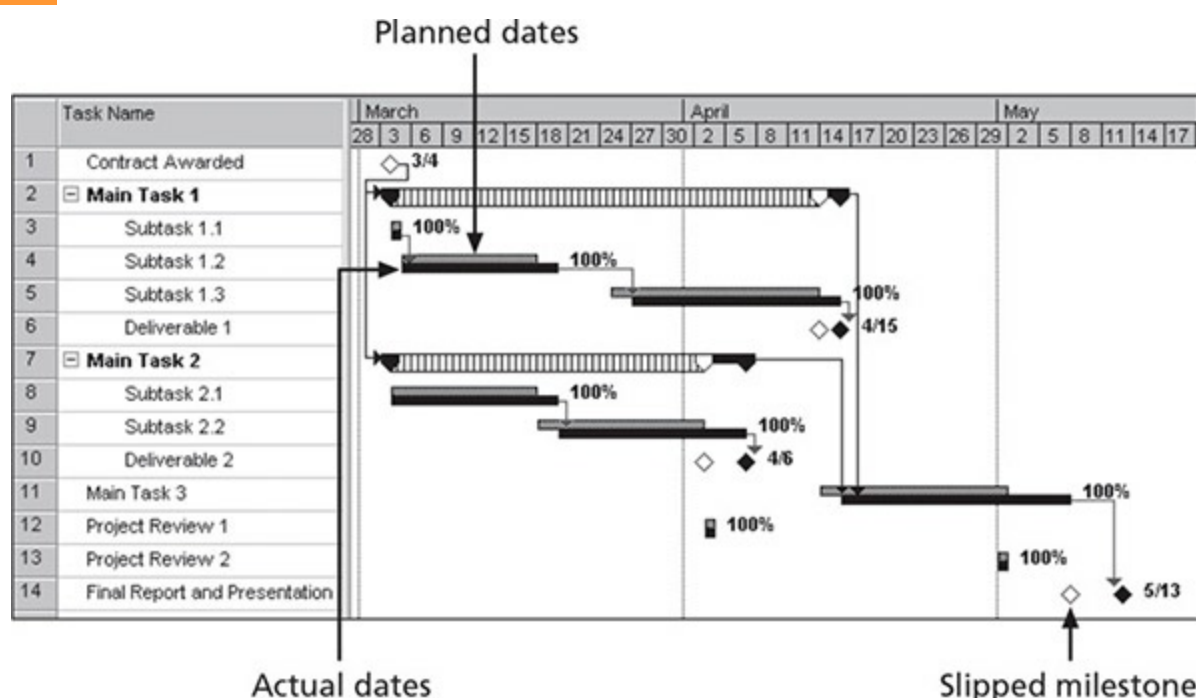
The 20-second rule capitalizes on people's preference for taking the path of least resistance. For example, you can more easily resist having an extra scoop of ice cream if you have to wait 20 seconds in line for it rather than have it served to you. Many people have trouble focusing on work because they are easily distracted by checking their e-mail, stocks, news, social media, etc. Achor recommends making it more difficult for yourself to be distracted. For example, keep e-mail closed while you are working. Don't

leave your favorite non-work-related websites open or have your passwords saved in them. You can actually “save time by adding time”—but only to the distracting behaviors at work.*

Using Tracking Gantt Charts to Compare Planned and Actual Dates

You can use a special form of a Gantt chart to evaluate progress on a project by showing actual schedule information. **Figure 6-7** shows a **Tracking Gantt chart**—a Gantt chart that compares planned and actual project schedule information. The planned schedule dates for activities are called the **baseline dates**, and the entire approved planned schedule is called the **schedule baseline**. The Tracking Gantt chart includes columns labeled “Start” and “Finish” to represent actual start and finish dates for each task, as well as columns labeled “Baseline Start” and “Baseline Finish” to represent planned start and finish dates for each task. (These columns are hidden in Figure 6-7.) In this example, the project is completed, but several tasks missed their planned start and finish dates.

Figure 6-7. Sample tracking Gantt chart



To serve as a progress evaluation tool, a Tracking Gantt chart uses a few additional symbols:

- Notice that the Gantt chart in **Figure 6-7** often shows two horizontal bars for tasks. The top horizontal bar represents the planned or baseline duration for each task. The bar below it represents the actual duration. Subtasks 1.2 and 1.3 illustrate this type of

display. If these two bars are the same length, meaning they start and end on the same dates, then the actual schedule was the same as the planned schedule for that task. This scheduling occurred for Subtask 1.1, in which the task started and ended as planned on March 4. If the bars do not start and end on the same dates, then the actual schedule differed from the planned or baseline schedule. If the top horizontal bar is shorter than the bottom one, the task took longer than planned, as you can see for Subtask 1.2. If the top horizontal bar is longer than the bottom one, the task took less time than planned. A striped horizontal bar, as illustrated by Main Tasks 1 and 2, represents the planned duration for summary tasks. The black bar adjoining it shows progress for summary tasks. For example, Main Task 2 clearly shows that the actual duration took longer than planned.

- A white diamond on the Tracking Gantt chart represents a [slipped milestone](#). A slipped milestone means the milestone activity was completed later than originally planned. The last task provides an example of a slipped milestone because the final report and presentation were completed later than planned.
- Percentages to the right of the horizontal bars display the percentage of work completed for each task. For example, 100 percent means the task is finished, and 50 percent means the task is still in progress but is 50 percent completed.

A Tracking Gantt chart is based on the percentage of work completed for project tasks or the actual start and finish dates. It allows the project manager to monitor schedule progress on individual tasks and the whole project. For example, [Figure 6-7](#) shows that the project is completed. It started on time, but it finished a little late, on May 13 (5/13) instead of May 8.

The main advantage of using Gantt charts is that they provide a standard format for displaying planned and actual project schedule information. In addition, they are easy to create and understand. The main disadvantage of Gantt charts is that they do not *usually* show relationships or dependencies between tasks. If Gantt charts are created using project management software and tasks are linked, then the dependencies *will* be displayed, but differently than they would be displayed on a network diagram. Whether you view dependencies on a Gantt chart or network diagram is a matter of personal preference.

Critical Path Method

Many projects fail to meet schedule expectations. [Critical path method \(CPM\)](#) —also called [critical path analysis](#) —is a network diagramming technique used to predict total project duration. This important tool helps you combat project schedule overruns. A [critical path](#) for a project is the series of activities that determine the *earliest* time by which the project can be completed. It is the *longest* path through the network diagram and has the least amount of

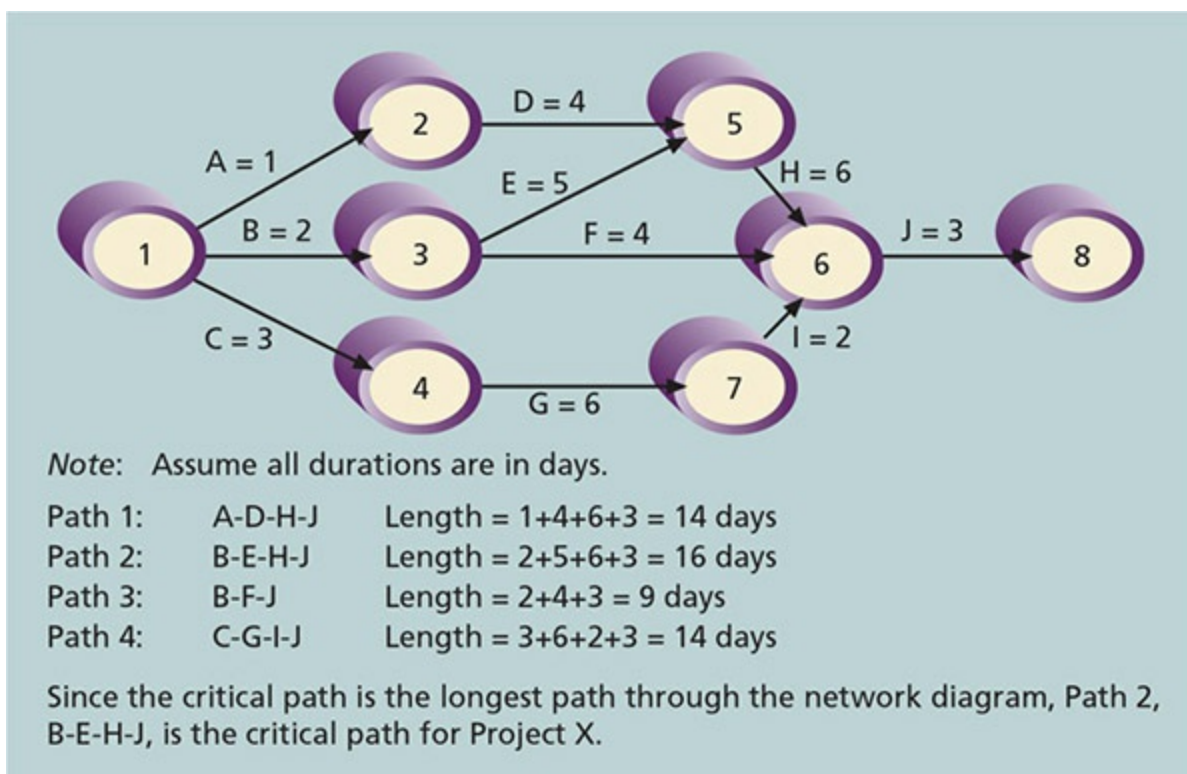
slack or float. **Slack** or **float** is the amount of time an activity may be delayed without delaying a succeeding activity or the project finish date. Normally, several tasks are done in parallel on projects, and most projects have multiple paths through a network diagram. The longest path or the path that contains the critical tasks is what drives the completion date for the project. You are not finished with the project until you have finished *all* the tasks.

Calculating the Critical Path

To find the critical path for a project, you must first develop a good network diagram, which in turn requires a good activity list based on the WBS. Once you create a network diagram, you must also estimate the duration of each activity to determine the critical path. Calculating the critical path involves adding the durations for all activities on each path through the network diagram. The longest path is the critical path.

Figure 6-8 shows the AOA network diagram for Project X again. Note that you can use either the AOA or precedence diagramming method to determine the critical path on projects. Figure 6-8 shows all of the paths—a total of four—through the network diagram. Note that each path starts at the first node (1) and ends at the last node (8) on the AOA network diagram. This figure also shows the length or total duration of each path through the network diagram. These lengths are computed by adding the durations of each activity on the path. Because path B-E-H-J has the longest duration at 16 days, it is the critical path for the project.

Figure 6-8. Determining the critical path for project X



What does the critical path really mean? Even though the critical path is the *longest* path, it represents the *shortest* time required to complete a project. If one or more activities on the critical path take longer than planned, the whole project schedule will slip *unless* the project manager takes corrective action.

Project teams can be creative in managing the critical path. For example, Joan Knutson, a well-known author and speaker in the project management field, often describes how a gorilla helped Apple Inc. complete a project on time. Team members worked in an area with cubicles, and whoever was in charge of the current task on the critical path had a stuffed gorilla on top of his or her cubicle. Everyone knew that person was under the most time pressure and did not need distractions. When a critical task was completed, the person in charge of the next critical task received the gorilla.

Growing Grass Can Be on the Critical Path

People are often confused about what a project's critical path really means. Some people think the critical path includes the most critical activities, but it is concerned only with the time dimension of a project. The fact that its name includes the word *critical* does not mean that it includes all critical activities. For example, Frank Addeman, Executive Project Director at Walt Disney Imagineering, explained in a keynote address at a PMI-ISSIG Professional Development Seminar that growing grass was on the critical path for building Disney's Animal Kingdom theme park! The 500-acre park required special grass for its animal inhabitants, and some of the grass took years to grow. Another misconception is that the critical path is the shortest path through the network diagram. In some areas, such as transportation modeling, identifying the shortest path in network diagrams is the goal. For a project, however, each task or activity on the critical path, as well as other paths, must be done in order to complete the project. It is not a matter of choosing the shortest path.

Other aspects of critical path analysis may cause confusion. Can there be more than one critical path on a project? Does the critical path ever change? In the Project X example, suppose that Activity A has a duration estimate of three days instead of one day. This new duration estimate would make the length of Path 1 equal to 16 days. Now the project has two longest paths of equal duration, so there are two critical paths. Therefore, a project *can* have more than one critical path. Project managers should closely monitor performance of activities on the critical path to avoid late project completion. If there is more than one critical path, project managers must keep their eyes on all of them.

The critical path on a project can change as the project progresses. For example, suppose everything is going as planned at the beginning of the project. Suppose that Activities A, B, C, D, E, F, and G all start and finish as planned. Then suppose that Activity I runs into problems. If Activity I takes more than four days, it will cause path C-G-I-J to be longer than the other

paths, assuming they progress as planned. This change would cause path C-G-I-J to become the new critical path. Therefore, the critical path can change on a project.

Using Critical Path Analysis to Make Schedule Trade-Offs

It is important to know the critical path throughout the life of a project so the project manager *can* make trade-offs. If a task on the critical path is behind schedule, the project manager must be aware of the problem and decide what to do about it. Should the schedule be renegotiated with stakeholders? Should more resources be allocated to other items on the critical path to make up for that time? Is it acceptable for the project to finish behind schedule? By keeping track of the critical path, the project manager and the team take a proactive role in managing the project schedule.

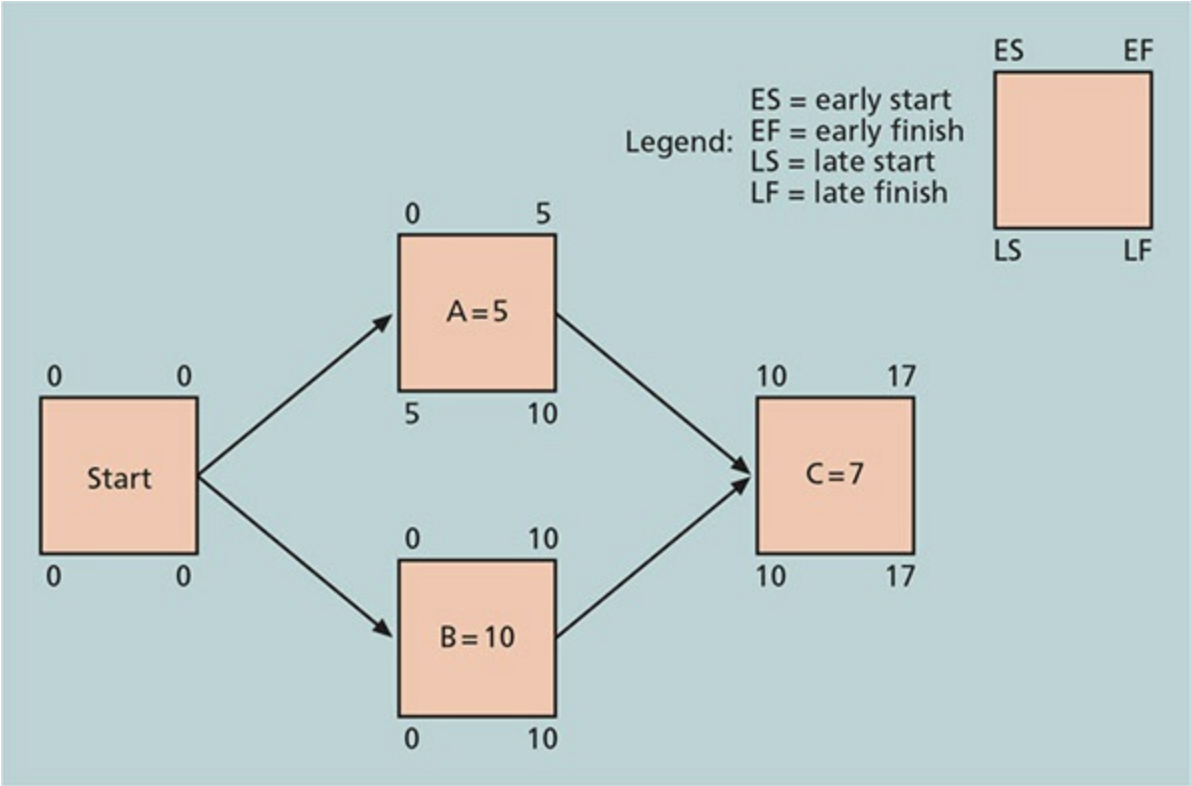
A technique that can help project managers make schedule trade-offs is determining the free slack and total slack for each project activity. **Free slack** or **free float** is the amount of time an activity can be delayed without delaying the early start date of any immediately following activities. The **early start date** is the earliest possible time an activity can start based on the project network logic. **Total slack** or **total float** is the amount of time an activity can be delayed from its early start without delaying the planned project finish date.

Project managers calculate free slack and total slack by doing a forward and backward pass through a network diagram. A **forward pass** determines the early start and early finish dates for each activity. The **early finish date** is the earliest possible time an activity can finish based on the project network logic. The project start date is equal to the early start date for the first network diagram activity. The early start plus the duration of the first activity is equal to the early finish date of the first activity. It is also equal to the early start date of each subsequent activity unless an activity has multiple predecessors. When an activity has multiple predecessors, its early start date is the latest of the early finish dates of those predecessors. For example, Tasks D and E immediately precede Task H in Figure 6-8. The early start date for Task H, therefore, is the early finish date of Task E, because it occurs later than the early finish date of Task D. A **backward pass** through the network diagram determines the late start and late finish dates for each activity in a similar fashion. The **late start date** is the latest possible time an activity might begin without delaying the project finish date. The **late finish date** is the latest possible time an activity can be completed without delaying the project finish date.

Project managers can determine the early and late start and finish dates of each activity by hand. For example, **Figure 6-9** shows a simple network diagram with three tasks, A, B, and C. Tasks A and B both precede Task C. Assume that all duration estimates are in days. Task A has an estimated duration of 5 days, Task B has an estimated duration of 10 days, and Task C has an estimated duration of 7 days. There are only two paths through this small network

diagram: Path A-C has a duration of 12 days (5 + 7), and path B-C has a duration of 17 days (10 + 7). Because path B-C is longer, it is the critical path. There is no float or slack on this path, so the early and late start and finish dates are the same. However, Task A has 5 days of float or slack. Its early start date is day 0, and its late start date is day 5. Its early finish date is day 5, and its late finish date is day 10. Both the free and total float amounts for Task A are 5 days.

Figure 6-9. Calculating early and late start and finish dates



Using project management software is a much faster and easier way to determine early and late start and finish dates and free and total slack amounts for activities. **Table 6-1** shows the free and total slack for all activities on the network diagram for Project X using the data from Figure 6-8 and assuming that Tasks A, B, and C started on March 26, 2018. (The network diagram is shown in Figure 6-4, which was created with Microsoft Project.) The data in this table was created by selecting the Schedule Table view in Microsoft Project.

Table 6-1. Free and total float or slack for project X

	Start	Finish	Late Start	Late Finish	Free Slack	Total Slack
A	3/26/2018	3/26/2018	3/28/2018	3/29/2018	0d	2d
B	3/26/2018	3/27/2018	3/26/2018	3/28/2018	0d	0d
C	3/26/2018	3/28/2018	3/28/2018	4/2/2018	0d	2d
D	3/27/2018	3/30/2018	3/29/2018	4/4/2018	2d	2d
E	3/28/2018	4/3/2018	3/28/2018	4/4/2018	0d	0d

F	3/28/2018	4/2/2018	4/6/2018	4/12/2018	7d	7d
G	3/29/2018	4/5/2018	4/2/2018	4/10/2018	0d	2d
H	4/4/2018	4/11/2018	4/4/2018	4/12/2018	0d	0d
I	4/6/2018	4/9/2018	4/10/2018	4/12/2018	2d	2d
J	4/12/2018	4/16/2018	4/12/2018	4/16/2018	0d	0d

Knowing the amount of float or slack allows project managers to know whether the schedule is flexible and how flexible it might be. For example, at 7 days (7d), Task F has the most free and total slack. The most slack on any other activity is only 2 days (2d). Understanding how to create and use slack information provides a basis for negotiating project schedules. See the Help information in Microsoft Project or research other resources for more detailed information on calculating slack.

Using the Critical Path to Shorten a Project Schedule

It is common for stakeholders to want to shorten a project schedule estimate. A project team may have done its best to develop a project schedule by defining activities, determining sequencing, and estimating resources and durations for each activity. The results of this work may have shown that the project team needs 10 months to complete the project, but the sponsor might ask if the project can be done in eight or nine months. (Rarely do people ask the project team to take longer than suggested.) By knowing the critical path, the project manager and the team can use several duration compression techniques to shorten the project schedule. One technique is to reduce the duration of activities on the critical path. The project manager can shorten the duration of critical-path activities by allocating more resources to those activities or by changing their scope. Several experts suggest that it is extremely difficult to shorten a reasonable project schedule by more than 25 percent.

Recall that Sue Johnson in the opening case was having schedule problems with the online registration project because several users missed important project review meetings and one of the senior programmers quit. If Sue and her team created a realistic project schedule, produced accurate duration estimates, and established dependencies between tasks, they could analyze their status in terms of meeting the May 1 deadline. If some activities on the critical path had already slipped and they did not build in extra time at the end of the project, then they would have to take corrective actions to finish the project on time. Sue could request that her company or the college provide more people to work on the project in an effort to make up time. She could also request that the scope of activities be reduced to complete the project on time. Sue could also use project schedule management techniques, such as crashing or fast tracking, to shorten the project schedule.

Crashing is a technique for making cost and schedule trade-offs to obtain the greatest amount of schedule compression for the least incremental cost. For example, suppose that one of the items on the critical path for the online registration project was entering course data for the new semester into the new system. If this task has not yet been done and was originally estimated to take two weeks based on the college providing a part-time data entry clerk, Sue could suggest that the college have the clerk work full time to finish the task in one week instead of two. This change would not cost Sue's company more money, and it could shorten the project end date by one week. If the college could not meet this request, Sue could consider hiring a temporary data entry person for one week to help get the task done faster. By focusing on tasks on the critical path that could be done more quickly for no extra cost or a small cost, the project schedule can be shortened.

The main advantage of crashing is shortening the time needed to finish a project. The main disadvantage of crashing is that it often increases total project costs. You will learn more about project costs in Chapter 7, Project Cost Management.

Another technique for shortening a project schedule is fast tracking. **Fast tracking** involves doing activities in parallel that you would normally do in sequence. For example, Sue Johnson's project team may have planned not to start any of the coding for the online registration system until *all* of the analysis was done. Instead, they could consider starting some coding activity before the analysis is completed. The main advantage of fast tracking, like crashing, is that it can shorten the time needed to finish a project. The main disadvantage is that it can increase cost and actually lengthen the project schedule because starting some tasks too soon often increases project risk and results in rework.

Importance of Updating Critical Path Data

In addition to finding the critical path at the beginning of a project, it is important to update the schedule with actual data. After the project team completes activities, the project manager should document the actual durations of those activities. The project manager should also document revised estimates for activities in progress or yet to be started. These revisions often cause a project's critical path to change, resulting in a new estimated completion date for the project. Again, proactive project managers and their teams stay on top of changes so they can make informed decisions and keep stakeholders involved in major project decisions.

Critical Chain Scheduling

Another technique that addresses the challenge of meeting or beating project finish dates is an application of the Theory of Constraints called critical chain scheduling. The **Theory of Constraints (TOC)** is a management philosophy developed by Eliyahu M. Goldratt and discussed in his books *The Goal* and *Critical Chain*. The Theory of Constraints is based on the

metaphor of a chain and its weakest link: Any complex system at any point in time often has only one aspect or constraint that limits the ability to achieve more of the system's goal. For the system to attain any significant improvements, that constraint must be identified, and the whole system must be managed with it in mind. [Critical chain scheduling](#) is a method that considers limited resources when creating a project schedule and includes buffers to protect the project completion date.

An important concept in critical chain scheduling is the availability of scarce resources. Some projects cannot be done unless a particular resource is available to work on one or several tasks. For example, if a television network wants to produce a show centered around a particular celebrity, it must first check the availability of that celebrity. As another example, if a particular piece of equipment is needed full time to complete each of two tasks that were originally planned to occur simultaneously, critical chain scheduling acknowledges that you must either delay one of those tasks until the equipment is available or find another piece of equipment in order to meet the schedule. Other important concepts related to critical chain scheduling include multitasking and time buffers.

Although many people are proud to say they are good at multitasking, multitasking is not a good thing to do if you want to finish a project in a timely manner. [Multitasking](#) occurs when a resource works on more than one task at a time. This situation occurs frequently on projects. People are assigned to multiple tasks within the same project or different tasks on multiple projects. For example, suppose someone is working on three different tasks—Task 1, Task 2, and Task 3—for three different projects, and each task takes 10 days to complete. If the person did not multitask, and instead completed each task sequentially starting with Task 1, then Task 1 would be completed after day 10, Task 2 would be completed after day 20, and Task 3 would be completed after day 30, as shown in Figure 6-10a. However, because many people in this situation try to please all three parties who need their tasks completed, they often work on the first task for some time, then the second, then the third, then go back to the first task, and so on, as shown in Figure 6-10b. In this example, the tasks were all half-done one at a time, then completed one at a time. Task 1 is now completed at the end of day 20 instead of day 10, Task 2 is completed at the end of day 25 instead of day 20, and Task 3 is still completed on day 30. This example illustrates how multitasking can delay task completion. Multitasking also often involves wasted setup time, which increases total duration.

Figure 6-10a. Three tasks without multitasking

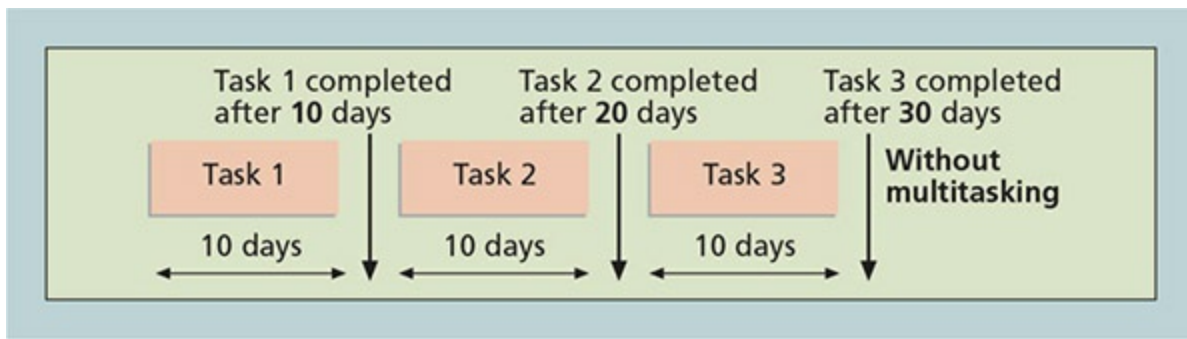
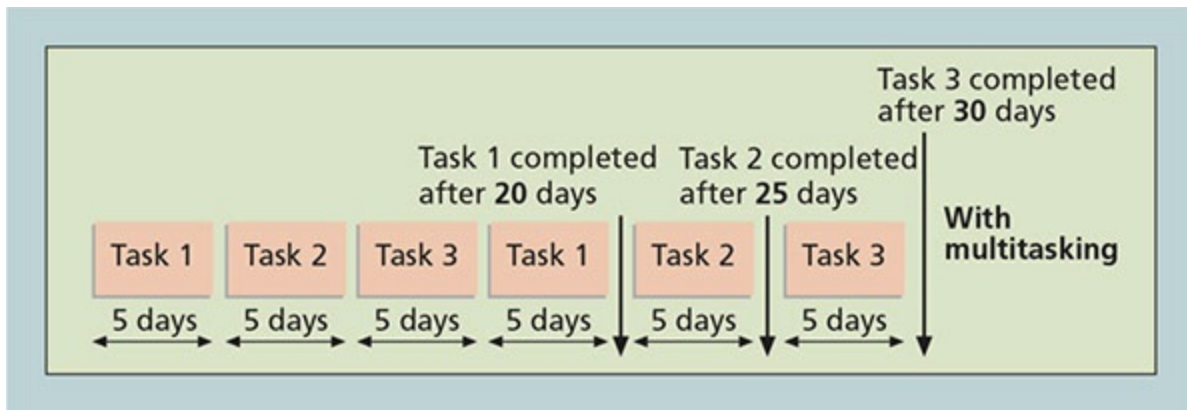


Figure 6-10b. Three tasks with multitasking



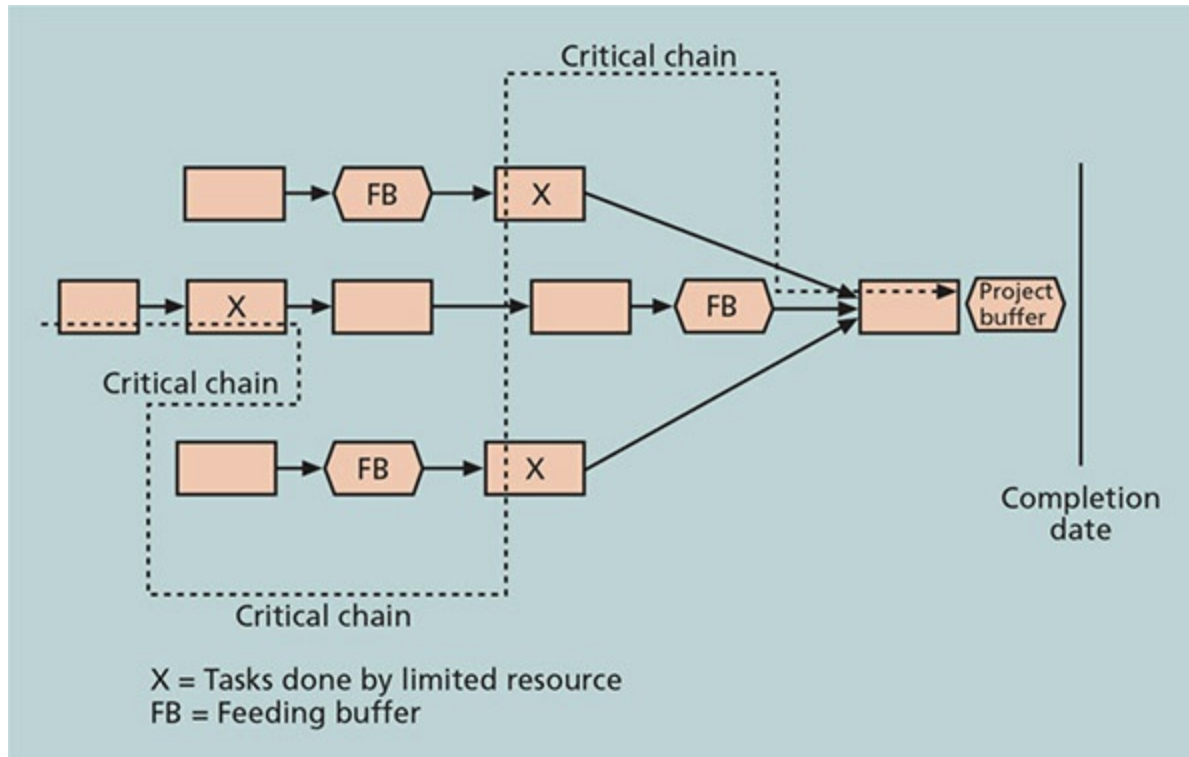
Critical chain scheduling assumes that resources do not multitask or at least minimize multitasking. Someone should not be assigned to two tasks simultaneously on the same project when critical chain scheduling is in effect. Likewise, critical chain theory suggests that projects be prioritized so that people who are working on more than one project at a time know which tasks take priority. Preventing multitasking avoids resource conflicts and wasted setup time caused by shifting between multiple tasks over time.

An essential concept to improving project finish dates with critical chain scheduling is to change the way people make task estimates. Many people add a safety or **buffer**—additional time to complete a task—to an estimate to account for various factors. These factors include the negative effects of multitasking, distractions, and interruptions, fear that estimates will be reduced, and Murphy's Law. **Murphy's Law** states that if something can go wrong, it will. Critical chain scheduling removes buffers from individual tasks and instead creates a **project buffer**, which is time added before the project's due date. Critical chain scheduling also protects tasks on the critical chain from being delayed by using **feeding buffers**, which consist of time added before tasks on the critical chain if they are preceded by other tasks that are not on the critical path.

Figure 6-11 provides an example of a network diagram constructed using critical chain

scheduling. Note that the critical chain accounts for a limited resource, X, and the schedule includes use of feeding buffers and a project buffer in the network diagram. The tasks marked with an X are part of the critical chain, which can be interpreted as being the critical path using this technique. The task estimates in critical chain scheduling should be shorter than traditional estimates because they do not include their own buffers. Not having task buffers should mean fewer occurrences of [Parkinson's Law](#), which states that work expands to fill the time allowed. The feeding and project buffers protect the date that really needs to be met—the project completion date.

Figure 6-11. Example of critical chain scheduling*



* Eliyahu Goldratt, *Critical Chain* (Great Barrington, MA: The North River Press, 1997), p. 218.

Several organizations have reported successes with critical chain scheduling. For example, the What Went Right? feature shows how the healthcare industry is learning to think differently by using the Theory of Constraints.

What Went Right?

Experts in healthcare quality management have thought carefully about how scheduling works at a doctor's office or healthcare clinic and have considered how it might be improved:

Consider a relatively simple system of a physician's office or clinic. The steps in the process could be patients checking in, filling out forms, having vital signs taken by a nurse, seeing the physician, seeing the nurse for a prescribed procedure such as vaccination, and so forth. These steps could take place in a simple linear sequence or chain. . . . Each link in this chain has the ability to perform its tasks at different average rates. In this example, the first resource can process 13 patients, charts, or blood samples per hour; the second can process 17, and so forth. One may think that this process can produce 13 per hour, the average of all resources. In fact, this process or chain only can produce an average of eight per hour. The chain is only as strong as its weakest link and the rate of the slowest resource in this example, the weakest link, is eight. This is true regardless of how fast each of the other resources can process individually, how much work is stuffed into the pipeline, or how complex the process or set of interconnected processes is to complete. Moreover, improving the performance of any link besides the constraint does nothing to improve the system as a whole.*

In fact, the Avraham Y. Goldratt Institute (www.goldratt.com) applied this thinking at a client's site and showed just how well critical chain scheduling can work. The National University Hospital in Singapore decreased patient admission times by more than 50 percent. Improved scheduling lowered average wait times, which went from six to eight hours to less than three hours. By applying critical chain techniques, the hospital admitted 63 percent of patients in less than 1.5 hours. Although these times still seem long, they show significant improvement.*

As you can see, critical chain scheduling is a fairly complicated yet powerful tool that involves critical path analysis, resource constraints, and changes in how buffers are used as part of task estimates. Some people consider critical chain scheduling one of the most important new concepts in the field of project management.

Program Evaluation and Review Technique (PERT)

When there is a high degree of uncertainty about the individual activity duration estimates, the network analysis [Program Evaluation and Review Technique \(PERT\)](#) can be used to estimate project duration. PERT applies the critical path method (CPM) to a weighted average duration estimate. This approach was developed at about the same time as CPM, in the late 1950s, and it also uses network diagrams, which are still sometimes referred to as PERT charts.

PERT uses **probabilistic time estimates**—duration estimates based on using optimistic, most likely, and pessimistic estimates of activity durations—instead of one specific or discrete duration estimate, as CPM does. To use PERT, you calculate a weighted average for the duration estimate of each project activity using the following formula:

$$\text{PERT weight average} = \frac{\text{optimistic time} + 4 * \text{most likely time} + \text{pessimistic time}}{6}$$

By using the **PERT weighted average** for each activity duration estimate, the total project duration estimate takes into account the risk or uncertainty in the individual activity estimates. Note that the above formula assumes a normal distribution or bell curve. You can also use other distributions. Consult other sources for more details.

Suppose that Sue Johnson’s project team in the opening case used PERT to determine the schedule for the online registration system project. The team would have to collect numbers for the optimistic, most likely, and pessimistic duration estimates for each project activity. Suppose that one of the activities was to design an input screen for the system. Someone might estimate that it would take about two weeks or 10 workdays to do this activity. Without using PERT, the duration estimate for that activity would be 10 workdays. Using PERT, the project team would also need to estimate the pessimistic and optimistic times for completing this activity. Suppose an optimistic estimate is that the input screen can be designed in eight workdays, and a pessimistic time estimate is 24 workdays. Applying the PERT formula, you get the following:

$$\begin{aligned}\text{PERT weight average} &= \frac{\text{optimistic time} + 4 * \text{most likely time} + \text{pessimistic time}}{6} \\ &= 12 \text{ workdays}\end{aligned}$$

Instead of using the most likely duration estimate of 10 workdays, the project team would use 12 workdays when doing critical path analysis. These additional two days could help the project team get the work completed on time.

The main advantage of PERT is that it attempts to address the risk associated with duration estimates. Because many projects exceed schedule estimates, PERT may help in developing schedules that are more realistic. PERT’s main disadvantages are that it involves more work than CPM because it requires several duration estimates, and there are better probabilistic methods for assessing schedule risk. (See the information on Monte Carlo simulations in Chapter 11, Project Risk Management.)

Agile and Schedule Management

Recall from **Chapter 2** that two of the core values of the Manifesto for Agile Software Development are “customer collaboration over contract negotiation” and “responding to

change over following a plan.” These values seem to contradict some of the structured project schedule management processes and tools described in this chapter. Agile methods like Scrum were designed to address collaboration and flexibility, especially on projects with a complex scope of work. For example, the product owner defines and prioritizes the work to be done within a sprint, so customer collaboration is designed into the process. The short period of each sprint (normally two to four weeks) and daily Scrum meetings provide an environment where team members collaborate to focus on completing specific tasks.

An adaptive or agile approach can be used on large projects, even after they are started using a more prescriptive approach. For example, the FBI says that switching to an agile approach is what helped them complete the Sentinel project described in the What Went Wrong? feature earlier in the chapter. The system went live “thanks to the fact that it was rescued by following an agile software development methodology. . . . The primary lesson? Agile actually works. It’s not perfect, and many people in government and in the contractor community still struggle with it, but it’s succeeding where it counts—enabling the rollout of large-scale IT projects that are on time, on budget and actually do what stakeholders want them to do. Imagine that.”*

How was project schedule management different after moving to an agile approach? After changing to a Scrum method in 2010, the work for the Sentinel project was organized into user stories based on the original systems requirements specifications. Each user story was allocated a number of story points based on difficulty. At the beginning of each two-week sprint, the development team identified which user stories to do during each sprint. At the end of each sprint the team demonstrated the system, regardless of whether all of the work was completed, but only the user stories that passed tests (approved by the customers) were reported as complete. If user stories were not complete, they were moved to the product backlog. This approach helped the team focus on delivering a working system that met customer needs in a specified timeframe. In other words, the emphasis was on completing some useful work for the customer in short time increments versus trying to define all the work required first and then scheduling when it could be done.*

Controlling the Schedule

The final process in project schedule management is controlling the schedule. Like scope control, schedule control is a portion of the integrated change control process in project integration management. The goal of schedule control is to know the status of the schedule, influence the factors that cause schedule changes, determine that the schedule has changed, and manage changes when they occur.

The main inputs to schedule control are the project management plan, project documents (like the lessons-learned register, project calendars, project schedule, resource calendars, and

schedule data), work performance data, and organizational process assets. Some of the tools and techniques include the following:

- Data analysis tools, including
 - Earned value analysis, described in detail in Chapter 7, Project Cost Management
 - Iteration burndown charts, as described in **Chapter 3**
 - Performance reviews
 - Trend analysis
 - Variance analysis
 - What-if-scenario analysis
- Critical path method, described earlier in this chapter
- Project management information systems
- Resource optimization, such as resource leveling, described in Chapter 9, Project Resource Management
- Leads and lags, described earlier
- Schedule compression, such as crashing and fast tracking

The main outputs of schedule control include work performance information, schedule forecasts, change requests, project management plan updates, and project documents updates.

Many issues are involved in controlling changes to project schedules. It is important first to ensure that the project schedule is realistic. Many projects, especially in IT, have very unrealistic schedule expectations. It is also important to use discipline and leadership to emphasize the importance of following and meeting project schedules. Although the various tools and techniques assist in developing and managing project schedules, project managers must handle several personnel-related issues to keep projects on track. “Most projects fail because of people issues, not from failure to draw a good PERT chart.”* Project managers can perform a number of reality checks that will help them manage changes to project schedules. Several soft skills can help project managers to control schedule changes.

Reality Checks on Scheduling and the Need for Discipline

It is important for projects to have realistic schedule goals and for project managers to use discipline to help meet those goals. One of a project manager's first reality checks is to review the draft schedule that is usually included in the project charter. Although this draft schedule might include only a project start and end date, the project charter sets some initial schedule expectations for the project. Next, the project manager and project team should prepare a more detailed schedule and get stakeholders' approval. To establish the schedule, it is critical to get involvement and commitment from all project team members, top management, the customer, and other key stakeholders.

Another type of reality check comes from progress meetings with stakeholders. The project manager is responsible for keeping the project on track, and key stakeholders like to stay informed, often through high-level periodic reviews. Project managers often illustrate progress with a Tracking Gantt chart that shows key deliverables and activities. The project manager needs to understand the schedule, including why activities are or are not on track, and take a proactive approach to meeting stakeholder expectations. It is also important to verify schedule progress, as Sue Johnson from the opening case discovered (see the [Case Wrap-Up](#) later in this chapter). Just because a team member says a task was completed on time does not always mean that it was. Project managers must review the actual work and develop a good relationship with team members to ensure that work is completed as planned or changes are reported as needed.

Top management hates surprises, so the project manager must be clear and honest in communicating project status. By no means should project managers create the illusion that the project is going fine when it is actually having serious problems. When serious conflicts arise that could affect the project schedule, the project manager must alert top management and work with them to resolve the conflicts.

Project managers must also use discipline to control project schedules. Several IT project managers have discovered that setting firm dates for key project milestones helps minimize schedule changes. It is very easy for scope creep to raise its ugly head on IT projects. Insisting that important schedule dates be met and that proper planning and analysis be completed up front helps everyone focus on doing what is most important for the project. This discipline results in meeting project schedules.

Using Software to Assist in Project Schedule Management

Several types of software are available that assist with project schedule management. Software for facilitating communications helps project managers exchange schedule-related information with project stakeholders. Decision support models can help project managers analyze various trade-offs that can be made to address schedule issues.

However, project management software was designed specifically for performing project management tasks. You can use project management software to draw network diagrams, determine the critical path for a project, create Gantt charts, and report, view, and filter specific project schedule management information. For example, **Figures 6-4, 6-5, 6-6, 6-7,** and Table 6-1 were all created using Microsoft Project; Appendix A (available on the Companion website for this text) shows many more examples of using this software to assist in project schedule management.

Many projects involve hundreds of tasks with complicated dependencies. After you enter the necessary information, project management software automatically generates a network diagram and calculates the critical path(s) for the project. It also highlights the critical path in red on the network diagram.

Project management software also calculates the free and total float or slack for all activities. Using project management software eliminates the need to perform cumbersome calculations manually and allows for “what if” analysis as activity duration estimates or dependencies change. Recall that knowing which activities have the most slack allows the project manager to reallocate resources or make other changes to compress the schedule or help keep it on track.

Project 2016 easily creates Gantt charts and Tracking Gantt charts, which make it easier for project teams to track actual schedule performance versus the planned or baseline schedule. However, for a project to benefit from using Tracking Gantt charts, actual schedule information must be entered in a timely manner. Some organizations use e-mail or other communications software to send updated task and schedule information to the person responsible for updating the schedule. He or she can then quickly authorize these updates and enter them directly into the project management software. This process provides an accurate and up-to-date project schedule in Gantt chart form.

Project 2016 also includes many built-in reports, views, and filters to assist in project schedule management. For example, a project manager can quickly run a report to list all tasks that are scheduled to start soon, and then send a reminder to the people responsible for these tasks. A project manager who was presenting project schedule information to top management could create a Gantt chart showing only summary tasks or milestones. You can also create custom reports, views, tables, and filters. See Appendix A (available on the Companion website for this text) to learn how to use the project schedule management features of Project 2016.

Global Issues

Most software companies use case studies or customer testimonials to showcase how

their software helps customers solve business problems. Microsoft tells the customer story of Mexico's Secretary of Economy, who wanted to ensure that IT initiatives aligned with business goals and improved project management efficiency. Internal IT project requests often took several weeks to fulfill because the organization lacked a centralized method of capturing project information and used several diagramming tools.

The Director General of Information Technology decided to establish an Office of Project Management and search for a solution that combined professional diagramming capabilities with project management and collaboration tools. Assistant Director General of the Office of Information Technology, Carlos Benítez Gonzales, worked with other agencies to find the best solution. After deploying several of Microsoft's tools (Visio, Project, and SharePoint), the organization was able to handle more internal IT projects without adding staff, while fulfilling requests 60 percent faster.

"By documenting and incorporating governance procedures and project lifecycle stages into a standardized workflow process, the Secretary's IT team can finish projects in less time. Benítez estimates that it previously took an average of five business days to fulfill requests. The team can now complete the same work in two business days on average. . . . 'We now have a single platform for managing process information that used to be scattered throughout the organization and inaccessible because it was stored on local computers,' says Benítez. The IT team can now handle four times the number of concurrent projects without the need to hire additional staff."*

Words of Caution on Using Project Management Software

Many people misuse project management software because they do not understand the concepts behind creating a network diagram, determining the critical path, or setting a schedule baseline. They might also rely too heavily on sample files or templates in developing their own project schedules. Understanding the underlying concepts (or even being able to work with the tools manually) is critical to successful use of project management software, as is understanding the specific needs of your project.

Many top managers, including software professionals, have made blatant errors using various versions of Microsoft Project and similar tools. For example, one top manager did not know how to establish dependencies among project activities and entered every single start and end date for hundreds of activities. When asked what would happen if the project started a week or two late, she responded that she would have to reenter all of the dates.

This manager did not understand the importance of establishing relationships among the

tasks, which allows the software to update formulas automatically when the inputs change. If the project start date slips by one week, the project management software will update all the other dates automatically, as long as they are not manually entered into the software. If one activity cannot start before another ends, and the first activity's actual start date is two days late, the start date for the succeeding activity will automatically be moved back two days. To achieve this type of functionality, tasks that have relationships must be linked in project management software.

Another top manager on a large IT project did not know how to set a baseline in Microsoft Project. He spent almost one day every week copying and pasting information from Microsoft Project into a spreadsheet and using complicated "IF" statements to figure out what activities were behind schedule. He had never received training on Microsoft Project and did not know about many of its capabilities. To use any software effectively, users must have adequate training in the software and an understanding of its underlying concepts.

Many project management software programs also come with templates or sample files. It is very easy to use these files without considering unique project needs. For example, a project manager for a software development project can use the Microsoft Project Software Development template file, the files from similar projects done in the past, or sample files purchased from other companies. All of these files include suggested tasks, durations, and relationships. There are benefits to using templates or sample files, such as less setup time and a reality check if the project manager has never led a certain type of project before. However, there are drawbacks to this approach. Many assumptions made in these template files might not apply to the project, such as a design phase that takes three months to complete or the performance of certain types of testing. Project managers and their teams should be careful not to rely too much on templates or sample files and ignore the unique concerns of their particular projects.

Considerations for Agile/Adaptive Environments

The *PMBOK® Guide – Sixth Edition* provides the following information for project schedule management:

Adaptive approaches use short cycles to undertake work, review the results, and adapt as necessary. These cycles provide rapid feedback on the approaches and suitability of deliverables, and generally manifest as iterative schedule and on-demand, pull-based scheduling, as discussed in the section on Key Trends and Emerging Practices in Project Schedule Management.

In large organizations, there may be a mixture of small projects and large initiatives requiring long-term roadmaps to manage the development of these programs using scaling factors (e.g., team size, geographical distribution, regulatory compliance, organizational complexity, and technical complexity). To address the full delivery life cycle for larger, enterprise-wide systems, a range of techniques utilizing a predictive approach, adaptive approach, or a hybrid of both may need to be adopted. The organization may need to combine practices from several core methods, or adopt a method that has already done so, and adopt a few principles and practices of more traditional techniques.

The role of the project manager does not change based on managing projects using a predictive development life cycle of managing projects in adaptive environments. However, to be successful in using adaptive approaches, the project manager will need to be familiar with the tools and techniques to understand how to apply them effectively.*

Schedule management is radically different using Agile and Scrum in particular. Projects that rely heavily on the critical path method consider meeting the project's estimated completion date as a crucial component of success. Agile projects, on the other hand, may not even need to estimate activity durations or project schedules at all. Why? Because overall project completion time is not important. The customer/product owners are empowered to demand what they want in each and every short iteration (in the product backlog), and they get to see and touch a working product, and approve of it, by the end of each iteration. Customer satisfaction and project progress are extremely transparent at each iteration.

Project schedule management is challenging for all types of projects. This chapter highlights important concepts, processes, and tools and techniques in this knowledge area.

Case Wrap-Up

It was now March 15, just a month and a half before the new online registration system was supposed to go live. The project was in total chaos. Sue Johnson thought she could handle all of the conflicts that kept occurring on the project, and she was too proud and scared to admit to her top management or the college president that things were not going well. She spent a lot of time preparing a detailed schedule for the project, and she thought she was using the project management software well enough to keep up with project status. However, the five main programmers on the project all figured out a way to generate automatic updates for their tasks every week, saying that everything was completed as planned. They paid very little attention to the actual plan and hated filling

out status information. Sue did not check most of their work to verify that it was actually completed. In addition, the head of the Registrar's Office was uninterested in the project and delegated sign-off responsibility to one of his clerks, who did not completely understand the registration process. When Sue and her team started testing the new system, she learned they were using last year's course data, which caused additional problems because the college was moving from quarters to semesters in the new term. How could they have missed that requirement? Sue hung her head in shame and anguish as she walked into a meeting with her manager to ask for help. She learned the hard way how difficult it was to keep a project on track. She wished she had spent more time talking face-to-face with key project stakeholders, especially her programmers and the Registrar's Office representatives, to verify that the project was on schedule and that the schedule was updated accurately.

Chapter Summary

Project schedule management is often cited as the main source of conflict on projects. Most IT projects exceed time estimates. The main processes involved in project schedule management include planning schedule management, defining activities, sequencing activities, estimating activity durations, developing the schedule, and controlling the schedule.

Planning schedule management involves determining the policies, procedures, and documentation that will be used for planning, executing, and controlling the project schedule. The main output is a schedule management plan.

Defining activities involves identifying the specific activities that must be completed to produce the project deliverables. It usually results in a more detailed WBS.

Sequencing activities determines the relationships or dependencies between activities. Three reasons for creating relationships are that they are mandatory based on the nature of the work, discretionary based on the project team's experience, or external based on non-project activities. Activity sequencing must be done in order to use critical path analysis.

Network diagrams are the preferred technique for showing activity sequencing. The two methods used to create these diagrams are the arrow diagramming method and the precedence diagramming method. There are four types of relationships between tasks: finish-to-start, finish-to-finish, start-to-start, and start-to-finish.

Estimating activity durations creates estimates for the amount of time it will take to complete each activity. These time estimates include the actual amount of time worked plus elapsed time.

Developing the schedule uses results from all of the other project schedule management processes to determine the start and end dates for the project. Project managers often use Gantt charts to display the project schedule. Tracking Gantt charts show planned and actual schedule information.

The critical path method predicts total project duration. The critical path for a project is the series of activities that determines the earliest completion date for the project. It is the longest path through a network diagram. If any activity on the critical path slips, the whole project will slip unless the project manager takes corrective action.

Crashing and fast tracking are two techniques for shortening project schedules. Project managers and their team members must be careful about accepting unreasonable schedules,

especially for IT projects. There is a practical limit to shortening schedules of about 25 percent, and it almost always increases project cost and risk.

Critical chain scheduling is an application of the Theory of Constraints (TOC) that uses critical path analysis, resource constraints, and buffers to help meet project completion dates.

The Program Evaluation and Review Technique (PERT) is a network analysis technique used to estimate project duration when there is a high degree of uncertainty about the individual activity duration estimates. PERT uses optimistic, most likely, and pessimistic estimates of activity durations. PERT is seldom used today.

Agile methods like Scrum take a different approach to project schedule management by providing more flexibility. The short time period of each sprint (normally two to four weeks) and daily Scrum meetings provide an environment where team members collaborate to focus on completing specific tasks within that time period. The product owner identifies and prioritizes tasks to be done during each sprint.

Controlling the schedule is the final process in project schedule management. Even though scheduling techniques are very important, most projects fail because of personnel issues, not from a poor network diagram. Project managers must involve all stakeholders in the schedule development process. It is critical to set realistic project schedules and use discipline to meet schedule goals.

Project management software can assist in project scheduling if used properly. With project management software, you can avoid the need to perform cumbersome calculations manually and perform “what if” analysis as activity duration estimates or dependencies change. Many people misuse project management software because they do not understand the concepts behind creating a network diagram, determining the critical path, or setting a schedule baseline. Project managers must also avoid relying too much on sample files or templates when creating their unique project schedules.

Be sure to consider how project schedule management can differ in agile/adaptive environments.

Discussion Questions

1. Why do you think schedule issues often cause the most conflicts on projects?
2. Why is defining activities a process of project schedule management instead of project scope management?
3. Why is it important to determine activity sequencing on projects? Discuss diagrams you

have seen that are similar to network diagrams. Describe their similarities and differences.

4. Explain the difference between estimating activity durations and estimating the effort required to perform an activity.
5. Explain the following schedule development tools and concepts: Gantt charts, critical path method, PERT, critical chain scheduling, and sprints.
6. What do you think about adding slack to individual task estimates (sometimes called *padding estimates*)? What do you think about adding a project buffer for the entire project, as critical chain scheduling suggests? What are some ethical considerations when using slack and buffers?
7. How can you minimize or control changes to project schedules?
8. List some of the reports you can generate with Project 2016 to assist with project schedule management.
9. Why is it difficult to use project management software well?
10. How is schedule management different when using an agile approach?

Quick Quiz

1. Which of the following processes involves determining the policies, procedures, and documentation that will be used for planning, executing, and controlling the project schedule?
 - a. Planning schedule management
 - b. Defining activities
 - c. Estimating activity resources
 - d. Activity sequencing
2. Predecessors, successors, logical relationships, leads and lags, resource requirements, constraints, imposed dates, and assumptions are all examples of ____ .
 - a. items in an activity list
 - b. items on a Gantt chart
 - c. milestone attributes

- d. activity attributes
3. As the project manager for a software development project, you are helping to develop the project schedule. You decide that writing code for a system should not start until users sign off on the analysis work. What type of dependency is this?
- a. Technical
 - b. Mandatory
 - c. Discretionary
 - d. External
4. You cannot start editing a technical report until someone else completes the first draft. What type of dependency does this represent?
- a. Finish-to-start
 - b. Start-to-start
 - c. Finish-to-finish
 - d. Start-to-finish
5. Which of the following statements is false?
- a. A burndown chart is a tool for schedule control.
 - b. Duration and effort are synonymous terms.
 - c. A three-point estimate includes an optimistic, a most likely, and a pessimistic estimate.
 - d. A Gantt chart is a common tool for displaying project schedule information.
6. What symbol on a Gantt chart represents a slipped milestone?
- a. A black arrow
 - b. A white arrow
 - c. A black diamond
 - d. A white diamond

7. What type of diagram shows planned and actual project schedule information?
- a. A network diagram
 - b. A Gantt chart
 - c. A Tracking Gantt chart
 - d. A milestone chart
8. ____ is a network diagramming technique used to predict total project duration.
- a. PERT
 - b. A Gantt chart
 - c. Critical path method
 - d. Crashing
9. Which of the following statements is false?
- a. Growing grass was on the critical path for a large theme park project.
 - b. The critical path is the series of activities that determine the earliest time by which a project can be completed.
 - c. A forward pass through a project network diagram determines the early start and early finish dates for each activity.
 - d. Fast tracking is a technique for making cost and schedule trade-offs to obtain the greatest amount of schedule compression for the least incremental cost.
10. ____ is a method of scheduling that considers limited resources when creating a project schedule and includes buffers to protect the project completion date.
- a. Parkinson's Law
 - b. Scrum
 - c. Critical path analysis
 - d. Critical chain scheduling

Exercises

1. Using **Figure 6-2**, enter the activities, their durations (in days), and their relationships in Project 2016. Use a project start date of March 26, 2018 if you want the dates to match exactly. View the network diagram. Does it look like Figure 6-4? Print the network diagram on one page. Return to the Gantt chart view. To re-create Table 6-1, right-click the **Select All** button to the left of the Task Mode column heading and select **Schedule**. Alternatively, you can click the **View** tab and click the **Tables** button under the Data group and then select **Schedule**. You may need to move the split bar to the right to reveal all of the table columns. (See Appendix A—available on the Companion website for this text—for detailed information on using Project 2016.) Write a few paragraphs explaining what the network diagram and schedule table show about Project X's schedule.
2. Consider **Table 6-2**. All duration estimates or estimated times are in days, and the network proceeds from Node 1 to Node 9. (Note that you can easily change this table to create multiple exercises.)

Table 6-2. Network diagram data for a small project

Activity	Initial Node	Final Node	Estimated Duration
A	1	2	2
B	2	3	2
C	2	4	3
D	2	5	4
E	3	6	2
F	4	6	3
G	5	7	6
H	6	8	2
I	6	7	5
J	7	8	1
K	8	9	2

- a. Draw an AOA network diagram representing the project. Put the node numbers in circles and draw arrows from node to node, labeling each arrow with the activity letter and estimated time.
- b. Identify all of the paths on the network diagram and note how long they are, using **Figure 6-8** as a guide for how to represent each path.
- c. What is the critical path for this project and how long is it?
- d. What is the shortest possible time needed to complete this project?

3. Consider **Table 6-3**. All duration estimates or estimated times are in weeks, and the network proceeds from Node 1 to Node 8. (Note that you can easily change this table to create multiple exercises.)
- Draw an AOA network diagram representing the project. Put the node numbers in circles and draw arrows from node to node, labeling each arrow with the activity letter and estimated time.
 - Identify all of the paths on the network diagram and note how long they are, using **Figure 6-8** as a guide for how to represent each path.
 - What is the critical path for this project and how long is it?
 - What is the shortest possible time needed to complete this project?

Table 6-3. Network diagram data for a large project

Activity	Initial Node	Final Node	Estimated Duration
A	1	2	10
B	1	3	12
C	1	4	8
D	2	3	4
E	2	5	8
F	3	4	6
G	4	5	4
H	4	6	8
I	5	6	6
J	5	8	12
K	6	7	8
L	7	8	10

4. Enter the information from **Exercise 2** into Project 2016. View the network diagram and task schedule table to see the critical path and float or slack for each activity. Print the Gantt chart and network diagram views and the task schedule table. Write a short paper that interprets this information for someone unfamiliar with project schedule management.
5. Enter the information from **Exercise 3** into Project 2016. View the network diagram and task schedule table to see the critical path and float or slack for each activity. Print the Gantt chart and network diagram views and the task schedule table. Write a short paper

that interprets this information for someone unfamiliar with project schedule management.

6. Create a scenario and schedule for a 6-month or 12-month project. Include the five process groups, but focus on key tasks for executing the project. For example, you could develop a schedule for starting a business, renovating a house, developing an app, or writing a book. Give only your scenario information to a classmate to work on, and compare the results. Create a schedule based on your classmate's scenario. Write a short paper summarizing the similarities and differences in what each of you created and the challenges you faced.
7. Find at least three different sample schedules created in Microsoft Project, MindView, or other project management software. Analyze the schedules, focusing on how complete the task lists are, how realistic the durations are, whether dependencies are included, and so on. Document your results in a short paper, including screenshots of the schedules and references.
8. Interview someone who uses some of the techniques discussed in this chapter. How does the person feel about network diagrams, critical path analysis, Gantt charts, critical chain scheduling, Scrum, using project management software, and managing the personnel issues involved in project schedule management? Write a short paper describing the responses.
9. Review two different articles about critical chain scheduling. Write a short paper describing how this technique can help improve project schedule management.
10. Search for videos that explain how and why you would want to find the critical path for a project. Write a short paper describing your findings. Summarize the best two videos; provide screenshots and explain why you liked the videos. (Note: Instructors can change the topic from critical path to other concepts in this chapter as well.)
11. Research how schedule management is performed on agile projects. Summarize your findings in a short paper or presentation.

Running Case

You are the project manager for the Global Treps Project, sponsored by Dr. K. Team members include Bobby, your IT guy; Kim, a new college grad now working for a non profit group in Vietnam; Ashok, a business student in India; and Alfreda, a student in the United States originally from Ethiopia. You plan to outsource some of the work (e.g., purchasing laptops, developing a website feature for accepting donations, and creating videos for the website). Recall that your schedule and cost goals are to complete the project in six months for under \$120,000.

Tasks

- 1.** Review the WBS and Gantt chart you created for **Tasks 3** and 4 in Chapter 5. Propose three to five additional activities that would help you estimate resources and durations. Write a one-page paper describing these new activities.
- 2.** Identify at least five milestones for the Global Treps Project. Write a short paper describing each milestone using the SMART criteria. Discuss how determining the details of these milestones might add activities or tasks to the Gantt chart. Remember that milestones normally have no duration, so you must have tasks that will lead to completing them.
- 3.** Using the Gantt chart you created for **Task 4** in Chapter 5 and the new activities and milestones you proposed in Tasks 1 and 2 above, create a new Gantt chart using Project 2016 or another tool. Estimate the task durations and enter dependencies as appropriate. Remember that your schedule goal for the project is six months. Print the Gantt chart and network diagram, each on one page.

Key Terms

[activity](#) p.223

[activity attributes](#) p.226

[activity list](#) p.226

[activity-on-arrow \(AOA\)](#) p.229

[arrow diagramming method \(ADM\)](#) p.229

[backward pass](#) p.241

[baseline dates](#) p.238

[buffer](#) p.245

[burst](#) p.230

[crashing](#) p.243

[critical chain scheduling](#) p.244

[critical path](#) p.238

[critical path method \(CPM\) or critical path analysis](#) p.238

[dependency](#) p.228

[discretionary dependencies](#) p.228

[dummy activities](#) p.232

[duration](#) p.233

[early finish date](#) p.241

[early start date](#) p.241

[effort](#) p.233

[external dependencies](#) p.228

[fast tracking](#) p.243

[feeding buffers](#) p.245

[finish-to-finish dependency](#) p.231

[finish-to-start dependency](#) p.231

[float](#) p.239

[forward pass](#) p.241

[free slack \(free float\)](#) p.240

[Gantt chart](#) p.234

[internal dependencies](#)

[late finish date](#) p.241

[late start date](#) p.241

[mandatory dependencies](#) p.228

[merge](#) p.230

[milestone](#) p.226

[multitasking](#) p.244

[Murphy's Law](#) p.245

[network diagram](#) p.229

[node](#) p.229

[Parkinson's Law](#) p.245

[PERT weighted average](#) p.247

[precedence diagramming method \(PDM\)](#) p.230

[probabilistic time estimates](#) p.247

[Program Evaluation and Review Technique \(PERT\)](#) p.247

[project buffer](#) p.245

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start-to-start dependency p.231

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Theory of Constraints (TOC) p.243

three-point estimate p.234

total slack (total float) p.241

Tracking Gantt chart p.237